NATIONAL BOARD OF EXAMINERS IN OPTOMETRY

Patient Assessment & Management

Part II · PAM / TMOD

A case-driven review of every clinical presentation the Part II PAM examination can build a case around, with the diagnostic reasoning, the exact treatment, and the follow-up interval for each.

FORMAT	Image-intensive, case-based, Pearson centers
ITEMS	350 items in cases of 3–6 items each
SESSIONS	2 sessions × 175 items · 3.5 hours each
OPTIONS PER ITEM	3 to 10, single best or multiple-response
TMOD	~120 embedded items, separately scored
SCORING	Scaled 100–900 · 300 and above passes
DISEASE WEIGHTING	60–70% health / disease / trauma
REFRACTIVE WEIGHTING	30–40% refractive / sensory / oculomotor

Structure, item counts, and content weights follow the NBEO Part II PAM Content Outline and current published exam information. Candidates receive an overall PAM pass/fail decision and a separate TMOD breakout decision. Eponyms appear in the non-possessive form used on the exam.

How to use this guide

Part II PAM is not a knowledge test. It is a **decision** test. You are given a patient, an image, and a set of findings, and you are asked what this is, what supports it, what you would do, and what you would do next. That means the highest-yield unit of study is not a fact but a **complete clinical pattern**: presentation, distinguishing features, the differential you must exclude, the exact treatment with dose and duration, and the follow-up interval.

Every chapter in this guide is therefore built around that five-part pattern, and the tables are laid out so you can cover the right-hand columns and reconstruct the management from the presentation alone.

What the exam actually asks

OFFICIAL PART II PAM ITEM TYPES

Item type	What it tests, and how to prepare for it
Clinical correlation of basic science principles	Pathophysiology and aetiology, anatomy, biochemistry, physiology, immunology, microbiology, pathology, optics, pharmacology. This is where Part I material reappears, but always attached to a patient. Prepare by asking "why does this patient have this sign?" for every condition.
Diagnosis	The single most appropriate diagnosis. Prepare by drilling the *distinguishing* feature of each look-alike pair, not the classic description.
Related to diagnosis	Which data support the diagnosis, which additional data correlate, and which test to order next. Prepare by learning what each test *rules in or out*, not just what it measures.
Treatment and management	The single most appropriate treatment. Prepare with exact drug, concentration, frequency, and duration.
Related to treatment and management	Mechanism of the treatment, additional data needed to treat safely, patient education, follow-up interval, and prognosis.
Legal issues and ethics	Licensure and government regulation, professional ethics, the doctor-patient relationship, professional liability.
Public health	Epidemiology, biostatistics and measurement, environmental vision, health care policy and administration.

Blueprint: how the 350 items are distributed

NORMAL HEALTH / DISEASE / TRAUMA · 60–70% OF ITEMS

Condition area	Item count	Study priority
Conjunctiva / cornea / refractive surgery	33–44	Highest. Tied for the largest area.
Retina / choroid / vitreous	33–44	Highest. Tied for the largest area, and the most image-dependent.
Optic nerve / neuro-ophthalmic pathways	28–38	Very high, and notably heavier than on Part I.
Lens / cataract / IOL / pre- and post-operative care	17–28	High, and much heavier than on Part I. Know perioperative comanagement.

Condition area	Item count	Study priority
Glaucoma	16–27	High.
Lids / lashes / lacrimal / adnexa / orbit	11–22	Moderate.
Episclera / sclera / anterior uvea	11–22	Moderate, and heavily TMOD-weighted.
Emergencies / trauma	11–22	Moderate, and here it is a **named category** rather than embedded as on Part I.
Systemic health	11–22	Moderate as a standalone category, but systemic disease permeates other cases.

REFRACTIVE / SENSORY / OCULOMOTOR PROCESSES · 30–40% OF ITEMS

Condition area	Item count	Note
Contact lenses	17–28	**Roughly double the Part I weighting.** Fitting decisions, complications, and specialty lenses.
Accommodative / vergence / oculomotor anomalies	20–25	The single largest refractive area. Diagnosis and management, not norms alone.
Ametropia	11–22	Less weighted than on Part I; the emphasis shifts to clinical decisions.
Ophthalmic optics / spectacles	6–17	Dispensing problems and non-adapt troubleshooting.
Low vision	6–17	Device selection and rehabilitation planning.
Amblyopia / strabismus	6–17	Management pathways and trial evidence.
Perceptual function / color vision	6–17	Acquired defects and test selection.
Visual and human development	0–6	Often embedded in pediatric cases rather than asked separately.

TMOD: THE SEPARATELY SCORED EXAM INSIDE THE EXAM

About **120 of the 350 items** are classified as Treatment and Management of Ocular Disease and generate a **separate pass/fail decision**. You can pass PAM and fail TMOD, and then have to sit TMOD as a standalone examination. An item is TMOD if it concerns:

1. Formulation of the most appropriate disease **diagnosis that will be treated or managed**.
2. **Selection** of treatment or management, including systemic considerations.
3. **Dose, form, schedule, and duration** of treatment.
4. **Contraindications and side effects** of medication, including systemic considerations.
5. **Follow-up and prognosis**, including reassessment of the diagnosis after starting treatment.
6. Treatment and management of ocular **emergencies and urgencies**.

The practical consequence: knowing that bacterial keratitis is treated with a fluoroquinolone is not enough. You need *moxifloxacin 0.5%, one drop every hour around the clock for 24 to 48 hours, then taper, review in 24 hours, culture first if central or larger than 1 to 2 mm*. This guide gives doses at that resolution throughout, and Chapter 13 is a dedicated formulary.

Reading conventions the exam uses, and you must not misread

OFFICIAL CASE CONVENTIONS

Convention	What it means for you
"Review of systems"	Entries are **current symptoms the patient reports now**. It is not past history.
"Patient medical history"	Current medical conditions and diagnoses.
HgbA1c	Every patient with diabetes will have one. **No reference range will be given**, and interpretation is treated as an entry-level skill. Know that `≥6.5%` is diagnostic, `<7%` is a typical target, and that poor control changes both your urgency and your refractive advice.
"BVA"	**Best (best-corrected) visual acuity.** It already includes whatever was needed to achieve it, including refraction and pinhole. If BVA is reduced, **do not look for a pinhole entry and do not conclude that refraction was not attempted**: it is implied.
Near acuity distance	Assume near vision was tested at `16 inches (40 cm)` unless stated otherwise.
Visual fields side by side	**Right field on the right, left field on the left**, so the image numbers will look out of sequence, because images are numbered in the order referenced and the right eye is always referenced first. When fields are stacked vertically, **right on top, left below**.
Images	Photographs may carry normal *or* abnormal findings, and you are expected to interpret them. Images include colour photographs, visual fields, fluorescein and ICG angiography, OCT, ultrasonography, and radiologic imaging.
Multiple-response items	The stem states the number, and ends with **(Select 3)** or similar. **All-or-nothing scoring: you must choose all of the correct answers and only the correct answers.** There is no partial credit.
Scope of the item	Assume every item refers to **the patient depicted**. If it refers to the general population, the stem will say so explicitly ("in the general population", "in most patients").
Frequency and criticality	Cases are either **typical presentations of common conditions** or **pathognomonic presentations of rare but critical conditions**. If a case looks bizarre, it is probably the classic textbook picture of something dangerous, not a subtle variant of something common.

A REPEATABLE METHOD FOR WORKING A CASE

1. **Read the demographics and chief complaint first, and form a hypothesis before reading further.** Age, sex, race, and complaint duration eliminate most of the differential immediately.
2. **Read the review of systems and medications as clues, not background.** Every medication on the list is there for a reason: it is either the cause, a contraindication, or a diagnostic hint.
3. **Look at the image before reading the findings.** Form your own impression, then check it against the data. Reading the numbers first anchors you and makes you miss what is in the photograph.
4. **Name the diagnosis out loud in your head before looking at the options.** Then find your answer among them. Reading the options first is how you get talked into a plausible distractor.
5. **For each item, decide which of the seven item types it is.** A "related to treatment" item wants mechanism, education, follow-up, or prognosis, and you will find it faster if you know which of those it is asking for.
6. **Answer the whole case before moving on**, then do not revisit it. Later items in a case sometimes reveal the diagnosis; use that to check earlier answers *within* the case.
7. **On multiple-response items, count.** Select exactly the number requested. Then re-read each selected option and ask "is this definitely true for this patient?" One wrong inclusion loses the whole item.

Timeline for preparation

THREE SCHEDULES

Phase	10 weeks	5 weeks	2 weeks
Build	Weeks 1–5: one disease chapter per day, cold-testing the objectives first. Build a personal one-line-per-condition management sheet as you go.	Weeks 1–2: two chapters per day, high-yield and management tables only.	Days 1–5: the TMOD formulary (Ch 13), the follow-up interval tables (Ch 14), and every high-yield box.
Drill	Weeks 6–8: all retrieval blocks plus the case bank in Appendix A, closed book.	Weeks 3–4: case bank plus the formulary daily.	Days 6–10: case bank and the differential tables.
Simulate	Weeks 9–10: timed 175-item blocks at `72 seconds` per item; error-log by condition area *and* by item type.	Week 5: two timed blocks, then targeted repair.	Days 11–13: one timed block, repair the top three clusters.
Day before	Formulary, follow-up intervals, and the mnemonic and look-alike tables. Nothing new.		

Pace check. 175 items in 3.5 hours is `72 seconds` per item, and cases carry a reading overhead, so the effective budget on later items is tighter. Practise reading a case scenario in under 60 seconds.

Contents

CHAPTER 1 · WORKING THE CASE

Image interpretation and test selection

Related to diagnosis item type | All condition areas | Image-intensive format

ANSWER THESE COLD BEFORE READING

1. Name the four causes of hyperfluorescence and the two causes of hypofluorescence on angiography.
2. Give the B-scan features that separate retinal detachment, posterior vitreous detachment, and choroidal detachment.
3. Distinguish choroidal melanoma from metastasis and haemangioma by internal reflectivity.
4. State which visual field program you would order for glaucoma, for hydroxychloroquine screening, and for a suspected chiasmal lesion.
5. Give the OCT signature of diabetic macular oedema, central serous chorioretinopathy, and vitreomacular traction.
6. For each of six uveitis presentations, name the laboratory tests you would order.

WHY THIS CHAPTER

All

Every case includes at least one image, and "which test next" is one of the seven official item types.

Fluorescein and indocyanine green angiography

ANGIOGRAPHIC PHASES WITH TIMINGS

Phase	Timing	What is filling
Choroidal flush (prearterial)	8–12 s	Patchy choroidal filling from the short posterior ciliary arteries, plus any cilioretinal artery.
Arterial	10–15 s	Retinal arterioles fill.
Arteriovenous (capillary)	~15–20 s	Complete arteriolar filling with early laminar venous flow. Best phase for the foveal avascular zone and microaneurysms.
Venous	~20–30 s	Laminar then complete venous filling.
Recirculation	~30 s to 5 min	Declining intensity; leakage becomes obvious.
Late	5–10 min	Dye has cleared from normal vessels; residual signal means staining or pooling.

FOUR CAUSES OF HYPERFLUORESCENCE, TWO OF HYPOFLUORESCENCE

Pattern	Mechanism and examples
Leakage	Dye escapes from incompetent vessels into tissue, with increasing size and fuzzy margins over time. Choroidal neovascularisation, diabetic microaneurysms, cystoid macular oedema (**petalloid**), papilloedema, vasculitis.
Pooling	Dye collects in an anatomic space with **discrete margins that do not enlarge**. Subretinal fluid in central serous chorioretinopathy, pigment epithelial detachment (a smooth, sharply bordered, uniformly bright oval).
Staining	Dye binds tissue and persists late without expanding. Disciform scars, drusen, sclera, optic disc drusen.
Transmission (window) defect	**Loss of RPE pigment** unmasking normal choroidal fluorescence: appears **early, is bright, and fades with the choroid without enlarging**. Geographic atrophy, laser scars, RPE atrophy.
Blocking (masking)	Something anterior absorbs the excitation or emission light. Haemorrhage (preretinal blocks everything, subretinal blocks less), pigment, lipid exudate, media opacity.
Filling defect	Absent perfusion. Capillary non-perfusion in diabetes, retinal artery occlusion, choroidal watershed infarct, capillary dropout in the FAZ.

The discriminating question between pooling, leakage, and staining is what happens to the margins over time: leakage expands and blurs, pooling stays put with crisp borders, and staining neither expands nor fades.

- **Fluorescein cautions**: nausea in about `5-10%`, urine and skin discolouration for 24 to 36 hours, and rare anaphylaxis. It is not iodine-based. Crosses the placenta, so it is generally avoided in pregnancy though not absolutely contraindicated.
- **Indocyanine green** is `98%` protein-bound and images in the infrared (absorbs `~805 nm`, emits `~835 nm`), so it penetrates blood, pigment, and exudate and images the **choroid**. Use it for polypoidal choroidal vasculopathy, occult CNV, central serous chorioretinopathy, choroidal haemangioma, and the white dot syndromes. **Contains iodine: contraindicated in iodine or shellfish allergy and in significant liver disease.**

Optical coherence tomography

OCT SIGNATURES WORTH RECOGNISING INSTANTLY

Condition	OCT appearance
Diabetic macular oedema	Intraretinal **cystoid spaces predominantly in the inner nuclear and outer plexiform layers**, retinal thickening, hyperreflective foci (lipid), and sometimes subretinal fluid or a taut hyperreflective epiretinal membrane. Centre-involving means the central subfield is thickened.
Central serous chorioretinopathy	**Smooth dome of subretinal fluid** with an intact photoreceptor layer, often a small pigment epithelial detachment, and a **thickened choroid (pachychoroid)** on enhanced depth imaging.
Neovascular AMD	Subretinal or sub-RPE hyperreflective material, subretinal and intraretinal fluid, pigment epithelial detachment, and a double-layer sign in type 1 (occult) lesions.

Condition	OCT appearance
Dry AMD and geographic atrophy	Drusen as RPE elevations with variable internal reflectivity, then **complete RPE and outer retinal atrophy** with choroidal hypertransmission and outer retinal tubulation.
Epiretinal membrane	Hyperreflective line on the inner surface with retinal wrinkling, inner retinal thickening, and loss of the foveal contour.
Vitreomacular traction	Persistent vitreous attachment at the fovea with anteroposterior traction, foveal elevation, and often intraretinal cysts.
Macular hole	Full-thickness defect with intraretinal cysts at the edges (Gass stages 1 to 4; stage 4 has a complete PVD with a Weiss ring). A **lamellar** hole spares the outer retina; a **pseudohole** is an ERM with steep foveal walls and no tissue loss.
Retinal artery occlusion, acute	Inner retinal hyperreflectivity and thickening, later replaced by inner retinal thinning with a preserved outer retina.
Hydroxychloroquine toxicity	Parafoveal (or, in Asian patients, pericentral) **loss of the ellipsoid zone with outer nuclear layer thinning and a preserved foveal centre**: the "flying saucer" sign.
Glaucoma	RNFL thinning most often **inferotemporal then superotemporal** (a double-hump pattern with reduced peaks), macular ganglion cell complex loss, and increasing cup depth with lamina cribrosa posterior bowing.
Papilloedema versus pseudopapilloedema	Papilloedema shows a smooth, gradual, hyporeflective subretinal hyporeflective space and a "lazy V" pattern; optic disc drusen appear as discrete signal-poor ovoid masses with a hyperreflective margin above the lamina.

OCT ARTEFACTS THAT CREATE FALSE DIAGNOSES

- **Segmentation error**: the software mis-draws a boundary, and the thickness map turns red or green spuriously. Always look at the raw B-scan before believing a thickness map.
- **"Red disease"**: a normal eye flagged abnormal, usually a high myope, a tilted disc, a small disc, or an eye with an epiretinal membrane inflating the RNFL.
- **"Green disease"**: real glaucomatous loss hidden inside a normal-appearing map, usually a large disc, a very thick baseline RNFL, or loss outside the measured circle.
- **Floater or vessel shadowing** producing a focal false defect.
- **Poor signal strength** from dry eye, cataract, or poor fixation. Lubricate and repeat before interpreting.
- **Decentration** shifting the measurement circle and manufacturing an asymmetry.

Ultrasonography

B-SCAN DIFFERENTIATION OF POSTERIOR SEGMENT MEMBRANES AND MASSES

Entity	B-scan features
Rhegmatogenous retinal detachment	Thick, **highly reflective** membrane, **inserts at the optic disc and the ora serrata**, folded but does not cross the disc, restricted "after-movement" on kinetic scanning, remains attached at 100% gain.

Entity	B-scan features
Posterior vitreous detachment	Thin, **low reflective** membrane, **does not insert at the disc**, moves freely and continues moving after the eye stops (marked after-movement), and often disappears at reduced gain.
Tractional retinal detachment	Membrane tented toward vitreous adhesions, immobile, often with a "table-top" configuration in diabetes.
Choroidal detachment	Thick, **smooth, dome-shaped**, extends anteriorly to the ciliary body but **stops short of the disc** (the vortex veins tether it), immobile, and may be lobular with multiple domes.
Vitreous haemorrhage	Mobile, dispersed low-amplitude opacities without a defined membrane; layering inferiorly; look beneath it for a break or detachment.
Asteroid hyalosis	Highly reflective discrete points suspended in the vitreous with a clear space in front of the retina.
Retinoschisis	Thin, smooth, immobile, dome-shaped, does not extend to the disc, and typically inferotemporal.

CHOROIDAL MASSES BY A-SCAN INTERNAL REFLECTIVITY

Lesion	Reflectivity	Other features
Choroidal melanoma	**Low**, with acoustic hollowness	**Mushroom or collar-stud** shape when it breaks Bruch membrane, choroidal excavation, orange lipofuscin, subretinal fluid, and documented growth.
Choroidal metastasis	**High** and irregular	Creamy, plateau-shaped, often multifocal and bilateral, with more subretinal fluid than the tumour thickness would suggest.
Choroidal haemangioma	**High** and homogeneous	Orange-red, early intense fluorescein filling with late washout, and characteristic ICG appearance; diffuse form in Sturge-Weber.
Choroidal naevus	Variable, usually flat	Thickness under `2 mm`, overlying drusen, a halo, and no growth or fluid.
Choroidal osteoma	**Very high** with dense shadowing	Juxtapapillary, orange-yellow, scalloped edges, marked acoustic shadowing behind it.
Disciform scar or haemorrhage	High	Associated with drusen and AMD context.

Corneal topography and tomography

PATTERN RECOGNITION

Pattern	Interpretation
Inferior steepening with a skewed radial axis	**Keratoconus.** Supporting indices: maximum keratometry rising, inferior-superior asymmetry above about `1.4 D`, thinnest point displaced inferotemporally, and posterior elevation above the best-fit sphere. Belin-Ambrósio deviation and the Ectasia Risk Score formalise this.
Inferior "crab claw" or "kissing doves"	**Pellucid marginal degeneration**: peripheral inferior thinning with the cornea protruding *above* the thin zone, producing high against-the-rule astigmatism.
Central island or small optical zone	Post-refractive surgery complication causing glare and monocular diplopia.

Pattern	Interpretation
Decentred ablation	Asymmetric flattening with induced coma; the patient reports ghosting that spectacles do not fix.
Irregular, non-orthogonal astigmatism with warpage	Contact lens warpage; repeat after weeks out of lenses before any surgical planning.
Superior flattening with lipid line	Terrien marginal degeneration.

Perimetry: which program, and reading it

CHOOSING THE TEST

Program	Indication
24-2 threshold	**Glaucoma** workhorse: 54 points to 24 degrees. Use 24-2C or add a 10-2 when central points are involved.
30-2	Wider survey; useful for neurologic disease and for baseline documentation.
10-2	Dense central sampling: **hydroxychloroquine screening**, advanced glaucoma, macular disease, and central or paracentral scotomas.
Kinetic (Goldmann) perimetry	Very advanced field loss, functional visual loss, neurologic disease where full peripheral extent matters, and patients who cannot perform static testing.
Confrontation and Amsler	Bedside screening; Amsler detects metamorphopsia and small central scotomas in macular disease.
Frequency doubling technology	Screening, exploiting early magnocellular loss.
Short-wavelength automated perimetry	Early glaucomatous blue-yellow loss; heavily affected by lens opacity.

FIELD PATTERN TO LESION, IN ONE TABLE

Pattern	Localisation and first action
Monocular defect with RAPD	Prechiasmal. Examine the disc; if the disc is normal and the defect is central, think retrobulbar optic neuritis or compression.
Arcuate or nasal step respecting the horizontal midline	Nerve fibre bundle: glaucoma. Correlate with disc and OCT.
Altitudinal	Anterior ischaemic optic neuropathy. **Over 50, exclude giant cell arteritis today.**
Enlarged blind spot with generalised constriction, bilateral	Papilloedema. Measure blood pressure, then image the brain and consider lumbar puncture.
Central or centrocaecal, bilateral and symmetric	Toxic, nutritional, or hereditary optic neuropathy. Review medications (ethambutol, amiodarone, linezolid), alcohol and tobacco, B_{12} and folate.
Bitemporal	Chiasm. Order MRI of the brain and sella with contrast; ask about endocrine symptoms.
Junctional (ipsilateral central plus contralateral superotemporal)	Optic nerve at the chiasm. MRI urgently.
Incongruous homonymous	Optic tract or lateral geniculate.

Pattern	Localisation and first action
Superior homonymous quadrant	Temporal lobe, Meyer loop.
Inferior homonymous quadrant	Parietal radiations.
Congruous homonymous with macular sparing	Occipital cortex, posterior cerebral artery. Treat as a stroke.
Concentric constriction, bilateral, with normal discs	Retinitis pigmentosa, advanced glaucoma, panretinal photocoagulation, **vigabatrin**, or non-organic loss (a field that does not enlarge with distance).
Ring scotoma	Retinitis pigmentosa, or the optical ring scotoma of aphakic spectacles.

Laboratory test selection

UVEITIS AND INFLAMMATORY WORKUP, TARGETED RATHER THAN SHOTGUN

Presentation	Tests to order
First episode, unilateral, acute, non-granulomatous anterior uveitis, otherwise well	**Usually none.** Treat and observe. A targeted history is more valuable than a panel.
Recurrent or bilateral acute anterior uveitis, or with back pain	**HLA-B27**, sacroiliac imaging, and referral to rheumatology. Also RPR plus a treponemal test.
Granulomatous anterior or panuveitis	**ACE and lysozyme, chest radiograph or chest CT** (for sarcoid), **RPR plus treponemal antibody**, **interferon-gamma release assay or tuberculin skin test**. Add serum calcium.
Intermediate uveitis / pars planitis	MRI of the brain for demyelination, plus sarcoid, syphilis, tuberculosis, and Lyme serology where endemic.
Retinal vasculitis, occlusive	Consider **Behçet** (clinical, HLA-B51), tuberculosis, syphilis, sarcoid, lupus, antiphospholipid antibodies, and ANCA.
Focal necrotising retinochoroiditis next to a scar	**Toxoplasma IgG and IgM**; IgG positive supports it, IgM suggests recent acquisition. A negative IgG makes it very unlikely.
Acute retinal necrosis	Anterior chamber or vitreous **PCR for VZV, HSV, and CMV**; check HIV status.
Bilateral panuveitis with serous detachments plus hearing loss	**Vogt-Koyanagi-Harada**: clinical criteria plus fluorescein and ICG angiography and OCT; HLA-DR4 supports.
Uveitis in a child with arthritis	**ANA**, and rheumatology referral. RF is usually negative in the high-risk group.
Uveitis in an elderly patient with vitreous cells and no response to steroid	Consider **primary vitreoretinal lymphoma**: vitreous biopsy with cytology, IL-10 to IL-6 ratio, *MYD88*, and MRI of the brain.
Necrotising scleritis or peripheral ulcerative keratitis	**Systemic workup is mandatory**: **ANCA (PR3 and MPO)**, RF and anti-CCP, ANA, ESR and CRP, urinalysis, chest imaging, and immediate rheumatology referral.

Presentation	Tests to order
Sudden vision loss over 50 with disc swelling	**ESR, CRP, and platelet count immediately**, then temporal artery biopsy. Do not wait for results to start steroid.

NEUROIMAGING: CHOOSING THE STUDY

Clinical question	Study to order
Orbital fracture, foreign body, acute orbital trauma	**CT of the orbits** without contrast, thin cuts, axial and coronal. CT is superior for bone and for metallic foreign bodies.
Orbital cellulitis with proptosis or restricted motility	**CT of the orbits and sinuses with contrast** to identify subperiosteal abscess.
Optic neuritis, optic nerve or chiasmal lesion, demyelination	**MRI of the brain and orbits with fat suppression and gadolinium**. MRI is superior for soft tissue and for white matter.
Pupil-involving third nerve palsy	**Urgent CT angiography or MR angiography** (catheter angiography if non-invasive studies are negative) to exclude a posterior communicating artery aneurysm. This is same-day.
Papilloedema	**MRI of the brain with contrast plus MR venography** to exclude venous sinus thrombosis, then lumbar puncture with opening pressure.
Horner syndrome, painful or acute	**CT or MR angiography of the head and neck from the aortic arch to the circle of Willis** to exclude carotid dissection. Same-day.
Horner syndrome in a child	MRI of the brain, neck, chest, and abdomen plus urinary catecholamines, to exclude **neuroblastoma**.
Amaurosis fugax or retinal artery occlusion	**Carotid duplex ultrasound, ECG with rhythm monitoring, and echocardiography**, plus urgent referral to a stroke service.
Suspected metallic intraocular foreign body	CT. **Avoid MRI until a ferromagnetic foreign body is excluded.**
Thyroid eye disease with suspected compressive optic neuropathy	**CT or MRI of the orbits** showing apical crowding with tendon-sparing muscle enlargement.

Other tests, and what each decides

Test	Decides
Photostress recovery test	Macular versus optic nerve cause of reduced acuity. Prolonged recovery (over about `50 seconds`, or markedly asymmetric) indicates **macular** disease; optic nerve disease recovers normally.
Potential acuity meter / pinhole / laser interferometry	How much acuity is recoverable behind a cataract.
Brightness comparison and red desaturation	Sensitive bedside screens for optic nerve disease.
Contrast sensitivity	Functional loss when Snellen acuity is preserved: early cataract, glaucoma, optic neuritis, amblyopia, post-refractive surgery.
Pattern ERG or multifocal ERG	Macular versus ganglion cell versus optic nerve dysfunction when the VEP is abnormal.
Full-field ERG	Generalised photoreceptor disease: retinitis pigmentosa, cone dystrophy, congenital stationary night blindness, cancer-associated retinopathy.

Test	Decides
EOG	RPE function; **abnormal with a normal ERG means Best disease**.
Fundus autofluorescence	RPE lipofuscin: hyper in Stargardt flecks and in the leading edge of geographic atrophy, hypo in atrophy. Also confirms optic disc drusen.
OCT angiography	Non-invasive perfusion mapping: FAZ enlargement in diabetes, CNV networks, and radial peripapillary capillary dropout in glaucoma. It does not show leakage.
Anterior segment OCT	Angle configuration and plateau iris, corneal layer localisation of pathology, and post-graft interface assessment.
Ultrasound biomicroscopy	Ciliary body tumours and cysts, plateau iris, cyclodialysis cleft, and a retained anterior foreign body.
Corneal confocal microscopy	*Acanthamoeba* cysts and fungal filaments in vivo; subbasal nerve density.
Seidel test	Aqueous leak: fluorescein diluting into a green rivulet under cobalt blue. Do it in any suspected penetrating injury or wound leak.
Forced duction and force generation	Restriction versus paresis.
Ice pack or rest test	Improvement of ptosis supports myasthenia gravis.

RETRIEVAL PRACTICE

1. A round, sharply bordered, uniformly bright area that appears early and does not change size through the study. Leakage, pooling, staining, or window defect?
2. Which angiographic study images the choroid, and what is the contraindication?
3. A thick highly reflective membrane inserting at the disc and ora with restricted after-movement. Diagnosis?
4. A smooth dome that stops short of the disc and is immobile. Diagnosis?
5. A mushroom-shaped choroidal mass with low internal reflectivity. Diagnosis and the two worst prognostic pathology features?
6. Which visual field program for hydroxychloroquine screening, and what OCT sign accompanies it?
7. Give the OCT signature that distinguishes central serous chorioretinopathy from neovascular AMD.
8. Explain "red disease" and "green disease" on OCT.
9. A patient over 50 with an altitudinal field defect and disc swelling. Which three blood tests, and what do you do before they return?
10. Which imaging study for a pupil-involving third nerve palsy, and on what timescale?
11. A painful Horner syndrome. Which study, and why the urgency?
12. A child with a new Horner syndrome. What must be excluded and how?
13. Interpret a prolonged photostress recovery time.
14. Necrotising scleritis: name four laboratory tests you must order.
15. What does an abnormal EOG with a normal ERG indicate?

KEY

1. **Pooling** if the borders are anatomic and the area does not enlarge; a **window defect** if it appears early, is bright, and *fades with the choroid*. The discriminator is whether it fades late (window) or persists as a pool.
2. Indocyanine green angiography; contraindicated in iodine or shellfish allergy and in significant hepatic disease.
3. Rhegmatogenous retinal detachment.
4. Choroidal detachment (tethered by the vortex veins).
5. Choroidal melanoma; **epithelioid cell type** and **monosomy 3 with 8q gain or BAP1 loss** (with extrascleral extension and large size also adverse).
6. **10-2**; parafoveal (or pericentral in Asian patients) ellipsoid zone loss with a preserved foveal centre, the flying saucer sign.

7. CSC shows a smooth serous elevation with an intact photoreceptor layer and a thickened choroid; neovascular AMD shows subretinal or sub-RPE hyperreflective material with intraretinal fluid and drusen.
8. Red disease is a false abnormal flag in a normal eye (high myopia, tilted or small disc, epiretinal membrane). Green disease is real loss hidden by a normal-looking map (large disc, thick baseline, loss outside the scan circle).
9. ESR, CRP, and platelet count; **start high-dose corticosteroid immediately** without waiting.
10. CT angiography or MR angiography of the head, **same day**, to exclude a posterior communicating artery aneurysm.
11. CT or MR angiography of the head and neck including the aortic arch, same day, because a painful Horner syndrome is a carotid dissection until proven otherwise and carries a stroke risk.
12. **Neuroblastoma**; MRI of the brain, neck, chest and abdomen plus urinary catecholamine metabolites.
13. It indicates **macular** rather than optic nerve pathology.
14. ANCA (PR3 and MPO), rheumatoid factor and anti-CCP, ANA, and ESR with CRP; add urinalysis and chest imaging, and refer to rheumatology urgently.
15. Best vitelliform macular dystrophy (*BEST1*, bestrophin).

CHAPTER RECAP

1. Hyperfluorescence is leakage, pooling, staining, or a window defect; the margins over time tell you which.
2. ICG images the choroid, contains iodine, and is the study for polypoidal disease and occult CNV.
3. OCT: read the raw B-scan, not the thickness map; know red disease and green disease.
4. B-scan: RD inserts at the disc and ora, PVD does not insert and after-moves, choroidal detachment stops short of the disc.
5. Melanoma is acoustically hollow and mushroom-shaped; metastasis and haemangioma are highly reflective.
6. 24-2 for glaucoma, 10-2 for hydroxychloroquine and central disease, kinetic for advanced or functional loss.
7. Field pattern localises: arcuate for glaucoma, altitudinal for AION, bitemporal for chiasm, congruous with macular sparing for occipital.
8. Order laboratory tests to answer a question, not as a panel. First unilateral non-granulomatous anterior uveitis usually needs none.
9. CT for bone and metal, MRI for soft tissue and white matter, angiography for aneurysm and dissection.
10. Three same-day imperatives: pupil-involving third nerve palsy, painful Horner syndrome, and giant cell arteritis.

CHAPTER 2 · DISEASE AND MANAGEMENT

Lids, lashes, lacrimal system, adnexa, orbit

Lids / lashes / lacrimal / adnexa / orbit · 11–22 items | TMOD-heavy | Emergencies / trauma

ANSWER THESE COLD BEFORE READING

1. Separate preseptal from orbital cellulitis on four findings, and give the management of each with doses.
2. Give the full treatment protocol for meibomian gland dysfunction, in order of escalation.
3. Name six features that make a lid lesion a biopsy rather than an observation.
4. State the management of congenital nasolacrimal duct obstruction by age.
5. Give the clinical activity score components in thyroid eye disease and the indication for urgent decompression.
6. Name the orbital differential for a child versus an adult with proptosis.

BLUEPRINT WEIGHT

11–22

Scored items, but with two of the exam's clearest emergencies embedded: orbital cellulitis and mucormycosis.

Blepharitis and meibomian gland dysfunction

ANTERIOR VERSUS POSTERIOR LID MARGIN DISEASE

Feature	Anterior blepharitis	Posterior blepharitis (MGD)
Findings	**Staphylococcal**: hard, brittle, matted collarettes at the lash base, ulcerated margins, madarosis, trichiasis, poliosis, marginal infiltrates. **Seborrhoeic**: greasy, easily removed scurf; associated scalp dandruff. ***Demodex***: **cylindrical dandruff (collarettes) encircling the lash shaft**, lash misdirection.	Capped, plugged, or pouting meibomian orifices; thickened, notched, telangiectatic margins; turbid or toothpaste-like expressed meibum; foamy tear film; posterior margin displacement of the mucocutaneous junction; gland dropout on meibography.
Associations	Rosacea, seborrhoeic dermatitis, atopy.	Rosacea, seborrhoeic dermatitis, androgen deficiency, isotretinoin, chalazia.
Consequence	Staphylococcal exotoxin-driven marginal keratitis, phlyctenulosis, recurrent styes.	**Evaporative dry eye** with short tear break-up time; the commonest cause of dry eye overall.

MGD AND BLEPHARITIS: ESCALATE IN THIS ORDER

1. **Warm compresses** at `40–45°C` for `5–10 minutes`, twice daily, followed immediately by **lid massage and margin hygiene** with a commercial lid cleanser or dilute baby shampoo. Explain that this is a chronic condition requiring maintenance, not a course of treatment: non-adherence is the commonest cause of failure.
2. **Artificial tears**, preservative-free if used more than four times daily, plus a lipid-containing formulation for evaporative disease.
3. **Topical antibiotic-steroid** for an inflamed margin: azithromycin `1%` drops (used for its anti-inflammatory and meibum-normalising effect), or erythromycin or bacitracin ointment to the margins nightly for 2 to 4 weeks; add loteprednol `0.5%` twice daily for a short pulse if there is marginal keratitis.
4. **Oral doxycycline** `50–100 mg` **daily** (or minocycline, or oral azithromycin pulses) for `2 to 3 months`, exploiting **matrix metalloproteinase and lipase inhibition** rather than the antibacterial effect. **Contraindicated in pregnancy, breastfeeding, and children under 8.**
5. **Omega-3 supplementation**, though the DREAM study showed no benefit over placebo for dry eye symptoms, so it is optional rather than evidence-based.
6. **Target *Demodex*** where collarettes are present: tea tree oil derivative (terpinen-4-ol) lid scrubs, or topical **lotilaner** `0.25%` twice daily for 6 weeks.
7. **In-office procedures**: meibomian gland expression, thermal pulsation, intense pulsed light for rosacea-associated telangiectasia, and microblepharoexfoliation.
8. **Treat the skin**: refer for rosacea and seborrhoeic dermatitis management; ocular rosacea often precedes and exceeds the cutaneous disease.

LID MARGIN LUMPS: DISTINGUISHING AND TREATING

Lesion	Presentation	Management
External hordeolum (stye)	Acute, tender, pointing lesion at the lash line; gland of Zeis or Moll.	Warm compresses four times daily, lid hygiene, epilation of the associated lash to drain. Topical antibiotic ointment adds little but is commonly used. Systemic antibiotic only with surrounding cellulitis.
Internal hordeolum	Acute tender swelling within the tarsus, pointing to the conjunctival surface.	As above; may evolve into a chalazion.
Chalazion	**Non-tender, firm, rubbery** nodule; a lipogranuloma around extravasated meibum.	Warm compresses and massage for `4 to 6 weeks`; most resolve. Then intralesional triamcinolone (`5–10 mg/mL`, with a small risk of skin depigmentation and, rarely, retinal vascular occlusion) or incision and curettage through a **vertical conjunctival incision**. **A recurrent lesion in the same site, in a patient over 50, with madarosis, or with lid margin architectural loss, must be biopsied to exclude sebaceous cell carcinoma.**
Molluscum contagiosum	Umbilicated waxy nodule on the margin, causing a chronic follicular conjunctivitis by viral shedding.	Excision, curettage, or cryotherapy. Multiple lesions warrant HIV testing.

Lesion	Presentation	Management
Sebaceous cell carcinoma	Recurrent "chalazion" or unilateral chronic blepharoconjunctivitis unresponsive to treatment, favouring the **upper** lid, with madarosis.	Full-thickness biopsy with **fresh tissue for lipid stains** and map biopsies for pagetoid spread; refer to oculoplastics and oncology. High mortality if missed.
Basal cell carcinoma	**Lower lid and medial canthus**, pearly rolled edge, telangiectasia, central ulceration, painless, slowly enlarging.	Excision with margin control (Mohs for medial canthal or morpheaform lesions). Locally invasive; metastasis is rare.
Squamous cell carcinoma	Scaly, keratotic, ulcerated; may show perineural spread with numbness or a cranial neuropathy.	Excision with margins; imaging if perineural spread is suspected; true metastatic potential.

SIX FEATURES THAT CONVERT OBSERVATION INTO BIOPSY

Madarosis (lash loss), **loss of normal lid margin architecture**, **telangiectasia over the lesion**, **ulceration**, **induration or fixation to deeper tissue**, and **distortion or occlusion of the punctum**. Add three more clinical triggers: **recurrence in the same location**, **unilateral chronic blepharoconjunctivitis unresponsive to treatment**, and **pigmentation with irregular borders**.

Preseptal versus orbital cellulitis

THE DISTINCTION THAT CHANGES THE DISPOSITION

Feature	Preseptal cellulitis	Orbital (postseptal) cellulitis
Proptosis	**Absent**	**Present**
Ocular motility	Full and painless	**Restricted and painful**
Vision and pupil	Normal, no RAPD	May be reduced with an **RAPD** and dyschromatopsia
Chemosis and IOP	Minimal	Chemosis, raised IOP, resistance to retropulsion
Source	Skin trauma, insect bite, hordeolum, dacryocystitis, impetigo	**Ethmoid sinusitis** (via the lamina papyracea) in most children; dental infection, trauma, or surgery in adults
Organisms	*Staphylococcus aureus* including MRSA, *Streptococcus pyogenes*	Streptococci, *S. aureus*, *Haemophilus influenzae* in unvaccinated children, anaerobes and polymicrobial in adults
Imaging	Not routinely required	**CT of orbits and sinuses with contrast, urgently**, to identify a subperiosteal abscess

Feature	Preseptal cellulitis	Orbital (postseptal) cellulitis
Treatment	Oral antibiotic covering MRSA in an adult: **amoxicillin-clavulanate `875/125 mg` twice daily**, or clindamycin, or trimethoprim-sulfamethoxazole plus amoxicillin. Review in `24 to 48 hours`. Admit any child under 1 year, any patient who is toxic, or any failure to improve.	**Admission, intravenous antibiotics, and joint ophthalmology and otolaryngology care.** Typical regimen: intravenous vancomycin plus ceftriaxone (or ampicillin-sulbactam or piperacillin-tazobactam), adding metronidazole for intracranial extension or anaerobes. Surgical drainage for a large or medial subperiosteal abscess, for failure to improve in 24 to 48 hours, or for optic nerve compromise. Monitor vision, pupil, colour, and motility every few hours.
Complications	Rarely progresses if treated	Subperiosteal and orbital abscess, **cavernous sinus thrombosis**, meningitis, intracranial abscess, central retinal artery or vein occlusion, optic neuropathy, and death

THE ORBITAL INFECTION YOU MUST NOT MISS: MUCORMYCOSIS

A **diabetic in ketoacidosis**, or an immunocompromised or neutropenic patient, with **orbital apex signs, a black necrotic eschar on the palate or nasal mucosa, facial numbness, disproportionate pain, and rapidly progressive ophthalmoplegia with vision loss.** The fungus is angioinvasive, so tissue infarction and cranial neuropathies dominate.

Management: emergency admission, urgent biopsy for non-septate broad hyphae with right-angle branching, systemic liposomal amphotericin B, aggressive surgical debridement, and correction of the acidosis. Mortality remains high. Aspergillus produces a similar but more indolent picture in the less profoundly immunosuppressed.

Lacrimal disease

EPIPHORA AND LACRIMAL INFECTION

Condition	Presentation and testing	Management
Congenital nasolacrimal duct obstruction	Epiphora and mattering from the first weeks of life, with a normal cornea and no photophobia. Failure of canalisation at the **valve of Hasner**. Differentiate from congenital glaucoma (photophobia, blepharospasm, corneal enlargement and clouding, raised IOP).	**Crigler massage** (firm downward strokes over the sac) two to four times daily plus hygiene; topical antibiotic only for purulent discharge. `~90%` resolve by `12 months`. **Probing and irrigation** if unresolved at `12 months` (earlier for recurrent dacryocystitis or an amblyogenic factor), then balloon dacryoplasty or intubation, then dacryocystorhinostomy.
Congenital dacryocele / amniotocele	Bluish cystic swelling at the medial canthus at birth, sometimes with an intranasal cyst causing respiratory distress in a bilateral case.	Urgent decompression by massage, then probing; involve otolaryngology if there is airway compromise.

Condition	Presentation and testing	Management
Acquired nasolacrimal duct obstruction	Epiphora, mucoid reflux on sac pressure, and a wet eye with a normal ocular surface. **Dye disappearance test** abnormal; **irrigation** shows reflux; **Jones I and II** distinguish functional from anatomic. Middle-aged and older women predominate.	Dacryocystorhinostomy (external or endoscopic) with high success. Rule out a lacrimal sac tumour if there is bloody reflux, a mass above the medial canthal tendon, or unilateral disease in a younger patient.
Functional epiphora	Patent on irrigation but symptomatic; caused by **lid laxity, ectropion, punctal malposition, or orbicularis weakness (CN VII palsy)**: the lacrimal pump has failed.	Treat the lid, not the duct: lateral tarsal strip, punctoplasty, or medial spindle.
Acute dacryocystitis	Painful, red, tender swelling **below and medial to the medial canthal tendon**, with epiphora and often preseptal cellulitis. Distinguish from dacryoadenitis (superotemporal) and from an infected sebaceous cyst.	Warm compresses, **oral antibiotic covering staphylococci and streptococci** (amoxicillin-clavulanate, or clindamycin if MRSA is suspected), analgesia, and admission with intravenous therapy if systemically unwell or if there is orbital extension. **Do not irrigate or probe during acute infection.** Definitive dacryocystorhinostomy after the acute episode settles.
Canaliculitis	Localised swelling and pouting of the punctum with expressible cheesy concretions; classically *Actinomyces (Streptothrix)*.	Canalicular expression and curettage of concretions, plus topical antibiotic; punctoplasty if recurrent. It is frequently misdiagnosed as chronic conjunctivitis.
Dacryoadenitis	Painful **superotemporal** lid swelling with an S-shaped lid margin and possible motility restriction. Acute: viral (EBV, mumps) or bacterial. Chronic: sarcoid, IgG4-related disease, tuberculosis, lymphoma, Sjögren.	Acute viral: supportive. Acute bacterial: oral antibiotic. Chronic or bilateral: image and biopsy, and test ACE, lysozyme, serum IgG4, chest imaging, and interferon-gamma release assay.

Lid malposition and function

MALPOSITIONS: MECHANISM AND REPAIR

Condition	Mechanism	Management
Involutional ectropion	Horizontal lid laxity with medial canthal tendon and lateral canthal tendon stretching	Lubrication and taping short term; **lateral tarsal strip**, with a medial spindle for punctal eversion.
Cicatricial ectropion	Anterior lamellar shortening from burns, actinic damage, surgery, or trauma	Skin graft or flap plus horizontal tightening.
Paralytic ectropion	CN VII palsy	Aggressive lubrication, taping and moisture chamber at night, temporary tarsorrhaphy or a lid weight; treat the facial palsy cause.

Condition	Mechanism	Management
Involutional entropion	**Capsulopalpebral fascia disinsertion**, horizontal laxity, and overriding preseptal orbicularis	Temporary taping or Quickert sutures; definitive retractor reinsertion plus horizontal tightening.
Cicatricial entropion	Posterior lamellar (conjunctival and tarsal) scarring: trachoma, cicatrising conjunctivitis, chemical injury, chronic topical medication	Treat the underlying cicatrising disease systemically first; tarsal fracture or mucous membrane graft. **Surgery alone in active mucous membrane pemphigoid accelerates the disease.**
Trichiasis	Acquired lash misdirection	Epilation (recurs in weeks), electrolysis, cryotherapy, radiofrequency ablation, or lash follicle excision. Treat the cause (blepharitis, cicatrisation).
Distichiasis	An accessory row of lashes emerging from the meibomian orifices, congenital or acquired	Cryotherapy or lid splitting; congenital form associated with lymphoedema-distichiasis syndrome.
Floppy eyelid syndrome	Tarsal elastin loss, easily everted upper lid, papillary conjunctivitis, superior punctate keratopathy; the lid everts against the pillow overnight	**Ask about snoring and screen for obstructive sleep apnoea.** Nightly ointment and a shield or taping, side-sleeping avoidance; horizontal lid tightening if severe. CPAP treats the systemic driver.
Blepharospasm (essential)	Bilateral involuntary orbicularis contraction, a focal dystonia	Botulinum toxin injection, repeated at `3 to 4 month` intervals; treat concurrent dry eye, which lowers the threshold.
Hemifacial spasm	**Unilateral**, involving the whole hemiface including the lower face	MRI to exclude vascular compression of CN VII at the brainstem or a posterior fossa lesion; botulinum toxin or microvascular decompression.
Myokymia	Fine unilateral fasciculation of the lower lid, benign, provoked by fatigue, caffeine, and stress	Reassure; if it persists beyond months or spreads, image to exclude a pontine lesion.

PTOSIS: CLASSIFY BEFORE TREATING

Type	Distinguishing features
Aponeurotic (involutional)	Good levator function, high crease, deep sulcus, worse at the end of the day, often with contact lens history or eye rubbing.
Congenital myogenic	Poor levator function, **lid lag on downgaze** (the opposite of acquired ptosis), absent crease, chin-up head posture; risk of amblyopia and astigmatism.
Neurogenic: CN III palsy	Severe ptosis with ophthalmoplegia; **pupil status determines urgency**.
Neurogenic: Horner syndrome	Mild ptosis with miosis and lower lid elevation; **if painful, exclude carotid dissection today**.
Myasthenia gravis	**Variable and fatigable**, Cogan lid twitch, enhancement of the fellow lid on occlusion, ice pack test positive, **pupil always spared**. Test acetylcholine receptor and MuSK antibodies and image the chest for thymoma.
Chronic progressive external ophthalmoplegia	Slowly progressive, symmetric, **without diplopia**; with pigmentary retinopathy and heart block it is Kearns-Sayre and needs cardiology referral for pacing.
Mechanical	Lid mass, dermatochalasis, oedema, or scarring.

Type	Distinguishing features
Pseudoptosis	Contralateral lid retraction, hypotropia, enophthalmos, dermatochalasis, or brow ptosis.
Marcus Gunn jaw winking	Synkinetic lid elevation with jaw movement, from aberrant V3 innervation of the levator.

TWO THINGS TO DO AT EVERY PTOSIS VISIT

Measure and document MRD1, palpebral fissure height, levator function, and the crease position, and **perform a visual field with the lid taped and untaped** if surgery may be indicated: superior field loss with the lid down is what establishes functional rather than cosmetic indication for insurance and clinical purposes.

Oxymetazoline `0.1%` is approved for acquired blepharoptosis and produces about `1 mm` of elevation by stimulating Müller muscle; it is a useful non-surgical option and also a diagnostic demonstration for the patient.

Orbital disease

PROPTOSIS: THE DIFFERENTIAL BY PATTERN AND BY AGE

Presentation	Diagnoses to consider
Bilateral, axial, adult, chronic	**Thyroid eye disease** (by far the commonest cause of adult proptosis), IgG4-related disease, lymphoma, sarcoid, metastasis.
Unilateral, axial, adult, chronic	Thyroid eye disease (often asymmetric), cavernous haemangioma (the commonest benign adult orbital tumour: slowly progressive, well-encapsulated, intraconal), optic nerve sheath meningioma, glioma, lymphoma.
Unilateral, painful, acute or subacute, adult	**Orbital cellulitis**, idiopathic orbital inflammatory syndrome (orbital pseudotumour), **necrotising fungal infection**, ruptured dermoid, orbital haemorrhage, metastasis, carotid-cavernous fistula.
Pulsatile	Carotid-cavernous fistula, orbital arteriovenous malformation, sphenoid dysplasia in neurofibromatosis 1, orbital roof defect.
Worse on Valsalva or bending	**Orbital varix** or lymphangioma.
Child, acute, painful	**Orbital cellulitis from ethmoid sinusitis** (much the commonest), **rhabdomyosarcoma** (rapidly progressive over days to weeks: the commonest primary orbital malignancy of childhood, requiring urgent biopsy), ruptured dermoid cyst, orbital haemorrhage in a lymphangioma, leukaemic infiltration.
Child, chronic	Dermoid cyst (superotemporal, at a suture line, the commonest paediatric orbital lesion), capillary haemangioma (present in infancy, enlarges with crying, then involutes; treat with oral propranolol if amblyogenic), optic nerve glioma (consider neurofibromatosis 1), plexiform neurofibroma, lymphangioma.
Child, with periorbital ecchymosis	**Metastatic neuroblastoma** ("raccoon eyes"), also consider leukaemia and non-accidental injury.

THYROID EYE DISEASE: ACTIVITY, SEVERITY, AND THE DECISION POINTS

Clinical activity score (one point each, three or more indicates active disease): spontaneous retrobulbar pain, pain on eye movement, eyelid erythema, conjunctival redness, eyelid oedema, chemosis, and inflammation of the caruncle or plica.

Severity is graded separately (EUGOGO mild, moderate-to-severe, or sight-threatening). Treat *active* disease with immunomodulation; treat *inactive* disease with surgery.

1. **Every patient**: smoking cessation, achieve and maintain euthyroidism, aggressive lubrication, head elevation at night, selenium in mild disease, and prism for stable diplopia.
2. **Moderate-to-severe active disease**: intravenous methylprednisolone (a cumulative course, typically `4.5 g` over 12 weeks), and consider **teprotumumab** (anti-IGF-1R), mycophenolate, tocilizumab, or orbital radiotherapy.
3. **Sight-threatening disease**, meaning **compressive optic neuropathy** (reduced acuity or colour vision, RAPD, disc swelling or pallor, apical crowding on imaging) or corneal breakdown: **high-dose intravenous steroid immediately and urgent orbital decompression if there is no rapid response.**
4. **Rehabilitative surgery once inactive**, in this order: **orbital decompression, then strabismus surgery, then lid surgery**. Doing them out of order forces revision.

Radioactive iodine can worsen orbitopathy; give steroid cover in at-risk patients. **Ask about smoking at every visit**: it is the strongest modifiable predictor of progression and of treatment failure.

ORBITAL TRAUMA

Injury	Findings and management
Orbital floor blowout fracture	Enophthalmos or hypoglobus, **diplopia worse on upgaze with a positive forced duction** from inferior rectus or inferior oblique entrapment, infraorbital hypaesthesia, and orbital emphysema if the medial wall is also involved. Order **CT orbits**. **Repair urgently (within 24 to 48 hours) in a child with a "white-eyed" trapdoor fracture with an oculocardiac reflex (bradycardia, nausea, syncope)**; otherwise repair within 2 weeks for large fractures (over about half the floor), significant enophthalmos (over `2 mm`), or persistent diplopia in primary or reading gaze. Advise against nose-blowing, and give oral antibiotics if the sinus is involved.
Retrobulbar haemorrhage	**A sight-threatening emergency.** Proptosis, tense orbit, resistance to retropulsion, markedly raised IOP, reduced vision, RAPD, and a cherry red spot if the artery closes. **Immediate lateral canthotomy and inferior cantholysis** at the bedside, plus IOP-lowering agents; then imaging. Do not wait for a CT scan.
Traumatic optic neuropathy	Reduced vision with an RAPD after deceleration injury, often with a normal-looking disc initially. CT to exclude a canal fracture or haematoma. Steroid benefit is unproven and the CRASH trial showed harm in head injury; management is largely observation with surgical decompression reserved for specific cases.
Orbital foreign body	Organic material (wood, vegetable) must be removed because of infection and fistula risk; inert material may be observed. CT is preferred; **avoid MRI until metal is excluded**. Wood can be radiolucent on CT and simulate air.
Carotid-cavernous fistula	Pulsatile proptosis, orbital bruit, dilated corkscrew episcleral vessels, raised IOP, chemosis, and cranial neuropathies. Order CT or MR angiography and refer for neurointerventional treatment.

RETRIEVAL PRACTICE

1. A 7-year-old has fever, lid swelling, proptosis, and pain on eye movement after a week of nasal congestion. Give the diagnosis, imaging, and initial treatment.
2. Which four findings separate orbital from preseptal cellulitis?
3. A 62-year-old has a third "chalazion" in the same upper lid position with lash loss. What is your next step and what must you tell the pathologist?
4. Give the full escalation ladder for meibomian gland dysfunction, and the mechanism by which doxycycline helps.
5. A 4-month-old has had a watery, sticky eye since birth with a clear cornea. Diagnosis, first-line treatment, and when to intervene?
6. A patient has epiphora, patent irrigation, and a lax lower lid. Where is the problem and what is the treatment?
7. Painful red swelling below and medial to the medial canthal tendon. Diagnosis, treatment, and one thing not to do.
8. A patient with a chronic red eye has an easily everted upper lid and superior punctate staining. What must you ask about?
9. Give the seven components of the clinical activity score in thyroid eye disease.
10. State the surgical order in thyroid eye disease rehabilitation and the indication for urgent decompression.
11. A child is struck in the eye and has a quiet-looking eye, upgaze restriction, bradycardia and nausea. What is this and how urgently do you act?
12. A patient after retrobulbar anaesthesia has a tense proptosed orbit with IOP 58 mmHg and reduced vision. What do you do first?
13. Name the commonest primary orbital malignancy of childhood and its tempo.
14. Distinguish essential blepharospasm from hemifacial spasm and state what imaging is required.
15. Which ptosis type shows lid lag on downgaze, and why does that matter?

KEY

1. Orbital cellulitis secondary to ethmoid sinusitis; urgent **CT of the orbits and sinuses with contrast**; admit for **intravenous antibiotics** (vancomycin plus ceftriaxone or ampicillin-sulbactam), with otolaryngology involvement and drainage if there is a significant subperiosteal abscess or no improvement in 24 to 48 hours.
2. Proptosis, restricted and painful motility, reduced vision with an RAPD, and chemosis with resistance to retropulsion.
3. Full-thickness **biopsy** to exclude sebaceous cell carcinoma; tell the pathologist to keep **fresh, unfixed tissue for lipid stains (oil-red-O)** and to assess for pagetoid spread, because routine processing dissolves the lipid.
4. Warm compresses and hygiene, lubricants, topical azithromycin or antibiotic ointment with a short steroid pulse, oral doxycycline for 2 to 3 months, omega-3 optionally, *Demodex*-directed therapy, in-office expression or thermal pulsation, and treatment of the associated skin disease. Doxycycline works by **inhibiting matrix metalloproteinases and bacterial lipases**, not by killing organisms.
5. Congenital nasolacrimal duct obstruction; **Crigler massage** plus hygiene; probe at `12 months` if unresolved, or sooner for recurrent dacryocystitis.
6. The lacrimal **pump** and lid position, not the duct: this is functional epiphora, treated with a lateral tarsal strip and, if needed, punctoplasty or a medial spindle.
7. Acute dacryocystitis; warm compresses plus oral antibiotic covering staphylococci and streptococci, with admission if systemically unwell; **do not probe or irrigate during the acute infection.**
8. **Snoring and daytime somnolence**: floppy eyelid syndrome is strongly associated with obstructive sleep apnoea.
9. Spontaneous retrobulbar pain, pain on eye movement, lid erythema, conjunctival redness, lid oedema, chemosis, and caruncle or plica inflammation.
10. Decompression, then strabismus, then lids. Urgent decompression for **compressive optic neuropathy** (or corneal breakdown) that does not respond rapidly to high-dose intravenous steroid.
11. A **white-eyed trapdoor orbital floor fracture with muscle entrapment and an oculocardiac reflex**; this requires surgical release **within 24 to 48 hours**.

12. Perform an immediate **lateral canthotomy and inferior cantholysis** at the bedside, with IOP-lowering therapy; imaging comes after.
13. **Rhabdomyosarcoma**; it progresses over days to weeks and requires urgent biopsy.
14. Essential blepharospasm is bilateral and confined to the orbicularis; hemifacial spasm is unilateral and involves the whole hemiface. Hemifacial spasm requires **MRI** to exclude vascular compression of the facial nerve or a posterior fossa lesion.
15. **Congenital myogenic ptosis**; the lid lags on downgeaze because the fibrotic levator cannot relax, which distinguishes it from acquired ptosis and predicts poor levator function and the need for a different surgical approach, plus amblyopia risk.

CHAPTER RECAP

1. MGD is chronic and needs a maintenance plan, not a prescription; doxycycline works by MMP inhibition.
2. Proptosis, painful restricted motility, RAPD, and chemosis mean postseptal disease, CT, and admission.
3. Ketoacidosis plus orbital apex signs plus a black eschar means mucormycosis and an emergency.
4. Recurrent chalazion with madarosis in an older patient is sebaceous carcinoma until biopsied on fresh tissue.
5. Congenital NLDO: massage, 90% resolve by a year, probe at 12 months.
6. Patent irrigation with a wet eye means the pump or the lid, so fix the lid.
7. Floppy eyelid means ask about sleep apnoea.
8. Thyroid eye disease: score the activity, stop the smoking, decompress before you straighten before you lift.
9. White-eyed trapdoor fracture with an oculocardiac reflex is a 24 to 48 hour surgical problem.
10. Retrobulbar haemorrhage is treated with a canthotomy at the bedside, not with a scan.

CHAPTER 3 · DISEASE AND MANAGEMENT

Conjunctiva, ocular surface, and dry eye

Conjunctiva / cornea / refractive surgery · 33–44 items | TMOD-heavy | Public health (adenovirus, trachoma)

ANSWER THESE COLD BEFORE READING

1. Differentiate viral, bacterial, allergic, chlamydial, and toxic conjunctivitis on five features each.
2. Give the treatment of hyperacute purulent conjunctivitis with exact doses.
3. Give the neonatal conjunctivitis differential by day of onset.
4. State the TFOS DEWS II staged management of dry eye.
5. Distinguish vernal from atopic from giant papillary conjunctivitis.
6. Name the features that make a conjunctival pigmented or gelatinous lesion suspicious.

BLUEPRINT WEIGHT

33–44

Shared with cornea and refractive surgery: the largest disease area on the examination.

The red eye: differentiating conjunctivitis

FIVE PATTERNS OF CONJUNCTIVITIS

Feature	Viral (adenoviral)	Bacterial	Allergic	Chlamydial (adult)
Discharge	Watery, serous	**Purulent, mucopurulent**, lids stuck shut on waking	Stringy, mucoid, clear	Mucopurulent, scant
Conjunctival reaction	**Follicles** (inferior), petechial haemorrhage, membranes or pseudomembranes in severe cases	**Papillae**, diffuse hyperaemia	**Papillae** and chemosis; giant papillae in vernal	**Large follicles**, especially inferior, and superior limbal infiltrates
Symptom	Gritty, burning, foreign body sensation	Irritation, sticking	**Itching is the cardinal symptom**	Chronic low-grade irritation for weeks to months
Preauricular node	**Present**	Absent (except gonococcal)	Absent	**Present**
Laterality and course	Starts unilateral, becomes bilateral in days; peaks at 5 to 7 days; **subepithelial infiltrates at 7 to 14 days**	Often bilateral, self-limited in 7 to 10 days	Bilateral, seasonal or perennial, recurrent	Unilateral or bilateral, indolent, unresponsive to standard drops

Feature	Viral (adenoviral)	Bacterial	Allergic	Chlamydial (adult)
Key extra clue	Recent contact with a red eye, upper respiratory infection, or a clinic waiting room. **Highly contagious for 10 to 14 days.**	Purulence out of proportion to redness; in a child, concurrent otitis (*Haemophilus*)	Personal or family atopy, seasonal timing, itch	**Sexual history**; concurrent urethritis or cervicitis; treatment failure with topical antibiotics
Treatment	**Supportive**: cool compresses, preservative-free lubricants, hygiene education (separate towels, no shared linen, hand washing, stay off work or school while discharging). Remove membranes. Topical steroid (loteprednol `0.5%` four times daily, tapered) *only* for membranes, marked discomfort, or visually significant subepithelial infiltrates, understanding that it **prolongs viral shedding** and can cause infiltrate rebound. Povidone-iodine has some supportive evidence. **Antibiotics do nothing.**	Broad-spectrum topical antibiotic four times daily for 5 to 7 days: polymyxin-trimethoprim, erythromycin ointment, azithromycin `1%`, or a fluoroquinolone for contact lens wearers. Most cases are self-limited; treatment shortens the course and reduces transmission.	Allergen avoidance and cool compresses; **dual-acting antihistamine and mast cell stabiliser** (olopatadine, ketotifen, alcaftadine) once or twice daily; add a short pulse of loteprednol for severe flares; refrigerate the drops for symptomatic relief.	**Oral azithromycin `1 g` as a single dose** (or doxycycline `100 mg` twice daily for 7 days), **plus treatment of all sexual partners**, plus screening for other sexually transmitted infections. Topical therapy alone fails. Report as required by local law.

HYPERACUTE PURULENT CONJUNCTIVITIS: THE 24-HOUR DIAGNOSIS

Copious purulent discharge that reaccumulates within minutes, marked lid oedema, severe chemosis, and a tender preauricular node means *Neisseria gonorrhoeae* until proven otherwise. It can **penetrate an intact corneal epithelium and perforate within days**.

1. **Gram stain and culture immediately** (chocolate or Thayer-Martin agar) plus nucleic acid amplification testing.
2. **Systemic therapy is mandatory**: **ceftriaxone `1 g` intramuscularly or intravenously** as a single dose for conjunctivitis without corneal involvement; admit for intravenous therapy if the cornea is involved. Add treatment for presumed chlamydial co-infection.
3. **Copious saline lavage** of the discharge, and a topical fluoroquinolone or bacitracin as adjunct.
4. **Treat sexual partners**, screen for other sexually transmitted infections including HIV and syphilis, and report per public health requirements.
5. In a **neonate (onset day 2 to 5)**, admit for systemic ceftriaxone `25–50 mg/kg`, and lavage.

NEONATAL CONJUNCTIVITIS (OPHTHALMIA NEONATORUM) BY DAY OF ONSET

Onset	Cause	Features and treatment
Day 1	Chemical	Silver nitrate prophylaxis (largely historical); mild, self-limited.
Day 2–5	***Neisseria gonorrhoeae***	Hyperacute purulent, severe lid oedema, risk of corneal perforation. **Admit; systemic ceftriaxone plus lavage.**
Day 5–14	***Chlamydia trachomatis*** (the commonest identified cause)	**Papillary, not follicular**, because the neonate cannot form follicles; often with pseudomembranes. Risk of **pneumonitis**. Treat with **oral erythromycin `50 mg/kg/day` divided four times daily for 14 days**; treat the mother and her partner. Topical therapy alone is inadequate.
Day 3–14	Other bacteria (*S. aureus*, streptococci, *Haemophilus*)	Topical antibiotic; culture if severe.
Week 1–2	**Herpes simplex**	Vesicles on the lid, and possible keratitis or systemic disease. **Systemic aciclovir** plus topical antiviral; paediatric admission for evaluation of disseminated and CNS disease.

Allergic and immune ocular surface disease

THE ALLERGIC SPECTRUM

Condition	Distinguishing features	Management
Seasonal / perennial allergic conjunctivitis	Itching, watery discharge, chemosis, fine papillae, lid oedema; no corneal involvement. Type I.	Avoidance, cool compresses, dual-acting agent; oral antihistamine if there is rhinitis. Warn that topical vasoconstrictors cause rebound.

Condition	Distinguishing features	Management
Vernal keratoconjunctivitis	**Boys, warm dry climates, seasonal, ages 5 to 20**, usually resolving after puberty. **Giant cobblestone papillae on the superior tarsus**, limbal **Horner-Trantas dots** (eosinophil aggregates), **shield ulcer** and plaque on the superior cornea, thick ropy mucus. Eosinophils on conjunctival scraping.	Mast cell stabiliser as maintenance, a topical steroid pulse for flares, and **topical cyclosporine `0.05–2%` or tacrolimus as steroid-sparing therapy**. A shield ulcer requires steroid plus antibiotic cover and, if plaque persists, superficial keratectomy. Supratarsal steroid injection for refractory disease.
Atopic keratoconjunctivitis	**Adults** with atopic dermatitis, **perennial, chronic and progressive**. Papillae are **more inferior** than in vernal, with cicatrisation, symblepharon, punctate keratopathy, **keratoconus, anterior subcapsular (shield) cataract**, and a high risk of **herpes simplex** and staphylococcal infection.	Aggressive lubrication, dual-acting agent, cyclosporine or tacrolimus, steroid pulses, plus systemic management of the dermatitis (dupilumab has transformed this, though it can cause its own conjunctivitis). Watch for keratoconus.
Giant papillary conjunctivitis	Contact lens (or suture, prosthesis, or filtering bleb) related; papillae above `0.3 mm` on the superior tarsus, lens awareness, mucus, lens decentration and blur late in the day. Mechanical plus immunologic.	Temporarily stop lens wear, change to daily disposables and a different material, improve care and replacement compliance, and use a mast cell stabiliser. A short steroid pulse for severe cases.

Condition	Distinguishing features	Management
Contact dermatoblepharitis	Type IV reaction: **eczematous, scaly, itchy periocular skin** with conjunctival hyperaemia, appearing days after starting a drop. Commonest culprits: **neomycin, atropine, brimonidine and apraclonidine, preservatives, and glaucoma medications**.	Stop the offending agent. Cool compresses, mild topical steroid or a preservative-free lubricant; switch to a preservative-free alternative.
Toxic (medicamentosa) conjunctivitis	Chronic redness, inferior punctate keratopathy, and follicles from chronic drop use: aminoglycosides, antivirals, benzalkonium chloride, and over-the-counter vasoconstrictors.	Withdraw the agent and support the surface with preservative-free lubricants. Improvement takes weeks.
Superior limbic keratoconjunctivitis	Bilateral superior bulbar hyperaemia and staining with superior tarsal papillae and filaments, symptoms out of proportion to signs. **Associated with thyroid disease** and with lid-globe apposition.	Lubrication, bandage lens, silver nitrate `0.5%` application, or conjunctival resection or thermal cautery; test thyroid function.
Mucous membrane (ocular cicatricial) pemphigoid	Type II autoimmune; **progressive subconjunctival fibrosis, fornix foreshortening, symblepharon, entropion, trichiasis, keratinisation**, in an older patient, often bilateral and asymmetric, with oral involvement.	**Requires systemic immunosuppression**: dapsone, mycophenolate, methotrexate, or rituximab, coordinated with dermatology or rheumatology. Aggressive lubrication and lid surgery for trichiasis. **Surgery in active disease accelerates cicatrisation**: suppress first.

Condition	Distinguishing features	Management
Stevens-Johnson syndrome / toxic epidermal necrolysis	Acute mucocutaneous emergency after sulfonamides, anticonvulsants, allopurinol, NSAIDs, or *Mycoplasma*; pseudomembranous conjunctivitis with epithelial sloughing.	**Acute-phase intervention changes the outcome:** aggressive preservative-free lubrication, symblepharon lysis with a glass rod daily, topical steroid and antibiotic, and **amniotic membrane transplantation early**, which substantially reduces late cicatrisation and limbal stem cell deficiency. Coordinate with the burns or dermatology team.

Trachoma and public health

TRACHOMA: THE LEADING INFECTIOUS CAUSE OF BLINDNESS WORLDWIDE

Chlamydia trachomatis serotypes **A, B, Ba, and C**, transmitted by eye-seeking flies, fomites, and close contact in crowded, water-poor settings.

WHO simplified grading (FISTO): **TF** trachomatous inflammation, follicular (five or more follicles on the upper tarsus); **TI** intense inflammation (papillary thickening obscuring more than half the tarsal vessels); **TS** scarring (**Arlt line**, a linear tarsal scar); **TT** trichiasis; **CO** corneal opacity. Limbal follicle resolution leaves **Herbert pits**.

WHO SAFE strategy: **S**urgery for trichiasis, **A**ntibiotics (mass azithromycin `20 mg/kg` single dose), **F**acial cleanliness, and **E**nvironmental improvement (water and sanitation). Note that the last two, which are not medical interventions at all, are the ones that interrupt transmission durably.

Dry eye disease

CLASSIFICATION AND ITS CONSEQUENCE FOR TREATMENT

Subtype	Mechanism and findings	What it changes
Evaporative (lipid-deficient)	**Meibomian gland dysfunction**: capped orifices, thick meibum, gland dropout, short tear break-up time with a normal or high tear volume.	Treat the lids: heat, expression, doxycycline, azithromycin, thermal pulsation. **The commonest subtype by a wide margin.**
Aqueous-deficient	Reduced lacrimal secretion: low Schirmer, low tear meniscus, high osmolarity. Sjögren and non-Sjögren.	Add volume and retain it: frequent preservative-free tears, punctal occlusion, secretagogues, anti-inflammatory therapy. **Test for Sjögren** (anti-SS-A/Ro, anti-SS-B/La, RF, ANA) in a young or severe case.
Mixed	Both mechanisms; the usual clinical reality.	Treat both arms.

Subtype	Mechanism and findings	What it changes
Neurotrophic / neuropathic	**Symptoms far exceeding signs** (pain without staining), or the reverse in true neurotrophic disease with reduced corneal sensation. Post-LASIK, herpetic, diabetic, chronic preservative exposure.	Corneal sensation testing changes the plan: neurotrophic disease needs protection (autologous serum, cenegermin, tarsorrhaphy, scleral lens), while neuropathic pain needs systemic neuromodulation and often does not respond to more tears.

STAGED MANAGEMENT, FOLLOWING THE TFOS DEWS II STRUCTURE

1. **Stage 1.** Educate on the chronic nature and on modifiable factors: environment (humidity, fans, air conditioning), **screen use with reduced blink rate and the 20-20-20 habit**, dietary and fluid intake, review of systemic medications (antihistamines, anticholinergics, diuretics, isotretinoin, beta-blockers, hormone therapy), lid hygiene and warm compresses, and **preservative-free artificial tears** if used more than four times daily. Treat contributing lid and surface disease.
2. **Stage 2.** Add non-preserved tears if not already, **overnight ointment or gel**, tea tree derivative or lotilaner for *Demodex*, **punctal occlusion** (collagen or silicone plugs; occlude the inferior punctum first), a moisture chamber or humidifier, **in-office thermal expression or intense pulsed light**, and prescription anti-inflammatory therapy: **topical cyclosporine `0.05%` or `0.09%` twice daily** (allow 3 to 6 months), **lifitegrast `5%` twice daily** (faster onset, dysgeusia is common), **a short pulse of loteprednol `0.5%`**, **oral doxycycline or azithromycin**, or **varenicline nasal spray** as a secretagogue.
3. **Stage 3. Oral secretagogues** (pilocarpine, cevimeline for Sjögren), **autologous serum** or plasma-derived drops, **therapeutic contact lens options** (soft bandage lens, then a rigid scleral lens, which is often transformative in severe disease), and amniotic membrane.
4. **Stage 4.** Longer-term topical steroid, **surgical punctal occlusion (cautery)**, **tarsorrhaphy**, salivary gland transplantation, and management of any cicatrising disease.

Key sequencing point: never occlude the puncta in an eye with significant surface inflammation without first controlling the inflammation, because you are then retaining inflammatory tears. Similarly, treat the lids before adding volume in an evaporative case.

DRY EYE TESTS AND WHAT THEY ADD

Test	Value
Symptom questionnaire (OSDI, SPEED, DEQ-5)	Establishes the diagnosis and tracks change; symptoms and signs correlate poorly, so measure both.
Tear break-up time	Under `10 s` (and especially under `5 s`) supports evaporative disease. Non-invasive measurement is preferred.
Tear osmolarity	`>308 mOsm/L`, or an inter-eye difference above `8 mOsm/L`, supports the diagnosis and is one of the earliest changes.
Ocular surface staining	Fluorescein for the cornea, **lissamine green or rose bengal for the conjunctiva and lid margin**. Interpalpebral staining suggests exposure and evaporation; inferior staining suggests toxicity or exposure; superior staining suggests SLK or a mechanical cause. **Lid wiper epitheliopathy** is an early sign.
Schirmer I / phenol red thread	`<10 mm / 5 min` and `<10 mm / 15 s` respectively support aqueous deficiency; `<5 mm` is severe.

Test	Value
Meibography and expression	Quantifies gland dropout and secretion quality; useful for demonstrating the problem to the patient.
MMP-9 point-of-care testing	Confirms an inflammatory component, supporting anti-inflammatory therapy.
Corneal sensation	Separates neurotrophic from neuropathic from ordinary dry eye; changes the entire plan.

Conjunctival lesions and degenerations

LUMPS AND PIGMENT ON THE CONJUNCTIVA

Lesion	Features	Management
Pinguecula	Yellow-white elastotic nodule at the nasal or temporal limbus, not crossing onto the cornea; may become inflamed (pingueculitis).	Lubricants, UV protection, short topical steroid or NSAID pulse for inflammation; excision only for cosmesis or contact lens intolerance.
Pterygium	Fibrovascular wing crossing the limbus, destroying Bowman layer, with a **Stocker line** of iron at the leading edge; induces with-the-rule flattening and astigmatism.	UV protection and lubrication. Excise for encroachment on the visual axis (within `3–4 mm`), induced astigmatism, restriction of motility, recurrent inflammation, or contact lens intolerance. **Conjunctival autograft (or amniotic membrane) plus adjunctive mitomycin C reduces recurrence far more than bare sclera closure**, which has an unacceptable recurrence rate.
Conjunctival naevus	Pigmented, mobile with the conjunctiva, usually bulbar and juxtalimbal, with **intralesional cysts**, present since childhood, and may darken at puberty or in pregnancy.	Photograph and observe. **A pigmented lesion in the fornix, on the palpebral conjunctiva, or on the caruncle is atypical for a naevus and needs biopsy.**
Primary acquired melanosis	**Flat, patchy, acquired in middle age**, unilateral, indistinct borders. With atypia it progresses to melanoma in a substantial minority.	Map biopsy, then excision with cryotherapy or topical mitomycin C or interferon for diffuse disease; lifelong surveillance.
Racial (complexion-associated) melanosis	**Bilateral, symmetric, perilimbal, flat**, present from youth in darker-skinned individuals.	Benign; observe.
Conjunctival melanoma	Elevated, nodular, feeder vessels, may be amelanotic; arises from PAM with atypia, a naevus, or de novo.	Wide excision with no-touch technique, cryotherapy, and sentinel node consideration; **the conjunctiva has lymphatics**, so nodal metastasis to preauricular and submandibular nodes occurs.

Lesion	Features	Management
Ocular surface squamous neoplasia	Gelatinous, leukoplakic, or papilliform limbal lesion with **corkscrew feeder vessels**; UV and HPV related, and commoner in immunosuppression. Rose bengal stains it.	Excision with margins plus cryotherapy, or topical chemotherapy (interferon alfa-2b, mitomycin C, or 5-fluorouracil). **Consider HIV testing in a young patient.**
Conjunctival lymphoma	**Salmon-pink**, smooth, mobile, painless, often in the fornix; may be bilateral.	Biopsy with flow cytometry and clonality studies; systemic staging. MALT lymphoma is often treatable with local radiotherapy; consider *Chlamydia psittaci* association.
Papilloma	Pedunculated, fleshy, with a corkscrew vascular core; HPV 6 and 11; commoner in children at the caruncle and fornix.	Observation (many regress), excision with cryotherapy, or topical or oral cimetidine or interferon; excision alone can seed further lesions.
Subconjunctival haemorrhage	Flat, sharply bordered blood, painless, sudden.	Reassure and observe; resolves in 1 to 2 weeks. **Investigate if recurrent (check blood pressure, platelet count, coagulation, and anticoagulant use), if bilateral in a child (consider non-accidental injury), or if there is any chance of an occult perforation or of a 360-degree bullous haemorrhage after trauma.**
Conjunctival amyloid	Waxy, salmon-coloured, bleeds easily; Congo red positive with apple-green birefringence.	Biopsy; evaluate for systemic amyloidosis.
Concretions	Small yellow-white epithelial inclusions on the palpebral conjunctiva.	Remove only if eroding through and causing foreign body sensation.
Conjunctivochalasis	Redundant inferior conjunctiva riding over the lid margin, disrupting the tear meniscus and causing epiphora and irritation.	Lubrication; surgical resection or cautery if symptomatic and refractory. Frequently mistaken for isolated dry eye.

RED EYE FEATURES THAT MEAN IT IS NOT CONJUNCTIVITIS

- **Reduced visual acuity** that does not clear with a blink.
- **Ciliary flush**: circumlimbal violaceous injection that does not blanch with phenylephrine.
- **Photophobia** and true pain rather than grittiness.
- **Anterior chamber cells, flare, or hypopyon.**
- **Corneal infiltrate, ulcer, or oedema.**
- **Abnormal pupil**, either miotic and sluggish, or irregular from synechiae.
- **Raised or very low IOP.**
- **Unilateral and severe with a contact lens history.**

Any one of these reclassifies the case as keratitis, uveitis, scleritis, angle closure, or endophthalmitis, and changes the management entirely. Exam cases are frequently constructed exactly this way: a conjunctivitis-sounding history with one of these findings buried in the data.

RETRIEVAL PRACTICE

1. Watery discharge, inferior follicles, petechiae, a tender preauricular node, and a family member with a red eye. Diagnosis, treatment, and the single most important patient instruction.
2. When are subepithelial infiltrates expected in adenoviral disease, and what is the risk of treating them with steroid?
3. Copious purulent discharge reaccumulating within minutes in a sexually active adult. Diagnosis, and give the systemic treatment with dose and route.
4. A 9-day-old has a papillary conjunctivitis with pseudomembranes. Most likely organism, treatment with dose, and the systemic complication to warn about.
5. Why does the neonate not form follicles?
6. A 12-year-old boy has seasonal itching, giant superior tarsal papillae, and limbal white dots. Diagnosis, the name of the limbal finding, and the treatment ladder.
7. Distinguish vernal from atopic keratoconjunctivitis on three features.
8. Give the four WHO SAFE components and say which two interrupt transmission durably.
9. State the four stages of dry eye management and the one sequencing error to avoid.
10. A patient reports severe burning with essentially no staining and normal Schirmer. What must you test, and how does the answer change management?
11. Which stain do you use for the conjunctiva and lid margin, and what early sign does it reveal?
12. A flat, patchy, unilateral acquired pigmentation in a 55-year-old. Diagnosis, the risk, and management.
13. A gelatinous limbal lesion with corkscrew vessels in a 32-year-old. Diagnosis, treatment options, and what systemic test you should consider.
14. Name eight red eye features that exclude simple conjunctivitis.
15. Give three indications to excise a pterygium and the operation that minimises recurrence.

KEY

1. Adenoviral (epidemic) keratoconjunctivitis; supportive treatment with cool compresses and preservative-free lubricants; the key instruction is **hygiene and isolation**, because it is highly contagious for 10 to 14 days.
2. At **7 to 14 days**; steroid relieves them but **prolongs viral shedding** and causes rebound of the infiltrates on withdrawal.
3. *Neisseria gonorrhoeae* hyperacute conjunctivitis; **ceftriaxone `1 g` intramuscularly or intravenously** as a single dose (admit for intravenous therapy if the cornea is involved), plus chlamydial cover, lavage, partner treatment, and screening.
4. *Chlamydia trachomatis*; **oral erythromycin `50 mg/kg/day` in four divided doses for 14 days**; warn about **chlamydial pneumonitis**, and treat the mother and her partner.
5. The adenoid (lymphoid) layer of the conjunctival substantia propria does not develop until about 2 to 3 months of age.
6. Vernal keratoconjunctivitis; **Horner-Trantas dots**; mast cell stabiliser maintenance, steroid pulses for flares, cyclosporine or tacrolimus as steroid-sparing therapy, supratarsal steroid injection and superficial keratectomy for a persistent shield ulcer plaque.
7. Vernal: boys, seasonal, superior giant papillae, resolves after puberty. Atopic: adults with dermatitis, perennial and progressive, more inferior papillae with cicatrisation, plus keratoconus and shield cataract.
8. Surgery, Antibiotics, Facial cleanliness, Environmental improvement; **facial cleanliness and environmental improvement** interrupt transmission durably.
9. Stage 1 education and lubricants and lid care; stage 2 non-preserved tears, punctal occlusion, in-office procedures, and prescription anti-inflammatories; stage 3 secretagogues, autologous serum, therapeutic and scleral lenses; stage 4 chronic steroid, surgical occlusion, tarsorrhaphy. **Do not occlude the puncta before controlling surface inflammation.**
10. **Corneal sensation.** Reduced sensation means neurotrophic disease requiring protection (autologous serum, cenegermin, scleral lens, tarsorrhaphy); normal sensation with symptoms far exceeding signs suggests **neuropathic pain**, which needs neuromodulation rather than more tears.
11. **Lissamine green** (or rose bengal); it reveals **lid wiper epitheliopathy**, an early marker of ocular surface disease.
12. Primary acquired melanosis; with atypia it carries a significant risk of progression to melanoma;

management is map biopsy, excision with cryotherapy or topical chemotherapy for diffuse disease, and lifelong surveillance.

13. Ocular surface squamous neoplasia; excision with margins plus cryotherapy, or topical interferon alfa-2b, mitomycin C, or 5-fluorouracil; consider **HIV testing** given the young age.
14. Reduced acuity, ciliary flush that does not blanch, true pain and photophobia, anterior chamber cells or flare or hypopyon, corneal infiltrate or ulcer or oedema, abnormal pupil, abnormal IOP, and a contact lens history with severe unilateral disease.
15. Encroachment within 3 to 4 mm of the visual axis, significant induced astigmatism, restricted motility, recurrent inflammation, or contact lens intolerance; **excision with a conjunctival autograft** (with or without adjunctive mitomycin C) rather than bare sclera closure.

CHAPTER RECAP

1. Follicles plus a preauricular node plus watery discharge means viral; itch means allergic; hyperacute purulence means gonococcal.
2. Adenovirus is treated with hygiene, not antibiotics, and steroid prolongs shedding.
3. Gonococcal disease needs systemic ceftriaxone; chlamydial disease needs oral azithromycin and partner treatment.
4. Neonatal conjunctivitis is dated: day 2 to 5 gonococcal, day 5 to 14 chlamydial with pneumonitis risk.
5. Vernal is a seasonal boy with superior giant papillae and Horner-Trantas dots; atopic is a chronic adult with cicatrisation, keratoconus, and shield cataract.
6. Dry eye: identify the subtype, treat the lids in evaporative disease, and control inflammation before occluding puncta.
7. Corneal sensation separates neurotrophic from neuropathic disease and changes everything.
8. Acquired unilateral patchy pigmentation is PAM and needs mapping, not reassurance.
9. Corkscrew vessels with a gelatinous limbal lesion is ocular surface squamous neoplasia.
10. Eight red flags exclude simple conjunctivitis; exam cases hide one of them in the data.

CHAPTER 4 · DISEASE AND MANAGEMENT

Cornea, keratitis, ectasia, refractive surgery

Conjunctiva / cornea / refractive surgery · 33–44 items | TMOD-heavy | Emergencies / trauma | Contact lenses

ANSWER THESE COLD BEFORE READING

1. Give the empirical regimen for bacterial keratitis with drug, concentration, frequency, and review interval.
2. State when to culture a corneal ulcer and what you scrape.
3. Give the full HSV keratitis treatment by subtype, with the HEDS conclusions.
4. List the absolute and relative contraindications to LASIK.
5. Name the signs of corneal graft rejection and the immediate treatment.
6. Give the management of an alkali burn in order, and the prognostic sign.

BLUEPRINT WEIGHT

33–44

Tied for the largest disease area, and disproportionately represented in TMOD items.

Infectious keratitis

MATCHING ORGANISM TO PRESENTATION, AND THE EXACT TREATMENT

Organism	Presentation	Treatment
Bacterial (*Pseudomonas*, staphylococci, streptococci, *Moraxella*)	Rapid onset pain, photophobia, reduced vision, purulent discharge, **epithelial defect overlying a dense stromal infiltrate**, anterior chamber reaction with or without hypopyon, and lid oedema. *Pseudomonas*: contact lens wear, thick adherent mucopurulent discharge, rapid melting, ring infiltrate.	**Monotherapy for most cases**: moxifloxacin `0.5%` or besifloxacin `0.6%`, one drop **every hour day and night for 24 to 48 hours**, then taper by response. **Fortified dual therapy for severe, central, or vision-threatening ulcers**: fortified vancomycin `25-50 mg/mL` alternating with fortified tobramycin `14 mg/mL` every 30 to 60 minutes. Add a cycloplegic (homatropine `5%` two or three times daily) for pain and synechiae prevention. **Discontinue contact lenses. No steroid initially.** Review at **24 hours**, and consider adding steroid only after the organism is identified, the infiltrate is culture-proven bacterial, and it is clearly improving on antibiotic (the SCUT trial showed no overall benefit but suggested benefit in deep central ulcers and in non-*Nocardia* infections).

Organism	Presentation	Treatment
Fungal, filamentous (*Fusarium*, *Aspergillus*)	**Vegetative trauma** or contaminated lens solution; grey-white infiltrate with **feathery margins**, satellite lesions, an endothelial plaque, hypopyon, and a **relatively intact epithelium** with an indolent course over days to weeks.	**Natamycin `5%` hourly** (the Mycotic Ulcer Treatment Trial favoured natamycin over voriconazole, especially for *Fusarium*), plus epithelial debridement to aid penetration. Add oral voriconazole for deep or scleral involvement. Treat for weeks and taper slowly. **Steroids are contraindicated.**
Fungal, yeast (*Candida*)	Compromised surface: chronic steroid, dry eye, prior keratoplasty. Indolent, discrete, sometimes with a "cracked windshield" appearance.	Amphotericin B `0.15%` topically; consider oral fluconazole.
Acanthamoeba	Contact lens plus **water exposure**; **pain grossly out of proportion to the signs**, early **radial perineuritis**, then a **ring infiltrate**, pseudodendrites, and a protracted course frequently misdiagnosed as herpes simplex.	**Biguanide (PHMB `0.02–0.08%` or chlorhexidine `0.02–0.2%`) plus a diamidine (propamidine or hexamidine)**, hourly initially, then tapered over **6 to 12 months**. Debride epithelium early. Steroid worsens outcomes and should be avoided or used only late and cautiously for severe inflammation. Confocal microscopy and non-nutrient agar with an *E. coli* overlay confirm it.
Herpes simplex, epithelial	**Dendrite with terminal bulbs** and swollen epithelial borders; the ulcer bed stains with fluorescein and the borders with lissamine green or rose bengal. **Reduced corneal sensation.** Recurrent and unilateral.	**Ganciclovir `0.15%` gel five times daily** until healed then three times daily for a week, *or* trifluridine `1%` nine times daily, *or* **oral aciclovir `400 mg` five times daily for 7 to 10 days** (better tolerated, and preferable in children and in poor compliers). Debridement is optional. **No steroid for purely epithelial disease.**

Organism	Presentation	Treatment
Herpes simplex, stromal (immune, non-necrotising)	Stromal haze and oedema with an **intact epithelium**, a **Wessely immune ring**, neovascularisation, and no dendrite. Type III and IV immune reaction to viral antigen.	**Topical steroid (prednisolone acetate `1%`) with antiviral cover**: HEDS showed topical steroid plus trifluridine shortened resolution and reduced treatment failure. Add **oral aciclovir `400 mg` twice daily prophylaxis**, which HEDS showed **halves recurrence** (to about `19%` from `32%` at one year). Taper steroid very slowly over months.
Herpes simplex, endotheliitis (disciform)	Central disc of stromal oedema with keratic precipitates and an intact epithelium; often with raised IOP.	Topical steroid plus oral antiviral, and IOP control (avoid prostaglandins, which can reactivate herpes).
Herpes simplex, neurotrophic	Non-healing oval ulcer with smooth rolled margins, minimal inflammation, and **absent sensation**. It is not active infection.	Stop toxic drops, lubricate aggressively with preservative-free agents, bandage lens or amniotic membrane, autologous serum, **cenegermin (recombinant nerve growth factor)**, tarsorrhaphy. Antivirals and steroids do not heal it.
Herpes zoster ophthalmicus	Dermatomal V1 vesicles, **pseudodendrites without terminal bulbs** (elevated mucous plaques), stromal keratitis, uveitis with **sectoral iris atrophy** and raised IOP, scleritis, cranial neuropathies, and post-herpetic neuralgia. **Hutchinson sign** predicts ocular involvement.	**Oral antiviral within 72 hours**: valaciclovir `1 g` three times daily, or famciclovir `500 mg` three times daily, or aciclovir `800 mg` five times daily, for **7 to 10 days**. Topical steroid for stromal keratitis and uveitis with antiviral cover. Cycloplegia. Refer for pain management (gabapentin, pregabalin, amitriptyline) and consider low-dose antiviral prophylaxis for recurrent disease. Recommend the **recombinant zoster vaccine** for prevention in eligible patients.

WHEN TO CULTURE, AND HOW

Culture before treating if the infiltrate is **central, larger than about `1 to 2 mm`, involves the deep stroma or is associated with an anterior chamber reaction, is unresponsive to initial therapy, or occurs in an atypical host** (post-surgical, contact lens wearer, immunocompromised, or after trauma with organic matter).

Technique: topical anaesthetic, then scrape the **edge and base** of the ulcer with a sterile blade or spatula (not the centre of the necrotic slough), plate directly onto blood, chocolate, and Sabouraud agar plus thioglycollate broth, and make smears for Gram and Giemsa. Add non-nutrient agar for *Acanthamoeba* and PCR for viruses when suspected. Culture the contact lens, case, and solutions as well.

STERILE AND IMMUNE KERATITIS: DO NOT TREAT THESE AS INFECTION

Entity	Features and management
Staphylococcal marginal (catarrhal) infiltrate	Small peripheral infiltrate at **2, 4, 8, or 10 o'clock** with a **clear zone separating it from the limbus**, minimal pain, minimal or no staining, and associated blepharitis. A type III response to exotoxin. Treat the blepharitis plus a mild topical steroid with antibiotic cover; resolves in days.
Phlyctenulosis	Limbal nodule with intense vascularisation, a type IV response to staphylococcal or tuberculous antigen. Treat the source; topical steroid. Test for tuberculosis where indicated.
Thygeson superficial punctate keratitis	Bilateral, recurrent, **coarse elevated grey-white intraepithelial opacities** with minimal or no conjunctival injection, marked photophobia, and a course of years. Treat with a low-potency topical steroid, cyclosporine, or a bandage lens; it is exquisitely steroid-responsive and relapses on withdrawal.
Peripheral ulcerative keratitis	**Crescentic peripheral thinning with an epithelial defect, an inflammatory infiltrate, and often adjacent scleritis**, progressing circumferentially and centrally. **This is a systemic vasculitis presenting in the eye.** Order ANCA, RF and anti-CCP, ANA, ESR and CRP, urinalysis, and chest imaging, and refer urgently for **systemic immunosuppression** (cyclophosphamide or rituximab in ANCA disease). Topical steroid alone risks perforation. Doxycycline for MMP inhibition and a bandage lens or glue for impending perforation.
Mooren ulcer	Idiopathic peripheral ulcerative keratitis with an **undermined leading edge** and no scleritis and no systemic disease. Diagnosis of exclusion; treat with topical steroid or cyclosporine and systemic immunosuppression.
Interstitial keratitis	**Mid-stromal vascularisation and scarring with an intact epithelium**; **ghost vessels** when quiescent. Causes: congenital and acquired **syphilis** (classically bilateral, with Hutchinson teeth and deafness), tuberculosis (sectoral), Lyme, herpes, **Cogan syndrome** (with vestibuloauditory dysfunction, a rheumatologic emergency), and onchocerciasis. Investigate systemically; treat the cause plus topical steroid.
Exposure keratopathy	Inferior staining from lagophthalmos: CN VII palsy, proptosis, nocturnal lagophthalmos, sedation. Lubricate, ointment and taping or moisture chamber overnight, then tarsorrhaphy or a lid weight.

Epithelial trauma and dystrophy

ABRASION, EROSION, AND FOREIGN BODY

Problem	Management
Corneal abrasion	Anaesthetise, examine for a retained foreign body (**evert both lids**), rule out perforation with Seidel. Treat with **topical antibiotic (erythromycin ointment or a fluoroquinolone drop) four times daily**, a cycloplegic for a large or painful abrasion, oral analgesia, and a lubricant. **Do not patch a contact lens-related or vegetative-trauma abrasion** (infection risk), and **never dispense topical anaesthetic**. Topical NSAID may reduce pain but carries a melting risk in a compromised surface. Review in 24 hours if central or large.
Recurrent corneal erosion	Pain on waking, from failed epithelial adhesion (prior abrasion, or **epithelial basement membrane dystrophy**, so examine both eyes for maps, dots, and fingerprints). Manage with hypertonic saline `5%` ointment nightly for months, aggressive lubrication, doxycycline plus a short topical steroid pulse (an MMP-directed regimen), a bandage lens, then **anterior stromal micropuncture** or **phototherapeutic keratectomy**, or epithelial debridement with diamond burr polishing.
Corneal foreign body	Remove under slit lamp with a needle or spud; remove a **rust ring** with a burr, in one or two visits. Assess depth and rule out penetration (Seidel, gonioscopy, dilated examination, CT if a high-velocity mechanism). Then treat as an abrasion. **A hammering, grinding, or metal-on-metal mechanism mandates exclusion of an intraocular foreign body.**
UV photokeratitis	Welding without protection, snow, or tanning lamps; diffuse punctate staining with severe pain 6 to 12 hours later. Cycloplegia, lubricants, oral analgesia; resolves in 24 to 48 hours. This is an occupational and public health teaching moment.
Chemical burn	**Irrigation before anything else, including before measuring acuity.** Anaesthetise, evert lids and sweep the fornices for particulate matter, and irrigate with `1–2 L` of saline or lactated Ringer until the **pH normalises to 7.0 to 7.4**, rechecking after 5 to 10 minutes. Alkali (lye, ammonia, lime, cement) saponifies lipid and penetrates deeply and progressively; acid coagulates protein and self-limits, except hydrofluoric acid. **The extent of perilimbal blanching is the key prognostic sign** (Roper-Hall and Dua grading). Then topical antibiotic, cycloplegia, preservative-free lubricants, topical steroid (short course, for inflammation), **oral doxycycline and vitamin C** to inhibit collagenase and support collagen synthesis, IOP monitoring, and daily symblepharon lysis. Refer severe burns for amniotic membrane or limbal stem cell transplantation.

Ectasia

KERATOCONUS AND ITS RELATIVES

Condition	Signs	Management
Keratoconus	Progressive irregular myopic astigmatism, **scissors reflex** on retinoscopy, distorted keratometry mires, **Vogt striae** (vertical stromal lines that disappear on digital pressure), **Fleischer ring** (iron at the cone base), **Munson sign** (V-shaped lower lid indentation on downgaze), **Rizzuti sign**, corneal thinning at the cone apex, and apical scarring. Inferior steepening on topography with posterior elevation. Associations: **eye rubbing, atopy, Down syndrome, Leber congenital amaurosis, connective tissue disease, obstructive sleep apnoea, floppy eyelid syndrome**.	**Stop eye rubbing** and treat the allergy driving it: this is the single most important behavioural intervention. Spectacles then **rigid gas permeable, hybrid, piggyback, or scleral lenses** for optical correction. **Corneal collagen cross-linking** (riboflavin plus UV-A) halts progression and is indicated for documented progression, especially in patients under 30, ideally at a corneal thickness above `400 µm`. Intrastromal corneal ring segments for contact-lens-intolerant eyes. **Deep anterior lamellar or penetrating keratoplasty** for scarring or intolerance. **Keratoconus is an absolute contraindication to LASIK.**
Acute corneal hydrops	Sudden pain, photophobia, and vision loss with dramatic stromal oedema from a **break in Descemet membrane**.	Hypertonic saline `5%`, cycloplegia, a bandage lens for comfort, topical steroid cautiously, and time: it resolves over **2 to 4 months** leaving a scar that sometimes flattens the cone and improves the fit. Intracameral gas can shorten the course.
Pellucid marginal degeneration	**Inferior peripheral** thinning with the cornea protruding *above* the thin band; high against-the-rule astigmatism; "crab claw" topography.	Scleral or specialty lenses; surgery is difficult because the thinning is peripheral. Distinguish from keratoconus, where the thinning is at the apex of the protrusion.

Condition	Signs	Management
Keratoglobus	Global thinning with generalised protrusion, extreme fragility, and a high rupture risk; associated with Ehlers-Danlos type VI.	Protective eyewear at all times; surgery is high risk.
Post-refractive surgery ectasia	Progressive steepening and thinning months to years after LASIK, from insufficient residual stromal bed or unrecognised preoperative forme fruste keratoconus.	Rigid or scleral lenses, cross-linking, ring segments, or keratoplasty. This is why preoperative tomography with posterior elevation is mandatory.

Endothelial disease and corneal oedema

CAUSES AND MANAGEMENT OF CORNEAL OEDEMA

Cause	Features and management
Fuchs endothelial corneal dystrophy	**Central guttae, a beaten-metal endothelial appearance, pigment dusting, then stromal oedema and epithelial bullae**, with vision worst on waking that improves through the day. Female predominance; *TCF4* repeat expansion. Manage with hypertonic saline `5%` drops and ointment, a hair dryer at arm's length in the morning, IOP reduction, a bandage lens for ruptured bullae, then **DMEK or DSEK** (endothelial keratoplasty gives faster recovery and lower rejection risk than penetrating keratoplasty). **Warn before cataract surgery: pre-existing Fuchs greatly increases the risk of postoperative decompensation, and a combined or staged procedure may be needed.**
Pseudophakic and aphakic bullous keratopathy	Endothelial loss after surgery, particularly with a vitreous-touching or malpositioned IOL. Same medical measures, then endothelial keratoplasty; IOL exchange if it is the cause.
Posterior polymorphous dystrophy	Vesicular and band lesions with endothelial metaplasia; may cause iridocorneal adhesions and glaucoma. Often asymptomatic; monitor IOP.
Congenital hereditary endothelial dystrophy	Bilateral diffuse ground-glass oedema from birth with a thick cornea and no photophobia; distinguish from congenital glaucoma (raised IOP, buphthalmos, Haab striae) and from birth trauma (unilateral, vertical Descemet breaks).
Iridocorneal endothelial syndrome	**Unilateral**, in a middle-aged woman: abnormal endothelium migrating across the angle and iris, causing peripheral anterior synechiae, **corectopia and polycoria**, iris atrophy, and secondary glaucoma. Three variants: Chandler (oedema dominant), progressive iris atrophy, and Cogan-Reese (iris naevus). Manage the glaucoma; keratoplasty for oedema.
Acute IOP elevation	Microcystic epithelial oedema in acute angle closure; the oedema clears as pressure falls.

CORNEAL GRAFT REJECTION: RECOGNISE AND TREAT WITHIN HOURS

Symptoms: redness, pain or discomfort, photophobia, reduced vision, in a previously clear graft. **Signs**: circumcorneal injection, anterior chamber cells, **keratic precipitates on the donor endothelium**, a **Khodadoust line** (an advancing linear front of endothelial rejection), subepithelial infiltrates (epithelial or stromal rejection), graft oedema, and new vascularisation crossing the graft-host junction.

Treatment: **intensive topical steroid immediately** (prednisolone acetate `1%` hourly, or difluprednate), consider oral or periocular steroid for severe endothelial rejection, and contact the surgeon the same day. Prompt treatment usually reverses it; delay causes permanent endothelial loss.

Risk factors: corneal vascularisation, prior rejection or regraft, large or eccentric grafts, active inflammation at surgery, loose or broken sutures, herpetic disease, and young age. **Endothelial keratoplasty (DMEK) has a substantially lower rejection rate than penetrating keratoplasty.** Remember that graft failure without inflammation (primary donor failure or late endothelial decompensation) is a different problem and steroids will not fix it.

Refractive surgery: selection, complications, comanagement

CANDIDATE ASSESSMENT

Requirement	Detail
Stable refraction	No more than about `0.50 D` change over `12 months`; generally age `18` or older, and realistically 21 or older.
Adequate corneal thickness	Residual stromal bed at least `250 to 300 µm` and total residual thickness at least `400 µm`. Ablation is roughly `12 to 15 µm` per dioptre at a `6 mm` optical zone.
Normal tomography	No inferior steepening, no abnormal posterior elevation, symmetric and orthogonal astigmatism, normal thinnest-point location. This is the single most important screening step for ectasia risk.
Healthy ocular surface	Treat dry eye and blepharitis *before* surgery; untreated dry eye guarantees a dissatisfied postoperative patient.
Realistic expectations	Discuss presbyopia (a myope losing near comfort after correction), night vision, dry eye, enhancement rates, and the possibility of needing glasses.

CONTRAINDICATIONS

Absolute	Relative
• **Keratoconus, pellucid marginal degeneration, or any ectasia**, including forme fruste • Insufficient corneal thickness for a safe residual bed • Unstable refraction • Uncontrolled autoimmune or collagen vascular disease • Uncontrolled glaucoma with advanced field loss • Active infection or inflammation (herpetic keratitis, uveitis) • Severe untreated dry eye or exposure keratopathy • **Pregnancy and breastfeeding** • Cataract of visual significance (choose lens surgery instead)	• Diabetes, especially if poorly controlled • Prior herpetic keratitis (needs antiviral prophylaxis) • Large pupils (night vision symptoms) • Amblyopia or monocularity • Very high myopia or hyperopia (consider a phakic IOL) • Isotretinoin or amiodarone use • Epithelial basement membrane dystrophy (favour surface ablation) • Occupation with a high risk of blunt trauma (favour PRK or SMILE over LASIK) • Unrealistic expectations or obsessive personality traits

POSTOPERATIVE COMPLICATIONS: RECOGNITION AND MANAGEMENT

Complication	Timing and features	Management
Diffuse lamellar keratitis (sands of the Sahara)	Day 1 to 5. **Sterile**, fine white granular infiltrate in the flap interface, starting peripherally and moving centrally; minimal pain and no conjunctival injection. Graded 1 to 4.	Intensive topical steroid, escalating with grade; **grade 3 to 4 requires flap lift and irrigation** to prevent stromal melt and hyperopic shift. Distinguish from infectious keratitis, which is focal, painful, injected, and has an overlying epithelial defect.
Infectious keratitis	Focal infiltrate with pain, injection, and anterior chamber reaction. Early: staphylococci and streptococci. Late (weeks to months): **atypical mycobacteria** and fungi.	**Lift the flap, culture, and irrigate with antibiotic.** Fortified topical therapy; add amikacin or a macrolide for atypical mycobacteria. Stop steroid.

Complication	Timing and features	Management
Epithelial ingrowth	Weeks; a nest of epithelial cells growing under the flap edge, causing irregular astigmatism and, if progressive, flap melt.	Observe if peripheral and non-progressive; lift, scrape, and consider suturing, glue, or alcohol for progressive ingrowth.
Flap striae or wrinkles	Day 1; macrostriae (from flap misalignment) cause immediate blur and irregular astigmatism.	Early refloating and stretching within days; late striae become fixed.
Traumatic flap dislocation	Any time, even years later.	**Urgent refloat and repositioning.** Warn every LASIK patient that this risk is lifelong.
Dry eye	Universal to some degree from **subbasal nerve transection**; peaks in the first months and usually improves over 6 to 12 months.	Preservative-free lubricants, punctal occlusion, cyclosporine, and pre-emptive treatment before surgery. It is the commonest cause of postoperative dissatisfaction.
Haze and regression	Weeks to months, predominantly after **PRK** and with high corrections; Bowman layer does not regenerate.	Prophylactic mitomycin C at surgery, prolonged topical steroid, and UV protection; phototherapeutic keratectomy for established haze.
Night vision symptoms, glare, halo, starburst	Large pupils, small optical zones, spherical aberration, decentred ablation.	Time, brimonidine or dilute pilocarpine for pupil size, wavefront-guided retreatment for higher-order aberration. Counsel preoperatively.
Central toxic keratopathy	Week 1 to 2; central stromal opacity with hyperopic shift and striae, non-inflammatory.	Observation; steroids do not help and it improves over months.
Post-LASIK IOP measurement error	Thinner cornea means Goldmann **underestimates** IOP.	Document the preoperative pachymetry and IOP; use peripheral or alternative tonometry, and be alert for **pressure-induced stromal keratopathy**, in which a steroid-driven IOP rise causes interface fluid that makes the pressure read falsely low.

Complication	Timing and features	Management
IOL calculation later in life	Keratometry overestimates corneal power after myopic ablation, and underestimates after hyperopic ablation.	Use post-refractive-surgery formulas (Barrett True-K, Haigis-L, ASCRS calculator), supply the surgeon with **pre-refractive-surgery keratometry and refraction** if available, and set expectations for a lower predictability. **Documenting and giving the patient a copy of their pre-surgical data is a genuine clinical duty.**

RETRIEVAL PRACTICE

1. A soft contact lens wearer who sleeps in lenses has a `3 mm` central infiltrate with an epithelial defect and a hypopyon. Give the organism, the immediate steps, and the treatment regimen with frequency.
2. When do you culture a corneal ulcer, and exactly what do you scrape?
3. A gardener struck in the eye by a branch has a grey infiltrate with feathery margins and satellites. Diagnosis, first-line drug, and one thing you must not do.
4. A lens wearer who rinses lenses in tap water has pain far exceeding the signs and a ring infiltrate. Diagnosis, two drug classes, and the treatment duration.
5. A dendrite with terminal bulbs and reduced corneal sensation. Give three treatment options with doses.
6. Stromal haze with an intact epithelium and a Wessely ring in a patient with prior dendritic keratitis. Treatment, and what HEDS showed about prophylaxis?
7. Distinguish HSV from VZV corneal lesions, and give the VZV antiviral dose and time window.
8. A peripheral infiltrate at 4 o'clock with a clear limbal zone and minimal pain. Diagnosis and treatment.
9. Crescentic peripheral corneal thinning with an epithelial defect and adjacent scleritis. What is this, what do you order, and what is the danger of topical steroid alone?
10. Give the management of an alkali burn in order, and the single best prognostic sign.
11. Name five signs of keratoconus and the one behavioural intervention that matters most.
12. State the indications for corneal collagen cross-linking.
13. Name three absolute contraindications to LASIK.
14. Day 3 after LASIK there is diffuse fine granular infiltrate in the interface, minimal pain, no injection. Diagnosis, and what changes if it is grade 3?
15. List the signs of corneal graft rejection and the immediate treatment.
16. Why does Goldmann underestimate IOP after myopic LASIK, and what is pressure-induced stromal keratopathy?

KEY

1. *Pseudomonas aeruginosa*. Stop lens wear, **culture (scrape the edge and base)** and culture the lens, case and solution, then start **moxifloxacin `0.5%` hourly around the clock** (or fortified vancomycin alternating with fortified tobramycin every 30 to 60 minutes for this severity), add homatropine `5%`, no steroid initially, and review in **24 hours**.
2. Culture if central, larger than 1 to 2 mm, deep, with an anterior chamber reaction, unresponsive, or in an atypical host. Scrape the **edge and base**, not the central slough, and plate directly.
3. Filamentous fungal keratitis; **natamycin `5%` hourly** with epithelial debridement; **do not give steroid.**
4. *Acanthamoeba* keratitis; a **biguanide** (PHMB or chlorhexidine) plus a **diamidine** (propamidine or hexamidine), for **6 to 12 months**.
5. Ganciclovir `0.15%` gel five times daily; trifluridine `1%` nine times daily; or oral aciclovir `400 mg` five times daily for 7 to 10 days.
6. Topical prednisolone acetate `1%` with antiviral cover, tapered slowly. HEDS showed **oral aciclovir `400 mg` twice daily halves recurrence** (about 19% versus 32% at one year).

7. HSV gives a true dendrite with terminal bulbs and an ulcerated base; VZV gives an elevated mucous pseudodendrite without terminal bulbs. VZV: **valaciclovir 1 g three times daily for 7 to 10 days, started within 72 hours**.
8. Staphylococcal marginal (catarrhal) infiltrate, a type III reaction to exotoxin; treat the blepharitis plus a mild topical steroid with antibiotic cover.
9. **Peripheral ulcerative keratitis**, a systemic vasculitis. Order ANCA (PR3 and MPO), RF and anti-CCP, ANA, ESR and CRP, urinalysis, and chest imaging, and refer urgently. Topical steroid alone risks **perforation** because the underlying systemic vasculitis is untreated.
10. Irrigate first, before anything else, after anaesthesia and sweeping the fornices, with 1 to 2 litres until **pH is 7.0 to 7.4**; then antibiotic, cycloplegia, preservative-free lubricants, short-course steroid, oral doxycycline and vitamin C, IOP monitoring, and daily symblepharon lysis. The best prognostic sign is the **extent of perilimbal blanching**.
11. Scissors reflex, Vogt striae, Fleischer ring, Munson sign, apical thinning or scarring (also Rizzuti sign and distorted mires). The key intervention is **stopping eye rubbing** and treating the allergy driving it.
12. **Documented progression** of ectasia, especially in a patient under 30, with adequate corneal thickness (generally above 400 µm) and no significant central scarring.
13. Keratoconus or any ectasia including forme fruste, insufficient corneal thickness, and pregnancy (also unstable refraction, active herpetic keratitis, and uncontrolled autoimmune disease).
14. Diffuse lamellar keratitis; **grade 3 to 4 requires flap lift and irrigation** in addition to intensive steroid, to prevent stromal melt and hyperopic shift.
15. Injection, anterior chamber cells, keratic precipitates on the donor endothelium, a Khodadoust line, subepithelial infiltrates, graft oedema, and vessels crossing the junction. Treat with **intensive topical steroid immediately** and contact the surgeon the same day.
16. The cornea is thinner and less rigid, so applanation reads low. **Pressure-induced stromal keratopathy** is a steroid-induced IOP rise that drives fluid into the flap interface, which makes the applanated pressure read falsely low and can therefore mask dangerous hypertension.

CHAPTER RECAP

1. Bacterial keratitis: culture if central or large, fourth-generation fluoroquinolone hourly or fortified duotherapy, cycloplegia, no early steroid, review at 24 hours.
2. Vegetation means fungus and natamycin; water means *Acanthamoeba*, a biguanide plus a diamidine, and months of treatment.
3. HSV epithelial disease is antiviral only; stromal disease is steroid with antiviral cover plus prophylaxis that halves recurrence.
4. Peripheral ulcerative keratitis is a systemic vasculitis and needs systemic immunosuppression.
5. Chemical burn: irrigate before you do anything else, to a pH of 7.0 to 7.4, and read the limbal blanching.
6. Keratoconus: stop the rubbing, fit specialty lenses, cross-link documented progression, and never do LASIK.
7. Fuchs dystrophy changes cataract surgery planning and consent.
8. Graft rejection is a same-day intensive steroid diagnosis; look for a Khodadoust line.
9. DLK is sterile, diffuse, and painless; infectious keratitis is focal, painful, and injected.
10. After myopic LASIK, IOP reads low and IOL calculations need special formulas, so document and give the patient their pre-surgical data.

CHAPTER 5 · DISEASE AND MANAGEMENT

Lens, cataract, IOLs, and perioperative care

Lens / cataract / IOL / pre- and post-operative care · 17–28 items | TMOD-heavy | Ametropia

ANSWER THESE COLD BEFORE READING

1. Match cataract morphology to symptom pattern and to systemic cause.
2. State the indications for cataract surgery beyond a Snellen threshold.
3. Give the IOL formula variables and explain the sources of refractive surprise.
4. Give the postoperative visit schedule with what you check at each visit.
5. Distinguish the four causes of reduced vision in the first week after cataract surgery.
6. Give the presentation, timing, organisms, and management of postoperative endophthalmitis.

BLUEPRINT WEIGHT

17–28

Roughly double the Part I weighting. Comanagement and postoperative complications are the emphasis.

Cataract: morphology, symptoms, and cause

READING THE MORPHOLOGY BACKWARDS TO THE CAUSE

Type	Symptoms	Associations and refractive effect
Nuclear sclerosis	Gradual distance blur, reduced contrast, **colour desaturation** (blues suppressed), and better near vision without glasses.	**Myopic shift** and the "second sight" phenomenon; index change from crystallin aggregation and chromophore accumulation. Ageing, smoking, myopia.
Cortical (cuneiform) spokes	**Glare and starbursts, worse at night and in oncoming headlights**; monocular diplopia; symptoms often out of proportion to Snellen acuity.	Fibre hydration with Morgagnian globules. Ageing, diabetes, UV exposure, corticosteroids.
Posterior subcapsular	**Disproportionate near blur and glare in bright light** (because the pupil constricts onto the opacity), with better vision in dim light. Acuity drops early relative to lens appearance.	Bladder (Wedl) cells migrating posteriorly. **Corticosteroids (any route), diabetes, radiation, uveitis, high myopia, retinitis pigmentosa, vitrectomy.**

Type	Symptoms	Associations and refractive effect
Anterior subcapsular / shield	Central blur.	**Atopic dermatitis**, trauma, uveitis with anterior synechiae, alkali burn.
Snowflake (polychromatic)	Rapid onset in a young patient.	**Type 1 diabetes with poor control**, sometimes reversible with glycaemic control.
Oil droplet	Infantile.	**Galactosaemia** (galactitol accumulation), reversible with dietary restriction if early.
Rosette / stellate	Post-traumatic, may progress or stabilise.	Blunt trauma; look for a **Vossius ring** of imprinted iris pigment, angle recession, phacodonesis, and iridodialysis.
Christmas tree (polychromatic needles)	Glare.	**Myotonic dystrophy**; also age-related. Look for ptosis, frontal balding, and grip weakness.
Sunflower	Variable.	**Wilson disease** (with a Kayser-Fleischer ring) or intraocular copper (chalcosis).
Spoke-like (with cornea verticillata)	Variable.	**Fabry disease**.
Congenital nuclear, with salt-and-pepper retinopathy	Leukocoria, nystagmus.	**Congenital rubella**; also deafness and cardiac defects.
Posterior polar	Glare, and often better than expected acuity.	Autosomal dominant; **high risk of posterior capsule dehiscence at surgery**, so the surgeon must be warned.
Pseudoexfoliation-associated	Any morphology.	Target pattern on the anterior capsule, Sampaolesi line, **weak zonules and poor dilation**, and higher rates of capsule rupture, zonular dialysis, and late IOL subluxation. Also higher glaucoma risk.

WHEN TO OPERATE: THE INDICATION IS FUNCTION, NOT ACUITY

1. **The patient's visual function no longer meets their needs** and the cataract accounts for it. Ask about driving (particularly night driving and glare), reading, work, hobbies, falls, and independence. A 20/40 cortical cataract with disabling glare is a stronger indication than a 20/60 nuclear cataract in someone who does not drive.
2. **Anisometropia** from asymmetric progression producing intolerable aniseikonia.
3. **The lens is causing disease**: phacomorphic angle closure, phacolytic glaucoma, phacoantigenic uveitis, or a dislocating lens.
4. **The lens prevents management of posterior segment disease**: obscuring the view or the ability to treat diabetic retinopathy or a retinal tear.
5. **In a child**, any visually significant lens opacity is urgent because of amblyopia; a dense unilateral congenital cataract must be addressed within the first weeks of life.

Preoperative work you own as the comanaging optometrist: treat the ocular surface, identify and document Fuchs dystrophy, pseudoexfoliation, a small pupil, prior refractive surgery, **tamsulosin or other α-blocker use (floppy iris)**, amblyopia, macular pathology on OCT, and glaucoma status. Discuss target refraction and lens options honestly. Set expectations, because postoperative dissatisfaction is usually an expectations problem, not a surgical one.

IOL selection and the refractive target

IOL CALCULATION: WHAT DRIVES ERROR

Variable	Contribution and pitfalls
Axial length	The largest single source of error. Optical biometry (partial coherence or swept-source) is far more reproducible than contact ultrasound, which compresses the cornea and shortens the reading. `1 mm` of error produces roughly `2.5 to 3 D` of refractive error.
Keratometry	Errors from dry eye, contact lens warpage, or prior refractive surgery. **After myopic ablation, standard keratometry overestimates corneal power**, so the standard formula selects too weak a lens and the patient ends up hyperopic. Use post-refractive formulas (Barrett True-K, Haigis-L, the ASCRS calculator) and total corneal power measurements.
Effective lens position	Estimated by the formula from axial length, keratometry, and in newer formulas anterior chamber depth and lens thickness. Formula choice matters most at the extremes: modern formulas (Barrett Universal II, Hill-RBF, Kane, Olsen) outperform SRK/T, Holladay, and Hoffer Q, and older formulas each have known biases in short and long eyes.
Posterior corneal astigmatism	Typically about `0.3 D` against the rule, which is why toric calculations that ignore it overcorrect with-the-rule and undercorrect against-the-rule astigmatism. Modern toric calculators (Barrett Toric) account for it.
Surgically induced astigmatism	Incision location and size; a temporal clear corneal incision induces less than a superior one.

LENS OPTIONS AND WHAT TO TELL THE PATIENT

Option	Advantage	Trade-off to disclose
Monofocal, distance target	Best contrast and quality of vision; least dysphotopsia; lowest cost.	Reading glasses required.
Monofocal, monovision	Spectacle independence for many tasks without a multifocal optic.	Reduced stereopsis; trial with contact lenses first if possible; avoid in patients with a history of strabismus or amblyopia.
Toric	Corrects `>1.0–1.5 D` of regular corneal astigmatism.	Requires accurate axis alignment; rotation of `10°` loses about a third of the effect, and `30°` loses all of it. Not for irregular astigmatism.
Extended depth of focus	Good intermediate vision with fewer photic phenomena than a full multifocal.	Near vision often still needs help; some loss of contrast.
Diffractive multifocal / trifocal	Greatest spectacle independence.	**Halos and glare, reduced contrast sensitivity, and poor tolerance of any macular disease, irregular cornea, or significant dry eye.** Requires a near-perfect refractive outcome and a patient who understands neuroadaptation takes months. Not for monocular patients, established macular pathology, or glaucoma with field loss.
Light-adjustable lens	Postoperative refractive fine-tuning with UV light.	Requires multiple adjustment visits and strict UV protection until locked in.

Postoperative care and comanagement

THE STANDARD VISIT SCHEDULE AND WHAT TO CHECK

Visit	Assess	Typical medication
Day 1	Acuity (expect it to be modest, and warn the patient), IOP (a spike is common and matters in glaucoma), corneal clarity and any wound leak on Seidel, anterior chamber depth and inflammation, IOL position and centration, and pupil shape.	Topical antibiotic four times daily for a week; topical steroid four times daily tapering over 3 to 4 weeks; topical NSAID for pain and macular oedema prophylaxis.
Week 1	Improving acuity, reducing inflammation, IOP, wound integrity, corneal oedema resolving. **Reinforce shield use at night and no rubbing, no swimming, no heavy lifting, and hand hygiene before drops.**	Continue and begin taper.
Week 3–4	Refraction becoming stable, dilated fundus examination (look for cystoid macular oedema, a retinal break, and the untreated fellow eye), IOP, and posterior capsule clarity.	Steroid taper completed; continue NSAID in higher-risk eyes.
Week 4–6	**Final refraction and spectacle prescription** once the refraction is stable and the wound is healed. Plan second eye surgery. Discuss YAG timing if the capsule is already hazing.	Off drops in most eyes.

REDUCED VISION IN THE FIRST POSTOPERATIVE WEEK: FOUR CAUSES, FOUR ACTIONS

Cause	Findings	Action
Corneal oedema	Stromal thickening and folds, often with a high day-1 IOP or long surgery; worse with pre-existing Fuchs.	Continue steroid, add hypertonic saline, lower IOP, and reassure; most resolve over days to weeks.
Raised IOP	Retained viscoelastic (day 1), steroid response (weeks 2 to 6), pupillary block, or pigment. Corneal oedema and pain.	Aqueous suppressants; burp the wound or paracentesis for a very high pressure; reduce or switch steroid for a steroid response.
Excessive inflammation	Cells and flare beyond expected, fibrin, posterior synechiae. Consider retained lens fragment, uveitis history, or diabetes.	Intensify steroid, add cycloplegia, and look for a retained fragment on gonioscopy and dilated examination.
Endophthalmitis	**Pain, worsening vision, hypopyon, fibrin, vitritis, lid oedema, and a red eye getting worse rather than better.**	**Same-hour referral for tap and intravitreal antibiotics.** Do not observe.

The discriminating rule: **a postoperative eye should get better every day. Any eye that is getting worse, or that develops new pain, is an emergency until proven otherwise.**

POSTOPERATIVE COMPLICATIONS BY TIMING

Complication	Timing	Recognition and management
Acute endophthalmitis	Days 1–7 (peak 3–5)	Pain, reduced vision, hypopyon, vitritis. Commonest organism ***Staphylococcus epidermidis*** (better prognosis), then *S. aureus* and streptococci (worse). **Endophthalmitis Vitrectomy Study**: if presenting vision is **light perception only**, immediate **vitrectomy** plus intravitreal antibiotics improves outcomes; if **hand motions or better**, tap and inject alone is equivalent. Intravitreal **vancomycin** `1 mg/0.1 mL` **plus ceftazidime** `2.25 mg/0.1 mL` (or amikacin). Systemic antibiotics added no benefit in EVS. Incidence roughly `1 in 1000` or lower with intracameral antibiotic prophylaxis.
Chronic (delayed) endophthalmitis	Weeks to months	***Cutibacterium (Propionibacterium) acnes***: low-grade granulomatous uveitis with a **white capsular plaque**, initially steroid-responsive then relapsing. Also fungal. Requires vitreous biopsy with prolonged anaerobic culture, intravitreal antibiotics, and often capsulectomy or IOL removal.
Toxic anterior segment syndrome	Day 1–2	**Sterile**: diffuse "limbus to limbus" corneal oedema, marked fibrin and hypopyon, but **no pain and no vitritis**, from a contaminant or a sterilisation failure. Responds to intensive steroid. The pain and vitritis distinguish infectious endophthalmitis from TASS.
Pseudophakic cystoid macular oedema (Irvine-Gass)	Peak 4–6 weeks	Painless gradual blur with a petalloid pattern on angiography and cystoid spaces on OCT. Risk factors: capsule rupture, vitreous loss, diabetes, uveitis, retinal vein occlusion, prostaglandin use, and prior CMO in the fellow eye. Treat with a **topical NSAID plus topical steroid**, escalating to periocular or intravitreal steroid, or acetazolamide. Most resolve; prophylaxis with an NSAID is standard in higher-risk eyes.
Posterior capsule opacification	Months to years	Gradual blur and glare from residual epithelial migration (Elschnig pearls, Soemmering ring). Treat with **Nd:YAG capsulotomy** once symptomatic. Complications: IOP spike (pretreat with brimonidine or apraclonidine), IOL pitting, transient iritis, **retinal detachment (higher risk in myopes)**, and cystoid macular oedema. Wait at least `3 months` after surgery where possible.
Retinal detachment	Months to years	Risk elevated after cataract surgery, more so with axial myopia, capsule rupture with vitreous loss, young age, and after YAG capsulotomy. **Educate every patient about flashes, floaters, and a curtain, at every postoperative visit.**
Dysphotopsia	Weeks	**Positive** (arcs, streaks, halos) from the IOL edge or a multifocal optic; often improves with neuroadaptation. **Negative** (a temporal dark crescent) from shadowing by the anterior capsule edge on a square-edged in-the-bag lens; most improve, and options include a reverse optic capture, sulcus lens, or piggyback lens.
Refractive surprise	Weeks	Verify the biometry, keratometry, and the correct implanted lens power; check for a wrong lens or a wrong axis on a toric. Manage with spectacles, contact lens, IOL exchange, a piggyback lens, or corneal refractive surgery.

Complication	Timing	Recognition and management
IOL decentration or subluxation	Any time	Consider zonular weakness (**pseudoexfoliation**, trauma, Marfan, prior vitrectomy, high myopia). Late in-the-bag subluxation is classically pseudoexfoliation. Presents with blur, monocular diplopia, or a visible lens edge.
Wound leak or hypotony	Days	Seidel positive, shallow chamber, low IOP, choroidal folds. Bandage lens or suture; watch for choroidal detachment and endophthalmitis risk.
Suprachoroidal haemorrhage	Intraoperative or days	Sudden pain, raised IOP, and a dark dome on examination; risk with hypertension, anticoagulation, and axial myopia.

COMANAGEMENT POINTS THAT PRODUCE EXAM ITEMS

- **Tamsulosin and other α_1-blockers** cause intraoperative floppy iris syndrome, and the effect persists for years after stopping, so **stopping the drug does not remove the risk**. Your job is to identify it and inform the surgeon.
- **Anticoagulants and antiplatelet agents are generally continued** for modern phacoemulsification with topical or intracameral anaesthesia; the thromboembolic risk of stopping them exceeds the bleeding risk.
- **Do not prescribe glasses until the refraction is stable**, generally 3 to 6 weeks after surgery, and later after a complicated case.
- **Warn about YAG timing**: capsulotomy before the capsule and IOL are fully stable increases complication risk.
- **A patient with a multifocal IOL and an unhappy outcome** needs the ocular surface, residual refractive error, posterior capsule, and macula checked in that order before concluding the lens is the problem.
- **Diabetic patients** have higher rates of macular oedema and progression of retinopathy after surgery; optimise control and consider prophylactic or perioperative anti-VEGF or steroid in eyes with existing macular oedema.
- **Pediatric cataract surgery** requires management of aphakia (contact lens or IOL by age and laterality), amblyopia therapy, and glaucoma surveillance for life. The Infant Aphakia Treatment Study found no acuity advantage for primary IOL implantation under 7 months and more adverse events, so contact lens correction is standard in early infancy.

Lens position and lens-induced glaucoma

LENS-RELATED SECONDARY GLAUCOMA

Entity	Mechanism, angle status, and treatment
Phacomorphic glaucoma	An intumescent lens pushes the iris forward: **angle closed**. Treat like acute angle closure medically, then **cataract extraction is definitive**. Iridotomy alone is insufficient because the mechanism is lens bulk, not pupillary block.
Phacolytic glaucoma	High-molecular-weight protein leaks from a hypermature cataract and macrophages block the meshwork: **angle open**, heavy cell and flare, no synechiae. Medical IOP control plus steroid, then **urgent cataract extraction**.
Phacoantigenic (phacoanaphylactic) uveitis	Granulomatous reaction to exposed lens protein after capsule rupture or surgery. Steroid plus removal of residual lens material.

Entity	Mechanism, angle status, and treatment
Lens particle glaucoma	Retained cortical fragments after surgery or trauma obstruct outflow: open angle with visible fragments. Medical control plus surgical removal.
Ectopia lentis with pupillary block	Anterior subluxation or microspherophakia. **Use cycloplegia, not miotics**: a miotic worsens block in microspherophakia by allowing the lens to move forward. Iridotomy, then lensectomy.
Pseudoexfoliation glaucoma	Fibrillar material obstructs the meshwork with heavy trabecular pigment and a Sampaolesi line. Higher and more fluctuant IOP than primary open-angle glaucoma, with faster progression and a better-than-average response to **laser trabeculoplasty**. Surgery is riskier because of zonular weakness and poor dilation.

RETRIEVAL PRACTICE

1. A patient reports the worst vision in bright sunlight and reading trouble disproportionate to their distance acuity. Which cataract type, and which drug class should you ask about?
2. A 42-year-old with a myopic shift of `1.50 D` over a year. Which cataract type, and what does "second sight" mean?
3. Name three cataract morphologies with a specific systemic cause and give each cause.
4. Give four indications for cataract surgery that are not based on Snellen acuity.
5. A patient had myopic LASIK 12 years ago and now needs cataract surgery. What goes wrong with standard IOL calculation, in which direction, and what do you do about it?
6. A toric IOL rotates `15°`. Roughly how much of its effect is lost?
7. Give three groups of patients in whom you would advise against a diffractive multifocal IOL.
8. Day 1 after surgery the IOP is 42 mmHg with corneal oedema. Give two likely causes and the management.
9. Day 4 after surgery the eye is more painful, vision is worse, and there is a hypopyon. Diagnosis, the most likely organism, and what the EVS tells you to do.
10. Day 1 after surgery there is limbus-to-limbus corneal oedema with fibrin and a hypopyon but no pain and no vitritis. Diagnosis and treatment.
11. A patient is 5 weeks post-surgery with gradually blurring vision and a petalloid pattern on angiography. Diagnosis, risk factors, and treatment.
12. Name three complications of Nd:YAG capsulotomy and the pretreatment you give.
13. A patient reports a temporal dark crescent after uncomplicated surgery. What is this and what is the natural history?
14. A hypermature cataract with IOP 44 mmHg, an open angle, and heavy cell and flare. Diagnosis and definitive treatment.
15. Why is a miotic the wrong choice in microspherophakia with pupillary block?
16. What did the Infant Aphakia Treatment Study conclude?

KEY

1. **Posterior subcapsular** cataract; ask about **corticosteroids by any route** (also diabetes, radiation, uveitis).
2. Nuclear sclerosis; "second sight" is the temporary improvement in uncorrected near vision produced by the myopic index shift.
3. Oil droplet in galactosaemia; Christmas tree in myotonic dystrophy; sunflower in Wilson disease (also snowflake in poorly controlled type 1 diabetes, spoke-like in Fabry, anterior subcapsular shield in atopic dermatitis).
4. Anisometropia with intolerable aniseikonia; lens-induced disease (phacomorphic, phacolytic, phacoantigenic); the lens preventing management of posterior segment disease; and any visually significant opacity in a child because of amblyopia.
5. Standard keratometry **overestimates** corneal power after myopic ablation, so the formula selects a lens that is too weak and the patient ends up **hyperopic**. Use post-refractive-surgery formulas (Barrett True-K,

Haigis-L, the ASCRS calculator) and supply pre-surgical keratometry and refraction.

6. About one third of the cylindrical effect.
7. Patients with macular disease, an irregular cornea or significant dry eye, glaucoma with field loss, monocularity, or unrealistic expectations.
8. Retained viscoelastic or pupillary block (also pre-existing glaucoma). Give aqueous suppressants, and burp the wound or perform a paracentesis if the pressure is dangerously high; recheck within hours.
9. Acute postoperative endophthalmitis; ***Staphylococcus epidermidis***. The EVS: if presenting vision is **light perception only**, immediate vitrectomy plus intravitreal antibiotics; if **hand motions or better**, tap and inject alone. Systemic antibiotics added no benefit.
10. Toxic anterior segment syndrome; intensive topical steroid. The absence of pain and vitritis distinguishes it from endophthalmitis.
11. Pseudophakic cystoid macular oedema (Irvine-Gass). Risk factors: capsule rupture with vitreous loss, diabetes, uveitis, vein occlusion, prostaglandin use, prior CMO in the fellow eye. Treat with topical NSAID plus steroid, escalating to periocular or intravitreal steroid or acetazolamide.
12. IOP spike, IOL pitting, iritis, retinal detachment (especially in myopes), and cystoid macular oedema. Pretreat with **brimonidine or apraclonidine**.
13. **Negative dysphotopsia**, from shadowing at the anterior capsule edge of a square-edged in-the-bag IOL; most improve spontaneously over months.
14. **Phacolytic glaucoma**; definitive treatment is **urgent cataract extraction** after medical IOP control and steroid.
15. A miotic relaxes the zonules and allows the small spherical lens to move forward, **worsening** the block; cycloplegia tightens the zonules and flattens the lens, moving it backwards.
16. Primary IOL implantation in infants under 7 months gave **no acuity advantage** over contact lens correction and produced **more adverse events and reoperations**, so contact lens correction of aphakia is standard in early infancy.

CHAPTER RECAP

1. Morphology predicts symptom pattern and often names the systemic cause.
2. The indication for surgery is functional need, not a Snellen number.
3. Your preoperative job is the surface, the macula, the endothelium, the pupil, the α-blocker history, and the expectations.
4. Prior myopic LASIK makes standard biometry produce a hyperopic surprise; use the right formula and supply old data.
5. A postoperative eye should improve daily; new pain or worsening vision is an emergency.
6. Endophthalmitis is painful with vitritis; TASS is painless without vitritis. EVS decides vitrectomy by presenting vision.
7. Irvine-Gass peaks at 4 to 6 weeks and responds to NSAID plus steroid.
8. YAG capsulotomy causes pressure spikes and raises detachment risk in myopes; pretreat and counsel.
9. Phacomorphic closes the angle and needs the lens out; phacolytic is open-angle and also needs the lens out.
10. Pseudoexfoliation is the answer to late in-the-bag IOL subluxation, poor dilation, and a difficult case.

CHAPTER 6 · DISEASE AND MANAGEMENT

Uveitis, episclera, and sclera

Episclera / sclera / anterior uvea · 11–22 items | TMOD-heavy | Systemic health

ANSWER THESE COLD BEFORE READING

1. Give the SUN grading for anterior chamber cells and for vitreous haze.
2. Distinguish granulomatous from non-granulomatous uveitis, and name three causes of each.
3. Give the exact treatment of acute anterior uveitis with drug, concentration, and taper.
4. Separate episcleritis from scleritis on four findings and state the management of each.
5. Name the four posterior uveitis entities you must not miss and the feature that identifies each.
6. State when uveitis requires a systemic workup and when it does not.

BLUEPRINT WEIGHT

11–22

Scored items, but with a very high TMOD density because almost every item involves treatment selection and follow-up.

Grading and describing uveitis

STANDARDIZATION OF UVEITIS NOMENCLATURE (SUN) GRADING

Grade	Anterior chamber cells (1 × 1 mm field)	Vitreous haze / flare
0	<1 cell	No haze; no flare
0.5+	1–5 cells	–
1+	6–15 cells	Faint haze; faint flare
2+	16–25 cells	Moderate haze, disc and vessels visible; moderate flare (iris and lens details clear)
3+	26–50 cells	Marked haze, disc hazy; marked flare (iris and lens details hazy)
4+	>50 cells	Dense haze, disc obscured; intense flare with fibrin or plasmoid aqueous

Describe every uveitis in five dimensions: *anatomic location* (anterior, intermediate, posterior, or pan), *onset* (sudden or insidious), *duration* (limited under three months, or persistent), *course* (acute, recurrent, or chronic), and *laterality*. That five-part description, plus granulomatous versus non-granulomatous, narrows the differential more than any laboratory panel.

GRANULOMATOUS VERSUS NON-GRANULOMATOUS

Feature	Granulomatous	Non-granulomatous
Keratic precipitates	**Large, greasy, "mutton-fat"**, often in an Arlt triangle inferiorly	Fine, small, white
Iris nodules	**Koeppe** (pupil margin) and **Busacca** (stroma)	Absent
Onset and course	Insidious, chronic, often bilateral	Sudden, acute, often unilateral or alternating

Feature	Granulomatous	Non-granulomatous
Posterior involvement	Common	Uncommon
Causes	**Sarcoidosis, tuberculosis, syphilis, Vogt-Koyanagi-Harada, sympathetic ophthalmia, herpetic disease, lens-induced, Lyme, toxoplasmosis, fungal, juvenile idiopathic arthritis (may be either)**	**HLA-B27 spondyloarthropathies, idiopathic, trauma, postoperative, Behçet, Fuchs uveitis syndrome, drug-induced**

Anterior uveitis

ACUTE ANTERIOR UVEITIS: THE STANDARD REGIMEN

1. **Topical corticosteroid, dosed to the inflammation.** Prednisolone acetate `1%` **every 1 to 2 hours while awake** for severe (3 to 4+) inflammation, or every 2 to 4 hours for moderate. **Shake the bottle** (it is a suspension). Difluprednate `0.05%` four times daily is an alternative with better penetration and a higher IOP risk.
2. **Cycloplegia** for comfort (ciliary spasm) and to prevent posterior synechiae: **cyclopentolate** `1%` **two or three times daily** for mild disease, or **homatropine** `5%` **two or three times daily**, or atropine `1%` daily for severe fibrinous inflammation. Avoid keeping the pupil fixed and fully dilated for weeks, which risks a dilated posterior synechia.
3. **Review in 3 to 7 days** depending on severity, then taper the steroid slowly: reduce by roughly one drop per day per week, or halve the frequency each week, and **never stop abruptly** because rebound is common. Total course is often 4 to 6 weeks.
4. **Measure IOP at every visit.** Pressure can rise from the inflammation (trabeculitis), from steroid response, or from synechial closure, and it can be low from ciliary shutdown.
5. **Dilate and examine the fundus** at presentation and again during the course. Anterior uveitis with a posterior finding is not anterior uveitis.
6. **Escalate** to periocular or oral steroid, or to a systemic steroid-sparing agent, for severe, bilateral, recurrent, or steroid-dependent disease, in conjunction with rheumatology.

Red flags that change management: hypopyon (think HLA-B27 or Behçet, and exclude infection and masquerade), a white eye with heavy inflammation in a child (juvenile idiopathic arthritis), raised IOP with sectoral iris atrophy (herpetic), bilateral disease with serous detachments (VKH), and a poor or paradoxical steroid response (masquerade syndrome, or infection).

ANTERIOR UVEITIS: IDENTIFYING THE CAUSE FROM THE PATTERN

Entity	Identifying features	Management specifics
HLA-B27 associated	Recurrent, **sudden, unilateral or alternating, non-granulomatous**, with intense fibrin, sometimes a **hypopyon**, and often marked photophobia. Young adult, back pain or stiffness, or a history of reactive arthritis, psoriasis, or inflammatory bowel disease.	Standard aggressive topical therapy. Test HLA-B27, image the sacroiliac joints, and refer to rheumatology. Treat the eye, but also start the systemic diagnosis: recurrent uveitis is often the presenting feature of spondyloarthropathy.
Juvenile idiopathic arthritis associated	**Chronic, bilateral, insidious, and *asym ptomatic* with a white eye.** Highest risk: **oligoarticular, ANA-positive, young girls with early-onset arthritis**. Complications: band keratopathy, posterior synechiae, cataract, glaucoma, hypotony, macular oedema, amblyopia.	Topical steroid at the lowest effective dose plus early **methotrexate**, escalating to **adalimumab** (which has strong trial support). **Screening every 3 months** in the high-risk group, because there are no symptoms to bring the child in. Coordinate with paediatric rheumatology.
Herpetic (HSV or VZV) anterior uveitis	**Raised IOP** with active inflammation (trabeculitis), **sectoral iris atrophy**, reduced corneal sensation, keratic precipitates that may be granulomatous, prior dendritic or dermatomal disease, and unilateral recurrence.	Topical steroid **with systemic antiviral cover** (aciclovir `400 mg` five times daily for HSV, or valaciclovir `1 g` three times daily for VZV), plus IOP control. **Avoid prostaglandin analogues, which can reactivate herpes.**

Entity	Identifying features	Management specifics
Cytomegalovirus anterior uveitis	Immunocompetent middle-aged or older adult, unilateral, **markedly raised IOP**, **linear or coin-shaped endothelial lesions**, and corneal oedema; frequently misdiagnosed as glaucomatocyclitic crisis.	Aqueous PCR confirms; treat with oral valganciclovir or topical ganciclovir plus steroid and IOP control.
Fuchs uveitis syndrome (heterochromic iridocyclitis)	Unilateral, chronic, **minimal symptoms and no injection**, **stellate keratic precipitates spread over the whole endothelium**, diffuse iris stromal atrophy with **heterochromia (the affected eye is lighter)**, **no posterior synechiae**, fine angle vessels with a filiform haemorrhage on gonioscopy (Amsler sign), early cataract and glaucoma. Rubella association.	**Steroids are largely ineffective and should be minimised.** Manage the cataract and the glaucoma, which are the real problems. Recognising it prevents years of unnecessary steroid.
Posner-Schlossman (glaucomatocyclitic crisis)	Recurrent unilateral episodes of **very high IOP (often 40 to 60 mmHg) with minimal inflammation** and an open angle, mild discomfort, and corneal oedema. CMV is implicated in many cases.	Aqueous suppressants plus a short topical steroid course; avoid miotics. Consider aqueous PCR for CMV in recurrent cases. Long-term glaucomatous damage can accumulate.
Traumatic iritis	History of blunt trauma, cells and flare within 24 to 72 hours, often with a hyphaema, sphincter tears, or angle recession.	Topical steroid and cycloplegia; document angle recession on gonioscopy and arrange **lifelong glaucoma surveillance**.

Entity	Identifying features	Management specifics
Drug-induced	Bisphosphonates, sulfonamides, rifabutin (with hypopyon), cidofovir, moxifloxacin (with pigment dispersion and iris transillumination defects), immune checkpoint inhibitors, prostaglandin analogues.	Withdraw the agent where possible and treat conventionally.
Masquerade syndromes	**Consider when the "uveitis" does not behave.** Retinoblastoma in a child (pseudohypopyon), leukaemia, primary vitreoretinal lymphoma in an older patient, retinal detachment, intraocular foreign body, juvenile xanthogranuloma, and pigment dispersion.	Reassess the diagnosis rather than escalating steroid. Image, and biopsy where indicated.

Intermediate uveitis

- **Presentation**: floaters and painless blur with minimal redness, in a young adult. Findings are **vitreous cells, snowballs** (vitreous aggregates inferiorly), and **snowbanking** (a fibroglial plaque over the inferior pars plana), plus peripheral periphlebitis and, commonly, **cystoid macular oedema, which is the main cause of vision loss**. Peripheral neovascularisation can occur, especially in children.
- **Pars planitis** is the idiopathic subset with snowbanking and no systemic association. **Associations to exclude**: **multiple sclerosis** (order MRI of the brain), sarcoidosis, tuberculosis, syphilis, and Lyme disease. Also consider lymphoma in an older patient.
- **Treatment ladder**: observation if vision is good and there is no macular oedema; **periocular or intravitreal steroid** (posterior sub-Tenon triamcinolone) for macular oedema or reduced vision; **oral steroid** for bilateral disease; then **steroid-sparing immunosuppression (methotrexate, mycophenolate) or adalimumab**; and **peripheral cryotherapy or laser and vitrectomy** for neovascularisation or dense vitreous opacity. Topical steroid does not reach the vitreous and is largely ineffective for this.

Posterior uveitis and the sight-threatening entities

POSTERIOR UVEITIS: THE IDENTIFYING FEATURE IS THE DIAGNOSIS

Entity	Identifying feature	Treatment
Toxoplasmosis	**Focal necrotising retinochoroiditis adjacent to a pigmented scar**, with overlying vitritis ("headlight in the fog"). Recurrences occur at the edge of old scars. Congenital form is bilateral and macular.	Treat if the lesion threatens the macula or nerve, if there is dense vitritis, or if the patient is immunocompromised or pregnant. **Classic triple therapy**: pyrimethamine (loading then maintenance) plus sulfadiazine plus **folinic acid**, or **trimethoprim-sulfamethoxazole** as a simpler alternative, or clindamycin. Add **oral prednisone 24 to 48 hours after starting antimicrobials**, never before. Treat for 4 to 6 weeks.
Acute retinal necrosis	**Peripheral confluent white retinal necrosis, occlusive arteritis, and dense vitritis, progressing circumferentially over days**, usually in an **immunocompetent** patient. VZV most often, then HSV. Retinal detachment risk approaches `50%`.	**Emergency.** Immediate high-dose intravenous aciclovir (or oral valaciclovir `2 g` three times daily) plus **intravitreal foscarnet or ganciclovir**, then prolonged oral antiviral prophylaxis to protect the fellow eye (which is involved in up to a third without treatment). Aspirin and systemic steroid are adjuncts after antivirals are started. Prophylactic laser and vitreoretinal referral.
Progressive outer retinal necrosis	Advanced AIDS; **minimal or no vitritis** with rapidly progressive deep multifocal outer retinal opacification and early macular involvement.	Combination intravitreal and systemic antivirals; restore immune function. Prognosis is poor.
CMV retinitis	**CD4 below** `50 cells/µL` or other profound immunosuppression; **indolent** confluent necrosis with haemorrhage ("pizza pie" or "brushfire") and **mild vitritis**, spreading slowly along vessels.	Oral valganciclovir induction then maintenance, with intravitreal ganciclovir or foscarnet for zone 1 disease; restore immunity with antiretroviral therapy, and watch for **immune recovery uveitis**.

Entity	Identifying feature	Treatment
Ocular syphilis	**Any pattern**. Characteristic forms: **acute syphilitic posterior placoid chorioretinitis** (a yellowish placoid outer retinal lesion at the posterior pole), interstitial keratitis with ghost vessels, salt-and-pepper fundus, and optic neuritis.	**Treat as neurosyphilis**: intravenous penicillin G `18–24 million units` daily for 10 to 14 days. Test HIV status, perform a lumbar puncture, and report to public health. Test with a treponemal-specific assay plus RPR or VDRL for activity.
Ocular tuberculosis	Choroidal tubercles, serpiginous-like choroiditis, occlusive retinal vasculitis (Eales-like), and granulomatous anterior uveitis.	Multidrug antituberculous therapy with an infectious disease specialist, plus steroid cover. **Watch for ethambutol optic neuropathy during treatment.**
Sarcoid posterior uveitis	**Candle-wax drippings** (segmental periphlebitis), snowballs, multifocal choroiditis, optic nerve granuloma, and often bilateral disease.	Steroid, then methotrexate or adalimumab; investigate with ACE, lysozyme, chest imaging, and biopsy of an accessible lesion.
Vogt-Koyanagi-Harada	Bilateral granulomatous panuveitis with **multifocal serous retinal detachments and disc oedema**, plus **headache, meningismus, tinnitus, hearing loss**, and later **sunset glow fundus**, vitiligo, poliosis, and alopecia. Commoner in pigmented populations; HLA-DR4.	**Prompt, prolonged, high-dose systemic corticosteroid plus early steroid-sparing immunosuppression; under-treatment produces the chronic recurrent form with sunset glow and severe visual loss.**
Behçet disease	Recurrent **occlusive retinal vasculitis affecting both arteries and veins**, panuveitis with a **shifting hypopyon**, plus oral and genital ulceration; HLA-B51.	Systemic immunosuppression from the outset; **anti-TNF therapy (infliximab or adalimumab)** and interferon are established. Steroid monotherapy is inadequate.

Entity	Identifying feature	Treatment
Birdshot chorioretinopathy	Multiple cream-coloured ovoid lesions radiating from the disc, especially nasally and inferiorly, with vitritis and retinal vasculitis; **HLA-A29** in the great majority.	Immunomodulation for vision-threatening disease; macular oedema drives the visual outcome.
Sympathetic ophthalmia	Bilateral granulomatous panuveitis **after penetrating trauma or surgery to the fellow eye**, with Dalen-Fuchs nodules; latency two weeks to decades, most within three months.	Aggressive systemic immunosuppression. Prevention is prompt primary repair.
Primary vitreoretinal lymphoma	Older patient, vitreous cells, **subretinal pigment epithelial creamy infiltrates**, poor or transient steroid response.	Vitreous biopsy with cytology, IL-10 to IL-6 ratio, and *MYD88*; MRI of the brain and cerebrospinal fluid examination. Intravitreal methotrexate or rituximab plus systemic therapy.
Endogenous endophthalmitis	Systemically unwell patient, intravenous drug use, indwelling line, diabetes, or recent bacteraemia; unilateral or bilateral vitritis with chorioretinal infiltrates. *Candida* gives a "string of pearls".	**Blood cultures, systemic antimicrobials, and search for the primary source (endocarditis, liver abscess, urinary source).** Intravitreal therapy and vitreoretinal referral. This is a systemic emergency, not just an eye problem.

Episcleritis and scleritis

THE DISTINCTION THAT DECIDES WHETHER YOU ORDER BLOOD TESTS

Feature	Episcleritis	Scleritis
Pain	Mild discomfort or none; "it looks worse than it feels"	**Severe, boring, radiating to the brow, jaw, or temple, waking the patient at night**, and tender to palpation
Colour and depth	Bright salmon-pink, superficial radial vessels	**Deep violaceous or bluish hue**, best seen in daylight, with scleral oedema on slit lamp with the beam angled

Feature	Episcleritis	Scleritis
Phenylephrine `2.5%`	**Blanches**	**Does not blanch**
Vessel mobility	Vessels move with a cotton bud	Deep plexus does not move
Systemic association	Uncommon (about `30%` at most, usually mild)	**Up to about half have an associated systemic disease**: rheumatoid arthritis, granulomatosis with polyangiitis, relapsing polychondritis, polyarteritis nodosa, lupus, inflammatory bowel disease, gout, sarcoid, and infection (herpes zoster, tuberculosis, syphilis, *Pseudomonas* after surgery)
Complications	None significant	Scleral thinning and staphyloma, keratitis, uveitis, glaucoma, macular oedema, exudative detachment, and **vision loss**
Treatment	Often self-limited over 1 to 3 weeks. Lubricants and cool compresses; oral NSAID for symptoms; a short course of a mild topical steroid (fluorometholone or loteprednol) for nodular or troublesome cases. Investigate only if recurrent or bilateral.	**Oral NSAID at full anti-inflammatory dose** (for example ibuprofen `600-800 mg` three times daily, or indomethacin, or naproxen), escalating to **oral prednisone** `1 mg/kg`, then to **systemic immunosuppression (methotrexate, mycophenolate) or a biologic**. **Necrotising scleritis requires immediate cyclophosphamide or rituximab and carries significant mortality from the underlying vasculitis.** Topical steroid alone is inadequate. Order ANCA, RF and anti-CCP, ANA, ESR and CRP, urinalysis, chest imaging, syphilis serology, and an interferon-gamma release assay, and refer to rheumatology.

SCLERITIS SUBTYPES

Subtype	Features
Diffuse anterior	Commonest and most benign; widespread injection and oedema.
Nodular anterior	Immobile, tender, deep nodule; distinguish from a mobile episcleral nodule.
Necrotising with inflammation	**The most destructive and the most strongly associated with systemic vasculitis and with mortality.** Avascular patches, scleral melt, and possible perforation. Requires immunosuppression today.
Necrotising without inflammation (scleromalacia perforans)	Painless progressive thinning with visible uvea, in long-standing rheumatoid arthritis. Deceptively quiet.
Posterior scleritis	**Pain with a normal-looking anterior segment**: the great mimic. Findings include disc oedema, exudative retinal detachment, choroidal folds, a T-sign on B-scan (fluid in the sub-Tenon space at the optic nerve insertion), proptosis, and restricted motility. Consider it in any painful eye with posterior findings and no anterior signs.

THREE SEQUENCING ERRORS THAT APPEAR IN EXAM ITEMS

- **Giving oral steroid before antimicrobials in toxoplasmosis.** Start the antimicrobial first, then add steroid 24 to 48 hours later.
- **Treating necrotising scleritis or peripheral ulcerative keratitis topically.** These are systemic vasculitides; topical therapy risks perforation and misses a life-threatening diagnosis.
- **Escalating steroid in a uveitis that is not responding.** Non-response, paradoxical worsening, or steroid dependence in an unusual demographic means reconsider the diagnosis: infection (herpetic, CMV, syphilis, tuberculosis, fungal), masquerade (lymphoma, retinoblastoma, leukaemia), or a retained foreign body.

RETRIEVAL PRACTICE

1. Give the SUN cell grade for 20 cells in a 1 by 1 mm field, and describe 2+ flare.
2. Name four features of granulomatous uveitis and three causes.
3. Give the full topical regimen for 3+ anterior uveitis, including the cycloplegic and the taper.
4. Why must you dilate a patient with anterior uveitis?
5. A 28-year-old man has his fourth episode of unilateral painful uveitis with fibrin, plus low back stiffness. Diagnosis, test, and referral.
6. A 5-year-old girl with oligoarticular arthritis has a white eye and 2+ cells. What is the diagnosis, the screening interval, and the systemic treatment ladder?
7. Unilateral uveitis with IOP 38 mmHg and sectoral iris atrophy. Diagnosis, treatment, and one drug to avoid.
8. Unilateral chronic uveitis with stellate diffuse keratic precipitates, iris heterochromia, and no synechiae. Diagnosis, and what does the treatment consist of?
9. A young adult with floaters, snowballs, and inferior snowbanking. Diagnosis, the main cause of vision loss, and the one systemic investigation you must order.
10. Focal white retinitis at the edge of an old pigmented scar with vitritis. Diagnosis, drug regimen, and the sequencing rule for steroid.
11. An immunocompetent adult with rapidly progressive peripheral confluent retinal whitening, arteritis, and dense vitritis. Diagnosis, urgency, and treatment.
12. A yellow placoid outer retinal lesion at the posterior pole with vitritis. Diagnosis, treatment regimen, and two mandatory additional actions.
13. Give four findings that separate episcleritis from scleritis.
14. A patient has severe boring eye pain that wakes them, a normal-looking anterior segment, disc oedema, and choroidal folds. Diagnosis and the confirming test.
15. A rheumatoid patient develops an avascular white scleral patch with severe pain. Diagnosis, treatment, and what is at stake beyond the eye.

KEY

1. 20 cells is **2+**. 2+ flare is moderate flare in which iris and lens details remain clear.
2. Mutton-fat keratic precipitates, Koeppe and Busacca iris nodules, insidious chronic bilateral course, and posterior involvement. Causes: sarcoidosis, tuberculosis, syphilis (also VKH, sympathetic ophthalmia, herpetic).
3. Prednisolone acetate **1%** every 1 to 2 hours while awake (shaken), plus homatropine **5%** two or three times daily; review in 3 to 7 days and taper by halving the frequency weekly over 4 to 6 weeks, never stopping abruptly.
4. To detect posterior involvement, because anterior uveitis with a posterior finding is a different disease with a different workup, and to prevent and break posterior synechiae.
5. HLA-B27-associated acute anterior uveitis; test **HLA-B27** and image the sacroiliac joints; refer to rheumatology for probable ankylosing spondylitis.
6. Juvenile idiopathic arthritis associated chronic anterior uveitis; screen **every 3 months**; treat with

the lowest effective topical steroid plus early **methotrexate**, escalating to **adalimumab**.

7. Herpetic anterior uveitis with trabeculitis; topical steroid **with systemic antiviral cover** plus IOP control; **avoid prostaglandin analogues**.
8. Fuchs uveitis syndrome; treatment is essentially **not steroid**, but management of the resulting cataract and glaucoma.
9. Intermediate uveitis or pars planitis; **cystoid macular oedema** causes the visual loss; order **MRI of the brain** to exclude multiple sclerosis.
10. Ocular toxoplasmosis; pyrimethamine plus sulfadiazine plus **folinic acid** (or trimethoprim-sulfamethoxazole); **add oral steroid only 24 to 48 hours after starting the antimicrobial**.
11. Acute retinal necrosis; a **same-day emergency**; high-dose intravenous aciclovir or oral valaciclovir `2 g` three times daily **plus intravitreal foscarnet or ganciclovir**, with prolonged prophylaxis to protect the fellow eye and vitreoretinal referral for detachment risk.
12. Acute syphilitic posterior placoid chorioretinitis; treat as **neurosyphilis with intravenous penicillin G `18–24 million units` daily for 10 to 14 days**; also **test for HIV** and **report to public health** (and perform a lumbar puncture).
13. Severe boring night-waking pain, a deep violaceous hue, failure to blanch with phenylephrine, and immobility of the deep vessels (plus tenderness to palpation).
14. **Posterior scleritis**; B-scan ultrasonography showing scleral thickening and a **T-sign**.
15. **Necrotising scleritis with inflammation**; immediate systemic immunosuppression with **cyclophosphamide or rituximab** plus high-dose steroid; what is at stake is **life**, because it signals active systemic vasculitis with significant mortality.

CHAPTER RECAP

1. Describe every uveitis in five dimensions plus granulomatous status before ordering anything.
2. Acute anterior uveitis: hourly prednisolone acetate, cycloplegia, slow taper, IOP at every visit, dilate the fundus.
3. HLA-B27 gives recurrent fibrinous unilateral disease; JIA gives a silent white bilateral eye needing three-monthly screening.
4. Raised IOP with sectoral iris atrophy means herpetic; stellate precipitates with heterochromia and no synechiae mean Fuchs, and Fuchs does not want steroid.
5. Intermediate uveitis needs an MRI and loses vision to macular oedema.
6. Toxoplasmosis sits beside an old scar; give antimicrobial first and steroid second.
7. ARN is an emergency in the immunocompetent; CMV retinitis is indolent below a CD4 of 50.
8. Syphilis imitates everything and is treated as neurosyphilis with intravenous penicillin.
9. Phenylephrine blanches episcleritis and not scleritis; scleritis needs systemic therapy and a systemic workup.
10. Non-response to steroid means rethink the diagnosis, not raise the dose.

CHAPTER 7 · DISEASE AND MANAGEMENT

Retina, choroid, and vitreous

Retina / choroid / vitreous · 33–44 items | TMOD-heavy | Systemic health | Emergencies / trauma

ANSWER THESE COLD BEFORE READING

1. Give the AREDS categories and the exact AREDS2 formulation with its indication.
2. State the DRCR Protocol T and Protocol S conclusions.
3. Give the management of CRAO in the first 24 hours and the mandatory exclusion over age 50.
4. Give the follow-up schedule for ischaemic CRVO and the month at which neovascular glaucoma peaks.
5. State which peripheral retinal lesions need prophylactic treatment and which do not.
6. Give the OCT-based staging and management of a macular hole and of vitreomacular traction.

BLUEPRINT WEIGHT

33–44

Tied for the largest disease area and the most image-dependent section of the exam.

Age-related macular degeneration

CLASSIFICATION AND MANAGEMENT BY CATEGORY

Category	Findings	Management
No AMD (AREDS 1)	No or few small drusen (under `63 µm`)	Routine care; lifestyle counselling.
Early AMD (AREDS 2)	Multiple small drusen or a few intermediate drusen (`63–125 µm`), or mild pigment change	Annual review, Amsler grid, and lifestyle counselling. **Supplements are not indicated at this stage.**
Intermediate AMD (AREDS 3)	Extensive intermediate drusen, at least one large druse (`>125 µm`), or geographic atrophy not involving the centre	**AREDS2 supplementation indicated.** Review every 6 to 12 months with OCT; daily home Amsler or a home monitoring device; educate on urgent presentation for new distortion.
Advanced AMD (AREDS 4)	Centre-involving geographic atrophy, or any neovascular AMD	Supplement to protect the fellow eye; treat neovascular disease; low vision rehabilitation for atrophy.

AREDS2: THE EXACT FORMULATION AND ITS LIMITS

Vitamin C `500 mg`, vitamin E `400 IU`, **lutein** `10 mg` **and zeaxanthin** `2 mg` (replacing β-carotene, which increased lung cancer risk in current and former smokers), zinc oxide `80 mg` (a `25 mg` dose was non-inferior), and cupric oxide `2 mg` (to prevent zinc-induced copper deficiency anaemia).

- **Indication**: intermediate AMD in one or both eyes, or advanced AMD in one eye. It reduces progression to advanced disease by roughly `25%` over five years.
- **It does not prevent AMD** in people without it, and it does not restore vision.
- **It does not prevent geographic atrophy**; that indication now belongs to intravitreal complement inhibitors (pegcetacoplan, avacincaptad pegol), which slow atrophy growth but have not demonstrated a functional visual benefit and carry a risk of converting to exudative disease.
- **Smoking cessation is the single most effective intervention**, roughly doubling to quadrupling risk while continued. Also address blood pressure, diet (leafy greens and oily fish), and UV protection.

NEOVASCULAR AMD: LESION TYPES AND TREATMENT

Type	Features and treatment
Type 1 (occult, sub-RPE)	Fibrovascular proliferation beneath the RPE; a "double-layer sign" on OCT; ill-defined late leakage on angiography. ICG and OCT angiography help.
Type 2 (classic, subretinal)	Above the RPE; a well-defined early lacy hyperfluorescent net with late leakage.
Type 3 (retinal angiomatous proliferation)	Intraretinal neovascularisation descending toward the RPE; often bilateral and associated with reticular pseudodrusen; higher atrophy risk.
Polypoidal choroidal vasculopathy	Orange-red nodules with serosanguineous detachments, commoner in Asian and pigmented populations; **ICG angiography is the diagnostic test**. May respond to photodynamic therapy in addition to anti-VEGF.
Treatment	**Intravitreal anti-VEGF is standard**: aflibercept, ranibizumab, brolucizumab, faricimab, or bevacizumab. Loading with monthly injections (usually three, or four for some agents), then a treat-and-extend or as-needed regimen. **CATT** showed ranibizumab and bevacizumab equivalent for acuity at one and two years. Faricimab and high-dose aflibercept allow longer intervals for many patients. **Photodynamic therapy** retains a role in polypoidal disease. Risks: endophthalmitis (about `1 in 2000 to 5000` injections), IOP rise, retinal tear, and intraocular inflammation (higher with brolucizumab).
What you must teach the patient	**Daily monocular Amsler or home monitoring, and same-week presentation for new distortion, a new grey patch, or reduced reading vision.** The fellow eye converts at roughly `5 to 10%` per year in high-risk eyes, and delayed presentation is the main determinant of a poor outcome.

Diabetic retinopathy and macular oedema

LANDMARK TRIALS YOU SHOULD BE ABLE TO CITE

Trial	Conclusion
DRS	Panretinal photocoagulation reduces severe visual loss in high-risk proliferative disease by about half.
ETDRS	Focal laser reduces moderate visual loss in clinically significant macular oedema by about half; defined the severity scale and the 4-2-1 rule; early PRP is beneficial in type 2 diabetes with severe non-proliferative disease.

Trial	Conclusion
DCCT and UKPDS	Intensive glycaemic control reduces incidence and progression; UKPDS additionally showed that **blood pressure control independently reduces progression and visual loss**. Beware transient **early worsening** after rapid improvement in control.
DRCR Protocol T	For centre-involving macular oedema, **aflibercept was superior at one year when baseline acuity was 20/50 or worse**; with better baseline acuity the agents were similar. Differences narrowed by two years.
DRCR Protocol S	**Ranibizumab was non-inferior to panretinal photocoagulation** for proliferative disease at two and five years, with less field loss and less vitrectomy but more injections and a heavy dependence on follow-up adherence. PRP remains preferable when follow-up is unreliable.
DRCR Protocol W	Prophylactic aflibercept in moderate-to-severe non-proliferative disease reduced progression to proliferative disease and centre-involving oedema, but did not improve visual acuity at four years, so it is not routine.

PRACTICAL DIABETIC MANAGEMENT, WITH INTERVALS

Stage	Review	Action
No retinopathy	12 months	Optimise A1c, blood pressure, lipids; smoking cessation; document.
Mild NPDR	12 months	As above.
Moderate NPDR	6–12 months	OCT if any macular thickening is suspected.
Severe NPDR (4-2-1)	3–4 months	Consider early PRP or anti-VEGF, particularly in type 2 diabetes, poor adherence, pregnancy, or before cataract surgery.
PDR, not high-risk	2–3 months	Treat if progressing.
High-risk PDR	Treat now	PRP and/or anti-VEGF; vitrectomy for non-clearing haemorrhage or macula-threatening traction.
Non-centre-involving DMO	3–4 months	Observe or focal laser.
Centre-involving DMO	4–6 weeks	**Anti-VEGF first-line**; steroid implant for pseudophakic or refractory eyes; adjunctive focal laser.
Pregnancy with diabetes	Each trimester	Examine before conception or in the first trimester, then each trimester and post partum; retinopathy accelerates in pregnancy. Gestational diabetes alone does not need retinal screening.

Rules that get tested: type 1 first screened **5 years** after diagnosis or at puberty, type 2 **at diagnosis**. Do not prescribe new glasses during unstable glycaemic control. Panretinal photocoagulation can worsen macular oedema, so treat the oedema first or concurrently. Anti-VEGF is contraindicated in active ocular infection and used cautiously within months of a stroke or myocardial infarction.

Retinal vascular occlusion

ARTERIAL OCCLUSION

Entity	Presentation	Management
Central retinal artery occlusion	Sudden painless profound monocular loss, RAPD, whitened inner retina with a **cherry red spot**, boxcarring, and later optic atrophy. Look for a **Hollenhorst plaque**, calcific emboli (higher mortality), or platelet-fibrin emboli.	**Treat as an acute stroke: immediate transfer to a stroke centre or emergency department.** Neurological risk in the following days is substantial. Ocular measures (ocular massage, paracentesis, IOP lowering, carbogen, hyperbaric oxygen) have weak evidence and must not delay stroke evaluation. **Over age 50, obtain ESR, CRP, and platelet count and exclude giant cell arteritis before anything else**; if suspected, start high-dose steroid immediately. Longer term: antiplatelet therapy, carotid imaging, echocardiography, rhythm monitoring, and vascular risk factor control. Watch for neovascularisation, which occurs in roughly `15 to 20%`.
Branch retinal artery occlusion	Sectoral field loss with corresponding retinal whitening along a distribution.	Identical systemic workup; the visual prognosis is better.
Cilioretinal artery occlusion	Isolated occlusion of a cilioretinal artery, sometimes with central macular involvement.	Strongly associated with **giant cell arteritis** when isolated, so investigate urgently.
Amaurosis fugax	Transient monocular loss like a descending curtain, resolving in minutes.	**A transient ischaemic attack.** Urgent (same day to 24 hours) stroke workup and antiplatelet therapy; giant cell arteritis screen over 50.
Ocular ischaemic syndrome	Mid-peripheral dot-blot haemorrhages, dilated but *not* tortuous veins, narrowed arteries, neovascularisation, **pain**, and a paradoxically low IOP despite rubeosis.	Implies `≥90%` carotid stenosis with about a `40%` five-year mortality; urgent vascular surgery referral and panretinal photocoagulation for neovascularisation.

VENOUS OCCLUSION

Entity	Presentation	Management
Central retinal vein occlusion, non-ischaemic	Dilated tortuous veins, haemorrhage in all four quadrants, disc oedema, macular oedema, few cotton wool spots, and no or mild RAPD; `<10` disc areas of non-perfusion.	**Anti-VEGF for macular oedema** (CRUISE, GALILEO, COPERNICUS support ranibizumab and aflibercept; SCORE supported triamcinolone but with more cataract and glaucoma). Address systemic risk: blood pressure, lipids, glucose, and **check IOP because glaucoma is a risk factor**. Review at **monthly intervals for the first 6 months**. About a third convert to ischaemic.

Entity	Presentation	Management
Central retinal vein occlusion, ischaemic	Extensive haemorrhage, marked cotton wool spots, poor acuity, **RAPD**, and `≥10` disc areas of capillary non-perfusion.	**Neovascular glaucoma risk is high and peaks at 3 to 4 months ("90-day glaucoma").** Examine **monthly with undilated gonioscopy and iris examination for 6 months**. **CVOS**: prophylactic panretinal photocoagulation is not indicated, but treat promptly at the first sign of iris or angle neovascularisation. Anti-VEGF for oedema and as an adjunct to control neovascularisation.
Branch retinal vein occlusion	Sectoral haemorrhage in a wedge respecting the horizontal raphe, at an arteriovenous crossing; hypertension is the dominant risk factor.	**Anti-VEGF for macular oedema** (BRAVO, VIBRANT). **BVOS**: grid laser helps macular oedema after haemorrhage clears, and sector scatter laser is used for neovascularisation. Treat hypertension.
Occlusion in a patient under 50	–	Add a **thrombophilia and inflammatory screen**: antiphospholipid antibodies, homocysteine, protein C and S, antithrombin III, factor V Leiden, full blood count with viscosity, ANA, ANCA, syphilis serology, and a review of oral contraceptive and hormone use. Consider Behçet and sarcoid for vasculitic occlusion.

Retinal breaks, detachment, and the peripheral retina

ACUTE POSTERIOR VITREOUS DETACHMENT: THE VISIT THAT PREVENTS BLINDNESS

History: sudden floaters, photopsia, and a Weiss ring. Ask specifically about **a curtain or shadow, and about a shower of floaters versus a single one**.

1. **Dilate both eyes and examine the peripheral retina with scleral indentation** (or at minimum a thorough three-mirror or wide-field examination). A retinal break is found in roughly `8 to 15%` of acute symptomatic PVDs, and in a higher proportion when there is vitreous haemorrhage or pigment in the vitreous.
2. **Vitreous haemorrhage or pigment granules (tobacco dust, Shafer sign) markedly raise the probability of a break.** If the view is obscured, obtain B-scan ultrasonography and re-examine.
3. **Treat symptomatic horseshoe (flap) tears and operculated tears with symptoms** by laser retinopexy or cryotherapy, urgently.
4. **Re-examine at 2 to 6 weeks** even if the initial examination is clear, because new breaks develop during ongoing vitreous separation. Re-examine sooner for new symptoms.
5. **Educate explicitly**: return immediately for an increase in floaters, flashes, a curtain, or any loss of field.

PERIPHERAL LESIONS: WHICH TO TREAT

Lesion	Risk and management
Symptomatic horseshoe (flap) tear	**High risk.** **Treat** with laser or cryo retinopexy.
Symptomatic operculated hole	Lower risk than a flap tear because the traction has released; treat if there is subretinal fluid, persistent traction, or high risk otherwise.

Lesion	Risk and management
Asymptomatic flap tear	Lower risk; often observed, but treat in a fellow eye of a detachment, in an aphakic or pseudophakic eye, with a family history, or before cataract or refractive surgery.
Retinal dialysis	**Treat.** Usually traumatic; classically inferotemporal (or superonasal after trauma). Often causes a slowly progressive detachment.
Lattice degeneration without a hole	Present in `6 to 10%` of the population; **generally observed**. Consider treatment in a fellow eye of a phakic detachment or with a strong family history.
Lattice with atrophic holes	Usually observed in an asymptomatic patient; the risk of detachment from atrophic holes alone is low, and treatment carries its own risk.
Asymptomatic atrophic round holes	Observe.
Pavingstone (cobblestone) degeneration	Benign; never causes detachment because the retina is adherent to the RPE. Observe.
Peripheral cystoid degeneration	Universal after childhood; benign.
Degenerative retinoschisis	Bilateral inferotemporal, smooth immobile dome, **absolute scotoma** (a detachment gives a relative one), inner layer breaks are large and round while outer layer breaks are irregular. Observe unless there are both inner and outer layer breaks producing a progressive detachment.
White with or without pressure	Benign; document to distinguish from a break on later examination.
Meridional folds and enclosed oral bays	Benign anatomic variants; a break may occur at the posterior end of a meridional fold.

RETINAL DETACHMENT TYPES

Type	Features and treatment
Rhegmatogenous	A break plus liquefied vitreous plus traction. Corrugated, mobile, convex surface, tobacco dust, low IOP, and a relative field defect. Risk factors: myopia, aphakia and pseudophakia, trauma, lattice, family history, Stickler syndrome. **Treat surgically**: pneumatic retinopexy, scleral buckle, or vitrectomy with tamponade. **Macula-on detachments are treated within 24 hours where possible; macula-off within a few days.** Refer urgently and instruct the patient to lie with the break dependent (positioning the detachment to limit progression toward the fovea).
Tractional	Concave, immobile, tented toward fibrovascular membranes; proliferative diabetic retinopathy, sickle cell disease, ROP, penetrating trauma. Vitrectomy with membrane delamination when the macula is threatened or involved.
Exudative (serous)	Smooth convex surface, **shifting subretinal fluid** with position, no break, and usually normal IOP. Causes: tumour (choroidal melanoma, metastasis, haemangioma), inflammation (VKH, posterior scleritis, sympathetic ophthalmia), central serous chorioretinopathy, hypertensive or eclamptic choroidopathy, Coats disease, nanophthalmos. **Treat the cause, not the retina.**

Macular disorders

NON-NEOVASCULAR MACULOPATHY

Condition	Presentation	Management
Central serous chorioretinopathy	Young to middle-aged man, **type A personality, steroid exposure (any route, including inhaled and topical), stress, obstructive sleep apnoea, pregnancy**; metamorphopsia, micropsia, and a hyperopic shift with a smooth serous elevation and a **thickened choroid**. Angiography shows an expansile dot or smokestack leak.	**Stop exogenous steroid** where possible; most acute cases resolve in 3 to 4 months. For persistent or recurrent disease: **half-dose or half-fluence photodynamic therapy** (best evidence, per PLACE), subthreshold micropulse laser, or an oral mineralocorticoid antagonist (evidence is weak). Anti-VEGF only if secondary choroidal neovascularisation develops. Screen for sleep apnoea.
Epiretinal membrane	Metamorphopsia and gradual blur; cellophane sheen then retinal striae; often after PVD, vein occlusion, uveitis, or surgery.	Observe if mild; **vitrectomy with membrane peel** for symptomatic distortion or acuity around 20/50 or worse.
Vitreomacular traction	Persistent foveal vitreous adhesion with distortion and foveal elevation.	Observe (many release spontaneously), pharmacologic vitreolysis with ocriplasmin in selected small focal adhesions, or vitrectomy.
Full-thickness macular hole	Central scotoma and marked distortion; a positive **Watzke-Allen sign** (a break in a slit beam across the fovea). Gass stages 1 (impending, foveal cyst) to 4 (full hole with complete PVD).	Observation for stage 1 (about half resolve). **Vitrectomy with internal limiting membrane peel and gas tamponade with face-down positioning** closes over `90%` of holes under `400 µm`; smaller and more recent holes do better.
Lamellar hole and pseudohole	Partial-thickness defect (lamellar) versus an ERM with steep foveal walls and no tissue loss (pseudohole).	Usually observed; OCT distinguishes them and prevents unnecessary surgery.

Condition	Presentation	Management
Myopic maculopathy	High axial myopia with lacquer cracks, posterior staphyloma, chorioretinal atrophy, myopic CNV (a small pigmented Fuchs spot), and a myopic macular hole with schisis.	Anti-VEGF for myopic CNV; monitor with OCT. Discuss lifelong detachment risk and symptom education.
Cystoid macular oedema	Petalloid on angiography and cystoid on OCT. Causes: postoperative (Irvine-Gass), diabetes, vein occlusion, uveitis, retinitis pigmentosa, epinephrine and prostaglandin analogues, and **nicotinic acid (which does not leak on angiography)**.	Treat the cause; topical NSAID plus steroid, then periocular or intravitreal steroid, or acetazolamide (particularly effective in retinitis pigmentosa).
Macular telangiectasia type 2	Middle-aged, bilateral, temporal parafoveal greying with blunted right-angled venules, superficial retinal crystals, and inner retinal cavitation on OCT without leakage-driven thickening.	No effective treatment for the non-proliferative form; anti-VEGF for the proliferative stage. Low vision support.

INHERITED RETINAL AND MACULAR DYSTROPHIES

Disease	Recognition and management
Retinitis pigmentosa	**Nyctalopia first, then progressive ring scotoma and field constriction**, with bone-spicule pigment, arteriolar attenuation, waxy disc pallor, posterior subcapsular cataract, and cystoid macular oedema. ERG shows **reduced rod responses early**. Syndromic forms: **Usher** (deafness), **Bardet-Biedl** (obesity, polydactyly, renal disease), **Refsum** (phytanic acid, neuropathy), **Kearns-Sayre** (ophthalmoplegia, heart block), **abetalipoproteinaemia** (treatable with vitamins A and E). Management: genetic testing and counselling, low vision rehabilitation, **acetazolamide for macular oedema**, cataract surgery, UV protection, avoidance of vitamin A in *ABCA4* disease, and gene therapy where a treatable genotype exists (voretigene neparvovec for biallelic *RPE65*).
Stargardt disease	*ABCA4*, autosomal recessive; central vision loss in youth with yellow pisciform flecks, a beaten-bronze macula, and a **dark choroid** on fluorescein angiography. **Avoid vitamin A supplementation.** Low vision rehabilitation and genetic counselling.

Disease	Recognition and management
Best vitelliform dystrophy	*BEST1*, autosomal dominant; an egg-yolk macular lesion progressing to scrambled and atrophic stages. **Normal ERG with a markedly abnormal EOG**. Monitor for choroidal neovascularisation.
Congenital stationary night blindness	Non-progressive nyctalopia; a **negative ERG** (normal a-wave, reduced b-wave); Schubert-Bornschein type. Fundus albipunctatus is a variant with delayed dark adaptation.
X-linked retinoschisis	Boys with reduced central vision, a **spoke-wheel foveal schisis**, and a **negative ERG**; peripheral schisis with vitreous veils.
Choroideremia	X-linked; progressive scalloped choroidal and RPE atrophy from the periphery inward with nyctalopia in boys.
Gyrate atrophy	Ornithine aminotransferase deficiency; scalloped chorioretinal atrophy, high myopia, and cataract; hyperornithinaemia. **Arginine-restricted diet and pyridoxine.**
Cone and cone-rod dystrophy	Photophobia, reduced acuity, and **colour vision loss early**, with a bull's-eye maculopathy; photopic ERG affected first.

Drug and toxic retinopathy

RECOGNITION AND MONITORING PROTOCOLS

Agent	Finding and monitoring
Hydroxychloroquine	**Parafoveal (or pericentral in Asian patients) ellipsoid zone loss with a preserved foveal centre**, progressing to bull's-eye maculopathy. **Risk above `5.0 mg/kg/day` of real body weight and after 5 years**; renal impairment and tamoxifen co-use increase risk. Baseline within a year, then **annual screening after 5 years** with **10-2 automated fields plus SD-OCT** (adding multifocal ERG and fundus autofluorescence). **Damage is irreversible and can progress after stopping.**
Thioridazine	Nummular then diffuse pigmentary retinopathy; dose-dependent and can progress after cessation. Chlorpromazine causes lens and corneal pigment instead.
Tamoxifen	Refractile crystalline deposits and cystoid macular oedema; annual dilated review.
Pentosan polysulfate	Distinctive pigmentary maculopathy with hyperpigmented spots; baseline and periodic OCT and autofluorescence in long-term users.
MEK inhibitors	Serous retinal detachments, usually self-limited; OCT surveillance and oncology liaison.
Nicotinic acid	Cystoid maculopathy that **does not leak on fluorescein**; reverses on stopping.
Interferon	Retinal haemorrhages and cotton wool spots; usually reversible.
Fingolimod	Macular oedema in the first 3 to 4 months; baseline and follow-up OCT.
Methanol	Acute bilateral vision loss with disc oedema from formic acid; a medical emergency needing fomepizole or ethanol and dialysis.

Choroidal lesions

NAEVUS VERSUS MELANOMA, AND THE REST

Lesion	Features and action
Choroidal naevus	Flat or minimally elevated, slate-grey, with overlying drusen and often a halo. **Risk factors for malignancy (TFSOM-DIM)**: **T**hickness over `2 mm`, subretinal **F**luid, **S**ymptoms, **O**range pigment, **M**argin within `3 mm` of the disc, ultrasound hollowness, and absence of a halo or drusen. Photograph, measure with ultrasound and OCT, and review at **6 months then annually**; refer any lesion with two or more risk factors or documented growth.

Lesion	Features and action
Choroidal melanoma	Elevated pigmented dome or mushroom shape with subretinal fluid, orange lipofuscin, low internal reflectivity, and growth. **Refer to an ocular oncology centre.** Treatment is plaque brachytherapy, proton beam, or enucleation. Prognostication by cytogenetics and gene expression profile; metastasis is **hepatic**, so liver surveillance is lifelong.
Choroidal metastasis	Creamy, plateau-shaped, often multifocal and bilateral, with disproportionate subretinal fluid and high internal reflectivity. Breast in women, lung in men. Systemic staging and oncology referral; external beam radiotherapy or systemic therapy.
Choroidal haemangioma	Orange-red, high and homogeneous reflectivity, early intense filling with late washout. Photodynamic therapy for exudation. Diffuse form in Sturge-Weber.
Choroidal osteoma	Juxtapapillary orange-yellow plaque with dense acoustic shadowing; may decalcify and lose vision, and may develop CNV.
Congenital hypertrophy of the RPE	Flat, jet-black, sharply demarcated with lacunae; benign. **Multiple bear-track lesions or pigmented ocular fundus lesions raise the question of familial adenomatous polyposis**, warranting colonoscopy referral.

Vitreous and paediatric or systemic retinal disease

Entity	Key points
Vitreous haemorrhage	Causes in order of frequency: **proliferative diabetic retinopathy, PVD with or without a retinal tear, retinal vein occlusion with neovascularisation, trauma, and retinal macroaneurysm**. If no view: **B-scan ultrasonography**, examine the fellow eye for clues, review in days, and consider urgent vitrectomy if a detachment is present. Advise head elevation and avoidance of anticoagulant dose changes without the prescriber.
Asteroid hyalosis	Refractile calcium-lipid spheres, unilateral in two thirds, usually asymptomatic, and rarely requiring intervention; it complicates fundus imaging and biometry.
Retinopathy of prematurity	Zones I to III and stages 1 to 5 with **plus disease** (venous dilatation and arterial tortuosity at the posterior pole) driving treatment. Screening by gestational age and birth weight; treatment with laser or intravitreal anti-VEGF. Long-term myopia, strabismus, and amblyopia follow-up.
Coats disease	Unilateral, boys, retinal telangiectasia with massive lipid exudation and exudative detachment; a leukocoria differential from retinoblastoma (no calcification). Laser or cryotherapy.
Familial exudative vitreoretinopathy	Bilateral peripheral avascularity with dragged discs and falciform folds, in a term infant, distinguishing it from ROP. Genetic testing and family screening.
Sickle cell retinopathy	**SC and S-thalassaemia genotypes give the worst proliferative disease.** Goldberg stages 1 to 5 with **sea fan** neovascularisation; salmon patches, black sunbursts, iridescent spots, and angioid streaks. Scatter laser for proliferation; **avoid carbonic anhydrase inhibitors in sickle hyphaema.**
White dot syndromes	**Multiple evanescent white dot syndrome**: young woman, unilateral, foveal granularity, self-limited. **Acute posterior multifocal placoid pigment epitheliopathy**: bilateral placoid lesions after a viral illness, with a cerebral vasculitis risk. **Serpiginous choroiditis**: peripapillary, chronic, sight-threatening, needing immunosuppression and exclusion of tuberculosis. **Punctate inner choroidopathy**: myopic young woman, CNV risk. **Birdshot**: HLA-A29. **Multifocal choroiditis with panuveitis**: chronic, CNV risk.

Entity	Key points
Traumatic retinal findings	**Commotio retinae** (photoreceptor outer segment disruption, not oedema; usually resolves), **choroidal rupture** (yellow-white streaks concentric to the disc, CNV risk), **traumatic macular hole**, **retinal dialysis**, **sclopetaria**, **Purtscher retinopathy** (polygonal whitening after chest compression, long bone fracture, or pancreatitis), **Valsalva retinopathy** (sub-internal limiting membrane haemorrhage, resolves), **Terson syndrome** (vitreous haemorrhage with subarachnoid haemorrhage), and **shaken baby syndrome** (multilayered haemorrhages to the periphery with perimacular folds: a **mandatory report**).

RETRIEVAL PRACTICE

1. Give the exact AREDS2 formulation and its precise indication.
2. Why was β-carotene removed, and why is copper included?
3. What did DRCR Protocol T conclude, and for which baseline acuity did it matter?
4. What did Protocol S conclude, and when would you still choose panretinal photocoagulation?
5. A 68-year-old has sudden painless loss with a cherry red spot. Give the first three actions.
6. An ischaemic CRVO. Give the follow-up interval, the examination technique, and the month of peak neovascular risk.
7. A 38-year-old with a CRVO. What extra tests do you order?
8. A patient with acute floaters and a Weiss ring. What is the probability of a break, what raises it, and when do you re-examine?
9. Which four peripheral lesions do you treat, and which four do you observe?
10. Distinguish degenerative retinoschisis from a rhegmatogenous detachment on three findings.
11. A macula-on rhegmatogenous detachment. What is the time frame for surgery and what positioning advice do you give?
12. A 44-year-old man on inhaled steroid has metamorphopsia and a smooth serous elevation with a thick choroid. Diagnosis, first action, and the best-evidence treatment for persistence.
13. Give the Watzke-Allen sign and the surgical success rate for a small full-thickness macular hole.
14. A young man with nyctalopia, ring scotoma, bone spicules, and a reduced rod ERG. Diagnosis, four syndromic associations, and the drug that treats his macular oedema.
15. State the hydroxychloroquine screening protocol precisely.
16. Name the risk factors that convert a choroidal naevus into a referral.
17. Which sickle genotype has the worst retinopathy, and what drug do you avoid in sickle hyphaema?
18. A child has multilayered retinal haemorrhages to the periphery with perimacular folds and a subdural haematoma. Diagnosis and your obligation.

KEY

1. Vitamin C `500 mg`, vitamin E `400 IU`, lutein `10 mg`, zeaxanthin `2 mg`, zinc oxide `80 mg`, cupric oxide `2 mg`; for **intermediate AMD in one or both eyes or advanced AMD in one eye**.
2. β-carotene increased lung cancer risk in current and former smokers; copper prevents zinc-induced copper deficiency anaemia.
3. Aflibercept was superior at one year for centre-involving diabetic macular oedema when baseline acuity was **20/50 or worse**; with better baseline acuity the agents performed similarly.
4. Ranibizumab was non-inferior to PRP for proliferative disease at two and five years, with less field loss. Choose PRP when **follow-up adherence is unreliable**, or when injections are impractical.
5. Recognise CRAO as a stroke: send immediately to a stroke centre or emergency department; obtain **ESR, CRP, and platelet count** to exclude giant cell arteritis; and if GCA is suspected, start high-dose corticosteroid immediately.
6. **Monthly for 6 months**, with **undilated gonioscopy and iris examination** at each visit; neovascular glaucoma peaks at **3 to 4 months**.
7. Thrombophilia and inflammatory screen: antiphospholipid antibodies, homocysteine, protein C and S, antithrombin III, factor V Leiden, full blood count and viscosity, ANA, ANCA, syphilis serology, and review of oral contraceptive use.

8. A break is present in roughly **8 to 15%**; **vitreous haemorrhage or pigment (Shafer sign)** markedly raises it; re-examine at **2 to 6 weeks** and sooner for new symptoms.
9. Treat: symptomatic horseshoe tear, retinal dialysis, symptomatic operculated hole with fluid or traction, and a break in the fellow eye of a detachment or before intraocular surgery. Observe: lattice without symptoms, atrophic round holes, pavingstone degeneration, peripheral cystoid degeneration, white with or without pressure, and meridional folds.
10. Retinoschisis is smooth, immobile, and gives an **absolute** scotoma; a detachment is corrugated, mobile, and gives a **relative** scotoma, usually with tobacco dust and a lower IOP.
11. Surgery **within 24 hours where possible**; advise the patient to position so that the **break is dependent**, limiting progression of fluid toward the fovea, and to avoid exertion.
12. Central serous chorioretinopathy; **stop the exogenous steroid** in consultation with the prescriber; the best-evidence treatment for persistent disease is **half-dose or half-fluence photodynamic therapy**.
13. A break or thinning of a vertical slit beam projected across the fovea; vitrectomy with internal limiting membrane peel and gas closes over `90%` of holes under `400 µm`.
14. Retinitis pigmentosa; Usher, Bardet-Biedl, Refsum, Kearns-Sayre (also abetalipoproteinaemia); **acetazolamide** for the macular oedema.
15. Risk above `5.0 mg/kg/day` real body weight; baseline within a year of starting; then **annual screening after 5 years** with **10-2 visual fields plus SD-OCT**, adding multifocal ERG and fundus autofluorescence where available, and screening earlier with renal impairment, high dose, or tamoxifen use.
16. Thickness over `2 mm`, subretinal fluid, symptoms, orange pigment, margin within `3 mm` of the disc, ultrasonographic hollowness, and absence of a halo or drusen; refer with two or more, or with documented growth.
17. **SC** (and S-thalassaemia); avoid **carbonic anhydrase inhibitors**, which cause acidosis and promote sickling in the anterior chamber.
18. Abusive head trauma (shaken baby syndrome); you have a **mandatory legal obligation to report** to child protective services, and to document findings carefully with imaging where possible.

CHAPTER RECAP

1. AREDS2 is for intermediate or unilateral advanced AMD only, and smoking cessation beats it.
2. Anti-VEGF is first-line for neovascular AMD, centre-involving diabetic oedema, and vein occlusion oedema.
3. Protocol T favoured aflibercept at worse baseline acuity; Protocol S made anti-VEGF a legitimate alternative to PRP when follow-up is reliable.
4. CRAO is a stroke, and over 50 it is giant cell arteritis until excluded.
5. Ischaemic CRVO needs monthly undilated gonioscopy for six months; the peak is at three to four months.
6. Acute PVD needs indented peripheral examination, Shafer sign assessment, and a repeat visit at two to six weeks.
7. Treat flap tears and dialyses; observe lattice, atrophic holes, pavingstone, and schisis without both-layer breaks.
8. Exudative detachment shifts with position and means find the cause.
9. CSC: stop the steroid, wait, then half-dose photodynamic therapy.
10. Hydroxychloroquine screening is 10-2 fields plus SD-OCT, annually after five years, with a 5.0 mg/kg threshold.

CHAPTER 8 · DISEASE AND MANAGEMENT

Optic nerve and neuro-ophthalmic disease

Optic nerve / neuro-ophthalmic pathways · 28–38 items | Emergencies / trauma | Systemic health

ANSWER THESE COLD BEFORE READING

1. Give the ONTT conclusions including the treatment that is contraindicated.
2. Separate arteritic from non-arteritic AION on six features and give the immediate management of each.
3. Give the modified Dandy criteria and the IIHTT conclusion.
4. Name the four disc appearances that mimic papilloedema and how to distinguish each.
5. State the workup for an isolated third, fourth, and sixth nerve palsy by age and pupil status.
6. Give the three same-day neuro-ophthalmic emergencies.

BLUEPRINT WEIGHT

28–38

Notably heavier than on Part I: this is the third largest disease area on the examination.

Optic neuropathies

OPTIC NEURITIS

Aspect	Detail
Presentation	Young adult, usually a woman aged 20 to 45. **Subacute monocular vision loss over hours to days** with **pain on eye movement in over 90%**, a central or diffuse field defect, an **RAPD**, red desaturation, and reduced contrast sensitivity. The disc is **normal in about two thirds (retrobulbar)** and swollen in one third. **Uhthoff phenomenon** (worsening with heat or exercise) and **Pulfrich phenomenon** may be reported.
Optic Neuritis Treatment Trial	**Intravenous methylprednisolone `250 mg` every 6 hours for 3 days, then oral prednisone `1 mg/kg` for 11 days with a taper, accelerated visual recovery but did not change final acuity.** **Oral prednisone `1 mg/kg` alone is contraindicated: it increased the recurrence rate.** Intravenous therapy delayed the onset of clinically definite multiple sclerosis at two years, but the effect was not sustained at longer follow-up. **MRI risk stratification**: with **two or more typical white matter lesions**, the 15-year risk of MS is about `72%`; with a **normal MRI**, about `25%`. Visual prognosis is good: most recover to 20/40 or better, though contrast, colour, and stereopsis often remain subtly abnormal.
When it is not typical optic neuritis	**Atypical features demand a different workup**: age under 12 or over 50, bilateral simultaneous involvement, no pain, severe loss (worse than counting fingers), a very swollen disc with haemorrhage, macular exudate, no recovery by 4 weeks, progressive worsening beyond 2 weeks, or steroid dependence. Test for **aquaporin-4 antibodies (neuromyelitis optica spectrum disorder)** and **MOG antibodies**, and consider sarcoid, syphilis, tuberculosis, Lyme, Bartonella, and compressive or infiltrative disease. NMOSD requires aggressive acute treatment (steroid then plasma exchange) and long-term immunosuppression, because attacks are severe and steroid-dependent.

Aspect	Detail
Management in practice	MRI of the brain and orbits with gadolinium and fat suppression. Offer intravenous steroid for rapid recovery, occupational need, or bilateral or severe loss; otherwise observation is acceptable for typical cases. Refer to neurology for MS risk counselling and consideration of disease-modifying therapy if MRI shows demyelination. Follow with fields and OCT for retinal nerve fibre layer thinning at 3 to 6 months.

ARTERITIC VERSUS NON-ARTERITIC ANTERIOR ISCHAEMIC OPTIC NEUROPATHY

Feature	Arteritic (giant cell arteritis)	Non-arteritic
Age	Usually `>70`, essentially never under 50	`50–70`
Vision	Often **counting fingers or worse**	Usually better than 20/200, altitudinal
Disc	**Chalky white, pallid swelling**, sometimes with cotton wool spots or a concurrent cilioretinal or central retinal artery occlusion	**Hyperaemic segmental swelling** with splinter haemorrhages, and a crowded "disc at risk" in the fellow eye
Symptoms	**Jaw claudication** (most specific), scalp tenderness, new headache, temporal artery tenderness, polymyalgia, weight loss, fever, and preceding amaurosis fugax or diplopia	Usually none; often noticed on waking; nocturnal hypotension and sleep apnoea are risk factors
Laboratory	**ESR often >50 mm/h**, raised CRP, thrombocytosis, normochromic anaemia	Normal inflammatory markers
Fellow eye	**Very high risk within days to weeks if untreated**	About `15%` over five years
Management	**Start high-dose corticosteroid immediately, before biopsy.** Intravenous methylprednisolone `1 g` daily for 3 days for visual loss, then oral prednisone `1 mg/kg`; or oral `1 mg/kg` if no visual loss. Arrange **temporal artery biopsy within about 2 weeks** (steroid does not immediately abolish histologic findings), and involve rheumatology. Tocilizumab is steroid-sparing. Bone protection and glucose monitoring for prolonged steroid.	No proven treatment. Manage vascular risk factors, **screen for obstructive sleep apnoea**, avoid nocturnal antihypertensive dosing where possible, and counsel on fellow eye risk. Aspirin is commonly given though the evidence is weak.

OTHER OPTIC NEUROPATHIES

Entity	Recognition and management
Posterior ischaemic optic neuropathy	Acute loss with an initially **normal disc** and later pallor; perioperative (prone spinal surgery, cardiac bypass with hypotension and anaemia), or arteritic. Exclude giant cell arteritis and compression by MRI.
Compressive optic neuropathy	**Slowly progressive painless loss with disc pallor, a field defect, and preserved acuity until late.** Optic nerve sheath meningioma (with optociliary shunt vessels and progressive pallor), sphenoid meningioma, pituitary lesion, thyroid eye disease. Any unexplained progressive optic neuropathy needs an MRI.
Infiltrative optic neuropathy	Lymphoma, leukaemia, sarcoid, and metastasis; disc swelling with rapid loss and often other neurologic signs. MRI plus systemic workup.

Entity	Recognition and management
Toxic and nutritional optic neuropathy	**Bilateral, symmetric, painless, progressive, with centrocaecal scotomas and dyschromatopsia** and initially normal discs. Causes: **ethambutol** (dose-dependent, above `15–25 mg/kg/day`), isoniazid, linezolid, amiodarone, chloramphenicol, **methanol**, tobacco and alcohol, and **B_{12}, folate, thiamine, and copper deficiency** (including after bariatric surgery). **Stop the offending agent immediately**, check B_{12}, folate, thiamine, and copper, and supplement. Recovery is possible but often incomplete.
Leber hereditary optic neuropathy	Mitochondrial (maternally inherited; mtDNA 11778, 3460, 14484), **young men, sequential bilateral painless severe central loss over weeks**, with peripapillary telangiectasia that does **not leak** on angiography. Genetic testing and counselling; idebenone and gene therapy are emerging; advise against smoking and alcohol.
Dominant optic atrophy	*OPA1*, autosomal dominant, insidious bilateral childhood-onset loss with **temporal wedge pallor** and a **blue-yellow (tritan) colour defect**; mild to moderate severity. Family examination.
Traumatic optic neuropathy	Deceleration injury transmitting force to the **intracanalicular** segment; RAPD with an initially normal disc. CT to exclude a canal fracture or haematoma. Steroid benefit is unproven and CRASH showed harm in head injury; management is generally observation.
Radiation optic neuropathy	Months to years after orbital or skull base radiotherapy; progressive loss with disc swelling then pallor and enhancement on MRI. Hyperbaric oxygen and anti-VEGF have been tried with limited success.
Optic disc drusen	Lumpy disc margin with an anomalous vascular pattern, **autofluorescent**, hyperreflective on B-scan and OCT, and **no obscuration of vessels**. Causes field defects and, rarely, non-arteritic AION. Benign, but the important job is distinguishing it from true papilloedema.
Optic nerve hypoplasia	Small disc with a **double ring sign**. **Screen for septo-optic dysplasia (de Morsier)** with MRI and endocrine evaluation, because pituitary insufficiency can be fatal.
Optic nerve pit and morning glory disc	Pit is associated with serous macular detachment; morning glory anomaly with basal encephalocele and moyamoya, warranting imaging.
Tilted disc	Inferonasal tilt with situs inversus and a crescent; produces a **superotemporal field defect that does not respect the vertical meridian and often improves with refractive correction**: a classic pseudo-bitemporal defect.

Disc swelling and raised intracranial pressure

PAPILLOEDEMA: IDENTIFY IT, THEN ACT

Signs: bilateral disc elevation with blurred margins, **obscuration of vessels as they cross the disc**, hyperaemia, peripapillary haemorrhages and cotton wool spots, **Paton lines** (circumferential retinal folds), **loss of spontaneous venous pulsation** (present in only about `80%` of normals, so its absence is soft evidence), and an enlarged blind spot with generalised constriction on perimetry. Acuity is often **normal until late**, so **acuity is not the monitoring parameter: fields are**.

Symptoms: headache worse on waking and with Valsalva, **transient visual obscurations** lasting seconds, **pulsatile tinnitus**, diplopia from a sixth nerve palsy, and nausea.

1. **Measure blood pressure immediately** to exclude malignant hypertension.
2. **Same-day neuroimaging**: MRI of the brain with contrast plus **MR venography** to exclude a mass and cerebral venous sinus thrombosis.
3. **Lumbar puncture with opening pressure** and cerebrospinal fluid analysis once a mass is excluded.
4. **Modified Dandy criteria for idiopathic intracranial hypertension**: papilloedema; a normal neurological examination except cranial nerve abnormalities; normal neuroimaging apart from typical findings (empty sella, transverse sinus stenosis, optic nerve sheath distension, posterior globe flattening); normal cerebrospinal fluid composition; and an **elevated opening pressure above `250 mm` H_2O in adults** (or `280` in children).
5. **Treat**: **weight loss** (the IIHTT showed acetazolamide plus a weight-reduction diet improved field outcomes more than diet alone), **acetazolamide** titrated upward as tolerated, topiramate as an alternative, and stop any contributing drug (tetracyclines, vitamin A and retinoids, oral contraceptives, lithium, growth hormone, and steroid withdrawal).
6. **Surgery for progressive field loss or fulminant disease**: optic nerve sheath fenestration (best for vision when headache is not dominant), ventriculoperitoneal or lumboperitoneal shunt, or venous sinus stenting.
7. **Monitor with serial automated perimetry and OCT**, initially every few weeks, and refer any progression urgently.

BILATERAL DISC ELEVATION: THE DIFFERENTIAL

Entity	Distinguishing features
True papilloedema	Vessels obscured, hyperaemia, haemorrhages, Paton lines, normal acuity early, headache and transient obscurations, raised opening pressure.
Optic disc drusen (pseudopapilloedema)	Lumpy margin, vessels **not** obscured, anomalous branching, **autofluorescence**, hyperreflectivity on B-scan and OCT, no hyperaemia, no haemorrhage, and often a family history.
Hyperopic crowded disc	Small disc with a full but not elevated appearance; normal fields; refractive context.
Myelinated nerve fibres	Feathery white patches obscuring vessels but continuous with the retina rather than confined to the disc.
Malignant hypertension	Grade 4 changes with haemorrhage, exudate, and a macular star, plus a blood pressure at or above `180/120` with target organ damage.
Bilateral AION or optic neuritis	Reduced acuity and colour vision, RAPD if asymmetric, and appropriate field defects.
Infiltrative or inflammatory (sarcoid, lymphoma, leukaemia)	Reduced acuity with other systemic features; MRI enhancement.
Diabetic papillopathy	Bilateral disc swelling with surprisingly preserved acuity in a diabetic; self-limited.

Pupil abnormalities

THE ANISOCORIA ALGORITHM

Finding	Interpretation and action
Anisocoria greater in bright light	The **larger** pupil is abnormal: parasympathetic problem. Consider **third nerve palsy** (look for ptosis and motility deficit: **same-day angiography**), **Adie tonic pupil** (sector palsy, vermiform movements, light-near dissociation, slow redilation, **constricts to pilocarpine** `0.125%`), **pharmacologic mydriasis** (does not constrict to pilocarpine `1%`), or **traumatic iris damage** (sphincter tears visible).
Anisocoria greater in dim light	The **smaller** pupil is abnormal: sympathetic problem. **Horner syndrome** with ptosis, reverse ptosis, and anhidrosis. Confirm with **apraclonidine** `0.5%` **(reversal of anisocoria)** or cocaine; localise with **hydroxyamphetamine** `1%`. **A painful Horner syndrome is a carotid dissection until proven otherwise: same-day CT or MR angiography from the arch to the circle of Willis.** In a **child**, exclude **neuroblastoma** with imaging of the neck, chest, and abdomen and urinary catecholamines. In an adult with a second-order pattern, exclude a **Pancoast tumour** with chest imaging.
Anisocoria equal in light and dark, under `1 mm`	**Physiologic anisocoria**, present in about `20%` of people. Review old photographs. No workup needed.
Light-near dissociation	Dorsal midbrain (Parinaud) syndrome with upgaze palsy and convergence-retraction nystagmus, Argyll Robertson pupils (neurosyphilis), Adie pupil, aberrant third nerve regeneration, severe bilateral afferent disease, and diabetic autonomic neuropathy.
Relative afferent pupillary defect	Indicates **asymmetric optic nerve or extensive retinal disease**, or an optic tract lesion. **A cataract does not cause an RAPD**, and neither does refractive error or non-organic loss. Bilateral symmetric optic neuropathy gives no RAPD.

Ocular motor cranial neuropathies

WORKUP BY NERVE, AGE, AND PUPIL

Nerve	Presentation	Workup and management
CN III	Ptosis, exotropia and hypotropia ("down and out"), limited adduction, elevation, and depression, with or without pupil involvement.	**Pupil-involving, or any pain, or any age under 50: same-day CT angiography or MR angiography to exclude a posterior communicating artery aneurysm.** Pupil-sparing complete palsy in a vasculopath over 50 may be observed with weekly review for two weeks (the pupil can become involved), then monthly; expect resolution in **8 to 12 weeks**. **Aberrant regeneration** (lid elevation on adduction or downgaze, pupil constriction on adduction) implies compression and mandates imaging. Any partial or progressive palsy is imaged.
CN IV	Vertical diplopia and torsion worse on downgaze and on ipsilateral head tilt, with a compensatory head tilt away from the affected side.	Commonest causes: **congenital** (large vertical fusional amplitudes above `3 Δ`, long-standing head tilt, facial asymmetry) and **closed head trauma** (often bilateral, with excyclotorsion over `10°` and a V pattern). Microvascular in a vasculopath. Image if there are other neurologic signs, if it is progressive, or if the patient is young without trauma. Manage with prism, then occlusion, then strabismus surgery once stable for months.

Nerve	Presentation	Workup and management
CN VI	Esotropia worse at distance with limited abduction and an incomitant deviation; head turn toward the affected side.	In a vasculopath over 50 with no other signs, observe with review; expect recovery in **8 to 12 weeks**. **Image if under 50, bilateral, progressive, associated with other cranial nerve signs, with papilloedema, or with headache**: a sixth nerve palsy is the classic **false localising sign of raised intracranial pressure**, and it also occurs in Gradenigo syndrome, cavernous sinus disease, nasopharyngeal carcinoma, and demyelination. In a child, always image.
Multiple cranial neuropathies	Combinations of III, IV, V1, V2, VI with proptosis, chemosis, or a Horner syndrome.	Localise to the **cavernous sinus or orbital apex** and image urgently: tumour, aneurysm, fistula, thrombosis, or Tolosa-Hunt syndrome. Painful ophthalmoplegia with V1 numbness and normal imaging suggests Tolosa-Hunt, which is exquisitely steroid-responsive; a rapid steroid response does not, however, exclude lymphoma.

THE MIMICS THAT ARE NOT CRANIAL NERVE PALSIES

- **Myasthenia gravis.** Can imitate any pattern of ophthalmoplegia, including an apparent third nerve palsy. The pupil is **always spared**. Look for variability, fatigability, Cogan lid twitch, enhancement of ptosis, and a positive ice pack or rest test.
- **Thyroid eye disease.** Restrictive, with positive forced duction, IOP rise on attempted gaze into the restricted direction, tendon-sparing muscle enlargement, and lid retraction.
- **Orbital fracture with entrapment.** Restrictive, with a trauma history and infraorbital hypaesthesia.
- **Chronic progressive external ophthalmoplegia.** Symmetric, slowly progressive, and typically *without diplopia*.
- **Skew deviation and internuclear ophthalmoplegia.** Brainstem, not nerve; look for other brainstem signs and the ocular tilt reaction.
- **Decompensated phoria.** Comitant, long-standing, with large fusional amplitudes.

Chiasmal, retrochiasmal, and higher visual disorders

LOCALISE, THEN ACT

Presentation	Localisation and management
Bitemporal hemianopia	Chiasm. **MRI of the brain and sella with contrast.** Pituitary adenoma (superior quadrants first, with endocrine symptoms and possible galactorrhoea or acromegaly), craniopharyngioma (inferior first, in children and young adults), meningioma, aneurysm, glioma. Look for **hemifield slide** and loss of stereopsis. Prolactinoma responds to a dopamine agonist; other adenomas need transsphenoidal surgery. **Pituitary apoplexy** (sudden headache, ophthalmoplegia, visual loss, hypotension) is a neurosurgical and endocrine emergency requiring steroid replacement.
Junctional scotoma	Optic nerve at the chiasm; urgent MRI.
Homonymous hemianopia	Retrochiasmal. **Congruity localises**: incongruous suggests tract or lateral geniculate, congruous suggests occipital. Any new homonymous defect is a **stroke workup**: urgent imaging, carotid and cardiac evaluation, and vascular risk management. Counsel about **driving**, which is legally restricted in most jurisdictions, and refer for rehabilitation with prism, scanning training, and occupational therapy.

Presentation	Localisation and management
Cortical blindness	Bilateral occipital infarction; normal pupils and normal discs. With denial of blindness it is **Anton syndrome**.
Balint syndrome	Bilateral parieto-occipital: simultanagnosia, optic ataxia, and oculomotor apraxia.
Visual hallucinations	**Charles Bonnet syndrome**: formed, complex, non-threatening hallucinations with *preserved insight* in a patient with vision loss; reassure and explain, and optimise vision and lighting. Distinguish from occipital seizure (brief, stereotyped, coloured shapes), migraine aura, delirium, and psychosis.
Non-organic (functional) visual loss	Suspect when findings are internally inconsistent: **no RAPD with claimed severe monocular loss**, spiral or tunnel fields that do not enlarge with distance, normal OCT and electrodiagnostics, and better performance on tests the patient believes are harder. Use the fogging test, the four-dioptre base-out prism test, stereoacuity (good stereo excludes severe monocular loss), and mirror or optokinetic testing. Manage with a positive, non-confrontational explanation and follow-up; exclude organic disease first, and consider psychological or social stressors, including in children where school or family stress is common.

THE THREE SAME-DAY NEURO-OPHTHALMIC EMERGENCIES

1. **Giant cell arteritis** (any acute optic neuropathy or artery occlusion over 50): ESR, CRP, platelets, and **start high-dose steroid before biopsy**.
2. **Pupil-involving third nerve palsy**: same-day angiography for a posterior communicating artery aneurysm.
3. **Painful Horner syndrome**: same-day angiography of the head and neck for carotid dissection.

Two more that belong on the same list: **papilloedema** (same-day imaging and blood pressure) and **a new homonymous hemianopia** (acute stroke pathway).

RETRIEVAL PRACTICE

1. Give the ONTT steroid regimen, what it changed, what it did not, and what is contraindicated.
2. Give the 15-year MS risk after optic neuritis with two or more MRI lesions and with a normal MRI.
3. Name seven atypical features of optic neuritis and the two antibodies you would order.
4. A 78-year-old has vision of counting fingers with a chalky white swollen disc and pain on chewing. Give the three tests and the first action.
5. Distinguish arteritic from non-arteritic AION on the disc appearance and the fellow eye risk.
6. Give the modified Dandy criteria and the opening pressure threshold in an adult.
7. State the IIHTT conclusion and four drugs that can cause the syndrome.
8. Which parameter do you monitor in papilloedema, and why not acuity?
9. Distinguish optic disc drusen from true papilloedema on four findings.
10. Anisocoria greater in dim light with mild ptosis. Which pupil is abnormal, and what is your next test?
11. A 55-year-old has a painful pupil-involving third nerve palsy. What do you order and when?
12. Bilateral progressive painless central loss with centrocaecal scotomas and colour loss in a patient on tuberculosis therapy. Diagnosis and management.
13. A young man loses central vision in one eye then the other over six weeks, with peripapillary telangiectasia that does not leak. Diagnosis and inheritance.
14. A patient reports formed, complex, non-threatening hallucinations and knows they are not real. Diagnosis and management.
15. Which findings suggest non-organic visual loss, and name three tests you would use?

16. Name the three same-day neuro-ophthalmic emergencies.

KEY

1. Intravenous methylprednisolone `250 mg` every 6 hours for 3 days, then oral prednisone `1 mg/kg` for 11 days with a taper. It **accelerated recovery** but did **not change final acuity**. **Oral prednisone `1 mg/kg` alone is contraindicated because it increased recurrence.**
2. About `72%` with two or more lesions, and about `25%` with a normal MRI.
3. Age under 12 or over 50, bilateral simultaneous onset, absence of pain, severe loss worse than counting fingers, marked disc swelling with haemorrhage or macular exudate, no recovery by 4 weeks, progression beyond 2 weeks, and steroid dependence. Order **aquaporin-4** and **MOG** antibodies.
4. **ESR, CRP, and platelet count**; first action is to **start high-dose corticosteroid immediately**, then arrange temporal artery biopsy within about two weeks.
5. Arteritic: chalky white pallid swelling, often with a concurrent artery occlusion, and a **very high** fellow eye risk within days to weeks. Non-arteritic: hyperaemic segmental swelling with splinter haemorrhages and a crowded fellow disc, with about `15%` fellow eye involvement over five years.
6. Papilloedema; a normal neurological examination apart from cranial nerve findings; normal imaging apart from typical IIH features; normal cerebrospinal fluid composition; and an opening pressure above $250\ \mathrm{mm}\ H_2O$.
7. Acetazolamide plus a weight-reduction diet improved visual field outcomes more than diet alone. Causative drugs: tetracyclines including minocycline, vitamin A and retinoids (isotretinoin), oral contraceptives, lithium, growth hormone, and steroid withdrawal.
8. **Automated visual fields** (with OCT), because acuity is preserved until very late and therefore fails to detect progression.
9. Drusen: lumpy margin, vessels **not** obscured, autofluorescent, hyperreflective on B-scan or OCT, no hyperaemia or haemorrhage, and often familial.
10. The **smaller** pupil is abnormal: a Horner syndrome. Next test is **apraclonidine `0.5%`** looking for reversal of anisocoria (or cocaine), then hydroxyamphetamine to localise.
11. **Same-day CT angiography or MR angiography** (with catheter angiography if negative) to exclude a posterior communicating artery aneurysm.
12. **Ethambutol optic neuropathy**; stop the ethambutol immediately in consultation with the treating physician, check B_{12}, folate, and copper, and monitor acuity, colour vision, and fields; recovery is possible but often incomplete.
13. Leber hereditary optic neuropathy; **mitochondrial, maternally inherited** (mtDNA 11778, 3460, 14484).
14. Charles Bonnet syndrome; reassure and explain the mechanism, optimise vision and lighting, and follow up. No antipsychotic is needed.
15. Internal inconsistency: absence of an RAPD with claimed severe monocular loss, tunnel or spiral fields that do not enlarge with distance, and normal OCT and electrodiagnostics. Tests: fogging, four-dioptre base-out prism, stereoacuity, and mirror or optokinetic testing.
16. Giant cell arteritis, a pupil-involving third nerve palsy, and a painful Horner syndrome (with papilloedema and a new homonymous hemianopia close behind).

CHAPTER RECAP

1. ONTT: intravenous steroid speeds recovery, changes nothing long term, and oral-alone is contraindicated. MRI stratifies MS risk 72% versus 25%.
2. Atypical optic neuritis means test aquaporin-4 and MOG and rethink the diagnosis.
3. Chalky white swelling with jaw claudication over 70 means steroid before biopsy.
4. Non-arteritic AION: manage vascular risk and screen for sleep apnoea; there is no treatment.
5. Papilloedema: blood pressure, MRI with venography, lumbar puncture, acetazolamide plus weight loss, and monitor with fields.
6. Optic disc drusen do not obscure vessels and are autofluorescent.
7. Anisocoria greater in light means the big pupil; greater in dark means the small one; equal and small means physiologic.
8. Pupil-involving third nerve palsy and painful Horner syndrome are same-day angiography.
9. Myasthenia mimics everything and always spares the pupil.
10. Toxic optic neuropathy is bilateral, symmetric, and centrocaecal; stop the drug and check B_{12}, folate, thiamine, and copper.

CHAPTER 9 · DISEASE AND MANAGEMENT

Glaucoma

Glaucoma · 16–27 items | TMOD-heavy | Emergencies (acute angle closure)

ANSWER THESE COLD BEFORE READING

1. State the conclusions of OHTS, EMGT, AGIS, CIGTS, CNTGS, and LiGHT and what each changed in practice.
2. Give the OHTS risk factors for conversion from ocular hypertension.
3. Give the full acute angle closure protocol in order, including when pilocarpine is added.
4. Distinguish six secondary open-angle and six secondary closed-angle mechanisms.
5. Give a target IOP framework by disease stage.
6. State how you decide that a field is progressing rather than fluctuating.

BLUEPRINT WEIGHT

16–27

Scored items, with a high TMOD density because nearly every item concerns treatment selection and follow-up.

The trials, and what each one changed

LANDMARK GLAUCOMA TRIALS

Trial	Population	Conclusion and clinical consequence
OHTS	Ocular hypertension	Reducing IOP by `20%` halved the 5-year conversion rate (from about `9.5%` to `4.4%`). **Baseline risk factors for conversion**: older **age**, higher **IOP**, **thinner central corneal thickness**, larger **vertical cup-to-disc ratio**, and higher **pattern standard deviation**. *Consequence*: risk-stratify rather than treat everyone; CCT became a standard measurement.
EMGT	Early manifest glaucoma	Treatment (laser plus betaxolol) reduced progression from `62%` to `45%` at 6 years. **Every `1 mmHg` of IOP reduction lowered progression risk by about `10%`.** Risk factors: higher baseline IOP, bilateral disease, exfoliation, worse baseline field, older age, and disc haemorrhage. *Consequence*: lower is better, and there is no safe threshold.
AGIS	Advanced glaucoma failing medication	Eyes with IOP **consistently below `18 mmHg`** (mean about `12.3`) had essentially no field progression over 6 years. Race influenced the optimal surgical sequence. *Consequence*: advanced disease needs a low target, often in the low teens.

Trial	Population	Conclusion and clinical consequence
CIGTS	Newly diagnosed glaucoma	Initial medication and initial trabeculectomy gave similar field outcomes at 5 years; surgery achieved lower IOP but with more cataract and local discomfort. *Consequence*: either is defensible; individualise.
CNTGS	Normal-tension glaucoma	A `30%` IOP reduction slowed progression (progression in `12%` of treated versus `35%` of untreated), though many untreated eyes progressed slowly or not at all. *Consequence*: normal-tension glaucoma is still pressure-dependent, and a 30% reduction is the conventional target.
LiGHT	Newly diagnosed open-angle glaucoma and ocular hypertension	**Selective laser trabeculoplasty as first-line** gave equal or better quality of life and disease control at 3 and 6 years, with about three quarters of eyes drop-free at 3 years and fewer patients needing surgery. *Consequence*: SLT is now a legitimate and often preferred first-line treatment.
ZAP	Narrow angles without glaucoma	Prophylactic laser iridotomy reduced the incidence of angle closure disease but the absolute event rate was low, so **universal prophylactic iridotomy in all narrow angles is not justified**; target higher-risk eyes.
EAGLE	Primary angle closure with raised IOP	**Early lens extraction** was more effective and more cost-effective than iridotomy plus medication in patients over 50. *Consequence*: consider cataract or clear lens extraction as a primary treatment in angle closure.
UKGTS	Newly diagnosed glaucoma	Latanoprost preserved visual field measurably against placebo, validating field-based endpoints in trials.

Primary open-angle glaucoma: diagnosis and monitoring

WHAT YOU DOCUMENT AT EVERY GLAUCOMA VISIT

Domain	Detail
IOP	Goldmann applanation with the time of day recorded, plus CCT at baseline. Consider diurnal or modified diurnal curves in patients progressing at apparently acceptable pressures.
Gonioscopy	**At baseline and periodically thereafter**, in dim illumination with a short beam avoiding the pupil, using indentation to distinguish appositional from synechial closure. Record the most posterior structure seen in each quadrant, iris configuration and insertion, pigment, blood vessels, and any synechiae or recession.

Domain	Detail
Optic disc	Disc size (a small disc makes a normal cup look suspicious and a large disc makes glaucoma look normal), **rim thickness by the ISNT rule** (Inferior > Superior > Nasal > Temporal), notching, vertical cup elongation, **disc haemorrhage** (a strong predictor of progression), peripapillary atrophy, bayoneting and nasalisation of vessels, and the laminar dot sign. **Photograph or image, because a drawing is not a baseline.**
Retinal nerve fibre layer and ganglion cell analysis	OCT with attention to raw B-scans and to red and green disease. RNFL thinning typically inferotemporal then superotemporal; macular ganglion cell complex thinning detects central and early damage.
Visual field	Threshold perimetry, usually 24-2, with 10-2 when central points are involved or the disease is advanced. Establish **two reliable baselines**. Assess reliability indices and artefacts before believing a defect.
Risk factors	Age, family history, race (African ancestry for open-angle, Asian and Inuit ancestry for angle closure), CCT, myopia (open-angle) or hyperopia (closure), diabetes, systemic hypotension and nocturnal dips, obstructive sleep apnoea, migraine and vasospasm, steroid use, and prior trauma or uveitis.

DECIDING THAT A FIELD IS REALLY PROGRESSING

1. **Require two reliable baselines.** The first field is often worse than the second because of a learning effect.
2. **Confirm any new defect with a repeat test** before acting. A single abnormal field changes nothing.
3. **Look at the pattern deviation plot and the GHT**, not just the global indices, and check that the defect is *anatomically plausible* (respects the horizontal midline, corresponds to the disc and OCT).
4. **Exclude artefacts**: lens rim, ptosis, miosis, uncorrected refractive error, cataract progression (which depresses MD diffusely), and fatigue.
5. **Use both event and trend analysis.** Guided progression analysis flags repeated worsening at the same points; the rate of MD or VFI change per year estimates the trajectory. A loss of more than about `1 dB per year` of MD is rapid.
6. **Correlate with structure.** Progression in both structure and function is convincing; progression in one alone warrants closer follow-up rather than immediate escalation, unless the disease is advanced or the patient is young.
7. **Consider life expectancy and stage.** A 45-year-old with early damage progressing slowly needs more aggressive treatment than an 88-year-old with the same rate.

TARGET IOP AS A STARTING FRAMEWORK, NOT A RULE

Stage	Initial target	Note
Ocular hypertension, higher risk	`20%` reduction	Per OHTS; treat only after risk stratification, and consider observation in low-risk patients.
Early glaucoma	`20–30%` reduction, often < `18 mmHg`	Reassess and lower the target if progression occurs.
Moderate glaucoma	`30%` reduction, often < `15 mmHg`	–
Advanced glaucoma	Often < `12 mmHg`	Per AGIS; consistently low pressure essentially arrests progression.
Normal-tension glaucoma	`30%` reduction	Per CNTGS; also evaluate for nocturnal hypotension, sleep apnoea, anaemia, and vasospasm, and exclude compressive and other non-glaucomatous causes with imaging when the disc, field, or acuity pattern is atypical.

WHEN NORMAL-TENSION GLAUCOMA SHOULD MAKE YOU ORDER AN MRI

Image when the picture does not fit: **reduced acuity or dyschromatopsia disproportionate to the field loss, disc pallor exceeding cupping, a field defect respecting the vertical meridian, rapid or asymmetric progression, an age under 50, neurologic symptoms, or a field defect that does not match the disc.** Compressive lesions, previous ischaemic events, and hereditary optic neuropathies all masquerade as normal-tension glaucoma.

Treatment

CHOOSING AND SEQUENCING THERAPY

Option	Role, and what to counsel
Prostaglandin analogue	**The default first-line medication**: once nightly, `25–33%` reduction, and it retains nocturnal efficacy. Counsel on hyperaemia, lash growth, iris and periocular pigmentation, and prostaglandin-associated periorbitopathy (which matters if trabeculectomy or lid surgery is later planned, and which is often asymmetric and cosmetically distressing). Avoid or use with caution in uveitis, in eyes at risk of macular oedema, and in herpetic disease.
Selective laser trabeculoplasty	**A legitimate first-line alternative** per LiGHT: about `20–30%` reduction, repeatable, no adherence problem, and no chronic surface toxicity. Best in eyes with pigmented angles including pseudoexfoliation and pigmentary glaucoma. Less effective in angle closure and in previously failed laser. Warn about transient IOP spike, mild inflammation, and the possibility of needing repeat treatment.
β-blocker	`20–25%` reduction, cheap, once or twice daily. **Contraindicated in asthma, COPD, bradycardia, and heart block; masks hypoglycaemia in insulin-treated diabetes.** Minimal nocturnal effect. Betaxolol is β_1-selective but not risk-free.
Carbonic anhydrase inhibitor, topical	`15–20%` reduction; useful adjunct; watch for surface stinging and for corneal decompensation in compromised endothelium; sulfonamide moiety.
α_2-agonist	`20–25%` reduction; up to `15%` develop follicular allergy over time. **Contraindicated under age 2 and avoided in young children**; avoid with monoamine oxidase inhibitors.
Rho kinase inhibitor (netarsudil)	About `20%` reduction by increasing trabecular outflow and lowering episcleral venous pressure; hyperaemia in most, conjunctival haemorrhage, and reversible cornea verticillata.
Fixed combinations	Improve adherence and reduce preservative load and washout. Use once two agents are established.
Minimally invasive glaucoma surgery	Trabecular bypass stents, canaloplasty, goniotomy, and subconjunctival microshunts, usually combined with cataract surgery in mild to moderate disease. Modest IOP reduction with a good safety profile; they reduce drop burden more than they reduce pressure.
Trabeculectomy with antimetabolite	The reference standard for substantial reduction in moderate to advanced disease. Risks: hypotony and maculopathy, bleb leak, blebitis and **bleb-related endophthalmitis (a lifelong risk requiring patient education about a red, painful eye with discharge)**, cataract, and encapsulation.

Option	Role, and what to counsel
Glaucoma drainage device	For failed trabeculectomy, neovascular and uveitic glaucoma, and eyes with conjunctival scarring. Risks include tube erosion, diplopia, corneal decompensation, and hypertensive phase.
Cyclodestruction	Transscleral or endoscopic cyclophotocoagulation for refractory disease or eyes with poor visual potential; micropulse variants are used earlier.
Lens extraction	Per EAGLE, an effective primary treatment in angle closure; also lowers IOP modestly in open-angle disease.

ADHERENCE IS THE COMMONEST CAUSE OF APPARENT TREATMENT FAILURE

Roughly **a third to a half of patients are non-adherent** within a year. Before escalating, ask non-judgementally about missed doses, cost, instillation technique, and side effects; watch the patient instil a drop; simplify the regimen; use fixed combinations; and consider laser or a sustained-release option. **Also confirm the diagnosis and the target**: progression at a "good" pressure can mean the pressure is only good in the clinic, or that the disease is not glaucoma.

Angle closure

ACUTE PRIMARY ANGLE CLOSURE: THE PROTOCOL IN ORDER

Presentation: severe pain, nausea and vomiting, blurred vision with haloes, a red eye, corneal microcystic oedema, a **mid-dilated vertically oval unreactive pupil**, a shallow chamber, a closed angle on gonioscopy or indentation, and IOP often `50 to 80 mmHg`. Risk factors: hyperopia, short axial length, Asian or Inuit ancestry, female sex, age over 50, family history, and a shallow chamber. Precipitants: dim light, prolonged near work, anticholinergics, sympathomimetics, and topiramate (though topiramate closure is a different, non-pupillary-block mechanism).

1. **Topical aqueous suppressants at once**: one drop each of a β-blocker, an α_2-agonist, and a topical carbonic anhydrase inhibitor.
2. **Oral or intravenous acetazolamide** `500 mg`, provided there is no sickle cell disease, sulfa allergy, or renal failure.
3. **Topical corticosteroid** (prednisolone acetate `1%`) to reduce inflammation and clear the cornea.
4. **Position the patient supine** to let the lens fall back, and consider **corneal indentation with a four-mirror goniolens** to force aqueous into the angle.
5. **Pilocarpine `1–2%` only once IOP has fallen below roughly** `40 to 50 mmHg`, because an ischaemic sphincter will not respond and pilocarpine can worsen block by shifting the lens forward.
6. **Hyperosmotic agent** (oral glycerin, or intravenous mannitol `20%`) if still uncontrolled, remembering that glycerin raises blood glucose and mannitol overloads the circulation.
7. **Definitive treatment: laser peripheral iridotomy** once the cornea clears, and **prophylactic iridotomy in the fellow eye**, which has roughly a `50%` risk over five years.
8. **Reassess the angle after iridotomy.** Persistent closure implies plateau iris or a lens-related or posterior mechanism, and **consider lens extraction (EAGLE)**. Monitor for chronic angle closure from residual synechiae and for glaucomatous damage.

ANGLE CLOSURE MECHANISMS: IRIDOTOMY DOES NOT FIX ALL OF THEM

Mechanism	Recognition and treatment
Pupillary block	Iris bombé with a convex iris; deepens after iridotomy. **Iridotomy is definitive.**
Plateau iris	Anteriorly positioned ciliary processes; the angle remains closed after a patent iridotomy, with a flat central iris and a **double hump sign** on indentation gonioscopy; confirmed on ultrasound biomicroscopy or anterior segment OCT. Treat with **argon laser peripheral iridoplasty**, pilocarpine, or lens extraction.
Phacomorphic	An intumescent lens; treat medically then **extract the lens**.
Lens subluxation or microspherophakia	**Use cycloplegia, not miotics.** Iridotomy then lensectomy.
Aqueous misdirection (malignant glaucoma)	Shallow chamber with **high IOP and a patent iridotomy**, usually after intraocular surgery. **Miotics worsen it.** Treat with cycloplegia, aqueous suppressants, and hyperosmotics, then Nd:YAG disruption of the anterior hyaloid or vitrectomy.
Ciliochoroidal effusion	**Topiramate, sulfonamides, acetazolamide**, uveitis, panretinal photocoagulation, nanophthalmos, scleral buckle. **Iridotomy does not help.** Stop the drug, give cycloplegia, aqueous suppressants, and steroid.

Mechanism	Recognition and treatment
Neovascular	Fibrovascular membrane pulling the angle closed; treat the ischaemia with anti-VEGF and panretinal photocoagulation, plus aqueous suppressants and often a drainage device. **Miotics and prostaglandins are relatively contraindicated.**
Inflammatory with peripheral anterior synechiae	Control the uveitis and use cycloplegia to break synechiae; iridotomy may be needed for seclusio pupillae with bombé.
Posterior segment mass or effusion	Tumour, uveal effusion, posterior scleritis, or a large choroidal detachment pushing the lens-iris complex forward. Image and treat the cause.

Secondary open-angle glaucoma

MECHANISM DETERMINES MANAGEMENT

Entity	Recognition	Management specifics
Pseudoexfoliation glaucoma	Fibrillar material on the anterior capsule in a **target pattern with a clear intermediate zone**, on the pupil margin and zonules; heavy irregular trabecular pigment with a **Sampaolesi line**; poor dilation; weak zonules; often asymmetric.	Higher and more fluctuant IOP with faster progression than primary open-angle glaucoma, so target lower and follow more closely. **Responds well to laser trabeculoplasty.** Warn about surgical risk (capsule rupture, zonular dialysis, late in-the-bag IOL subluxation) and about systemic elastosis associations.
Pigmentary glaucoma	Young myopic men; **Krukenberg spindle** (vertical endothelial pigment), mid-peripheral radial **iris transillumination defects**, dense homogeneous trabecular pigment, a deep chamber with a posteriorly bowed iris. Pressure spikes after exercise or dilation.	Reverse pupillary block is the mechanism, so **laser trabeculoplasty works well** (use lower energy because of the heavy pigment) and iridotomy is sometimes used. Burns out in later decades as pigment liberation declines.

Entity	Recognition	Management specifics
Steroid-induced glaucoma	Rising IOP weeks after starting steroid by any route, including inhaled, nasal, dermatologic, intra-articular, and intravitreal implants. About `5%` are high responders, and the rate is far higher in existing glaucoma, first-degree relatives, high myopes, diabetics, and children.	Stop or reduce the steroid, switch to loteprednol or fluorometholone, and treat the pressure. An intravitreal implant may keep the pressure up for months, so plan accordingly. **Check IOP at every steroid visit.**
Uveitic glaucoma	Multiple mechanisms: trabeculitis, inflammatory debris, steroid response, peripheral anterior synechiae, and pupillary block from seclusio pupillae.	Control the inflammation first, then the pressure. **Avoid prostaglandins and miotics.** Use aqueous suppressants; surgery often needs a drainage device rather than trabeculectomy.
Angle recession (traumatic) glaucoma	An abnormally wide ciliary body band with a torn iris insertion on gonioscopy, unilateral, often years after blunt trauma.	Standard medical therapy, but **trabeculoplasty responds poorly**. **Document angle recession and arrange lifelong annual surveillance**, because the glaucoma can appear decades later, and examine the fellow eye.
Neovascular glaucoma	Iris and angle neovascularisation with a fibrovascular membrane; pain, high IOP, and often a red eye. Causes: **proliferative diabetic retinopathy, ischaemic CRVO, ocular ischaemic syndrome**, and less commonly tumour or chronic uveitis.	**Treat the ischaemia**: anti-VEGF for rapid regression plus panretinal photocoagulation for durable control, alongside aqueous suppressants and steroid for comfort. Cyclodestruction or a drainage device for established angle closure. **Prevention is the real treatment: monthly undilated gonioscopy for 6 months after an ischaemic CRVO.**

Entity	Recognition	Management specifics
Lens-related	Phacolytic (macrophages), lens particle, phacoantigenic.	Remove the lens material.
Ghost cell and haemolytic glaucoma	Degenerated erythrocytes after vitreous haemorrhage obstruct the meshwork; khaki-coloured cells in the anterior chamber.	Medical control, anterior chamber washout, or vitrectomy.
Raised episcleral venous pressure	**Blood in Schlemm canal** on gonioscopy; carotid-cavernous fistula, Sturge-Weber, thyroid eye disease, superior vena cava obstruction, idiopathic.	Treat the cause; surgery carries a high risk of choroidal effusion because of the pressure gradient.
Iridocorneal endothelial syndrome	Unilateral, middle-aged woman, corectopia and polycoria, peripheral anterior synechiae, abnormal endothelium.	Medical therapy and drainage surgery; keratoplasty for oedema.
Posner-Schlossman	Recurrent very high IOP with minimal inflammation; CMV implicated.	Aqueous suppressants plus short-course steroid; consider aqueous PCR for CMV. Damage accumulates over years.

Childhood glaucoma

PRIMARY CONGENITAL GLAUCOMA: THE CLASSIC TRIAD AND THE URGENCY

Epiphora, photophobia, and blepharospasm in an infant, with **corneal enlargement (above `12 mm` in the first year), corneal clouding, horizontal Haab striae, buphthalmos, a deep chamber, and raised IOP**. *CYP1B1* in many cases.

Differentiate from: congenital nasolacrimal duct obstruction (tearing without photophobia, a clear cornea, and normal corneal diameter), megalocornea, birth trauma with vertical Descemet breaks, congenital hereditary endothelial dystrophy, and mucopolysaccharidosis.

Management is surgical and urgent: goniotomy or trabeculotomy first-line, with medical therapy only as a bridge (and **brimonidine is contraindicated in infants**). Then lifelong follow-up, with refractive correction and amblyopia therapy for the induced myopia and anisometropia, which cause much of the final visual disability.

Secondary childhood glaucoma: aphakic or pseudophakic glaucoma after paediatric cataract surgery (a lifelong risk requiring annual surveillance), Axenfeld-Rieger, aniridia, Sturge-Weber, neurofibromatosis 1, Peters anomaly, and uveitic glaucoma in juvenile idiopathic arthritis.

RETRIEVAL PRACTICE

1. Give the five OHTS baseline risk factors for conversion.
2. What did EMGT quantify about each mmHg of IOP reduction?
3. What IOP did AGIS associate with essentially no progression?
4. What did CNTGS show, and what is the conventional target in normal-tension glaucoma?
5. What did LiGHT change about first-line treatment?
6. What did EAGLE change about angle closure treatment?
7. State the ISNT rule and why disc size matters when applying it.
8. Give six steps you take before concluding that a visual field is progressing.
9. Give the acute angle closure protocol in order and state when pilocarpine is added and why.
10. The angle remains closed after a patent iridotomy with a flat central iris and a double hump on indentation. Diagnosis and two treatments.
11. A postoperative eye has a shallow chamber, high IOP, and a patent iridotomy. Diagnosis, and which drug class must you avoid?
12. A patient on topiramate has bilateral acute closure with a myopic shift. Mechanism, and why is iridotomy unhelpful?
13. Krukenberg spindle with mid-peripheral iris transillumination defects in a young myopic man. Diagnosis and the preferred laser treatment with one technical caveat.
14. Target pattern on the anterior capsule with a Sampaolesi line. Diagnosis, three surgical warnings, and the treatment that works particularly well.
15. Unilateral glaucoma with a widened ciliary body band. Diagnosis, the treatment that responds poorly, and your long-term plan.
16. An infant has tearing, photophobia, blepharospasm, and a corneal diameter of `13 mm`. Diagnosis, the definitive treatment, and one drug that is contraindicated.
17. Name five circumstances in which normal-tension glaucoma should prompt an MRI.

KEY

1. Older age, higher IOP, thinner central corneal thickness, larger vertical cup-to-disc ratio, and higher pattern standard deviation.
2. Each `1 mmHg` of IOP reduction reduced the risk of progression by about `10%`.
3. Consistently below `18 mmHg`, with a mean around `12.3 mmHg`.

4. A **30%** IOP reduction slowed progression (12% versus 35%), though many untreated eyes progressed slowly; the conventional target is a **30% reduction**.
5. It established **selective laser trabeculoplasty as an effective first-line treatment** with equal or better quality of life and disease control and fewer patients needing surgery.
6. It showed **early lens extraction** is more effective and more cost-effective than iridotomy plus medication in patients over 50 with primary angle closure and raised IOP.
7. Rim thickness normally Inferior > Superior > Nasal > Temporal. Disc size matters because a **small disc makes a normal cup look pathological** and a **large disc can hide real rim loss**.
8. Two reliable baselines; confirm with a repeat; read the pattern deviation and GHT; exclude artefacts; use both event and trend analysis; and correlate with structure, with the patient's age and stage informing the threshold for action.
9. Topical β-blocker, α_2-agonist and carbonic anhydrase inhibitor; oral or intravenous acetazolamide **500 mg**; topical steroid; supine positioning and corneal indentation; then **pilocarpine once IOP is below about 40 to 50 mmHg**, because an ischaemic sphincter will not respond and pilocarpine can worsen block by moving the lens forward; hyperosmotic if needed; then **laser iridotomy in both eyes**.
10. **Plateau iris**; argon laser peripheral iridoplasty or pilocarpine, and lens extraction.
11. **Aqueous misdirection (malignant glaucoma)**; avoid **miotics**, which worsen it. Treat with cycloplegia, aqueous suppressants, and hyperosmotics, then YAG hyaloidotomy or vitrectomy.
12. **Ciliochoroidal effusion** with forward rotation of the ciliary body and anterior lens movement; there is **no pupillary block**, so iridotomy does not address the mechanism. Stop the drug and give cycloplegia, aqueous suppressants, and steroid.
13. Pigmentary glaucoma from pigment dispersion syndrome; **laser trabeculoplasty works well**, but use **lower energy** because the heavy pigment absorbs more and risks a pressure spike.
14. Pseudoexfoliation glaucoma. Surgical warnings: poor dilation, zonular weakness with capsule rupture and zonular dialysis risk, and late in-the-bag IOL subluxation. **Laser trabeculoplasty** is particularly effective.
15. Angle recession glaucoma from prior blunt trauma; **trabeculoplasty responds poorly**; plan lifelong annual surveillance and examine the fellow eye, because the glaucoma can present decades later.
16. Primary congenital glaucoma; definitive treatment is **goniotomy or trabeculotomy**; **brimonidine is contraindicated** in infants because of apnoea and CNS depression.
17. Acuity or colour vision loss out of proportion to the field, disc pallor exceeding cupping, a field defect respecting the vertical meridian, rapid or asymmetric progression, age under 50, neurologic symptoms, or a field that does not match the disc.

CHAPTER RECAP

1. OHTS risk-stratifies, EMGT says every mmHg counts, AGIS says advanced disease needs the low teens, CNTGS says 30% even at normal pressure, LiGHT makes SLT first-line, EAGLE makes lens extraction a primary option in closure.
2. Gonioscopy and disc imaging at baseline are not optional.
3. Never diagnose progression on one field; use two baselines, confirmation, structure, and rate.
4. Target by stage, then lower it if the disease progresses.
5. Acute closure: suppress, steroid, position, indent, then pilocarpine, then bilateral iridotomy, then reassess the angle.
6. Iridotomy fixes pupillary block only; plateau iris, aqueous misdirection, ciliochoroidal effusion, and phacomorphic closure each need something else.
7. Pseudoexfoliation and pigmentary glaucoma both respond well to trabeculoplasty; angle recession does not.
8. Neovascular glaucoma is prevented by surveillance after an ischaemic CRVO, and treated by treating the ischaemia.
9. Check IOP at every visit in anyone on steroid by any route.
10. Congenital glaucoma is a surgical emergency with a lifelong refractive and amblyopia burden.

CHAPTER 10 · DISEASE AND MANAGEMENT

Ocular emergencies and trauma

Emergencies / trauma · 11–22 items | TMOD bullet 6: emergencies and urgencies | Legal issues and ethics

ANSWER THESE COLD BEFORE READING

1. List the conditions that must be managed within minutes, within hours, and within a day.
2. Give the management of a suspected open globe in order, and what you must not do.
3. Give the hyphaema management protocol and the two populations needing special caution.
4. Give the emergency management of retrobulbar haemorrhage.
5. State the eight-point examination sequence in blunt ocular trauma.
6. Give the reporting obligations for abusive injury and for reportable infections.

BLUEPRINT WEIGHT

11–22

A named category on Part II (unlike Part I, where it is embedded), and every item is effectively a TMOD item.

Triage: what must happen when

URGENCY TIERS

Timeframe	Conditions
Minutes act before completing the examination	**Chemical burn** (irrigate before measuring acuity), **retrobulbar haemorrhage with an orbital compartment syndrome** (canthotomy and cantholysis at the bedside), **central retinal artery occlusion** (activate the stroke pathway).
Same hour to same day	**Open globe or suspected penetrating injury**, **endophthalmitis**, **giant cell arteritis** (steroid before biopsy), **acute angle closure**, **orbital cellulitis**, **necrotising fungal orbital infection**, **pupil-involving third nerve palsy**, **painful Horner syndrome**, **papilloedema**, **acute retinal necrosis**, **gonococcal keratoconjunctivitis**, **a new homonymous hemianopia**, **hypertensive emergency with grade 4 retinopathy**, **a white-eyed trapdoor orbital fracture with an oculocardiac reflex in a child**, and **corneal graft rejection**.
Within 24 to 72 hours	**Macula-on rhegmatogenous retinal detachment** (ideally within 24 hours), **symptomatic retinal tear**, **amaurosis fugax** (stroke workup within 24 hours), **acute uveitis**, **infectious keratitis**, **hyphaema**, **traumatic iritis with raised IOP**, **acute optic neuritis**, **herpes zoster ophthalmicus within the 72-hour antiviral window**, and **neovascular AMD with new distortion**.
Within days to a week	Macula-off detachment (a few days), macular hole assessment, non-arteritic AION, vein occlusion with macular oedema, and routine postoperative complications that are improving.

Trauma: the systematic examination

EIGHT STEPS IN EVERY OCULAR TRAUMA ASSESSMENT

1. **Mechanism and timing.** Ask specifically about high-velocity metal on metal, hammering, grinding, glass, vegetation, chemical, blunt, and whether eye protection was worn. Establish tetanus status. **Document the mechanism verbatim**: it has medicolegal weight and it determines the differential.
2. **Visual acuity in both eyes, documented before any intervention** (except in a chemical burn, where you irrigate first). This is the single most important medicolegal and prognostic datum.
3. **Pupils and an RAPD**, checked before dilating.
4. **External and lid examination**, including a search for a canalicular laceration (any laceration medial to the punctum) and for orbital rim step-offs and infraorbital hypaesthesia.
5. **Motility and forced duction** if there is any restriction.
6. **Slit lamp with Seidel testing**: look for a shallow or deep chamber, hyphaema, iris transillumination defects, sphincter tears, phacodonesis, lens opacity, and vitreous in the anterior chamber. **Evert both lids and sweep the fornices.**
7. **IOP**, unless an open globe is suspected, in which case **do not apply anything to the eye**.
8. **Dilated fundus examination with scleral indentation** once an open globe is excluded, plus **gonioscopy** when the eye is quiet, to document angle recession. Order **CT of the orbits** for any suspicion of a fracture or an intraocular or intraorbital foreign body.

TRAUMA ENTITIES: MANAGEMENT

Injury	Signs	Management
Open globe / penetrating injury	Full-thickness wound, **positive Seidel**, a peaked or irregular pupil, iris or uveal prolapse, a shallow or very deep chamber, hyphaema, low IOP, subconjunctival haemorrhage that is bullous and 360 degrees, vitreous haemorrhage, or a visible entry track. A **high-velocity mechanism with a normal-looking eye still requires exclusion**.	**Do not press on the eye, do not measure IOP, do not instil ointment, do not attempt to remove a protruding foreign body, and give nothing by mouth.** Place a **rigid shield** (not a patch), elevate the head, give analgesia and an antiemetic (vomiting can extrude contents), start systemic antibiotic prophylaxis, update tetanus, obtain **CT of the orbits** (**avoid MRI until a ferromagnetic foreign body is excluded**), and arrange **immediate surgical referral** for repair within 24 hours. Warn about **endophthalmitis** and about **sympathetic ophthalmia** in the fellow eye.

Injury	Signs	Management
Intraocular foreign body	Entry wound, focal lens opacity, iris hole with transillumination, or a visible intraretinal body. Siderosis (iron) gives heterochromia, mydriasis, cataract, and pigmentary retinopathy; chalcosis (copper) gives a Kayser-Fleischer-like ring and sunflower cataract, and pure copper causes suppurative endophthalmitis.	CT, surgical removal, and intravitreal or systemic antibiotic prophylaxis. Vegetable matter and copper must come out; inert glass and plastic may sometimes be observed.
Hyphaema	Layered blood in the anterior chamber; grade by the fraction filled. Look for the cause (angle recession, iris root tear, ciliary body tear) and for associated injury.	Elevate the head, restrict activity, **shield**, **topical steroid** (reduces inflammation and rebleed risk) plus **cycloplegia**, and **daily review for the first 5 days** because rebleeding peaks at **2 to 5 days**. Avoid aspirin and NSAIDs. Treat raised IOP; consider tranexamic or aminocaproic acid in selected cases. Surgical washout for a total ("eight-ball") hyphaema, corneal blood staining, uncontrolled IOP, or failure to clear. **Two special populations**: in **sickle cell disease or trait, avoid carbonic anhydrase inhibitors and hyperosmotics** (acidosis and haemoconcentration promote sickling in the anterior chamber, which raises IOP dramatically and damages the nerve at lower pressures); and in **children**, watch for amblyopia and consider abuse. **Gonioscopy once the eye is quiet**, then lifelong glaucoma surveillance.
Traumatic iritis	Cells and flare within 24 to 72 hours; pain and photophobia.	Topical steroid and cycloplegia; check IOP (it can be low from ciliary shutdown or high from trabeculitis or debris) and document the angle.
Corneal abrasion and foreign body	See Chapter 4.	Antibiotic, cycloplegia if painful, no patching in contact lens or vegetative injuries, never dispense anaesthetic.
Chemical burn	Pain, blepharospasm, epithelial loss, and **perilimbal blanching**, which grades the severity.	**Irrigate immediately and copiously, before acuity, with anaesthetic and lid eversion and fornix sweeping, until pH is 7.0 to 7.4 on recheck.** Then antibiotic, cycloplegia, preservative-free lubricants, short-course topical steroid, **oral doxycycline and vitamin C**, IOP monitoring, and daily symblepharon lysis. Refer severe burns for amniotic membrane or limbal stem cell transplantation. Alkali penetrates progressively; acid self-limits.

Injury	Signs	Management
Lid laceration	Assess depth, margin involvement, levator involvement, and **canalicular involvement in any laceration medial to the punctum**.	Rule out an occult globe injury and an orbital foreign body first. Marginal, canalicular, levator, and canthal tendon injuries need oculoplastic repair with stenting; simple superficial lacerations can be repaired locally. Tetanus and antibiotic prophylaxis, especially for animal or human bites.
Orbital floor fracture	Enophthalmos, hypoglobus, diplopia worse on upgaze with positive forced duction, infraorbital hypaesthesia, orbital emphysema.	CT orbits. **Urgent (24 to 48 hours) repair for a white-eyed trapdoor fracture in a child with an oculocardiac reflex**; otherwise repair within 2 weeks for large fractures, enophthalmos over `2 mm`, or persistent functional diplopia. Advise no nose-blowing; add oral antibiotics if the sinus is involved. Warn about the small risk of blindness from the repair.
Retrobulbar haemorrhage	**Proptosis, a tense orbit resistant to retropulsion, markedly raised IOP, reduced vision, an RAPD**, and severe pain. Follows blunt trauma, retrobulbar injection, or orbital or lid surgery, or anticoagulation.	**Immediate lateral canthotomy and inferior cantholysis at the bedside, plus IOP-lowering agents. Do not wait for imaging.** Then image and refer. Retinal ischaemia becomes irreversible within roughly 60 to 100 minutes.
Traumatic optic neuropathy	Reduced vision with an RAPD after deceleration injury, often with an initially normal disc.	CT to exclude a canal fracture or haematoma. Steroid benefit is unproven and CRASH showed harm in head injury; management is largely observation with surgical decompression in selected cases.
Commotio retinae	Grey-white retinal opacification after blunt trauma, from outer segment disruption rather than oedema.	Observation; most recover, though a macular defect can persist.
Choroidal rupture	Yellow-white streaks concentric to the disc.	Observation with education about **choroidal neovascularisation risk** and Amsler monitoring; anti-VEGF if CNV develops.
Traumatic retinal dialysis and detachment	Dialysis at the vitreous base, classically inferotemporal or superonasal.	Retinopexy or surgery; examine both eyes with indentation, and re-examine at intervals because presentation can be delayed by months.

Injury	Signs	Management
Purtscher and Valsalva retinopathy	Polygonal retinal whitening after chest compression, long bone fracture, or pancreatitis; or sub-internal limiting membrane haemorrhage after straining.	Observation, plus treatment of the systemic cause. Valsalva haemorrhage may be treated with YAG membranotomy if dense and persistent.
Abusive head trauma (shaken baby)	**Multilayered retinal haemorrhages extending to the periphery, retinoschisis with perimacular folds, vitreous haemorrhage, and a subdural haematoma with no external injury.**	**Mandatory report to child protective services.** Document meticulously with drawings and, where possible, retinal imaging; admit the child; involve paediatrics and safeguarding. This is the single most important non-accidental injury finding in ophthalmology.
Solar and laser retinopathy	Foveal outer retinal disruption after eclipse viewing or laser pointer exposure.	Observation; educate on prevention. Public health messaging around eclipses is an examinable optometric responsibility.

SIX TRAUMA ERRORS THAT APPEAR IN EXAM STEMS

- **Measuring IOP or pressing on a suspected open globe.**
- **Ordering MRI before excluding a metallic foreign body.**
- **Giving a carbonic anhydrase inhibitor for hyphaema in a patient with sickle cell trait.**
- **Waiting for a CT scan before decompressing an orbital compartment syndrome.**
- **Measuring acuity before irrigating a chemical burn.**
- **Failing to document the mechanism, the initial acuity, and whether eye protection was worn.** All three are medicolegally decisive.

Non-traumatic emergencies: quick reference

PRESENTATION TO ACTION

Presentation	Immediate action
Sudden painless monocular vision loss	Differentiate CRAO (cherry red spot, whitened retina, RAPD: stroke pathway plus giant cell arteritis screen over 50), CRVO, vitreous haemorrhage, retinal detachment, ischaemic optic neuropathy, and occipital infarct. In every case over 50, **consider giant cell arteritis**.
Transient monocular vision loss	Amaurosis fugax: urgent stroke workup within 24 hours, antiplatelet therapy, and giant cell arteritis screen over 50.
Sudden onset flashes and floaters	Dilate both eyes with indentation, look for Shafer sign, treat any break, and re-examine at 2 to 6 weeks with explicit symptom education.

Presentation	Immediate action
A curtain or shadow in the field	Assume retinal detachment: same-day dilated examination and, if confirmed, urgent vitreoretinal referral with positioning advice.
Severe pain with nausea, vomiting, haloes, and a red eye	Acute angle closure: begin the protocol immediately and arrange bilateral iridotomy.
Painful red eye with reduced vision after intraocular surgery or injection	Endophthalmitis until proven otherwise: same-hour referral for tap and inject.
New binocular diplopia	Localise the nerve, check the pupil, look for other cranial nerve or brainstem signs, and image urgently for pupil-involving third nerve palsy, any young patient, progressive or multiple palsies, papilloedema, or associated pain.
Bilateral disc swelling with headache	Measure blood pressure, then same-day MRI with venography, then lumbar puncture.
Proptosis with fever and restricted motility	Orbital cellulitis: CT and admission for intravenous antibiotics.
Proptosis with a black nasal or palatal eschar in a diabetic	Mucormycosis: emergency admission, biopsy, systemic antifungal, and surgical debridement.
Leukocoria in a child	Same-week paediatric ophthalmology referral for retinoblastoma exclusion; also consider Coats disease, PFV, toxocariasis, cataract, and ROP.
A red painful eye in a patient with a filtering bleb	Blebitis or bleb-related endophthalmitis: same-day referral. This is a lifelong risk that every trabeculectomy patient must be taught to recognise.

Reporting and documentation obligations

WHAT YOU ARE LEGALLY REQUIRED TO DO

Situation	Obligation
Suspected child abuse or neglect	**Mandatory report** to child protective services in every United States jurisdiction. You report suspicion, not proof, and reporting in good faith carries immunity. Failure to report is itself an offence.
Suspected elder or vulnerable adult abuse	Mandatory reporting in most jurisdictions; know your state statute.
Reportable infectious disease	Gonorrhoea, chlamydia, syphilis, tuberculosis, and HIV are reportable to public health in the United States, with specifics by state. Partner notification is part of management for the sexually transmitted infections.
Vision and driving	Know your state's acuity and field standards and its reporting rules, which range from mandatory physician reporting to voluntary. Counsel and document the counselling in every case of vision loss that may affect driving, including homonymous field defects, advanced glaucoma, and bioptic telescope candidates.
Adverse drug or device events	Report to the FDA MedWatch programme; report vaccine events to VAERS.
Documentation in trauma	Record the mechanism in the patient's words, the time of injury, whether protective eyewear was used, initial acuity in both eyes, pupil findings, and a diagram. Photograph where possible. This record is the case if litigation follows.

RETRIEVAL PRACTICE

1. Name the three conditions that must be treated within minutes, before completing the examination.
2. A man was hammering metal and now has mild irritation with 20/20 vision and a normal-looking eye. What must you do and why?
3. List six things you must *not* do with a suspected open globe.
4. Give the hyphaema regimen, the review interval, and the reason for that interval.
5. Which two drug classes must you avoid in a hyphaema in a patient with sickle cell trait, and why?
6. A patient has proptosis, a tense orbit, IOP 60 mmHg, and an RAPD after a retrobulbar injection. What do you do first, and what is the time window?
7. A child is struck by a ball, the eye looks quiet, upgaze is limited, and he is bradycardic and nauseated. Diagnosis and time frame.
8. Give the eight steps of the trauma examination in order.
9. Which laceration location mandates canalicular assessment?
10. Give the management sequence for an alkali burn.
11. A 3-month-old has multilayered retinal haemorrhages extending to the periphery and perimacular folds. Diagnosis and your obligations.
12. A trabeculectomy patient presents with a red painful eye and discharge. What is the concern, and what do you do?
13. Which imaging modality is contraindicated until a metallic foreign body is excluded?
14. Name four situations creating a mandatory reporting obligation.
15. What three data points in a trauma record are most medicolegally decisive?

KEY

1. Chemical burn (irrigate first), retrobulbar haemorrhage with orbital compartment syndrome (canthotomy and cantholysis), and central retinal artery occlusion (activate the stroke pathway).
2. Exclude an **occult open globe and intraocular foreign body**: examine for a tiny entry wound and Seidel positivity, look for an iris transillumination defect and focal lens opacity, dilate and examine the retina, and obtain a **CT of the orbits**. A high-velocity metal-on-metal mechanism can penetrate with almost no external sign and normal acuity.
3. Do not press on the eye, do not measure IOP, do not instil ointment, do not remove a protruding foreign body, do not patch (use a rigid shield), and do not give anything by mouth (also avoid MRI until metal is excluded).
4. Head elevation, activity restriction, rigid shield, topical steroid plus cycloplegia, avoid aspirin and NSAIDs, treat raised IOP, and review **daily for 5 days** because rebleeding peaks at **2 to 5 days**.
5. **Carbonic anhydrase inhibitors** (systemic acidosis promotes sickling) and **hyperosmotic agents** (haemoconcentration promotes sickling); sickled cells obstruct the meshwork and the optic nerve tolerates high pressure poorly in sickle disease.
6. Perform an immediate **lateral canthotomy and inferior cantholysis** at the bedside with IOP-lowering agents; retinal ischaemia becomes irreversible within roughly **60 to 100 minutes**.
7. A **white-eyed trapdoor orbital floor fracture with muscle entrapment and an oculocardiac reflex**; surgical release within **24 to 48 hours**.
8. Mechanism and timing; acuity in both eyes; pupils and RAPD before dilating; external and lid examination including canalicular assessment; motility and forced duction; slit lamp with Seidel and lid eversion; IOP unless an open globe is suspected; and dilated indented fundus examination with gonioscopy and CT as indicated.
9. Any laceration **medial to the punctum**.
10. Anaesthetise, evert lids and sweep fornices, irrigate copiously until pH is 7.0 to 7.4 on recheck, then antibiotic, cycloplegia, preservative-free lubricants, short-course topical steroid, oral doxycycline and vitamin C, IOP monitoring, daily symblepharon lysis, and referral for amniotic membrane in severe cases.
11. Abusive head trauma; **mandatory report to child protective services**, meticulous documentation with imaging where possible, admission, and involvement of paediatrics and safeguarding.
12. **Blebitis or bleb-related endophthalmitis**; same-day referral for assessment and treatment. This risk is lifelong and every trabeculectomy patient should be taught the warning signs.
13. **MRI.**

14. Suspected child abuse, suspected elder or vulnerable adult abuse, reportable infectious disease (gonorrhoea, chlamydia, syphilis, tuberculosis, HIV), and adverse drug or device events to MedWatch. Driving reporting varies by state.
15. The **mechanism in the patient's own words**, the **initial visual acuity in both eyes**, and **whether protective eyewear was worn**.

CHAPTER RECAP

1. Three conditions are treated before the examination is finished: chemical burn, orbital compartment syndrome, and CRAO.
2. A high-velocity mechanism with a normal-looking eye still needs CT.
3. Shield, do not patch; do not press, do not measure, do not remove.
4. Hyphaema is reviewed daily for five days because rebleeding peaks at two to five.
5. Sickle status changes the hyphaema drug list.
6. Canthotomy and cantholysis happen at the bedside, not after a scan.
7. A quiet eye with restricted upgaze and bradycardia in a child is a 24 to 48 hour surgical problem.
8. Any laceration medial to the punctum involves the canaliculus until proven otherwise.
9. Multilayered peripheral retinal haemorrhages in an infant are a mandatory report.
10. Mechanism, initial acuity, and eyewear use are the three data points that decide a lawsuit.

CHAPTER 11 · DISEASE AND MANAGEMENT

Systemic disease: ocular findings and comanagement

Systemic health · 11–22 items | Embedded across all case categories | Legal issues and ethics (referral, communication)

ANSWER THESE COLD BEFORE READING

1. For eight systemic diseases, state the ocular finding that would make you the first clinician to diagnose it.
2. State what you write in a referral letter and to whom, for four urgent systemic findings.
3. Give the ocular screening interval mandated by each of six systemic conditions or drugs.
4. Name the systemic drugs whose ocular effects require you to contact the prescriber.
5. State the ocular manifestations you must know for HIV by CD4 count.
6. Give the ocular considerations in pregnancy.

BLUEPRINT WEIGHT

11–22

As a named category, plus systemic disease is the connective tissue of most other case categories.

When the eye makes the systemic diagnosis

OCULAR FINDINGS THAT SHOULD TRIGGER A SYSTEMIC DIAGNOSIS OR REFERRAL

Ocular finding	Systemic diagnosis	What you do
Cotton wool spots, dot-blot haemorrhages, or hard exudates in an asymptomatic adult	**Diabetes mellitus** or hypertension, previously undiagnosed	Check blood pressure in the room; refer for fasting glucose and HgbA1c with a written note describing the retinal findings and the urgency. Never simply advise the patient to "get checked".
Grade 4 hypertensive retinopathy with disc oedema	**Hypertensive emergency**	Measure the blood pressure and send the patient to the emergency department the same day. Telephone the receiving clinician.
Hollenhorst plaque, or any retinal embolus	**Carotid or cardiac embolic source**	Urgent stroke or vascular referral for carotid imaging, ECG and rhythm monitoring, echocardiography, lipids, and glucose; antiplatelet therapy.
Chalky white pallid disc swelling with jaw claudication over 50	**Giant cell arteritis**	ESR, CRP, and platelets, then **start high-dose corticosteroid immediately** and arrange same-day rheumatology or medical involvement plus temporal artery biopsy.
Bilateral disc oedema with headache and transient obscurations	**Raised intracranial pressure**	Blood pressure, then same-day MRI with venography, then lumbar puncture. Refer to neurology or the emergency department.

Ocular finding	Systemic diagnosis	What you do
Necrotising scleritis or peripheral ulcerative keratitis	**Systemic vasculitis**, typically granulomatosis with polyangiitis or rheumatoid arthritis	ANCA, RF and anti-CCP, ANA, ESR and CRP, urinalysis, chest imaging; urgent rheumatology referral for systemic immunosuppression. Mortality is at stake, not just the eye.
Lisch nodules	**Neurofibromatosis 1**	Examine for café-au-lait macules and axillary freckling; refer for genetic and paediatric or neurology assessment and family screening.
Retinal capillary haemangioblastoma	**Von Hippel-Lindau disease**	Refer for *VHL* genetic testing and lifelong surveillance for renal cell carcinoma, phaeochromocytoma, and CNS haemangioblastoma; screen the family.
Retinal astrocytic hamartoma	**Tuberous sclerosis**	Examine the skin for ash-leaf macules, shagreen patch, and facial angiofibromas; refer for neurology, cardiology, and renal assessment.
Multiple bear-track CHRPE or pigmented fundus lesions	**Familial adenomatous polyposis**	Refer for colonoscopy and genetic counselling; this can be lifesaving.
Kayser-Fleischer ring	**Wilson disease**	Refer for serum copper and caeruloplasmin, liver function, and hepatology or neurology assessment.
Angioid streaks	**Pseudoxanthoma elasticum**, Paget disease, sickle cell disease, Ehlers-Danlos	Examine flexural skin, refer for cardiovascular and haematologic assessment, and counsel about CNV risk and contact sports.
Superotemporal lens subluxation with tall stature	**Marfan syndrome**	**Urgent cardiology referral for aortic root imaging**: aortic dissection is the cause of death.
Inferonasal lens subluxation	**Homocystinuria**	Refer for metabolic testing; warn about thromboembolic risk under general anaesthesia.
Optically empty vitreous with high myopia in a child	**Stickler syndrome**	Genetic testing, family screening, and prophylactic retinal assessment; this is the leading inherited cause of childhood retinal detachment.
Floppy eyelid with chronic irritation	**Obstructive sleep apnoea**	Ask about snoring and daytime somnolence; refer for sleep study. Untreated apnoea carries cardiovascular, NAION, papilloedema, and glaucoma risk.
Bilateral lacrimal gland enlargement	**Sarcoidosis, IgG4-related disease, lymphoma, tuberculosis**	ACE, lysozyme, serum IgG4, chest imaging, interferon-gamma release assay, and biopsy.

Ocular finding	Systemic diagnosis	What you do
Cherry red spot in an infant with developmental regression	**Sphingolipidosis (Tay-Sachs, Niemann-Pick)**	Urgent paediatric neurology and metabolic referral.
Bilateral optic disc pallor with a small disc and a double ring sign in a child	**Septo-optic dysplasia with pituitary insufficiency**	MRI and **urgent endocrine evaluation**: undiagnosed adrenal insufficiency can be fatal.
Unilateral proptosis with periorbital ecchymosis in a toddler	**Metastatic neuroblastoma**	Same-day paediatric oncology referral.

Screening intervals mandated by systemic disease and by drugs

THE INTERVALS YOU ARE EXPECTED TO KNOW WITHOUT PROMPTING

Condition or drug	Interval	Detail
Type 1 diabetes	Annually from `5 years` after diagnosis or from puberty	Dilated examination; more often with retinopathy.
Type 2 diabetes	Annually, starting **at diagnosis**	Retinopathy may already be present.
Diabetes in pregnancy (pre-existing)	Pre-conception or first trimester, then **each trimester** and post partum	Retinopathy accelerates in pregnancy. Gestational diabetes alone does not require retinal screening.
Hydroxychloroquine	Baseline within a year, then **annually after 5 years**	`10-2` fields plus SD-OCT; earlier and more often above `5.0 mg/kg/day`, with renal impairment, or with tamoxifen.
Ethambutol	Baseline, then **monthly** during treatment at higher doses	Acuity, colour vision, and fields; stop the drug at the first sign of optic neuropathy.
Vigabatrin	Baseline and **every 3 months**	Formal perimetry; a black box warning for irreversible concentric field constriction.
Pentosan polysulfate	Baseline and periodic	OCT and fundus autofluorescence in long-term users.
Fingolimod	Baseline and at `3 to 4 months`	OCT for macular oedema, particularly in diabetics and prior uveitis.
Tamoxifen	Annually	Crystalline retinopathy and macular oedema.
Amiodarone	Annually	Verticillata needs no action; watch for optic neuropathy.

Condition or drug	Interval	Detail
Juvenile idiopathic arthritis, high risk	**Every 3 months**	ANA-positive oligoarticular young girls; the uveitis is silent.
Systemic corticosteroid, any route	At every visit while on treatment	IOP and lens; roughly 5% are high responders and the rate is far higher in glaucoma, high myopia, diabetes, and children.
Sickle cell disease (SC, S-thal)	Annually from about age 10	Dilated peripheral examination for proliferative retinopathy.
Down syndrome	Regularly through childhood, then periodically	Refractive error, accommodative insufficiency (bifocals are often indicated), strabismus, keratoconus, cataract, and blepharitis.
Marfan syndrome	Annually	Lens position, refraction, retinal periphery; ensure cardiology follow-up is in place.
Neurofibromatosis 1	Annually in childhood	Optic pathway glioma surveillance with acuity, colour, fields, and OCT.
Prematurity (ROP survivors)	Per screening protocol, then annually	Myopia, strabismus, amblyopia, and late retinal complications.

HIV and immunosuppression

OCULAR DISEASE BY CD4 COUNT

CD4 (cells/μL)	Findings
Any count	Dry eye, conjunctival microvasculopathy, **Kaposi sarcoma** of the lid or conjunctiva, **molluscum contagiosum** (multiple lesions), **ocular surface squamous neoplasia in a young patient**, herpes zoster ophthalmicus (which in a young adult should prompt HIV testing), and syphilis with any uveitis pattern.
<200	**HIV retinopathy** (cotton wool spots and haemorrhages without infection), *Pneumocystis* choroiditis, cryptococcal choroiditis, toxoplasmic retinochoroiditis (often bilateral, multifocal, and without an adjacent scar).
<100	**Progressive outer retinal necrosis** (minimal vitritis, rapid outer retinal opacification, poor prognosis).
<50	**CMV retinitis**: indolent confluent necrosis with haemorrhage, mild vitritis, spreading along vessels. Treat with valganciclovir plus intravitreal therapy for zone 1 disease; restore immunity, then watch for **immune recovery uveitis**.

IMMUNOSUPPRESSED PATIENTS: THREE RULES

1. **Assume atypical presentations.** Infection may cause minimal inflammation, and the absence of vitritis does not exclude retinitis.
2. **Order the test that identifies the organism**, because empirical treatment is far more likely to be wrong: aqueous or vitreous PCR, blood cultures, and imaging.
3. **Coordinate care.** An ocular opportunistic infection is a marker of systemic immunological status and must be communicated to the treating physician the same day.

Pregnancy

OCULAR CONSIDERATIONS IN PREGNANCY AND LACTATION

Issue	Management
Physiologic change	Corneal thickening and reduced sensitivity, contact lens intolerance, transient refractive change, reduced tear production, and a **fall in IOP**. **Defer new spectacle and contact lens prescribing and elective refractive surgery until several months post partum.**
Diabetic retinopathy	Accelerates; examine pre-conception or in the first trimester, then each trimester and post partum. Poor pre-pregnancy control and rapid intensification of control both increase risk.
Pre-eclampsia and eclampsia	Blurred vision, photopsia, serous retinal detachment, and cortical blindness with hypertensive choroidopathy. **Visual symptoms in a hypertensive pregnant patient are an obstetric emergency.**
Central serous chorioretinopathy	Occurs in pregnancy, often in the third trimester, and usually resolves post partum; avoid treatment where possible.
Glaucoma medication	IOP falls physiologically, so some patients can reduce therapy. **Prostaglandin analogues are usually avoided** (theoretical uterine effect); β-blockers are used with punctal occlusion but cross the placenta and enter milk; **carbonic anhydrase inhibitors are generally avoided**; brimonidine is avoided near delivery. Laser trabeculoplasty is a good alternative.
Antibiotics	Prefer erythromycin, azithromycin, and β-lactams. **Avoid tetracyclines (dental and skeletal effects) and generally avoid fluoroquinolones and sulfonamides near term.**
Diagnostic agents	Fluorescein crosses the placenta and is generally avoided; ICG is also avoided. Dilation with tropicamide and phenylephrine `2.5%` is generally acceptable with punctal occlusion; **avoid phenylephrine `10%`**.
Pituitary adenoma	Can enlarge in pregnancy; monitor visual fields in a known adenoma.

Communicating with the rest of medicine

WHAT A GOOD REFERRAL LETTER CONTAINS, AND WHY IT IS AN EXAMINABLE SKILL

1. **The urgency, stated explicitly and at the top**: "please see today", "within 24 hours", "within 2 weeks", or "routine". Ambiguous urgency is the commonest failure.
2. **The specific question you are asking**, not just the finding. "Please assess for giant cell arteritis; I have started prednisone 60 mg today" is actionable. "Abnormal disc" is not.
3. **The ocular findings in plain clinical language**, including acuity, pupil findings, and the relevant images.
4. **What you have already done**, including any treatment started and any tests ordered with results if available.
5. **Your contact details and a request for a reply.**
6. **A telephone call in addition to the letter for anything same-day.** A letter alone does not discharge your duty in an emergency.
7. **Documentation that the patient was told the urgency and understood it**, including the words used, and any follow-up call made if the patient did not attend. Failure to communicate urgency and to follow up is a recurring source of liability.

Referrals that must be specific about destination and timeframe: "refer to a retinal specialist for repair of a macula-on retinal detachment within 24 hours" rather than "refer to ophthalmology". Part III PEPS grades this explicitly, and Part II tests the same judgement.

DRUGS WHOSE OCULAR EFFECT REQUIRES YOU TO CONTACT THE PRESCRIBER

Drug	Reason
Hydroxychloroquine	Toxicity found on screening: the drug should be stopped or reduced in discussion with rheumatology, since it is often essential for disease control. Damage is irreversible.
Ethambutol	Optic neuropathy: **stop immediately** in discussion with the infectious disease team, who will substitute another agent.
Vigabatrin	Field constriction: discuss with neurology, weighing seizure control against irreversible visual loss.
Systemic or inhaled corticosteroid	Raised IOP or cataract: discuss dose reduction or an alternative; do not simply stop a systemic steroid.
Amiodarone	Optic neuropathy: cardiology must weigh the arrhythmia risk of stopping.
Tamsulosin and other α-blockers	Floppy iris: inform the cataract surgeon; stopping the drug does not remove the risk.
Isotretinoin with a tetracycline	Combined risk of pseudotumour cerebri: this combination should be avoided.
Topiramate	Acute angle closure with myopic shift: stop the drug in discussion with the prescriber; iridotomy will not help.
Anticholinergics, antihistamines, antidepressants	Reduced accommodation, dry eye, and rarely angle closure: review the whole list before attributing symptoms to age.
Immune checkpoint inhibitors and MEK inhibitors	Uveitis, serous detachments, and thyroid eye disease: coordinate with oncology rather than stopping cancer therapy unilaterally.

RETRIEVAL PRACTICE

1. You find cotton wool spots and dot-blot haemorrhages in an asymptomatic 52-year-old. Give three actions.
2. A tall 19-year-old has superotemporal lens subluxation. What is the urgent referral and why?
3. A patient has multiple bear-track pigmented fundus lesions. What must you refer for?
4. Give the diabetic retinal screening rules for type 1, type 2, and pregnancy.
5. Give the hydroxychloroquine screening protocol and the dose threshold.
6. Which anticonvulsant requires three-monthly perimetry and why?
7. A 6-year-old ANA-positive girl with oligoarticular arthritis. Screening interval and why it must be that frequent?
8. Match CD4 counts of <50, <100, and <200 to their characteristic ocular infections.
9. A young adult presents with herpes zoster ophthalmicus. What test should you consider offering?
10. Which glaucoma medication classes are usually avoided in pregnancy, and what is a good alternative?
11. A pregnant patient with hypertension reports photopsia and blurred vision. What is this and what do you do?
12. Name six elements of an adequate referral letter.
13. Why is "refer to ophthalmology" an inadequate plan?
14. Name four drugs whose ocular toxicity requires you to contact the prescriber rather than stopping the drug yourself.
15. A patient with an easily everted upper lid and superior punctate keratopathy. What systemic condition, and what are its risks?

KEY

1. Measure blood pressure in the room; refer with a written note describing the findings for fasting glucose and HgbA1c (and lipid profile); and arrange your own ocular follow-up rather than discharging.
2. Marfan syndrome; **urgent cardiology referral for aortic root imaging**, because aortic dissection is the cause of death.
3. Colonoscopy and genetic counselling for **familial adenomatous polyposis**.
4. Type 1: annually from **5 years** after diagnosis or from puberty. Type 2: annually from **diagnosis**. Pregnancy with pre-existing diabetes: pre-conception or first trimester, then each trimester and post partum; gestational diabetes alone needs no retinal screening.
5. Baseline within a year, then **annually after 5 years** with `10-2` fields plus SD-OCT; risk rises above **`5.0 mg/kg/day` of real body weight**.
6. **Vigabatrin**, because it causes irreversible concentric visual field constriction in up to a third of users.
7. **Every 3 months**, because the uveitis is completely asymptomatic with a white eye, so there is no symptom to bring her in before irreversible damage occurs.
8. <50: CMV retinitis. <100: progressive outer retinal necrosis. <200: HIV retinopathy, *Pneumocystis* and cryptococcal choroiditis, and atypical toxoplasmosis.
9. **HIV testing**: zoster in a young adult suggests immunosuppression.
10. **Prostaglandin analogues and carbonic anhydrase inhibitors** are usually avoided, and brimonidine is avoided near delivery; β-blockers are used cautiously with punctal occlusion. **Laser trabeculoplasty** is a good alternative, and IOP often falls physiologically anyway.
11. Pre-eclampsia with hypertensive choroidopathy or impending eclampsia; **this is an obstetric emergency**: contact the obstetric team and send the patient immediately.
12. Explicit urgency, a specific question, the ocular findings including acuity and images, what you have already done, your contact details with a request for reply, and a telephone call for anything same-day.
13. Because it specifies neither the subspecialty, the reason, nor the timeframe, so it cannot be triaged. Part III PEPS scores referrals on exactly these three elements, and the same judgement is tested here.
14. Hydroxychloroquine, ethambutol, vigabatrin, and systemic corticosteroid (also amiodarone and cancer immunotherapies).
15. **Obstructive sleep apnoea** (floppy eyelid syndrome); risks include cardiovascular disease, non-arteritic AION, papilloedema, glaucoma, and central serous chorioretinopathy.

CHAPTER RECAP

1. Some ocular findings make you the first clinician to diagnose a systemic disease; know which, and act rather than advise.
2. Blood pressure belongs in the examination room, not in a recommendation letter.
3. Marfan means cardiology, VHL means renal surveillance, bear-track CHRPE means colonoscopy, Lisch nodules mean NF1 screening.
4. Diabetic screening starts at 5 years in type 1 and at diagnosis in type 2, and every trimester in pregnancy.
5. Hydroxychloroquine, ethambutol, vigabatrin, and pentosan polysulfate have formal monitoring protocols.
6. ANA-positive oligoarticular JIA needs three-monthly screening because the eye is silent.
7. CD4 below 50 means CMV retinitis; zoster in a young adult means offer HIV testing.
8. In pregnancy: defer refractive prescribing, screen diabetics each trimester, avoid prostaglandins and CAIs, and treat visual symptoms with hypertension as an obstetric emergency.
9. A referral must state urgency, destination, reason, and timeframe, and a same-day referral requires a phone call.
10. Do not stop a systemic drug unilaterally; contact the prescriber and share the decision.

CHAPTER 12 · REFRACTIVE AND SENSORY

Ametropia, spectacles, vergence, amblyopia, strabismus

Accommodative / vergence / oculomotor · 20–25 items | Ametropia · 11–22 items | Ophthalmic optics / spectacles · 6–17 items | Amblyopia / strabismus · 6–17 items

ANSWER THESE COLD BEFORE READING

1. Give the case findings that define each of the nine accommodative and vergence diagnoses.
2. State the CITT conclusion and the treatment it supports.
3. Give the decision rules for prescribing in a child, by refractive error and age.
4. Give the eight-step troubleshooting sequence for spectacle non-adaptation.
5. State the PEDIG conclusions on patching dose, atropine, and older children.
6. Give the management of intermittent exotropia and of accommodative esotropia.

BLUEPRINT WEIGHT

43–81

Combined scored items across these four areas: the accommodative and vergence area alone is the largest refractive category on the exam.

Accommodative and vergence anomalies: diagnosis by pattern

THE NINE DIAGNOSES, AND THE FINDINGS THAT DEFINE EACH

Diagnosis	Defining findings	Management, in order
Convergence insufficiency	Exophoria **greater at near**, **receded near point of convergence** (break beyond `6 cm`), **reduced base-out at near**, low calculated AC/A, high NRA relative to PRA, and symptoms of asthenopia, diplopia, loss of place, and sleepiness with reading.	**Office-based vergence and accommodative therapy with home reinforcement**, per CITT. Then base-in prism only if therapy is impractical, and surgery rarely. Also treat any associated accommodative insufficiency, and consider a history of concussion.
Convergence excess	Esophoria **greater at near**, **high AC/A**, reduced base-in at near, reduced PRA, high accommodative lag.	**Plus add at near**, sized from the gradient AC/A and confirmed by the effect on the near phoria and on NRA and PRA balance. Then negative fusional vergence therapy. Check for uncorrected hyperopia first.
Divergence insufficiency	Esophoria **greater at distance**, reduced base-in at distance, low AC/A, and diplopia at distance.	**First exclude a sixth nerve palsy, raised intracranial pressure, myasthenia, thyroid disease, and sagging eye syndrome.** Then **base-out prism at distance**, and vergence therapy.

Diagnosis	Defining findings	Management, in order
Divergence excess	Exophoria **greater at distance by more than** `10 Δ`, high AC/A.	Distinguish **true from simulated** divergence excess with a one-hour patch test or a `+3.00` add. Manage with vergence therapy, minus overcorrection in selected patients, prism, or surgery.
Basic exophoria or esophoria	Roughly equal magnitude at distance and near, normal AC/A.	Prism by **Sheard for exo** or Percival, plus vergence therapy in the deficient direction.
Fusional vergence dysfunction	**Normal phorias with reduced ranges in both directions** and reduced vergence facility, with symptoms disproportionate to the phoria.	Vergence therapy. Prism is unhelpful because there is no significant phoria to neutralise.
Accommodative insufficiency	Amplitude more than `2 D` below the Hofstetter minimum, **high lag on MEM**, reduced PRA, and failure of the minus lens on facility testing.	**Plus add** plus accommodative therapy. **Exclude systemic causes**: medication (anticholinergics, antihistamines, antidepressants, isotretinoin), **concussion or traumatic brain injury**, diabetes, thyroid disease, Adie pupil, mononucleosis or other recent viral illness, and anaemia.
Accommodative excess or spasm	Variable **pseudomyopia**, reduced NRA, failure of the plus lens on facility testing, headache and photophobia, and a large difference between manifest and cycloplegic refraction.	Full cycloplegic refraction; remove precipitants including excessive near work and stress; plus add for near; accommodative therapy; cycloplegia for a short period in severe cases. Consider spasm of the near reflex (with convergence and miosis) and a functional or neurologic component.
Accommodative infacility	Normal amplitude and lag but **reduced facility in both directions**.	Accommodative therapy with flippers, progressing to binocular flippers.

THE CITT RESULT, AND WHAT IT MEANS FOR HOW YOU ANSWER

The **Convergence Insufficiency Treatment Trial** compared office-based vergence and accommodative therapy with home reinforcement, home-based pencil push-ups, home-based computer vergence therapy, and office-based placebo therapy in symptomatic children aged 9 to 17.

Office-based vergence and accommodative therapy with home reinforcement was significantly superior on both symptom score and clinical measures. Pencil push-ups alone performed no better than placebo. The result has been replicated in adults.

So when a Part II item offers "home pencil push-ups" as a treatment for symptomatic convergence insufficiency, it is a distractor. The answer is office-based therapy with home reinforcement.

PRISM CRITERIA, AND WHICH DEVIATION EACH SUITS

Criterion	Formula and use
Sheard	Prism = (2 × phoria - compensating fusional reserve) / 3. Best for **exophoria**.

Criterion	Formula and use
Percival	Prism = (G - 2g) / 3, where G is the greater and g the lesser limit of the range. Does not need a phoria measurement; often preferred for **esophoria**.
1:1 rule	Prism = (near exophoria - base-in break) / 2. For **exophoria**, especially divergence excess.
Associated phoria (fixation disparity)	Prescribe the full associated phoria measured on a Mallett unit or Wesson card. **The method of choice for vertical deviations**, where fusional reserves are small.

Prescribing decisions

WHEN TO PRESCRIBE IN CHILDREN: A PRACTICAL FRAMEWORK

Error	Prescribe if	Notes
Hyperopia, infant to 1 year	Above about `+5.00 to +6.00 D`, or above `+2.50 D` of anisometropia	Prescribe less than the full cycloplegic finding to preserve the emmetropisation drive, unless there is esotropia.
Hyperopia, 1 to 3 years	Above about `+4.50 D`, or with esotropia, or with reduced acuity	**With accommodative esotropia, prescribe the full cycloplegic plus**, regardless of magnitude.
Hyperopia, over 3 years	Above about `+3.50 D`, or symptomatic, or with reduced acuity, or with esotropia or a high AC/A	Also prescribe for asthenopia and for reading difficulty even at lower powers.
Myopia	Above about `-3.00 D` in infancy; any myopia reducing function once school age	Prescribe full correction; undercorrection **accelerates** progression and should be avoided. Discuss **myopia control** in any progressing child.
Astigmatism	Above about `2.00 to 2.50 D` in infancy, `1.50 D` at 1 to 3 years, and `1.00 D` over 3 years	Meridional amblyopia risk; oblique axes are more amblyogenic.
Anisometropia	Above about `1.00 D` spherical or `1.50 D` cylindrical difference at any age	The main amblyogenic factor that produces no visible sign, which is why screening matters.

MYOPIA CONTROL: WHAT TO OFFER AND WHAT TO SAY

1. **Establish progression** with cycloplegic refraction and, ideally, axial length. Axial length is the better outcome measure because it is not confounded by accommodative tone.
2. **Behavioural advice**: at least `2 hours` of outdoor time daily, which has the best evidence for delaying onset; sensible near-work habits with breaks and an adequate working distance.
3. **Optical options**: **orthokeratology** (roughly `40 to 50%` slowing), **multifocal or dual-focus soft contact lenses** (MiSight has FDA approval on the basis of a randomised trial), and **defocus-incorporated spectacle lenses** (DIMS and highly aspherical lenslet designs).
4. **Pharmacologic**: **low-dose atropine**. LAMP showed `0.05%` is more effective than `0.025%` and `0.01%`, at the cost of more photophobia and near blur, and with greater rebound on cessation. Some trials of `0.01%` in non-Asian populations have been negative, so counsel honestly about uncertainty.
5. **Do not undercorrect**, and correct fully.
6. **Counsel about the reason**: the goal is not spectacle independence, it is reducing the lifetime risk of myopic maculopathy, retinal detachment, glaucoma, and cataract, all of which rise steeply with axial length.

PRESCRIBING DECISIONS IN ADULTS

Situation	Rule
Small change with no symptoms	Do not change the prescription. Explain why; an unnecessary change is a common source of non-adaptation.
Large cylinder axis change in an adapted patient	Change cautiously and partially, or keep the old axis with a compromised power, and warn about the adaptation period.
Unstable glycaemic control	**Do not prescribe.** Recheck once control is stable.
Pregnancy	Defer non-essential changes and refractive surgery until several months post partum.
Recent cataract or refractive surgery	Wait for refractive stability, generally `3 to 6 weeks` after cataract surgery and longer after a complicated case or surface ablation.
Contact lens wearer	Refract out of lenses after an adequate wash-out period, and repeat topography before any surgical planning, because **warpage** distorts both.
First presbyopic add	Determine by at least two methods, verify the range of clear vision brackets the actual task and distance, and ask about occupational distances (monitor, music, dashboard) before choosing a design.
Anisometropia with vertical imbalance at near	Above about `1.5 to 2 Δ`, use slab-off, dissimilar segment heights, contact lenses, or separate near glasses.

Spectacle non-adaptation: a troubleshooting algorithm

WORK THROUGH THESE IN ORDER BEFORE REMAKING THE LENSES

1. **Verify the spectacles against the prescription** and against ANSI tolerances: sphere, cylinder power and axis, add, prism, and optical centre placement. Most "wrong prescriptions" are actually dispensing errors.
2. **Measure the monocular PDs and the fitting heights on the patient**. Incorrect monocular centration and fitting height are the two commonest causes of progressive lens failure.
3. **Check vertex distance, pantoscopic tilt, and face form.** A change in any of these alters effective power and induces unwanted cylinder in aspheric designs.
4. **Compare with the previous pair**, including base curve, material, and lens design. A change from a `6` base to a `2` base, or from CR-39 to 1.67, changes magnification and peripheral aberration even at identical power.
5. **Re-refract, and specifically check the cylinder axis** and any change in add.
6. **Check the binocular status**: a newly induced or newly uncorrected vertical or horizontal imbalance, or a decompensating phoria.
7. **Examine the ocular surface and the media.** Dry eye and early cataract cause complaints that no lens change will fix, and are frequently the real answer.
8. **Ask what specifically is wrong, and when.** "Blurry at the computer" points to the corridor or the add; "the floor looks curved" points to base curve or tilt; "double" points to prism; "I feel pulled" points to unequal magnification; "worse in the evening" points to the surface or the add.

A useful discipline: **do not remake a pair of glasses until you can state the mechanism of the complaint.** Remaking without a mechanism produces a second failure.

Amblyopia

PEDIG AND AMBLYOPIA TREATMENT STUDY CONCLUSIONS

Question	Answer from the trials
Do glasses alone work?	Yes. **Refractive correction alone improves amblyopia in a large proportion**, with improvement continuing for up to `30 weeks`. Give an adequate spectacle trial before starting occlusion.
How much patching?	`2 hours` daily is as effective as `6 hours` for **moderate** amblyopia (20/40 to 20/80). `6 hours` is as effective as full-time for **severe** amblyopia (20/100 to 20/400).
Is atropine equivalent?	Yes for moderate amblyopia, and **weekend-only atropine equals daily**. It is a legitimate first-line choice, often with better adherence.
Does age exclude treatment?	No. Children aged `7 to 12` respond, and `13 to 17`-year-olds respond if previously untreated. **Age alone is not a reason to withhold treatment.**
Do near activities help?	They added little measurable benefit over patching alone, though they may aid engagement and adherence.
What happens after stopping?	Recurrence in roughly **a quarter**. Taper rather than stop abruptly and follow for at least a year.

Question	Answer from the trials
Do binocular treatments work?	Dichoptic and digital therapeutic approaches show benefit in some trials, particularly in older children, and are a reasonable adjunct.

AMBLYOPIA MANAGEMENT ERRORS

- **Patching before an adequate spectacle trial.**
- **Failing to cycloplege.** Latent hyperopia is the whole diagnosis in accommodative esotropia and in many anisometropic amblyopes.
- **Failing to examine for organic disease.** Any amblyopia that does not respond, or that presents with an afferent defect, nystagmus, or an abnormal fundus, needs a structural explanation: optic nerve hypoplasia, macular scar, retinoblastoma, cataract, or a chiasmal lesion.
- **Stopping abruptly at the endpoint** rather than tapering, which invites recurrence.
- **Occluding the sound eye without monitoring it**, risking occlusion amblyopia in a very young child.
- **Assuming a teenager cannot improve.**

Strabismus management

COMMON DEVIATIONS: MANAGEMENT PATHWAY

Diagnosis	Key findings	Management
Infantile esotropia	Onset before `6 months`, large stable angle, minimal hyperopia, cross-fixation, latent nystagmus, inferior oblique overaction, dissociated vertical deviation, asymmetric monocular optokinetic response.	Correct any significant refractive error, treat amblyopia, then **surgical alignment, generally before age 2**, to give the best chance of gross stereopsis and peripheral fusion. Expect monofixation syndrome as the realistic outcome, and warn about the need for further surgery.
Refractive accommodative esotropia	Onset 2 to 3 years, significant hyperopia (average around `+4.00 D`), normal AC/A, deviation equal at distance and near, and fully eliminated by full cycloplegic plus.	**Full cycloplegic hyperopic correction** and amblyopia treatment. Do not undercorrect. Review the refraction as the child grows, and warn that reducing the plus can decompensate the deviation.
Non-refractive accommodative esotropia (high AC/A)	Deviation much larger at near with low hyperopia and a high AC/A.	**Flat-top bifocal add set to bisect the pupil**, sized to eliminate the near deviation; alternatives are miotics or surgery. Reduce the add over time as the AC/A normalises.

Diagnosis	Key findings	Management
Partially accommodative esotropia	Plus lenses reduce but do not eliminate the deviation.	Full plus, amblyopia treatment, then surgery for the residual angle.
Sensory esotropia or exotropia	Unilateral vision loss with a deviation.	**Dilate and examine the fundus in any new unilateral strabismus in a child: retinoblastoma, cataract, optic nerve disease, and macular scarring all present this way.** Treat the cause, then consider surgery for alignment.
Acute acquired comitant esotropia	Sudden comitant esotropia in an older child or adult.	**Neuroimage** to exclude a posterior fossa lesion or Chiari malformation, even though many cases are benign or associated with heavy near-device use. Then prism or surgery.
Intermittent exotropia	The commonest childhood exodeviation; subtypes basic, divergence excess, and convergence insufficiency type. Grade control with the Newcastle or LACTOSE scale.	Correct refractive error (including full myopic correction), treat amblyopia, then observation with control monitoring, vergence therapy for the convergence insufficiency type, part-time occlusion, minus overcorrection, or surgery for deteriorating control, increasing frequency, or a large angle. Photographs from the family are useful documentation of control.
Adult strabismus with diplopia	Any acquired deviation.	Establish comitance, forced duction, and saccadic velocity to separate paresis from restriction; identify the cause; then **prism** (Fresnel for large or unstable deviations, ground-in when stable), occlusion or a Bangerter filter, botulinum toxin, or surgery once the deviation has been stable for several months. Never operate on a deviation that is still changing.
Superior oblique palsy	Hypertropia worse in contralateral gaze, worse on ipsilateral head tilt, and worse in downgaze, with excyclotorsion.	Congenital cases (large vertical fusional amplitudes, long-standing head tilt, facial asymmetry) may need surgery for the head posture. Acquired cases are usually traumatic or microvascular; observe for spontaneous recovery over months with prism, then surgery. Bilateral cases with more than `10°` of excyclotorsion suggest head trauma.
Duane retraction syndrome	Limited abduction with globe retraction and fissure narrowing on adduction.	Part of the congenital cranial dysinnervation disorders; surgery only for a significant head turn or primary-position deviation, and never a simple recession-resection.
Brown syndrome	Restricted elevation in adduction with a positive forced duction.	Observe congenital cases; treat acquired trochleitis with steroid and investigate for inflammatory arthritis.

Diagnosis	Key findings	Management
Restrictive strabismus (thyroid, fracture)	Positive forced duction, IOP rise on gaze into the restricted field, normal saccadic velocity.	Treat the underlying disease and wait for stability; prism for stable diplopia; surgery on an inactive process only, and in thyroid eye disease only after decompression if that is needed.
Myasthenic strabismus	Variable, fatigable, pupil-sparing.	Do not operate. Treat the myasthenia (pyridostigmine, immunosuppression, thymectomy where indicated) and use prism or occlusion for stable residual deviation.

RETRIEVAL PRACTICE

1. Exophoria greater at near with a receded NPC and reduced base-out at near. Diagnosis and first-line treatment, and what the trial evidence says about pencil push-ups.
2. Esophoria greater at near with a high AC/A and a high lag. Diagnosis and first-line treatment.
3. Esophoria greater at distance with diplopia in a 68-year-old. What must you exclude before treating?
4. Normal phorias with reduced ranges in both directions. Diagnosis, and why prism will not help.
5. A 15-year-old develops accommodative insufficiency after a sports injury. What is the likely cause and what else do you screen for?
6. Which prism criterion do you use for a vertical deviation, and why?
7. A 3-year-old with `+5.50 D` hyperopia and a `30 Δ` esotropia that resolves with full plus. Diagnosis and prescription rule.
8. Same child, but the deviation is `10 Δ` at distance and `35 Δ` at near with only `+1.00 D` hyperopia. Diagnosis and treatment.
9. A new unilateral esotropia in a 2-year-old. What must you do before anything else?
10. Give the PEDIG patching dose for moderate and severe amblyopia, and the atropine equivalence result.
11. Can a 15-year-old with previously untreated amblyopia improve?
12. Give five elements of myopia control advice with their approximate effect sizes.
13. Why must you not undercorrect a progressing myope?
14. Give the first four steps of spectacle non-adaptation troubleshooting.
15. A patient complains the floor looks curved in new glasses at the same power as their old pair. Name two likely mechanisms.
16. A 40-year-old develops acute comitant esotropia. What is the mandatory investigation?

KEY

1. Convergence insufficiency; **office-based vergence and accommodative therapy with home reinforcement**. CITT showed home pencil push-ups were **no better than placebo**.
2. Convergence excess; a **plus add at near** sized from the gradient AC/A, after correcting any uncorrected hyperopia.
3. A **sixth nerve palsy**, raised intracranial pressure, myasthenia, thyroid eye disease, and sagging eye syndrome.
4. Fusional vergence dysfunction; prism does not help because there is no significant phoria to neutralise. Treat with vergence therapy.
5. **Concussion or mild traumatic brain injury**; also screen for convergence insufficiency, saccadic and pursuit dysfunction, photophobia, and vestibular symptoms, and coordinate with the concussion team about return to play and to learning.
6. The **associated phoria (fixation disparity)**, prescribed in full, because vertical fusional reserves are small and the associated phoria is the most functionally relevant measure.

7. Refractive accommodative esotropia; prescribe the **full cycloplegic plus** and do not undercorrect.
8. Non-refractive accommodative esotropia with a high AC/A; treat with a **flat-top bifocal add set to bisect the pupil**, sized to eliminate the near deviation.
9. **Dilate and examine the fundus** to exclude retinoblastoma, cataract, optic nerve disease, and macular pathology causing a sensory deviation.
10. `2 hours` daily for moderate and `6 hours` for severe; atropine `1%` is equivalent to patching for moderate amblyopia, and **weekend-only dosing equals daily**.
11. **Yes.** Children aged 13 to 17 respond if previously untreated; age alone is not a reason to withhold treatment.
12. At least `2 hours` outdoors daily (best evidence for delaying onset); orthokeratology (about `40 to 50%` slowing); dual-focus or multifocal soft lenses; defocus-incorporated spectacle lenses; and low-dose atropine, with `0.05%` more effective than `0.01%` per LAMP but with more side effects and rebound.
13. Because undercorrection **accelerates** axial elongation rather than slowing it.
14. Verify the spectacles against the prescription and ANSI tolerances; measure monocular PDs and fitting heights on the patient; check vertex distance, pantoscopic tilt, and face form; and compare base curve, material, and design with the previous pair.
15. A **base curve change** and a change in **pantoscopic tilt or vertex distance** (or a switch to an aspheric or high-index design), all of which alter magnification and peripheral aberration at identical power.
16. **Neuroimaging** to exclude a posterior fossa lesion or Chiari malformation.

CHAPTER RECAP

1. Nine accommodative and vergence diagnoses, each defined by the near-distance relationship, the AC/A, and the deficient range.
2. CITT: office-based therapy with home reinforcement; pencil push-ups are a distractor.
3. High AC/A with near eso means plus add; low AC/A with near exo means therapy.
4. Distance esophoria in an older adult means exclude a sixth nerve palsy and raised pressure first.
5. Accommodative insufficiency in a teenager should make you ask about concussion.
6. Prescribe full cycloplegic plus in accommodative esotropia and never undercorrect a progressing myope.
7. Myopia control: outdoor time, ortho-k, dual-focus lenses, defocus spectacles, low-dose atropine, and honest counselling about axial length risk.
8. Glasses first in amblyopia, then 2 hours or 6 hours by severity, or atropine; taper, and treat teenagers.
9. New unilateral strabismus in a child means dilate the fundus.
10. Do not remake spectacles until you can state the mechanism of the complaint.

CHAPTER 13 · REFRACTIVE AND SENSORY

Contact lenses, low vision, colour, and pediatrics

Contact lenses · 17–28 items | Low vision · 6–17 items | Perceptual function / color vision · 6–17 items | Visual and human development · 0–6 items

ANSWER THESE COLD BEFORE READING

1. Give the tear lens rule and compute a rigid lens power from a spectacle prescription and K readings.
2. Differentiate CLARE, CLPU, infiltrative keratitis, and microbial keratitis, and give the action for each.
3. State the modality and material decisions for a keratoconic patient, a presbyope, and a child.
4. Give the required magnification calculation and the reserve factor for a low vision patient.
5. State the Köllner rule and the test you would choose for a suspected acquired colour defect.
6. Give the paediatric examination schedule and the milestones that trigger referral.

BLUEPRINT WEIGHT

29–68

Combined scored items. Contact lenses alone are roughly double the Part I weighting.

Contact lens fitting and problem solving

THE CALCULATIONS THAT RECUR

1. **Vertex conversion**: $F_{corneal}$ = F / (1 - dF), with d in metres. Matters above about `±4 D`.
2. **Tear lens** = base curve (D) - flattest K (D). **SAM and FAP**: fit steeper, add minus; fit flatter, add plus. `0.05 mm` is about `0.25 D`.
3. **Residual astigmatism** = total refractive astigmatism - corneal astigmatism. A spherical rigid lens neutralises the corneal component only.
4. **Toric soft lens axis**: **LARS**, left add and right subtract, and only when the rotation is stable and repeatable.
5. **Sagittal relationship**: increasing diameter increases sag and behaves like a steeper fit, so flatten the base curve by roughly `0.05 mm` for every `0.5 mm` increase in diameter.

Worked example. Spectacle Rx `-7.00 -1.00 × 180` at `12 mm`; K readings `44.00 @ 180 / 45.00 @ 090`. Corneal astigmatism is `1.00 D` with the rule, so the residual astigmatism is -1.00 - (-1.00) = **zero**: a spherical rigid lens will correct it fully. Spherical equivalent at the spectacle plane is `-7.50`; vertex converted, -7.50 / (1 - 0.012 × -7.50) = -7.50 / 1.09 = `-6.88`, so `-6.87`. Fit on flattest K (`44.00`): tear lens is plano, so the lens power is `-6.75 to -7.00 D`.

MODALITY AND MATERIAL DECISIONS BY PATIENT

Patient	Recommended approach and reasoning
Low-risk daily wearer	**Daily disposable silicone hydrogel** where feasible: the lowest rate of infiltrative and infectious complications, no solution exposure, and no compliance failure with replacement. Cost is the main barrier.
Astigmatism above `0.75 to 1.00 D`	Toric soft lens with an appropriate stabilisation design; assess rotation stability before compensating with LARS. Consider a rigid lens for high or irregular astigmatism.

Patient	Recommended approach and reasoning
Keratoconus and irregular cornea	Escalate: spectacles, then **corneal rigid gas permeable** (target three-point touch), then **hybrid**, **piggyback**, or **scleral**. Scleral lenses often give the best acuity and comfort in advanced disease. **Refer for cross-linking if there is documented progression**, and reinforce that eye rubbing must stop.
Post-surgical and post-graft cornea	Scleral or specialty rigid lenses; be alert to endothelial compromise and limit wear time. Avoid excessive central clearance, which causes hypoxic oedema.
Presbyope	Options: **multifocal soft** (centre-near or centre-distance aspheric or concentric), **monovision**, modified monovision, or **translating rigid designs**. Set expectations for reduced contrast and possible halos; give a realistic trial period. Monovision reduces stereopsis, which matters for occupational driving.
Severe dry eye or ocular surface disease	**Scleral lenses** are often transformative, providing a fluid reservoir over the cornea. Manage midday fogging, conjunctival prolapse, and hypoxia by adjusting clearance and haptic alignment.
Aphakic infant	High-plus silicone elastomer or specialty soft lens, per the Infant Aphakia Treatment Study, with intensive parental training and frequent power updates as the eye grows.
Progressing myopic child	**Orthokeratology** or a **dual-focus or multifocal soft lens** for myopia control, with explicit hygiene teaching and parental supervision.
Therapeutic indications	Bandage lens for recurrent erosion, bullous keratopathy, or a persistent epithelial defect; prosthetic tinted lens for aniridia, albinism, or a disfigured iris; and a lens as a drug reservoir in selected cases.

CONTACT LENS COMPLICATIONS: DIFFERENTIATION AND ACTION

Complication	Distinguishing features	Action
Microbial keratitis	**Pain, an epithelial defect over the infiltrate, anterior chamber reaction, discharge, and progression.**	**Stop lens wear, culture if central or larger than 1 to 2 mm, treat aggressively, and review in 24 hours.** Culture the lens, case, and solution. Ask specifically about water exposure (*Acanthamoeba*) and overnight wear (*Pseudomonas*).
Contact lens acute red eye (CLARE)	Sudden overnight unilateral pain and redness with **peripheral or mid-peripheral infiltrates and an intact epithelium**; gram-negative endotoxin under a tight lens.	Discontinue lens wear; topical steroid with or without antibiotic cover once infection is excluded; refit looser and reconsider overnight wear.

Complication	Distinguishing features	Action
Contact lens peripheral ulcer (CLPU)	Small, round, peripheral, **full-thickness epithelial loss over a dense sterile infiltrate**, staphylococcal exotoxin driven; heals with a circular scar.	Discontinue lens wear; prophylactic antibiotic; address lid disease. It is sterile, but it is often treated as presumed infection until the course confirms otherwise.
Infiltrative keratitis	Multiple small subepithelial infiltrates with variable staining and mild symptoms.	Stop lenses, treat conservatively, and refit with a more frequent replacement schedule and better hygiene.
Giant papillary conjunctivitis	Upper tarsal papillae above `0.3 mm`, mucus, itching, increasing lens awareness and decentration late in the day.	Temporary discontinuation, then change to daily disposables and a different material, improve replacement compliance and cleaning, and add a mast cell stabiliser. A short steroid pulse for severe cases.
Superior epithelial arcuate lesion (SEAL)	Arcuate superior epithelial split; mechanical, associated with high-modulus silicone hydrogels; often asymptomatic.	Change to a lower-modulus material or a different design; lubricate.
3 and 9 o'clock staining	Peripheral desiccation with a rigid lens from poor blink and inadequate edge lift.	Increase edge lift, reduce diameter, improve blink, lubricate; untreated it progresses to vascularised limbal keratitis.
Corneal neovascularisation and microcysts	Chronic hypoxia; microcysts appear at 2 to 6 months and paradoxically increase transiently after switching to a higher-Dk lens.	Increase oxygen transmissibility, reduce wear time, stop overnight wear.
Corneal warpage	Spectacle blur after removal and distorted topography.	Weeks out of lenses before refraction or surgical planning; refit in a rigid material with better alignment.

Complication	Distinguishing features	Action
Solution toxicity or hypersensitivity	Diffuse punctate staining and burning on insertion; hydrogen peroxide that has not neutralised, or preservative sensitivity in high-water ionic materials.	Change the care system to a peroxide or preservative-free option, or switch to daily disposables.
Lens-induced hypoxic oedema	Striae at about `5%` oedema and folds at about `8%`, with polymegathism over time.	Increase Dk/t; the Holden-Mertz targets are **≥24 for daily and ≥87 for extended wear**.

HYGIENE COUNSELLING: THE POINTS THAT REDUCE KERATITIS RISK

- **No water: no rinsing, no storing, no showering or swimming in lenses without appropriate protection.** Tap water carries *Acanthamoeba*.
- **Rub and rinse** even with a "no-rub" solution.
- **Replace the case every three months**, air-dry it face down, and never top off solution.
- **Do not sleep in lenses** unless specifically prescribed; overnight wear multiplies microbial keratitis risk several-fold, and smoking multiplies it again.
- **Adhere to the replacement schedule**; over-wear of a monthly or two-weekly lens is extremely common and is a leading cause of infiltrative events.
- **Remove lenses and present immediately for a red, painful, photophobic, or blurred eye.** Teach this explicitly and document that you did.

Low vision rehabilitation

BUILDING A LOW VISION PLAN IN SIX STEPS

1. **Define the goals with the patient in their own words.** "Read my mail and my medication bottles", "see my grandchildren's faces", "keep working". Goals determine devices; a general request for "better vision" produces a drawer full of unused magnifiers.
2. **Measure carefully**: distance and near acuity with a logMAR chart at a measured distance, **contrast sensitivity**, **visual fields** (including a central 10-2 or an Amsler in macular disease), reading acuity and reading rate with continuous text, and glare testing.
3. **Refract meticulously.** A significant proportion of "low vision" is uncorrected refractive error; use large-step just-noticeable-difference technique, trial frame, and a stenopaeic slit or keratometry to find the cylinder axis.
4. **Calculate the magnification requirement**: required magnification = current acuity denominator / target acuity denominator, then apply a **reserve factor of about 2×** for fluent rather than threshold reading. **Kestenbaum** gives the starting add as the reciprocal of the acuity fraction (20/200 gives `+10 D`).
5. **Trial devices in the order of least intrusive first**: relative size (large print) and lighting, then hand and stand magnifiers, then spectacle-mounted microscopes, then telescopes for distance, then electronic magnifiers. Compare devices on **equivalent viewing power**, not on manufacturers' magnification labels.
6. **Address the non-optical half**: task lighting (often the highest-yield single change), contrast enhancement and reverse polarity, glare control with absorptive filters and brimmed hats, typoscopes, eccentric viewing and steady-eye training, orientation and mobility referral, occupational therapy, state services for the blind, driving assessment, and depression screening.

Worked example. A patient with geographic atrophy sees `20/250` and needs `20/50` for a medication label. Required magnification = 250/50 = **5**×, and with a 2× reserve for sustained reading, plan for about **10**×, or an equivalent viewing power near `+40 D`. In practice that means an electronic magnifier for sustained reading and a `+12 to +16 D` hand magnifier for spot tasks, plus a task lamp.

DEVICE SELECTION BY PROBLEM

Problem	Best-fit devices
Central scotoma, needs sustained reading	Electronic magnifier or CCTV with contrast reversal; spectacle-mounted microscope for hands-free work; eccentric viewing training.
Central scotoma, spot tasks only	Illuminated hand magnifier; stand magnifier for tremor or arthritis.
Distance tasks (signs, faces, theatre)	Hand-held or spectacle-mounted telescope; bioptic telescope for driving where legally permitted.
Intermediate tasks (music, computer, cooking)	Telemicroscope, occupational progressive or single-vision intermediate spectacles, screen magnification software.
Peripheral field loss (retinitis pigmentosa, advanced glaucoma)	Orientation and mobility training first; minifying (reverse) telescope or field-expanding prisms; night-vision aids; explicit driving and mobility counselling.
Homonymous hemianopia	Field-expanding prisms with base toward the field loss, scanning training, and occupational therapy; counsel about driving law.

Problem	Best-fit devices
Photophobia (albinism, cone dystrophy, aniridia)	Deep absorptive filters, tinted or prosthetic contact lenses, brimmed hats, and indoor lighting control.
Contrast loss with good acuity	Lighting, contrast enhancement, and filters; explain why the Snellen number does not describe their difficulty.

Colour vision and perceptual function

CLINICAL DECISIONS IN COLOUR VISION

Question	Answer
Congenital or acquired?	**Symmetry between the eyes is the strongest single clue.** Congenital defects are symmetric, stable, red-green in most cases, with normal acuity. Acquired defects are often asymmetric, changing, accompanied by reduced acuity, and frequently blue-yellow.
Which test?	**Ishihara** screens congenital red-green only. **HRR** detects, classifies, and grades both red-green and blue-yellow, so it is the better choice when acquired disease is suspected. **D-15 or desaturated Lanthony D-15** classifies the axis quickly. **Farnsworth-Munsell 100-hue** quantifies for monitoring. The **anomaloscope** is the reference standard. **Red desaturation comparison** is the fastest bedside screen for optic nerve disease.
Köllner rule	Outer retinal disease gives blue-yellow defects; optic nerve disease gives red-green. **Exceptions**: glaucoma and dominant optic atrophy (optic nerve but blue-yellow), and Stargardt and cone dystrophy (outer retina but red-green).
Occupational counselling	Certain occupations have colour standards (aviation, maritime, rail, some emergency services, electrical trades). Counsel factually, document, and refer for the specific occupational test rather than extrapolating from an Ishihara result. Explain that colour-defective individuals adapt well but should know their limitation.
Children	Test around school entry with a numeral-free version. The purpose is not treatment but **informing the child, the family, and the school**, so that colour-coded teaching materials do not become an unrecognised barrier. Tinted lenses do not cure the defect, though they can aid some discrimination tasks.
Drug-related	Digoxin (xanthopsia), sildenafil (transient blue tinge), ethambutol (red-green with optic neuropathy), and chloroquine. A new colour complaint should prompt a medication review.

PERCEPTUAL AND FUNCTIONAL COMPLAINTS WITH NORMAL ACUITY

Complaint	Consider
"My vision is not right but the chart is fine"	Contrast sensitivity loss (early cataract, glaucoma, optic neuritis, post-refractive surgery), dry eye, higher-order aberration, and macular disease with preserved acuity. **Measure contrast and image the macula.**
Glare and haloes	Cataract, corneal scatter or oedema, post-refractive surgery, multifocal IOL, large pupil, and posterior capsule opacification. Measure disability glare.
Metamorphopsia	Macular disease: epiretinal membrane, CNV, CSC, macular hole. OCT is mandatory.
Photopsia	PVD and retinal traction, migraine aura, retinal or optic nerve ischaemia, and paraneoplastic retinopathy.
Formed hallucinations with insight	Charles Bonnet syndrome in a patient with vision loss; reassure, explain, optimise lighting and vision.
Reading difficulty in a child with normal acuity	Uncorrected hyperopia or astigmatism, accommodative or vergence dysfunction, and a **specific learning disability**. **Vision therapy does not treat dyslexia**: treat the measurable visual condition, state that clearly, and refer for educational assessment.

Complaint	Consider
Symptoms after concussion	Convergence insufficiency, accommodative insufficiency, saccadic and pursuit dysfunction, photophobia, and vestibular-ocular mismatch. This cluster is common and treatable, and vision therapy has reasonable support here.
Inconsistent findings	Non-organic visual loss. Use the fogging test, four-dioptre base-out prism, stereoacuity, and mirror or optokinetic testing; explain positively and follow up; exclude organic disease first.

Pediatric examination and development

EXAMINATION SCHEDULE AND WHAT TRIGGERS REFERRAL

Age	Assess	Refer or investigate if
Newborn to 6 months	Red reflex (Brückner), fixation behaviour, pupils, external appearance, and ocular alignment.	**Absent or asymmetric red reflex (leukocoria) at any age is an emergency referral.** Also: no fix-and-follow by `3 months`, nystagmus, a constant deviation at any age, corneal enlargement or clouding, epiphora with photophobia, or an abnormal pupil.
6 to 12 months	First comprehensive examination, including cycloplegic refraction, alignment, and dilated fundus.	Significant refractive error by the thresholds in Chapter 12, any deviation, or any media or fundus abnormality.
3 to 5 years	Acuity with an age-appropriate optotype (matching or naming), stereopsis, colour, alignment, cycloplegic refraction, and dilated fundus.	Acuity below `20/40`, an interocular difference of two lines or more, failed stereopsis, or any deviation.
School age	Annual or biennial examination with acuity, binocular status, accommodation, refraction, and ocular health.	Reading difficulty, headaches, deteriorating school performance, progressing myopia, or a new deviation.

MILESTONES AND THE AGE AT WHICH ABSENCE IS ABNORMAL

Milestone	Expected by	Note
Pupil light reflex	29–31 weeks gestation	A marker of gestational maturity.
Steady fix and follow	6–8 weeks	**Absence at 3 months requires investigation.**
Blink to threat	2–5 months	–
Accurate saccades and smooth pursuit	2–3 months	Saccadic pursuit before this is normal.

Milestone	Expected by	Note
Adult-like accommodation	3–4 months	–
Stable ocular alignment	3–4 months	**A constant deviation at any age, or any deviation persisting past 4 months, is abnormal.**
Stereopsis	3–6 months onset	Near-adult stereoacuity by 2 to 3 years.
Acuity 20/50	1 year	20/400 at birth, 20/30 at 2 to 3 years, 20/20 by 3 to 5 years.

LEUKOCORIA: THE DIFFERENTIAL YOU MUST BE ABLE TO PRODUCE INSTANTLY

1. **Retinoblastoma**: calcification on CT or ultrasound; may be bilateral and multifocal if heritable; strabismus is the second commonest presentation. **Same-week referral to a paediatric ocular oncology service.**
2. **Congenital cataract**: urgent, because a dense unilateral cataract must be removed within weeks to avoid profound deprivation amblyopia.
3. **Coats disease**: unilateral, boys, telangiectasia with massive lipid exudation, no calcification.
4. **Persistent fetal vasculature**: microphthalmia, retrolental membrane, elongated ciliary processes.
5. **Toxocariasis**: granuloma with a traction band, puppy exposure, no systemic eosinophilia in the ocular form.
6. **Retinopathy of prematurity** (stage 5), **familial exudative vitreoretinopathy**, **retinal detachment**, **coloboma**, and **myelinated nerve fibres**.

RETRIEVAL PRACTICE

1. Spectacle Rx `-6.00 DS` at `12 mm`; K `42.00/42.00`; rigid lens fitted at `42.50`. Give the lens power.
2. Refraction `-2.00 -2.50 × 180`; corneal astigmatism `2.00 D` with the rule. What residual astigmatism remains with a spherical rigid lens?
3. A toric lens marked at axis 180 sits rotated `20°` to the patient's left. What do you re-order, and what must be true before you compensate?
4. A lens wearer has peripheral infiltrates with an intact epithelium after sleeping in lenses. Diagnosis and action.
5. A lens wearer has a small round peripheral full-thickness epithelial defect over a dense infiltrate. Diagnosis and mechanism.
6. What single behavioural instruction most reduces *Acanthamoeba* risk?
7. Give the Dk/t thresholds for daily and extended wear and the oedema percentages at which striae and folds appear.
8. A keratoconic patient is rigid-lens intolerant with `470 µm` minimum thickness and documented progression. Give the plan.
9. A patient sees `20/400` and needs `20/60`. Give the required magnification with and without a reserve factor.
10. Which single non-optical intervention most often improves reading in macular disease?
11. Why are manufacturers' magnification labels unreliable, and what do you compare instead?
12. A 60-year-old reports asymmetric colour desaturation with reduced acuity in one eye. Congenital or acquired, and which test would you choose?
13. State the Köllner rule and two exceptions.
14. A parent asks whether vision therapy will treat their child's dyslexia. What do you say?
15. An infant has an absent red reflex in one eye. What do you do?
16. Give six items in the leukocoria differential.

17. At what age does absent fix-and-follow require investigation, and by what age should a deviation have resolved?

KEY

1. Vertex: -6.00 / (1 - 0.012 × -6) = -6.00 / 1.072 = `-5.60`, so `-5.50 to -5.62` at the cornea. Fitted `0.50 D` steeper gives a +**0.50** tear lens, so SAM: add minus. Lens power ≈ `-6.00 to -6.12 D`.
2. Residual = -2.50 - (-2.00) = **-0.50** × **180**: half a dioptre with-the-rule astigmatism remains.
3. Rotation to the patient's left means **add**: 180 + 20 = 200, so re-order axis **020**. The rotation must be **stable and repeatable** across visits before compensating.
4. Contact lens acute red eye (CLARE); discontinue lens wear, treat the inflammation once infection is excluded, refit looser, and stop overnight wear.
5. Contact lens peripheral ulcer (CLPU); a sterile reaction to staphylococcal exotoxin.
6. **No water contact with lenses or cases**, including rinsing, storing, showering, and swimming without protection.
7. Dk/t at least `24` for daily wear and `87` for extended wear; striae appear at about `5%` oedema and folds at about `8%`.
8. **Refer for corneal collagen cross-linking** (thickness is adequate and progression is documented), fit a **scleral** or hybrid lens for optical correction, reinforce cessation of eye rubbing and treat the driving allergy, and consider ring segments if lens intolerance persists.
9. 400/60 = **6.7**×; with a 2× reserve for fluent reading, plan for about **13**×.
10. **Task lighting.** Doubling task luminance often helps more than adding magnification.
11. Because rated magnification is calculated inconsistently across manufacturers and does not account for the emergent vergence or working distance. Compare **equivalent viewing power**.
12. **Acquired**, because it is asymmetric and accompanied by reduced acuity; choose **HRR** (or a desaturated D-15 for axis, and 100-hue for quantitative monitoring), and pursue an optic nerve or macular cause.
13. Outer retinal disease gives blue-yellow and optic nerve disease gives red-green; exceptions include **glaucoma and dominant optic atrophy** (blue-yellow despite optic nerve involvement) and **Stargardt or cone dystrophy** (red-green despite outer retinal involvement).
14. That **vision therapy does not treat dyslexia**. You will diagnose and treat any measurable visual condition (refractive error, accommodative or vergence dysfunction, ocular disease), and you will refer for educational and psychological assessment, because reading disability is a language-based learning disorder.
15. **Urgent referral** for examination under anaesthesia and imaging to exclude **retinoblastoma**, along with cataract and other causes; do not delay for a repeat visit.
16. Retinoblastoma, congenital cataract, Coats disease, persistent fetal vasculature, toxocariasis, and retinopathy of prematurity (also FEVR, retinal detachment, coloboma, myelinated nerve fibres).
17. Absent fix-and-follow requires investigation by **3 months**; any deviation should have resolved by **3 to 4 months**, and a constant deviation is abnormal at any age.

CHAPTER RECAP

1. Vertex, tear lens (SAM and FAP), residual astigmatism, LARS, and sagittal relationships cover most contact lens calculations.
2. Daily disposables have the lowest complication rate; scleral lenses solve irregular corneas and severe surface disease.
3. CLARE has an intact epithelium, CLPU has a small round sterile ulcer, microbial keratitis has a defect over an infiltrate with a chamber reaction.
4. No water, rub and rinse, replace on schedule, do not sleep in lenses, and present immediately for a red painful eye.
5. Low vision: define goals, refract properly, calculate magnification with a 2× reserve, compare devices by equivalent viewing power, and fix the lighting.
6. Symmetry separates congenital from acquired colour defects; Ishihara screens congenital red-green only.
7. Köllner with its four exceptions.
8. Vision therapy treats measurable visual dysfunction, including post-concussion, but does not treat dyslexia.
9. Leukocoria and an absent red reflex are urgent referrals at any age.
10. Fix-and-follow by 3 months, alignment by 4 months, 20/20 by 3 to 5 years.

CHAPTER 14 · TMOD CORE

Therapeutic formulary: drug, dose, schedule, duration

TMOD bullets 2, 3, and 4: selection, dose and duration, contraindications | ~120 separately scored items

TMOD items require the **dose, form, schedule, and duration**, not the drug class. This chapter is the reference for that resolution. Cover the right-hand column, name the regimen from the diagnosis, then check. Doses are typical adult regimens; adjust for weight, renal and hepatic function, pregnancy, and age.

Anti-infectives

BACTERIAL DISEASE

Indication	Regimen
Bacterial conjunctivitis	Polymyxin B-trimethoprim, or erythromycin `0.5%` ointment, or azithromycin `1%`, or a fluoroquinolone: **1 drop four times daily for 5 to 7 days** (ointment four times daily). Contact lens wearers: use a fluoroquinolone and stop lens wear.
Bacterial keratitis, empirical monotherapy	**Moxifloxacin `0.5%` or besifloxacin `0.6%`: 1 drop every hour day and night for 24 to 48 hours**, then taper by clinical response over 1 to 2 weeks. Add homatropine `5%` two or three times daily. Review at **24 hours**.
Bacterial keratitis, severe or central	**Fortified vancomycin `25–50 mg/mL` alternating with fortified tobramycin `14 mg/mL`, each every 30 to 60 minutes around the clock**, after culture. Consider ceftazidime `50 mg/mL` instead of tobramycin for less surface toxicity.
Adult chlamydial (inclusion) conjunctivitis	**Azithromycin `1 g` orally as a single dose**, or doxycycline `100 mg` twice daily for 7 days. **Treat all sexual partners**, screen for other sexually transmitted infections, and report.
Trachoma (mass treatment)	Azithromycin `20 mg/kg` (maximum `1 g`) orally as a single dose, within the SAFE strategy.
Gonococcal conjunctivitis, adult	**Ceftriaxone `1 g` intramuscularly or intravenously as a single dose**, plus chlamydial cover, plus copious saline lavage and a topical fluoroquinolone. **Admit for intravenous therapy if the cornea is involved.**
Gonococcal ophthalmia neonatorum	**Ceftriaxone `25–50 mg/kg` (maximum `125 mg`) intravenously or intramuscularly as a single dose**, plus lavage; admit.
Chlamydial ophthalmia neonatorum	**Oral erythromycin `50 mg/kg/day` in four divided doses for 14 days**; treat the mother and her partner; warn about pneumonitis.
Blepharitis / MGD	Lid hygiene plus **azithromycin `1%` drops** (twice daily for 2 days then once daily for 2 to 4 weeks) or erythromycin ointment to the margins nightly; **oral doxycycline `50–100 mg` daily for 2 to 3 months**; lotilaner `0.25%` twice daily for 6 weeks for *Demodex*.

Indication	Regimen
Preseptal cellulitis, adult	**Amoxicillin-clavulanate `875/125 mg` twice daily for 7 to 10 days**, or clindamycin `300 mg` four times daily, or trimethoprim-sulfamethoxazole plus amoxicillin if MRSA is suspected. Review in 24 to 48 hours.
Orbital cellulitis	**Admit.** Intravenous vancomycin plus ceftriaxone (or ampicillin-sulbactam or piperacillin-tazobactam), adding metronidazole for intracranial or anaerobic involvement.
Dacryocystitis, acute	Warm compresses plus **amoxicillin-clavulanate `875/125 mg` twice daily** (or clindamycin for MRSA); admit for intravenous therapy if systemically unwell. Do not probe acutely.
Endophthalmitis, postoperative	**Intravitreal vancomycin `1 mg / 0.1 mL` plus ceftazidime `2.25 mg / 0.1 mL`** (or amikacin `0.4 mg`), with vitrectomy if presenting vision is light perception only (EVS). Systemic antibiotics added no benefit in EVS.
Ocular syphilis	**Aqueous penicillin G `18–24 million units` daily intravenously for 10 to 14 days** (treat as neurosyphilis). Test HIV, perform a lumbar puncture, and report.
Lyme disease with ocular involvement	Doxycycline `100 mg` twice daily for 14 to 21 days; intravenous ceftriaxone for neurologic or severe disease.
Ocular tuberculosis	Multidrug therapy (isoniazid, rifampicin, pyrazinamide, ethambutol) for 6 to 9 months with an infectious disease specialist, plus corticosteroid cover. **Monitor for ethambutol optic neuropathy.**

VIRAL, FUNGAL, AND PARASITIC DISEASE

Indication	Regimen
HSV epithelial keratitis	**Ganciclovir `0.15%` gel five times daily until healed, then three times daily for 7 days**; or trifluridine `1%` nine times daily; or **oral aciclovir `400 mg` five times daily for 7 to 10 days**. **No steroid.**
HSV stromal keratitis	**Prednisolone acetate `1%` with antiviral cover**, tapered over months, plus **oral aciclovir `400 mg` twice daily** prophylaxis (HEDS: halves recurrence).
HSV prophylaxis	**Aciclovir `400 mg` twice daily**, or valaciclovir `500 mg` daily, long term.
Herpes zoster ophthalmicus	**Valaciclovir `1 g` three times daily**, or famciclovir `500 mg` three times daily, or aciclovir `800 mg` five times daily, **for 7 to 10 days, started within 72 hours**. Add topical steroid and cycloplegia for keratitis or uveitis, and refer for neuralgia management. Recommend recombinant zoster vaccine for prevention.
Acute retinal necrosis	**Intravenous aciclovir `10 mg/kg` every 8 hours for 5 to 10 days then oral, or oral valaciclovir `2 g` three times daily, plus intravitreal foscarnet or ganciclovir**, then prolonged oral prophylaxis. Vitreoretinal referral.
CMV retinitis	Oral valganciclovir `900 mg` twice daily for 21 days then `900 mg` daily maintenance, plus intravitreal ganciclovir or foscarnet for zone 1 disease; restore immunity.
Adenoviral conjunctivitis	**Supportive only**: cool compresses, preservative-free lubricants, hygiene education. Remove membranes. Loteprednol `0.5%` four times daily tapered, only for membranes or visually significant infiltrates.
Filamentous fungal keratitis	**Natamycin `5%` hourly** with epithelial debridement, tapering over weeks; add oral voriconazole `200 mg` twice daily for deep disease. **No steroid.**

Indication	Regimen
Yeast (*Candida*) keratitis	Amphotericin B `0.15%` hourly; consider oral fluconazole.
***Acanthamoeba* keratitis**	**PHMB `0.02–0.08%` or chlorhexidine `0.02–0.2%` plus propamidine or hexamidine, hourly initially, tapering over 6 to 12 months**, with epithelial debridement. Avoid or delay steroid.
Ocular toxoplasmosis	**Pyrimethamine `100 mg` loading then `25–50 mg` daily, plus sulfadiazine `1 g` four times daily, plus folinic acid `5–10 mg` three times weekly, for 4 to 6 weeks**; or trimethoprim-sulfamethoxazole one double-strength tablet twice daily. **Add oral prednisone `0.5–1 mg/kg` 24 to 48 hours after starting antimicrobials** for sight-threatening lesions.
Onchocerciasis	Ivermectin `150 µg/kg` as a single dose, repeated periodically. **Avoid diethylcarbamazine.**

Anti-inflammatory and immunomodulatory

CORTICOSTEROIDS AND NSAIDS

Indication	Regimen
Acute anterior uveitis, 3–4+ cells	**Prednisolone acetate `1%` every 1 to 2 hours while awake** (shake the bottle), plus **homatropine `5%` two or three times daily**. Review in 3 to 7 days, then taper by halving the frequency weekly over 4 to 6 weeks. Check IOP at every visit. Alternative: difluprednate `0.05%` four times daily.
Mild anterior uveitis	Prednisolone acetate `1%` four to six times daily, plus cyclopentolate `1%` two or three times daily; taper over 2 to 4 weeks.
Episcleritis	Lubricants; oral NSAID for symptoms; fluorometholone `0.1%` or loteprednol `0.5%` four times daily for a short course in nodular or troublesome cases.
Non-necrotising scleritis	**Oral NSAID at full dose** (ibuprofen `600–800 mg` three times daily, or indomethacin `25–50 mg` three times daily, or naproxen `500 mg` twice daily) with gastric protection; escalate to **oral prednisone `1 mg/kg`**, then to systemic immunosuppression.
Necrotising scleritis or peripheral ulcerative keratitis	**Systemic immunosuppression urgently**: high-dose steroid plus cyclophosphamide or rituximab, with rheumatology. Add doxycycline for MMP inhibition and a bandage lens or glue for impending perforation.
Intermediate uveitis with macular oedema	Posterior sub-Tenon triamcinolone `40 mg`, or intravitreal triamcinolone or a dexamethasone implant; oral prednisone `1 mg/kg` for bilateral disease; then methotrexate, mycophenolate, or adalimumab.
Postoperative inflammation	Prednisolone acetate `1%` four times daily tapering over 3 to 4 weeks, plus a topical NSAID and a topical antibiotic four times daily for a week.
Pseudophakic cystoid macular oedema	**Topical NSAID (ketorolac `0.5%` or bromfenac) plus prednisolone acetate `1%`, four times daily for 4 to 8 weeks**; escalate to sub-Tenon or intravitreal steroid, or oral acetazolamide `250 mg` twice daily.
Allergic conjunctivitis	**Olopatadine `0.2–0.7%` once daily**, or ketotifen, alcaftadine, epinastine, or bepotastine once or twice daily; add a mast cell stabiliser for maintenance in vernal or atopic disease, and loteprednol `0.5%` in short pulses for flares.

Indication	Regimen
Vernal or atopic keratoconjunctivitis	Mast cell stabiliser (lodoxamide, cromolyn) four times daily as maintenance, steroid pulses for flares, and **topical cyclosporine `0.05-2%` or tacrolimus `0.03%`** as steroid-sparing therapy; supratarsal triamcinolone for refractory shield ulcers.
Dry eye, inflammatory	**Cyclosporine `0.05%` or `0.09%` twice daily (allow 3 to 6 months)**, or **lifitegrast `5%` twice daily**; loteprednol `0.5%` for a 2 to 4 week induction; preservative-free lubricants as often as needed; punctal plugs after inflammation is controlled.
Non-infectious uveitis, steroid-sparing	Methotrexate `15-25 mg` weekly with folic acid; mycophenolate `1-1.5 g` twice daily; azathioprine (check TPMT); **adalimumab `80 mg` loading then `40 mg` every 2 weeks**. **Screen for latent tuberculosis and hepatitis B before a biologic. Etanercept is ineffective and can induce uveitis.**
Giant cell arteritis	**Intravenous methylprednisolone `1 g` daily for 3 days then oral prednisone `1 mg/kg`** for visual loss; oral `1 mg/kg` if no visual loss. Add tocilizumab as steroid-sparing. Bone protection, gastric protection, and glucose monitoring.
Optic neuritis (if treating)	**Intravenous methylprednisolone `1 g` daily (or `250 mg` every 6 hours) for 3 days, then oral prednisone `1 mg/kg` for 11 days with a taper. Never oral prednisone alone at `1 mg/kg`.**
Thyroid eye disease, moderate to severe active	Intravenous methylprednisolone, typically `500 mg` weekly for 6 weeks then `250 mg` weekly for 6 weeks (cumulative about `4.5 g`); teprotumumab `10 mg/kg` then `20 mg/kg` every 3 weeks for 8 infusions; selenium in mild disease.

Glaucoma

DOSES, MAGNITUDES, AND CAUTIONS

Agent	Dose	Reduction and cautions
Latanoprost `0.005%`, travoprost `0.004%`, bimatoprost `0.01%`, tafluprost `0.0015%`	Once nightly	`25-33%`. Hyperaemia, lash growth, iris and periocular pigmentation, periorbitopathy; caution in uveitis, aphakia with a ruptured capsule (macular oedema), and herpetic keratitis.
Latanoprostene bunod `0.024%`	Once nightly	Prostaglandin plus nitric oxide donation; slightly greater reduction than latanoprost.
Timolol `0.25-0.5%`	Once or twice daily	`20-25%`. **Contraindicated in asthma, COPD, bradycardia, heart block; masks hypoglycaemia.** Little nocturnal effect. Betaxolol `0.25%` twice daily is β_1-selective.
Brimonidine `0.1-0.2%`	Twice or three times daily	`20-25%`. Follicular allergy in up to `15%`; **contraindicated under age 2**; avoid with monoamine oxidase inhibitors.
Dorzolamide `2%` / brinzolamide `1%`	Two or three times daily	`15-20%`. Stinging, bitter taste, corneal decompensation risk in compromised endothelium; sulfonamide moiety.
Netarsudil `0.02%`	Once nightly	About `20%`. Hyperaemia in most, conjunctival haemorrhage, reversible verticillata.
Pilocarpine `1-4%`	Three or four times daily	`15-25%`. Brow ache, induced myopia, miosis, retinal detachment risk; contraindicated in uveitic and neovascular glaucoma.

Agent	Dose	Reduction and cautions
Acetazolamide, oral	`250–500 mg` two to four times daily, or `500 mg` sustained release twice daily	`30–40%`. Paraesthesia, malaise, metallic taste, weight loss, metabolic acidosis, hypokalaemia, renal stones, rare aplastic anaemia. **Avoid in sickle cell disease, sulfa allergy, renal or hepatic failure, and with high-dose salicylates.**
Mannitol `20%`	`1–2 g/kg` intravenously over 30 to 60 minutes	Acute closure or pre-operative. **Avoid in cardiac or renal failure.**
Acute angle closure	–	One drop each of a β-blocker, an α_2-agonist, and a topical carbonic anhydrase inhibitor; **acetazolamide** `500 mg`; topical prednisolone acetate `1%`; supine positioning and corneal indentation; **pilocarpine** `1–2%` **once IOP is below about 40 to 50 mmHg**; hyperosmotic if needed; then **bilateral laser iridotomy**.

Diagnostic and adjunctive agents

DIAGNOSTIC PHARMACOLOGY

Agent	Use and dose
Tropicamide `0.5–1%`	Routine dilation; onset 15 to 20 minutes, duration 4 to 6 hours; a weak cycloplegic.
Phenylephrine `2.5%`	Adjunct mydriatic; also differentiates conjunctival from deep episcleral injection, and tests Müller muscle in ptosis. **Avoid** `10%` **in cardiovascular disease and in infants.**
Cyclopentolate `0.5–2%`	Cycloplegic refraction: `1%` in children (`0.5%` in infants), two drops five minutes apart, refract at 30 to 45 minutes. CNS effects at higher concentrations.
Atropine `1%`	Strong cycloplegia for uveitis, dense irides, and amblyopia penalisation; duration 7 to 14 days. **Systemic toxicity risk in children.**
Atropine `0.01–0.05%`	Myopia control, once nightly; `0.05%` is more effective but with more photophobia, near blur, and rebound.
Proparacaine `0.5%`	Topical anaesthesia; onset under 30 seconds, duration 15 to 20 minutes. **Never dispensed.**
Fluorescein sodium	Epithelial staining, Seidel testing, tear break-up time, applanation. Discolours soft lenses: remove them first and wait at least 30 to 60 minutes before reinsertion.
Lissamine green / rose bengal	Conjunctival and lid margin staining, lid wiper epitheliopathy; rose bengal stings and stains rose bengal-sensitive cells including herpetic borders.
Pilocarpine `0.125%`	Diagnostic for a tonic (Adie) pupil: it constricts a denervated sphincter.
Apraclonidine `0.5%`	Diagnostic for Horner syndrome: reversal of anisokoria. Also used to blunt post-laser IOP spikes.
Hydroxyamphetamine `1%`	Localises a Horner syndrome: dilates in first- and second-order lesions, fails in third-order.
Oxymetazoline `0.1%`	Acquired blepharoptosis: about `1 mm` of elevation via Müller muscle.
Pilocarpine `1.25%`	Presbyopia: pupil-modulating pinhole effect; warn about brow ache, dim vision, and (theoretically) retinal detachment risk in high myopes.
Hypertonic saline `5%`	Corneal oedema and recurrent erosion: drops four times daily and ointment at night.

Agent	Use and dose
Cenegermin `0.002%`	Neurotrophic keratitis: six times daily for 8 weeks.
Autologous serum `20–50%`	Persistent epithelial defect, severe dry eye, neurotrophic keratopathy: four to eight times daily.
Doxycycline `50–100 mg` **daily**	MMP inhibition: MGD, ocular rosacea, recurrent erosion, corneal melting, chemical burn. **Not in pregnancy, breastfeeding, or under age 8.**
Vitamin C `1–2 g` **daily**	Chemical burn: supports collagen synthesis.
Acetazolamide `250–500 mg`	Also for idiopathic intracranial hypertension (titrated upward, often to `1 to 4 g` daily), macular oedema in retinitis pigmentosa, and pseudophakic CMO.

Intravitreal and injectable agents

Agent	Indication and regimen
Aflibercept `2 mg`	Neovascular AMD, DMO, RVO macular oedema, PDR: three or more monthly loading doses then treat-and-extend. High-dose `8 mg` formulations allow longer intervals.
Ranibizumab `0.3–0.5 mg`	Same indications plus ROP; monthly then as needed or treat-and-extend.
Bevacizumab `1.25 mg`	Off-label, equivalent acuity outcomes to ranibizumab in CATT; widely used for cost reasons.
Faricimab `6 mg`	Bispecific VEGF-A and angiopoietin-2; longer intervals for many patients.
Brolucizumab `6 mg`	Longer intervals but a higher rate of intraocular inflammation and occlusive vasculitis.
Pegcetacoplan, avacincaptad pegol	Geographic atrophy: monthly or every other month; slows atrophy growth without proven functional benefit, and can precipitate exudative conversion.
Dexamethasone intravitreal implant `0.7 mg`	DMO, RVO oedema, non-infectious posterior uveitis; lasts 3 to 6 months. Expect cataract and IOP rise.
Fluocinolone implants	Chronic DMO and uveitis; up to 3 years. High rates of cataract and IOP rise.
Triamcinolone `40 mg`	Posterior sub-Tenon or intravitreal for uveitic or postoperative macular oedema.
Botulinum toxin	Blepharospasm and hemifacial spasm (repeat every 3 to 4 months), acute sixth nerve palsy, and small-angle strabismus.
Ocriplasmin `0.125 mg`	Symptomatic vitreomacular adhesion; limited efficacy in focal small adhesions only.

CONTRAINDICATIONS YOU MUST BE ABLE TO STATE WITHOUT HESITATION

Drug	Contraindication or caution
Non-selective β-blocker	Asthma, COPD, bradycardia, heart block, decompensated heart failure; masks hypoglycaemia.
Brimonidine	Age under 2; monoamine oxidase inhibitors; caution in young children generally.
Carbonic anhydrase inhibitor (oral)	Sickle cell disease (including hyphaema in trait), sulfa allergy, renal or hepatic failure, high-dose salicylates.
Prostaglandin analogue	Active uveitis, herpetic keratitis, macular oedema risk (aphakia with capsule rupture); usually avoided in pregnancy.
Pilocarpine and other miotics	Uveitic and neovascular glaucoma; retinal detachment risk; **microspherophakia and aqueous misdirection, where they worsen the mechanism**.
Topical steroid	Untreated herpes simplex epithelial keratitis; fungal and *Acanthamoeba* keratitis; any undiagnosed infectious infiltrate.

Drug	Contraindication or caution
Topical NSAID	Compromised or dry ocular surface, diabetes, post-surgical melting risk; avoid prolonged unsupervised use.
Tetracyclines	Pregnancy, breastfeeding, children under 8; combined with isotretinoin (pseudotumour cerebri).
Fluoroquinolones (systemic)	Tendon rupture, aortic dissection, neuropathy, QT prolongation; avoid in children and pregnancy where alternatives exist.
Topical anaesthetic	**Never dispensed for home use.**
Atropine (systemic absorption)	Infants and small children; Down syndrome; document the "hot, dry, red, blind, mad" toxidrome for parents.
Adalimumab and other biologics	Untreated latent tuberculosis, hepatitis B, active infection, demyelinating disease; **etanercept is not used for uveitis**.

RETRIEVAL PRACTICE: GIVE THE FULL REGIMEN, NOT THE CLASS

1. Adult inclusion conjunctivitis.
2. Gonococcal conjunctivitis without corneal involvement.
3. Chlamydial ophthalmia neonatorum.
4. A central `2 mm` bacterial corneal ulcer in a contact lens wearer.
5. *Fusarium* keratitis.
6. *Acanthamoeba* keratitis.
7. HSV dendritic keratitis, and then HSV stromal keratitis with prophylaxis.
8. Herpes zoster ophthalmicus presenting at 48 hours.
9. Sight-threatening ocular toxoplasmosis.
10. Acute anterior uveitis with 4+ cells and fibrin.
11. Necrotising scleritis in a rheumatoid patient.
12. Giant cell arteritis with visual loss.
13. Acute angle closure with IOP 62 mmHg.
14. Pseudophakic cystoid macular oedema at 6 weeks.
15. Postoperative endophthalmitis with vision of hand motions.
16. Moderate dry eye with an inflammatory component.

KEY

1. Azithromycin `1 g` orally once (or doxycycline `100 mg` twice daily for 7 days), plus partner treatment, STI screening, and reporting.
2. Ceftriaxone `1 g` intramuscularly or intravenously once, plus chlamydial cover, saline lavage, and a topical fluoroquinolone; partner treatment and screening.
3. Oral erythromycin `50 mg/kg/day` in four divided doses for 14 days; treat the mother and partner; warn about pneumonitis.
4. Culture (scrape edge and base), then moxifloxacin `0.5%` hourly around the clock for 24 to 48 hours (or fortified vancomycin with tobramycin every 30 to 60 minutes for this severity), plus homatropine `5%`, no steroid, stop lens wear, review at 24 hours.
5. Natamycin `5%` hourly with epithelial debridement, tapering over weeks; oral voriconazole for deep disease; no steroid.
6. PHMB or chlorhexidine plus propamidine or hexamidine, hourly then tapering over 6 to 12 months, with debridement; avoid early steroid.
7. Dendritic: ganciclovir `0.15%` gel five times daily until healed then three times daily for a week, or oral aciclovir `400 mg` five times daily for 7 to 10 days, with no steroid. Stromal: prednisolone acetate `1%` with antiviral cover, tapered over months, plus aciclovir `400 mg` twice daily prophylaxis.
8. Valaciclovir `1 g` three times daily for 7 to 10 days (within the 72-hour window), plus topical steroid and cycloplegia if there is keratitis or uveitis, plus neuralgia management.

9. Pyrimethamine `100 mg` load then `25–50 mg` daily, sulfadiazine `1 g` four times daily, and folinic acid `5–10 mg` three times weekly for 4 to 6 weeks, with oral prednisone added 24 to 48 hours later.
10. Prednisolone acetate `1%` every hour while awake plus homatropine `5%` three times daily; review in 3 days; taper over 4 to 6 weeks; IOP at every visit; dilate the fundus.
11. High-dose systemic corticosteroid plus **cyclophosphamide or rituximab**, with urgent rheumatology involvement; add doxycycline and a bandage lens for corneal melting.
12. Intravenous methylprednisolone `1 g` daily for 3 days then oral prednisone `1 mg/kg`; start before biopsy; add tocilizumab as steroid-sparing; bone and gastric protection.
13. Topical β-blocker, α_2-agonist, and CAI; acetazolamide `500 mg`; topical prednisolone; supine positioning and indentation; pilocarpine once IOP is below 40 to 50 mmHg; hyperosmotic if needed; then bilateral laser iridotomy.
14. Ketorolac `0.5%` plus prednisolone acetate `1%` four times daily for 4 to 8 weeks; escalate to sub-Tenon or intravitreal steroid or oral acetazolamide `250 mg` twice daily.
15. Tap and intravitreal **vancomycin `1 mg / 0.1 mL` plus ceftazidime `2.25 mg / 0.1 mL`**; vitrectomy is *not* required at hand motions per the EVS, which reserved it for light perception only.
16. Preservative-free lubricants, lid therapy for any MGD, plus **cyclosporine `0.05%` twice daily** (3 to 6 months for effect) or **lifitegrast `5%` twice daily**, with a 2 to 4 week loteprednol induction; punctal plugs once inflammation is controlled.

CHAPTER 15 · TMOD CORE

Follow-up intervals and referral urgency

TMOD bullet 5: follow-up and prognosis | Related to treatment and management item type

"When do you see this patient again?" is one of the most reliably asked questions on the examination, because it is the single best proxy for whether a candidate understands the natural history of a disease. This chapter collects the intervals in one place. Cover the interval column and produce it from the diagnosis.

Anterior segment

Condition	Review	Reason for that interval
Bacterial keratitis	**24 hours**, then daily until improving	Progression or failure must be caught within a day; the infiltrate can perforate.
Fungal keratitis	Daily initially, then weekly for weeks to months	Slow response and a long treatment course; watch for endothelial plaque and scleral extension.
***Acanthamoeba* keratitis**	Daily initially, then weekly to monthly for 6 to 12 months	Cysts persist; relapse on premature cessation is characteristic.
HSV epithelial keratitis	2 to 3 days	Confirm the dendrite is healing and no stromal or endothelial component has emerged.
HSV stromal keratitis	Weekly, then monthly during the steroid taper	Long taper with IOP monitoring and rebound risk.
Herpes zoster ophthalmicus	Days 3 to 7, then weekly while active	Delayed keratitis, uveitis, raised IOP, and scleritis appear over weeks.
Corneal abrasion, large or central	24 hours	Confirm healing and exclude infection.
Recurrent corneal erosion	1 to 2 weeks, then monthly	Assess the response to hypertonic saline and the need for a procedure.
Chemical burn, mild	24 hours, then every few days	Alkali injury progresses; IOP and limbal ischaemia must be tracked.
Chemical burn, moderate to severe	Daily	Symblepharon lysis, IOP, and melting risk.
Adenoviral conjunctivitis	As needed, or 1 to 2 weeks if membranes or infiltrates	Otherwise self-limited; educate rather than schedule.
Gonococcal conjunctivitis	**Daily until improving**	Corneal perforation risk within days.
Bacterial conjunctivitis	As needed, or 5 to 7 days if not improving	Self-limited in most cases.
Allergic conjunctivitis	2 to 4 weeks, then seasonally	Assess the response and reinforce avoidance.

Condition	Review	Reason for that interval
Vernal or atopic keratoconjunctivitis	1 to 2 weeks in a flare, then every 3 months	Shield ulcer and steroid complications.
Blepharitis and MGD	4 to 6 weeks, then every 6 months	Confirm adherence and the response before escalating.
Chalazion	4 to 6 weeks	Most resolve; a recurrent or atypical lesion is a biopsy.
Preseptal cellulitis	**24 to 48 hours**	Detect progression to orbital involvement.
Dry eye, moderate	4 to 6 weeks initially, then every 4 to 6 months	Cyclosporine takes 3 to 6 months; lifitegrast is faster.
Pterygium, observed	6 to 12 months with photographs and keratometry	Document growth and induced astigmatism.
Primary acquired melanosis	3 to 6 months with photographs	Detect growth or nodularity indicating melanoma.
Corneal graft, stable	Per surgeon, typically 3 to 6 months	Endothelial cell loss and rejection surveillance; **rejection is a same-day presentation**.
Keratoconus, stable adult	12 months with tomography	Detect progression; a young patient needs 6 months.
Keratoconus, adolescent or documented progression	3 to 6 months	Cross-linking window; progression is fastest in the second and third decades.
Contact lens wearer, healthy	12 months	Sooner for extended wear, ortho-k, scleral, or specialty fits.
Ortho-k or myopia control	1 day, 1 week, 1 month, then every 6 months with axial length	Infection risk plus efficacy monitoring.

Uveitis, glaucoma, and lens

Condition	Review	Reason
Acute anterior uveitis, severe	3 days	Confirm response, IOP, and absence of posterior involvement.
Acute anterior uveitis, moderate	5 to 7 days, then weekly during taper	Rebound on rapid taper.
JIA-associated uveitis, high risk, quiescent	**Every 3 months**	The eye is asymptomatic and white; symptoms never bring the child in.
Intermediate uveitis	1 to 3 months	Macular oedema is the vision-limiting complication.
Toxoplasmosis on treatment	Weekly	Assess response and drug toxicity (blood count on pyrimethamine).
Acute retinal necrosis	Every 1 to 3 days initially	Detachment risk approaches `50%`; fellow eye risk is high.
Scleritis	3 to 7 days until controlled	Escalation is often needed, and necrotising disease is a systemic emergency.

Condition	Review	Reason
Ocular hypertension, on treatment	3 to 4 months, with fields and imaging annually	Confirm target achieved and detect conversion.
Glaucoma, newly diagnosed or uncontrolled	4 to 6 weeks after any change	Verify the treatment effect before accepting the target.
Glaucoma, stable at target	3 to 6 months, with fields every 6 to 12 months	More often in advanced disease, in young patients, and after a disc haemorrhage.
Glaucoma, advanced or rapidly progressing	2 to 3 months, with fields every 4 to 6 months	Little reserve; establish the rate quickly (aim for several fields in the first two years).
Narrow angle after iridotomy	1 to 2 weeks (patency and IOP), then annually with gonioscopy	Detect plateau iris and residual closure.
Post-trabeculectomy	Day 1, week 1, then weekly for 4 to 6 weeks	Bleb management, hypotony, and leak. **Lifelong blebitis education.**
Angle recession after trauma	**Annually for life**	Glaucoma can present decades later.
Steroid user, any route	Every 4 to 6 weeks while on treatment	Steroid response typically emerges at 2 to 6 weeks.
Cataract surgery, uncomplicated	Day 1, week 1, week 3 to 4, then refract at 4 to 6 weeks	IOP spike, wound integrity, inflammation, and CMO at 4 to 6 weeks.
After Nd:YAG capsulotomy	1 hour (IOP), then 1 to 2 weeks	IOP spike; retinal detachment and CMO risk.
Hyphaema	**Daily for 5 days**, then weekly	Rebleeding peaks at 2 to 5 days; then gonioscopy once quiet.

Retina and neuro-ophthalmology

Condition	Review	Reason
Acute PVD, no break	**2 to 6 weeks**, sooner for new symptoms	New breaks develop during ongoing separation.
Acute PVD with vitreous haemorrhage	1 to 2 weeks, or sooner	Much higher probability of an undetected break.
Treated retinal break	1 to 2 weeks, then 4 to 6 weeks, then 3 to 6 months	Confirm the barrier and detect new breaks.
Untreated lattice or atrophic hole	6 to 12 months, with symptom education	Low event rate; education matters more than frequency.
Retinal detachment, macula-on	**Refer for surgery within 24 hours**	Preserve the fovea.
Retinal detachment, macula-off	Refer within a few days	Outcome is less time-critical once the fovea is off, but still urgent.
Diabetes, no retinopathy	12 months	Type 2 from diagnosis, type 1 from 5 years or puberty.

Condition	Review	Reason
Mild NPDR	12 months	–
Moderate NPDR	6 to 12 months	–
Severe NPDR	**3 to 4 months**	About half progress to proliferative disease within a year.
PDR, not high-risk	2 to 3 months	–
High-risk PDR	Treat now, then per treatment	DRS: PRP halves severe visual loss.
Centre-involving DMO on anti-VEGF	4 to 6 weeks	Loading then treat-and-extend.
Diabetes in pregnancy	Each trimester and post partum	Retinopathy accelerates.
Non-ischaemic CRVO	Monthly for 6 months	About a third convert to ischaemic.
Ischaemic CRVO	**Monthly for 6 months with undilated gonioscopy**	Neovascular glaucoma peaks at 3 to 4 months.
BRVO	1 to 2 months	Macular oedema and, less often, neovascularisation.
CRAO or BRAO	Systemic workup **today**; ocular review at 4 weeks then 3 months	Stroke risk is immediate; neovascularisation occurs in `15 to 20%`.
Intermediate AMD	6 to 12 months, with daily home monitoring	Conversion is sudden and time-critical.
Neovascular AMD on anti-VEGF	4 to 12 weeks per regimen	Treat-and-extend individualises the interval.
Central serous chorioretinopathy	4 to 8 weeks	Most resolve in 3 to 4 months; treat if persistent.
Choroidal naevus, low risk	6 months, then annually with photographs and ultrasound	Detect growth; two or more risk factors means referral.
Retinitis pigmentosa	Annually	Cataract, macular oedema, refraction, and low vision needs.
Hydroxychloroquine screening	Baseline within a year, then annually after 5 years	10-2 fields plus SD-OCT.
Optic neuritis	2 to 4 weeks, then 3 months	Confirm recovery; failure to recover means rethink the diagnosis.
Non-arteritic AION	4 to 6 weeks, then 3 to 6 months	Fellow eye education and vascular risk management.
Giant cell arteritis on steroid	Days initially, then per rheumatology	Fellow eye risk is highest in the first days.
Papilloedema on treatment	2 to 4 weeks with perimetry	Fields, not acuity, detect progression.

Condition	Review	Reason
Pupil-sparing microvascular CN III palsy	Weekly for 2 weeks, then monthly	The pupil can become involved; expect resolution in 8 to 12 weeks.
Microvascular CN VI palsy	4 to 6 weeks	Expect resolution in 8 to 12 weeks; image if not resolving by 3 months.
Amblyopia on treatment	Every 6 to 12 weeks, then taper and follow for at least a year	Recurrence in about a quarter.
Intermittent exotropia, observed	4 to 6 months with control grading	Deteriorating control is the surgical trigger.
Vergence or accommodative therapy	Weekly or fortnightly office sessions, reassess at 12 weeks	CITT protocol duration.

REFERRAL URGENCY: THE PHRASING THAT EARNS CREDIT

A referral must state **to whom, for what, and by when**. Compare:

Inadequate	Adequate
"Refer to ophthalmology."	"Refer to a vitreoretinal surgeon for repair of a macula-on superior rhegmatogenous retinal detachment **within 24 hours**."
"Refer for blood tests."	"Refer **today** to the emergency department for ESR, CRP, and platelet count and admission for intravenous methylprednisolone for suspected giant cell arteritis; I have given the first dose of prednisone `60 mg` orally."
"Advise the patient to see their doctor."	"Refer to the primary care physician **within 1 week** for fasting glucose and HgbA1c; retinal findings consistent with previously undiagnosed diabetes, described and imaged in the enclosed report. Blood pressure measured today `152/94`."
"Refer for a scan."	"Refer **same day** for CT angiography of the head to exclude a posterior communicating artery aneurysm in a pupil-involving third nerve palsy; I have telephoned the on-call neurology team."

Also document: that you explained the urgency to the patient, in what words, and what you did if the patient could not or would not attend. Part III PEPS scores referral specificity explicitly, and Part II asks the same judgement in multiple-choice form.

RETRIEVAL PRACTICE: STATE THE INTERVAL

1. Bacterial keratitis, first review.
2. Hyphaema, and why.
3. Ischaemic CRVO, and what you do at each visit.
4. Severe NPDR.
5. Acute PVD with no break found.
6. Acute PVD with vitreous haemorrhage.
7. Macula-on retinal detachment.
8. ANA-positive oligoarticular JIA, quiescent.
9. Angle recession after blunt trauma.
10. Hydroxychloroquine screening.

11. Papilloedema on acetazolamide, and what you monitor.
12. Pupil-sparing microvascular third nerve palsy.
13. Patient on inhaled corticosteroid with ocular hypertension.
14. Uncomplicated cataract surgery, and when you refract.
15. Adolescent with documented keratoconus progression.
16. Choroidal naevus with no risk factors.

KEY

1. **24 hours**, then daily until clearly improving.
2. **Daily for 5 days**, because rebleeding peaks at 2 to 5 days; then gonioscopy once quiet and lifelong glaucoma surveillance.
3. **Monthly for 6 months**, with **undilated gonioscopy and iris examination** at each visit, because neovascular glaucoma peaks at 3 to 4 months.
4. **3 to 4 months.**
5. **2 to 6 weeks**, and immediately for new symptoms.
6. **1 to 2 weeks or sooner**, because haemorrhage markedly raises the probability of an undetected break.
7. **Refer for surgery within 24 hours.**
8. **Every 3 months**, because the disease is asymptomatic with a white eye.
9. **Annually for life.**
10. Baseline within a year, then **annually after 5 years**, with 10-2 fields plus SD-OCT.
11. **2 to 4 weeks**, monitoring with **automated perimetry** and OCT rather than acuity.
12. **Weekly for 2 weeks** (the pupil can become involved), then monthly, expecting resolution in 8 to 12 weeks.
13. **Every 4 to 6 weeks** while on the steroid, because the response typically emerges at 2 to 6 weeks.
14. Day 1, week 1, week 3 to 4; **refract at 4 to 6 weeks** once the refraction is stable.
15. **3 to 6 months** with tomography, and refer for cross-linking.
16. **6 months, then annually** with photographs and ultrasonography.

CHAPTER 16 · NON-CLINICAL ITEM TYPES

Legal issues, ethics, and public health

Legal issues / ethics item type | Public health item type | Epidemiology and biostatistics

ANSWER THESE COLD BEFORE READING

1. Name the four elements a plaintiff must prove in a negligence claim.
2. Give the elements of valid informed consent and when it is not required.
3. Define abandonment and state how to terminate a doctor-patient relationship properly.
4. Name the four principles of biomedical ethics and apply each to a refusal of referral.
5. Define sensitivity, specificity, PPV, and likelihood ratio, and say which depend on prevalence.
6. Give the Wilson and Jungner screening criteria and apply them to glaucoma versus diabetic retinopathy.

BLUEPRINT WEIGHT

2 of 7

Legal issues and ethics, and public health, are two of the seven official item types, and they appear in cases alongside clinical items.

Regulation of practice and scope

WHO REGULATES WHAT

Body or instrument	Role
State board of optometry	**Licensure, scope of practice, discipline, and continuing education**. Scope is set by **state statute and regulation**, so therapeutic privileges, injection, laser, and oral prescribing authority vary by state. **You are responsible for knowing your own state's scope, and practising beyond it is unlawful regardless of your competence.**
NBEO	Develops and administers the national licensure examinations. Passing NBEO does not itself confer a licence.
ACOE	Accredits schools and colleges of optometry and residency programmes.
DEA	Registration for controlled substance prescribing where state scope allows it.
FDA	Drug and device approval; MedWatch adverse event reporting; the contact lens rule and the eyeglass rule.
FTC	**Eyeglass Rule**: the prescription must be released to the patient automatically at the completion of the refraction, at no extra charge, without requiring purchase. **Contact Lens Rule**: the contact lens prescription must be released, and sellers must verify prescriptions with the prescriber (with an eight-business-hour passive verification window). Prescription expiry is set by state law, commonly one to two years.
CMS	Medicare and Medicaid rules, including what counts as a covered medical versus routine vision service, and the documentation required.
OSHA	Workplace safety, including eye protection requirements and bloodborne pathogen standards for the practice.
HIPAA	Privacy and security of protected health information.
ANSI	Z80.1 for dress ophthalmic lens tolerances; Z87.1 for occupational and educational eye protection.

Negligence, standard of care, and consent

THE FOUR ELEMENTS OF NEGLIGENCE

1. **Duty.** A doctor-patient relationship existed, creating a duty of care. It can be created by seeing the patient, by giving specific advice by telephone, or by covering for a colleague.
2. **Breach.** The care fell below the **standard of care**, meaning what a reasonably prudent practitioner with similar training would do in similar circumstances. The standard is set by expert testimony, clinical practice guidelines, and the literature, not by what is customary in one office.
3. **Causation.** The breach *caused* the harm, both factually and proximately.
4. **Damages.** The patient suffered actual injury.

All four are required. A breach with no harm is not actionable negligence, and harm without a breach is not either.

The commonest optometric malpractice allegations: failure to diagnose (glaucoma, retinal detachment, tumour, papilloedema), failure to dilate, failure to refer or to refer urgently enough, failure to follow up on an abnormal finding, contact lens related injury, and failure to communicate the urgency of a referral. **Almost all of them are documentation failures as much as clinical ones.**

INFORMED CONSENT

Element	Detail
Capacity	The patient must be able to understand and to decide. Minors require a parent or guardian, with exceptions for emancipated minors and, in many states, for specific categories of care.
Disclosure	The **diagnosis, the proposed intervention, its risks and benefits, reasonable alternatives including doing nothing, and the consequences of refusing**.
Understanding	Confirmed, ideally by asking the patient to state back the plan. Use an interpreter when needed, and document that you did.
Voluntariness	Free of coercion.
Documentation	A signed form is evidence, not consent. The **note describing the conversation** is what matters.
Exceptions	A true emergency where the patient cannot consent and delay would cause harm; a valid prior refusal or advance directive; and legally authorised surrogate decision-making.
Informed refusal	Equally important and equally documented: record that the patient declined dilation, referral, or treatment, that you explained the specific risk in specific terms, and that they understood.

THE DOCTOR-PATIENT RELATIONSHIP

Issue	Rule
Abandonment	Terminating care unilaterally while the patient still needs it, without adequate notice or alternative. To terminate properly: give **written notice**, provide **reasonable time to find another provider** (commonly 30 days), **remain available for emergencies** in the interim, offer to **transfer records**, and document. Non-payment alone does not justify abandoning a patient mid-treatment.

Issue	Rule
Records	The patient owns the *information*; the practice owns the *physical or electronic record*. Patients have a right of access and to a copy under HIPAA, with limited exceptions. Retention periods are set by state law, commonly `7 years` for adults and, for minors, until some years past the age of majority. **Never alter a record after the fact; make a dated addendum instead.**
HIPAA	Use and disclosure for **treatment, payment, and health care operations** is permitted without specific authorisation. Disclose the **minimum necessary**. Required disclosures include public health reporting, abuse reporting, and court orders. Family members may generally be given information relevant to their involvement in care unless the patient objects, but be cautious. Breaches must be reported.
Telehealth	Licensure is generally required in the state where the **patient** is located. Standard of care does not change with the medium. Document the modality, the limitations of the remote examination, and the plan for in-person assessment.
Delegation and supervision	You are responsible for the acts of staff performing delegated testing. Scope for technicians is state-defined.
Advertising and referral relationships	Claims must be truthful and substantiated. **Anti-Kickback Statute** and **Stark law** restrict payment for referrals and self-referral for certain services. Comanagement fee arrangements must reflect actual services provided.
Impaired or incompetent colleague	Ethical and often legal duty to report to the state board; patient safety outweighs collegiality.

Ethics

THE FOUR PRINCIPLES, APPLIED

Principle	Meaning and an applied example
Autonomy	Respect the patient's informed choice, including refusal. A competent patient may decline dilation. *Your obligation is to inform them specifically of what you cannot rule out, to document that, and to offer an alternative or an early review.*
Beneficence	Act in the patient's interest. *Recommending a treatment even when it reduces practice revenue, and declining to sell a device that will not help.*
Non-maleficence	Do no harm. *Not dispensing topical anaesthetic; not prescribing a steroid to an undiagnosed infiltrate; not prescribing new glasses during unstable glycaemic control.*
Justice	Fair distribution and non-discrimination. *Not allocating appointment access by ability to pay for the urgent problem, and not declining care on the basis of protected characteristics.*

COMMON ETHICAL SCENARIOS AND THE DEFENSIBLE ANSWER

- **The patient refuses a needed referral.** Explore why (cost, fear, transport, disbelief), address it, restate the specific risk in concrete terms, document the informed refusal, offer an interim plan and a definite review date, and follow up by telephone. Do not simply record "patient declined".
- **You made an error.** Disclose it promptly and honestly, explain what happened and what you will do, apologise, correct it, and document. Concealment converts an error into misconduct, and disclosure reduces rather than increases litigation.
- **A colleague's care was substandard.** Focus on the patient's current needs; avoid disparagement; document objective findings; report to the board only when patient safety requires it.
- **A patient asks for a prescription you consider unsafe** (for example, a driving certification they do not meet). Refuse, explain, document, and follow reporting law.
- **A commercial incentive conflicts with the clinical decision.** Disclose any financial interest, and let the clinical indication drive the recommendation.
- **The patient is a child whose parent declines treatment for amblyopia or a cataract.** Educate persistently; involve the paediatrician; in cases of clear serious harm from neglect, child protection reporting obligations apply.
- **Sharing information with a spouse or adult child.** Ask the patient's permission in front of them; do not assume.

Public health: epidemiology and biostatistics

TEST PERFORMANCE MEASURES

Measure	Formula	Prevalence dependence and use
Sensitivity	`TP / (TP + FN)`	**Independent of prevalence.** A very sensitive test, when negative, rules out (SnNout). Prioritise for screening.
Specificity	`TN / (TN + FP)`	**Independent of prevalence.** A very specific test, when positive, rules in (SpPin). Prioritise for confirmation.
Positive predictive value	`TP / (TP + FP)`	**Prevalence-dependent**, and falls sharply in low-prevalence screening.
Negative predictive value	`TN / (TN + FN)`	Prevalence-dependent; rises as prevalence falls.
Positive likelihood ratio	`Sn / (1 − Sp)`	Prevalence-independent; above 10 is strongly diagnostic.
Negative likelihood ratio	`(1 − Sn) / Sp`	Below 0.1 strongly argues against disease.
Prevalence	`cases / population`	A snapshot; reflects incidence and duration.
Incidence	`new cases / at-risk population / time`	Measures risk of developing disease.
Relative risk	`risk exposed / risk unexposed`	Cohort studies and trials.
Odds ratio	`ad / bc`	Case-control studies; approximates RR when the outcome is rare.

Measure	Formula	Prevalence dependence and use
Absolute risk reduction	$risk_{control} - risk_{treated}$	The clinically interpretable measure.
Number needed to treat	`1 / ARR`	How many treated to prevent one event.

STUDY DESIGNS AND BIASES

Design	Strength, weakness, and the bias it is prone to
Systematic review and meta-analysis	Highest level; limited by included study quality, heterogeneity, and **publication bias**.
Randomised controlled trial	Randomisation controls known and unknown confounding; masking controls observer and placebo effects. Limited external validity; prone to **attrition bias**.
Cohort study	Gives incidence and relative risk; prone to **confounding** and loss to follow-up.
Case-control study	Efficient for rare disease; gives an odds ratio; prone to **recall and selection bias**.
Cross-sectional study	Gives prevalence; cannot establish temporal sequence.
Case series or report	Hypothesis-generating only.
Screening-specific biases	**Lead-time bias** (survival appears longer only because diagnosis was earlier) and **length-time bias** (screening preferentially detects slowly progressive disease). Both make screening look better than it is unless mortality or a hard functional endpoint is measured.
Statistical concepts	Type I error (α) is a false positive conclusion; type II (β) is a false negative; **power** = **1 - β**, conventionally **0.80**. A 95% confidence interval excluding the null corresponds to $p < 0.05$, and additionally conveys precision. Use parametric tests for normally distributed interval data and non-parametric otherwise; chi-square for categorical data.

SCREENING: WILSON AND JUNGNER CRITERIA, APPLIED

1. The condition is an **important health problem**.
2. There is a **recognisable latent or early symptomatic stage**.
3. The **natural history is understood**.
4. A **suitable, acceptable test** exists.
5. There is an **accepted, effective treatment** for detected disease.
6. Facilities for diagnosis and treatment are **available**.
7. There is an **agreed policy on whom to treat**.
8. The programme is **cost-effective** and **continuous**, not a one-off.

Diabetic retinopathy screening satisfies all eight, which is why it is universally endorsed. **Population glaucoma screening does not**: no single test has adequate sensitivity and specificity in a low-prevalence population, and the positive predictive value of any one test is poor, which is why case-finding in higher-risk groups is preferred to mass screening. **Amblyopia and paediatric vision screening** satisfies the criteria and is endorsed, with the caveat that instrument-based screening must be followed by a comprehensive examination.

HEALTH POLICY AND DELIVERY CONCEPTS

Concept	What you need to know
Medicare	Federal, primarily age 65 and over and certain disabilities. **Part B covers medically necessary eye care** (glaucoma, diabetic retinopathy, cataract surgery), **not routine refraction or spectacles** (with a post-cataract-surgery exception for one pair). Part D covers outpatient drugs.

Concept	What you need to know
Medicaid	Joint federal and state, income-based; benefits vary substantially by state and often include paediatric vision.
Vision plan versus medical plan	A vision plan covers routine refractive care and materials; a medical plan covers disease. **Billing the correct plan for the correct service is a compliance issue, not a preference.** Documentation must support the chief complaint and the diagnosis billed.
Coding	**ICD-10** for diagnoses, **CPT** for procedures and services. Codes must be supported by the record. Upcoding and unbundling are fraud.
EPSDT	Medicaid's early and periodic screening, diagnostic and treatment benefit for children, which includes vision.
Health disparities	Higher rates of undetected and undertreated glaucoma and diabetic retinopathy in Black and Hispanic populations, driven by access, insurance, and structural factors as much as biology. Addressing access is a public health function of the profession.
Environmental and occupational vision	**ANSI Z87.1** eye protection for hazardous work, with the frame and lenses both certified and marked. **Polycarbonate or Trivex** for anyone monocular, for children, and for sports. UV protection for outdoor work and for aphakia. Ergonomic advice for computer work: viewing distance and angle, blink and break habits, illumination, and glare control. Sports-specific protection, and counselling that **most sports eye injuries are preventable**.
Driving standards	
Public health interventions in eye care	Vitamin A supplementation, trachoma SAFE, cataract surgical rate, uncorrected refractive error programmes, and school screening are the interventions with the largest global impact.

RETRIEVAL PRACTICE

1. Name the four elements of negligence and state which one a "no harm" error fails.
2. Define the standard of care and say who establishes it.
3. Give five elements of valid informed consent.
4. A patient declines dilation. What must you do and document?
5. Define abandonment and give four steps for proper termination.
6. Under the FTC Eyeglass Rule, when must you release a spectacle prescription?
7. Which measures of test performance are prevalence-independent?
8. A test has 95% sensitivity and 90% specificity. What is the positive likelihood ratio?
9. Event rates are 16% in controls and 10% with treatment. Give ARR, RR, and NNT.
10. Name the two biases specific to screening studies and explain each.
11. Give four of the Wilson and Jungner criteria and explain why population glaucoma screening fails them.
12. Does Medicare Part B cover a routine refraction?
13. Which four ethical principles apply when a patient refuses an urgent referral, and what does each require of you?
14. You realise you missed a retinal tear at a previous visit and the patient now has a detachment. What do you do?
15. Which lens material is mandatory for a monocular patient, and under which standard is occupational eyewear certified?
16. Where must you be licensed to provide telehealth?

KEY

1. Duty, breach, causation, and damages; a "no harm" error fails **damages**.
2. What a reasonably prudent practitioner with similar training would do in similar circumstances; established by expert testimony, clinical guidelines, and the literature, not by local custom.
3. Capacity, disclosure (diagnosis, intervention, risks and benefits, alternatives including no treatment, and consequences of refusal), understanding, voluntariness, and documentation of the conversation.
4. Explain specifically what cannot be excluded without dilation (retinal tear, detachment, tumour, diabetic changes), offer alternatives such as wide-field imaging or an early return, document the **informed refusal** in the patient's own terms, and arrange a definite review.
5. Terminating care unilaterally while care is still needed. Proper termination: written notice, reasonable time to find another provider (commonly 30 days), availability for emergencies in the interim, offer to transfer records, and documentation.
6. **Automatically at the completion of the refraction**, at no additional charge, and without requiring the patient to purchase anything.
7. **Sensitivity, specificity, and both likelihood ratios.**
8. LR+ = 0.95 / (1 - 0.90) = 0.95 / 0.10 = **9.5**.
9. ARR = **6%**; RR = 10/16 = **0.625** (a 37.5% relative reduction); NNT = 1/0.06 ≈ **17**.
10. **Lead-time bias**: survival appears longer only because the diagnosis was made earlier, without changing the date of death. **Length-time bias**: screening preferentially detects slowly progressive disease with a longer latent phase, which has a better prognosis anyway.
11. An important health problem; a recognisable latent stage; a suitable and acceptable test; an effective accepted treatment (also understood natural history, available facilities, an agreed treatment policy, and cost-effectiveness with continuity). Population glaucoma screening fails principally on the **test**: no single test has adequate sensitivity and specificity, and in a low-prevalence population the positive predictive value is poor, producing many false positives.
12. **No.** Part B covers medically necessary care, not routine refraction or spectacles, with a limited post-cataract-surgery exception for one pair of glasses.
13. Autonomy (respect the refusal but ensure it is informed), beneficence (persist in recommending what helps and address the barrier), non-maleficence (avoid harm from your own inaction, including by documenting and following up), and justice (do not allow ability to pay or other irrelevant factors to determine access to the urgent care).
14. Disclose promptly and honestly, explain what happened and what will now be done, apologise, ensure the patient receives correct urgent care, document contemporaneously without altering the earlier record, and notify your insurer. Concealment converts an error into misconduct.
15. **Polycarbonate** (or Trivex); occupational eyewear is certified under **ANSI Z87.1**, with both frame and lens marked.
16. In the state where the **patient** is located.

CHAPTER RECAP

1. Duty, breach, causation, damages. Most optometric claims are failures to diagnose, to dilate, to refer urgently, or to follow up.
2. Consent is a documented conversation, and informed refusal is documented the same way.
3. Terminate a relationship in writing with time, emergency cover, and record transfer.
4. Scope of practice is state statute, and it is your responsibility to know yours.
5. The Eyeglass Rule requires automatic prescription release at no charge.
6. Sensitivity and specificity are prevalence-independent; predictive values are not.
7. NNT is 1 over the absolute risk reduction, and lead-time and length-time bias flatter screening.
8. Diabetic retinopathy screening meets the Wilson and Jungner criteria; population glaucoma screening does not.
9. Medicare covers disease, not refraction; bill the plan that matches the service.
10. Disclose your errors, protect the patient first, and never alter a record.

APPENDIX A

Case bank: fifteen exam-format cases

Each case follows the Part II PAM structure: a scenario with demographics, chief complaint, review of systems, medications, history, and clinical findings, followed by items of the seven official types. Give yourself **60 seconds to read the scenario and 60 seconds per item**. Answer the whole case before checking the key, then log every error by *condition area* and by *item type*.

Case 1

SCENARIO

62-year-old man. Chief complaint: "curtain coming down over my right eye since yesterday afternoon, and lots of black specks for a week."
Review of systems: flashing lights in the right eye; no pain; no headache.
Medications: atorvastatin, lisinopril.
Ocular history: myopia, currently `-7.00 D` both eyes. **Medical history:** hypertension.
Findings: BVA OD `20/60`, OS `20/20`. Pupils equal, no RAPD. IOP OD `10 mmHg`, OS `16 mmHg`. Anterior segment unremarkable. OD dilated: pigment granules in the anterior vitreous, a Weiss ring, and a corrugated elevated retina superotemporally extending to within 1 disc diameter of the fovea, with a horseshoe tear at 11 o'clock.

ITEMS

1. What is the most appropriate diagnosis? (a) Retinoschisis (b) Rhegmatogenous retinal detachment (c) Exudative retinal detachment (d) Choroidal detachment (e) Tractional retinal detachment
2. Which finding most strongly supports the diagnosis? (a) Low IOP (b) Absence of an RAPD (c) Pigment granules in the vitreous with a horseshoe tear (d) Myopia of `-7.00 D` (e) The Weiss ring
3. What is the most appropriate management? (a) Observe and review in 1 week (b) Laser retinopexy in the office today (c) Refer to a vitreoretinal surgeon for repair within 24 hours (d) Refer to ophthalmology routinely (e) Start topical steroid and review in 48 hours
4. Which positioning advice is most appropriate while awaiting surgery? (a) Sit upright (b) Lie so the tear is dependent (c) Lie prone (d) Face-down positioning (e) No positioning advice is useful
5. Which additional examination is most important today? (a) Gonioscopy (b) Visual field (c) Scleral indented examination of the fellow eye (d) Corneal topography (e) Colour vision testing

KEY

1. **(b)** A corrugated mobile elevated retina with a break, tobacco dust, and low IOP in a myope after a PVD is rhegmatogenous. Schisis is smooth and immobile with an absolute scotoma; exudative fluid shifts and has no break; choroidal detachment stops short of the disc.
2. **(c)** The break plus pigment granules establishes the mechanism. Low IOP and myopia are supportive but non-specific, and the Weiss ring only tells you a PVD has occurred.
3. **(c)** This is macula-threatening (macula-on) and should be repaired within 24 hours. The referral must specify the subspecialty, the reason, and the timeframe.
4. **(b)** Positioning the break dependent uses gravity to limit fluid tracking toward the fovea.
5. **(c)** Fellow-eye examination with indentation is essential: bilateral pathology is common in myopes, and a break there may warrant prophylactic treatment.

Case 2

SCENARIO

78-year-old woman. Chief complaint: "I lost the vision in my left eye this morning, suddenly."
Review of systems: aching in the jaw when chewing for three weeks; scalp tenderness; shoulder and hip stiffness; unintentional weight loss; low-grade fever.
Medications: amlodipine, acetaminophen.
Findings: BVA OD `20/25`, OS **counting fingers**. Left RAPD. IOP normal. OS disc: **chalky white oedema with blurred margins** and a single cotton wool spot; no haemorrhage. OD disc: normal calibre cup with a healthy rim.

ITEMS

1. What is the most likely diagnosis? (a) Non-arteritic anterior ischaemic optic neuropathy (b) Arteritic anterior ischaemic optic neuropathy (c) Optic neuritis (d) Central retinal vein occlusion (e) Compressive optic neuropathy
2. Which symptom is most specific for the diagnosis? (a) Weight loss (b) Scalp tenderness (c) Jaw claudication (d) Fever (e) Shoulder stiffness
3. Which three tests should be obtained immediately? (Select 3) (a) ESR (b) MRI of the orbits (c) CRP (d) Fluorescein angiography (e) Platelet count (f) Carotid duplex
4. What is the most appropriate immediate action? (a) Order a temporal artery biopsy and start treatment when it confirms the diagnosis (b) Start oral prednisone `20 mg` daily and review in a week (c) Start high-dose corticosteroid now and arrange biopsy (d) Start aspirin `325 mg` and refer routinely (e) Observe and repeat testing in 48 hours
5. What is the most important reason for that urgency? (a) The affected eye may recover (b) The fellow eye is at high risk within days (c) The biopsy becomes negative within 24 hours (d) It prevents the headache (e) It reduces the ESR

KEY

1. **(b)** Age, profound loss, chalky white pallid swelling, and the constitutional symptoms are arteritic. Non-arteritic gives hyperaemic segmental swelling with better acuity and no systemic symptoms.
2. **(c)** Jaw claudication is the most specific symptom of giant cell arteritis.
3. **(a), (c), (e)** ESR, CRP, and platelet count.
4. **(c)** Treatment precedes biopsy. Intravenous methylprednisolone `1 g` daily for 3 days is appropriate given visual loss, followed by oral prednisone `1 mg/kg`.
5. **(b)** Fellow eye involvement is common within days to weeks without treatment. The biopsy remains informative for roughly two weeks after starting steroid, so it is not the limiting factor.

Case 3

SCENARIO

24-year-old woman. Chief complaint: "my right eye has been red and very painful for two days and I can't see well."
Review of systems: light hurts; watery discharge; no itching.
Medications: combined oral contraceptive.
Ocular history: monthly replacement soft contact lenses, worn overnight "sometimes", rinsed in tap water when travelling.
Findings: BVA OD `20/200`, OS `20/20`. Marked ciliary flush OD. Cornea OD: a `4 mm` paracentral **ring-shaped stromal infiltrate** with an overlying epithelial defect and prominent **radial keratoneuritis**; anterior chamber 2+ cells. Pain is described as far worse than the appearance suggests. She has already used a topical steroid prescribed elsewhere with initial improvement then worsening.

ITEMS

1. What is the most likely organism? (a) *Pseudomonas aeruginosa* (b) *Fusarium* species (c) *Acanthamoeba* species (d) Herpes simplex virus (e) *Staphylococcus aureus*
2. Which historical detail is most specific for this diagnosis? (a) Overnight wear (b) Monthly replacement (c) Rinsing lenses in tap water (d) Oral contraceptive use (e) Prior steroid use
3. Which culture medium is most appropriate? (a) Blood agar (b) Chocolate agar (c) Sabouraud dextrose agar (d) Non-nutrient agar with an *E. coli* overlay (e) Thayer-Martin agar
4. Which treatment combination is most appropriate? (Select 2) (a) Polyhexamethylene biguanide (b) Natamycin `5%` (c) Propamidine isethionate (d) Trifluridine `1%` (e) Fortified tobramycin
5. What is the expected duration of treatment? (a) 7 days (b) 2 weeks (c) 4 to 6 weeks (d) 6 to 12 months (e) Lifelong
6. Regarding the previously prescribed steroid, what is the most appropriate advice? (a) Increase the frequency (b) Continue unchanged (c) Discontinue or minimise it, because it worsens outcomes in this infection (d) Switch to difluprednate (e) Add an oral steroid

KEY

1. **(c)** Pain out of proportion to signs, radial keratoneuritis, a ring infiltrate, and a protracted course after apparent steroid response are the *Acanthamoeba* pattern.
2. **(c)** Water exposure is the specific risk factor. Overnight wear raises risk for keratitis generally, especially *Pseudomonas*.
3. **(d)** Non-nutrient agar with an *E. coli* overlay; confocal microscopy is a useful adjunct.
4. **(a) and (c)** A biguanide plus a diamidine.
5. **(d)** Six to twelve months, because cysts persist and relapse follows premature cessation.
6. **(c)** Steroid promotes amoebic proliferation and worsens outcomes; it should be avoided or used only late and cautiously.

Case 4

SCENARIO

56-year-old man. Chief complaint: routine diabetic review; "reading is a bit harder than last year."
Review of systems: nocturia; numb feet.
Medications: metformin, empagliflozin, lisinopril, atorvastatin.
Medical history: type 2 diabetes for 11 years. **HgbA1c 9.4%.** Blood pressure today `148/88`.
Findings: BVA OD `20/40`, OS `20/30`. Both eyes: numerous dot-blot haemorrhages in all four quadrants, venous beading in two quadrants, intraretinal microvascular abnormalities in one quadrant, no neovascularisation. OCT: OD central subfield thickness `412 µm` with intraretinal cystoid spaces involving the centre; OS `268 µm`.

ITEMS

1. What is the retinopathy stage? (a) Mild NPDR (b) Moderate NPDR (c) Severe NPDR (d) PDR without high-risk characteristics (e) High-risk PDR
2. Which finding establishes that stage? (a) Dot-blot haemorrhages in four quadrants (b) Venous beading in two quadrants (c) IRMA in one quadrant (d) Any one of the above satisfies the 4-2-1 rule (e) The central subfield thickness
3. What is the most appropriate treatment for the right eye? (a) Focal macular laser (b) Intravitreal anti-VEGF (c) Panretinal photocoagulation (d) Observation with a 6-month review (e) Topical NSAID
4. Which trial supports that choice, and what did it show? (a) DRS, that PRP halves severe visual loss (b) ETDRS, that focal laser halves moderate visual loss (c) DRCR Protocol T, that aflibercept was superior at one year at baseline acuity of 20/50 or worse (d) DRCR Protocol S, that ranibizumab is non-inferior to PRP (e) UKPDS, that blood pressure control reduces progression
5. What is the appropriate ocular review interval for the left eye? (a) 1 month (b) 3 to 4 months (c) 6 months (d) 12 months (e) 2 years
6. Which additional action is most important? (a) Prescribe new reading glasses today (b) Communicate the A1c and blood pressure findings and the retinopathy severity to the primary care physician (c) Order a fluorescein angiogram (d) Start oral acetazolamide (e) Advise the patient to reduce salt intake

KEY

1. **(c)** Severe NPDR.
2. **(d)** Any one of the three 4-2-1 criteria suffices, and this patient meets all three.
3. **(b)** Intravitreal anti-VEGF is first-line for centre-involving diabetic macular oedema.
4. **(c)** Protocol T; at a baseline acuity of 20/50 or worse aflibercept was superior at one year, and this eye is 20/40 so the agents perform similarly, but anti-VEGF remains the class of choice.
5. **(b)** Both eyes have severe NPDR, which is reviewed every 3 to 4 months regardless of the oedema status.
6. **(b)** Systemic communication changes outcomes: UKPDS showed both glycaemic and blood pressure control independently reduce progression. Note also that you should **not** prescribe new glasses at an A1c of 9.4% because the refraction is unstable.

Case 5

SCENARIO

34-year-old man. Chief complaint: "my right eye is red and aching, and light is painful. This is the fourth time in three years."
Review of systems: low back stiffness worst in the morning, improving with exercise; fatigue.
Medications: ibuprofen as needed.
Findings: BVA OD `20/40`, OS `20/20`. OD: ciliary flush that does not blanch with phenylephrine `2.5%`, 3+ cells and 2+ flare with a fibrinous strand, fine non-pigmented keratic precipitates, no iris nodules, posterior synechiae at two clock hours. IOP OD `11 mmHg`, OS `15 mmHg`. Dilated fundus examination normal both eyes.

ITEMS

1. How is this uveitis best classified? (a) Chronic granulomatous anterior uveitis (b) Acute non-granulomatous anterior uveitis, recurrent (c) Intermediate uveitis (d) Panuveitis (e) Acute granulomatous panuveitis
2. Which laboratory test is most likely to be informative? (a) ACE and lysozyme (b) HLA-B27 (c) HLA-A29 (d) Aquaporin-4 antibody (e) ANCA
3. Which topical regimen is most appropriate? (a) Prednisolone acetate `1%` four times daily (b) Prednisolone acetate `1%` every 1 to 2 hours while awake plus homatropine `5%` three times daily (c) Loteprednol `0.5%` twice daily (d) Fluorometholone `0.1%` four times daily (e) Topical NSAID four times daily
4. Which drug should be avoided in this patient's IOP management if pressure later rises? (a) Timolol (b) Dorzolamide (c) Brimonidine (d) Latanoprost (e) Netarsudil
5. Which referral is most appropriate? (a) None (b) Rheumatology, for evaluation of axial spondyloarthritis (c) Neurology, for demyelination (d) Gastroenterology, for coeliac screening (e) Dermatology
6. What is the most appropriate review interval? (a) 24 hours (b) 3 to 7 days (c) 2 weeks (d) 1 month (e) 3 months

KEY

1. **(b)** Sudden, unilateral, recurrent, non-granulomatous (fine precipitates, no iris nodules), and anterior only.
2. **(b)** HLA-B27, with sacroiliac imaging, given the inflammatory back pain pattern.
3. **(b)** Intensive prednisolone acetate with a cycloplegic for this level of inflammation and for the fibrin and synechiae.
4. **(d)** Prostaglandin analogues are relatively contraindicated in active uveitis.
5. **(b)** Recurrent HLA-B27 uveitis with inflammatory back pain is frequently the presenting feature of ankylosing spondylitis.
6. **(b)** Three to seven days for this severity, with a slow taper thereafter and IOP at every visit.

Case 6

SCENARIO

9-year-old girl. Chief complaint (mother reporting): "her right eye turns in when she is tired, and she says letters move when she reads."
Review of systems: headaches after homework; falls asleep reading.
Medications: none.
Findings: BVA `20/20` each eye. Cycloplegic refraction `+0.50 DS` each eye. Cover test: orthophoria at distance, `14 Δ exophoria` at 40 cm. Near point of convergence break `14 cm`, recovery `19 cm`. Near base-out `6 / 10 / 4`. Near base-in `14 / 20 / 14`. NRA `+2.50`, PRA `-1.25`. Amplitude of accommodation `10 D` each eye. MEM `+1.25 D`. Stereoacuity 40 arcsec. Ocular health normal.

ITEMS

1. Calculate the AC/A ratio (PD `56 mm`) and give the diagnosis. (a) 8:1, convergence excess (b) 5.6:1, basic exophoria (c) 0:1, convergence insufficiency (d) 12:1, divergence excess (e) 4:1, normal
2. Which additional finding is present? (a) Accommodative excess (b) Accommodative insufficiency (c) Fusional vergence dysfunction (d) Divergence insufficiency (e) Vertical phoria
3. What is the most appropriate first-line management? (a) Base-in prism at near (b) Home pencil push-ups (c) Office-based vergence and accommodative therapy with home reinforcement (d) Plus add at near (e) Surgery
4. Which trial supports that choice? (a) PEDIG ATS (b) CITT (c) COMET (d) LAMP (e) CIGTS
5. What did that trial show about the option you rejected in item 3? (a) It was superior (b) It was equivalent (c) Home pencil push-ups performed no better than placebo therapy (d) Prism was superior (e) Surgery was equivalent
6. What is the appropriate follow-up plan? (a) Review in 12 months (b) Weekly or fortnightly office sessions with reassessment at about 12 weeks (c) Review in 6 months (d) No follow-up needed (e) Refer to a paediatric ophthalmologist

KEY

1. **(c)** AC/A = 5.6 + 0.4(-14 - 0) = 5.6 - 5.6 = **0:1**: a very low AC/A with near exophoria, a receded NPC, and reduced base-out at near. Convergence insufficiency.
2. **(b)** Reduced PRA (`-1.25`) and a high lag (`+1.25`) with a borderline amplitude indicate an accompanying accommodative insufficiency.
3. **(c)** Office-based vergence and accommodative therapy with home reinforcement.
4. **(b)** The Convergence Insufficiency Treatment Trial.
5. **(c)** Home pencil push-ups were no better than office-based placebo therapy.
6. **(b)** Weekly or fortnightly sessions with reassessment at around 12 weeks, following the CITT protocol.

Case 7

SCENARIO

71-year-old woman. Chief complaint: "straight lines look bent in my left eye and there's a grey patch in the middle, since about five days ago."
Review of systems: nil.
Medications: AREDS2 supplement, amlodipine.
Ocular history: intermediate AMD both eyes, diagnosed 3 years ago. Former smoker, 40 pack-years, quit 2 years ago.
Findings: BVA OD `20/30`, OS `20/80`. Amsler OS: central metamorphopsia and a small relative scotoma. OD: numerous large soft drusen and pigment clumping. OS: large soft drusen with a subretinal greyish-green lesion and a small subretinal haemorrhage at the fovea. OCT OS: subretinal fluid with subretinal hyperreflective material and a small pigment epithelial detachment.

ITEMS

1. What is the most likely diagnosis in the left eye? (a) Central serous chorioretinopathy (b) Neovascular age-related macular degeneration (c) Geographic atrophy (d) Adult vitelliform lesion (e) Macular telangiectasia
2. Which imaging modality would best delineate the lesion type if the fluorescein angiogram is equivocal? (a) B-scan ultrasonography (b) Fundus autofluorescence (c) Indocyanine green angiography (d) Corneal topography (e) Electroretinography
3. What is the most appropriate treatment? (a) Observation (b) Photodynamic therapy alone (c) Intravitreal anti-VEGF (d) Focal laser (e) Increase the AREDS2 dose
4. Regarding the AREDS2 supplement, which statement is most accurate? (a) It should be stopped now that neovascular disease has developed (b) It is indicated to reduce progression risk, including in the fellow eye (c) It prevents geographic atrophy (d) It contains β-carotene, which should be avoided in her (e) It restores lost vision
5. Which two elements of patient education are most important? (Select 2) (a) Daily monocular Amsler monitoring of the right eye (b) Reduce dietary zinc (c) Present within days for new distortion in the right eye (d) Avoid all sunlight (e) Discontinue amlodipine
6. What is her approximate annual risk of the fellow eye converting to neovascular disease? (a) Under 1% (b) 5 to 10% (c) 30% (d) 50% (e) Over 70%

KEY

1. **(b)** Subretinal fluid with subretinal hyperreflective material, haemorrhage, and a PED in an eye with pre-existing drusen.
2. **(c)** ICG angiography, particularly to identify polypoidal choroidal vasculopathy or occult type 1 disease.
3. **(c)** Intravitreal anti-VEGF, with monthly loading then a treat-and-extend regimen.
4. **(b)** It remains indicated for advanced disease in one eye, to reduce progression in the fellow eye. AREDS2 replaced β-carotene with lutein and zeaxanthin, and does not prevent geographic atrophy or restore vision.
5. **(a) and (c)** Daily monocular monitoring of the better eye and immediate presentation for new distortion.
6. **(b)** Roughly 5 to 10% per year in a high-risk fellow eye.

Case 8

SCENARIO

29-year-old woman. Chief complaint: "my left eye aches when I move it and my vision has faded over three days."
Review of systems: a period of numbness in the right leg last year that resolved over weeks; heat makes her vision worse.
Medications: none.
Findings: BVA OD `20/20`, OS `20/80`. Left RAPD. Ishihara OD 14/14, OS 5/14. IOP normal. Discs: OD normal, OS normal. Visual field OS: central depression. Pain on eye movement.

ITEMS

1. What is the most likely diagnosis? (a) Non-arteritic AION (b) Retrobulbar optic neuritis (c) Central serous chorioretinopathy (d) Compressive optic neuropathy (e) Non-organic visual loss
2. Which investigation is most appropriate? (a) CT of the orbits without contrast (b) MRI of the brain and orbits with gadolinium and fat suppression (c) Carotid duplex ultrasonography (d) Fluorescein angiography (e) Electroretinography
3. If that study shows three periventricular white matter lesions, what is her approximate 15-year risk of clinically definite multiple sclerosis? (a) 10% (b) 25% (c) 50% (d) 72% (e) 95%
4. Which treatment is contraindicated? (a) Intravenous methylprednisolone (b) Oral prednisone `1 mg/kg` as monotherapy (c) Observation (d) Referral to neurology (e) Vitamin D supplementation
5. What did the ONTT show about intravenous corticosteroid? (a) It improved final acuity (b) It accelerated recovery without changing final acuity (c) It prevented multiple sclerosis (d) It had no effect (e) It increased recurrence
6. Which feature, if present, would make you order aquaporin-4 and MOG antibodies? (Select 2) (a) Pain on eye movement (b) Bilateral simultaneous severe loss (c) A left RAPD (d) Failure to recover by 4 weeks (e) Reduced colour vision

KEY

1. **(b)** Young woman, pain on eye movement, RAPD, central field loss, dyschromatopsia, normal disc, and a prior likely demyelinating event with Uhthoff phenomenon.
2. **(b)** MRI of the brain and orbits with gadolinium and fat suppression.
3. **(d)** About 72% with two or more typical lesions.
4. **(b)** Oral prednisone `1 mg/kg` alone increased recurrence in the ONTT.
5. **(b)** It accelerated visual recovery but did not change final acuity.
6. **(b) and (d)** Bilateral severe simultaneous loss and failure to recover are atypical features suggesting NMOSD or MOG-associated disease.

Case 9

SCENARIO

67-year-old woman. Chief complaint: "sudden severe pain in my right eye tonight, I'm vomiting, and there are rainbow rings around the lights."
Review of systems: nausea and vomiting; frontal headache.
Medications: sertraline, recently started; diphenhydramine for sleep.
Ocular history: hyperopia, `+3.50 D` both eyes; no prior eye disease.
Findings: BVA OD `20/200`, OS `20/25`. OD: conjunctival injection, microcystic corneal oedema, a shallow anterior chamber (van Herick grade 1), a **mid-dilated vertically oval unreactive pupil**. IOP OD `64 mmHg`, OS `18 mmHg`. Gonioscopy OS: angle open to the trabecular meshwork only, with a convex iris.

ITEMS

1. What is the diagnosis? (a) Acute anterior uveitis (b) Acute primary angle closure (c) Neovascular glaucoma (d) Phacomorphic glaucoma (e) Posner-Schlossman syndrome
2. Which two medications in her list are relevant? (Select 2) (a) Sertraline (b) Diphenhydramine (c) Neither (d) Both act as anticholinergics or sympathomimetics that can precipitate closure (e) Only sertraline
3. Place these in the correct initial order: pilocarpine `2%`; topical β-blocker, α_2-agonist and carbonic anhydrase inhibitor; oral acetazolamide `500 mg`. (a) Pilocarpine, then topicals, then acetazolamide (b) Topicals and acetazolamide first, then pilocarpine once IOP falls (c) Acetazolamide only (d) Pilocarpine only (e) All simultaneously
4. Why is pilocarpine deferred? (a) It raises IOP (b) It causes corneal oedema (c) An ischaemic iris sphincter will not respond, and it can worsen pupillary block by moving the lens forward (d) It is contraindicated in hyperopia (e) It causes vomiting
5. What is the definitive treatment? (a) Trabeculectomy (b) Laser peripheral iridotomy in the affected eye (c) Laser peripheral iridotomy in both eyes (d) Cyclophotocoagulation (e) Topical prostaglandin
6. What is the approximate 5-year risk to the fellow eye without prophylaxis? (a) 5% (b) 15% (c) 50% (d) 80% (e) 95%
7. Which trial supports considering lens extraction as a primary treatment in this age group? (a) LiGHT (b) EAGLE (c) ZAP (d) AGIS (e) CNTGS

KEY

1. **(b)** Acute primary angle closure: hyperopia, shallow chamber, mid-dilated oval unreactive pupil, very high IOP, and a convex iris in the fellow eye.
2. **(b) and (d)** Diphenhydramine is anticholinergic; the general principle is that anticholinergics and sympathomimetics can precipitate closure in an anatomically predisposed eye.
3. **(b)** Aqueous suppressants and acetazolamide first; pilocarpine once the pressure falls below roughly 40 to 50 mmHg.
4. **(c)** As stated.
5. **(c)** Iridotomy in both eyes.
6. **(c)** Roughly 50%.
7. **(b)** EAGLE showed early lens extraction was more effective and more cost-effective than iridotomy plus medication in patients over 50.

Case 10

SCENARIO

4-year-old boy. Chief complaint (father reporting): "his left eye looks white in photographs and he seems to bump into things on that side."
Review of systems: otherwise well; normal growth.
Family history: none known.
Findings: fixes and follows with the right eye; will not maintain fixation with the left. Left esotropia of `20 Δ`. **Brückner test: the left fundus reflex is markedly brighter and whiter.** Left leukocoria on direct examination. Anterior segment clear both eyes, no cataract. IOP normal.

ITEMS

1. Which diagnosis must be excluded first? (a) Congenital cataract (b) Coats disease (c) Retinoblastoma (d) Persistent fetal vasculature (e) Toxocariasis
2. Which imaging finding would most support that diagnosis? (a) Absence of calcification (b) Intralesional calcification (c) High internal reflectivity (d) A retrolental membrane (e) Peripheral telangiectasia
3. What is the most appropriate action today? (a) Review in 3 months (b) Prescribe patching for amblyopia (c) Same-week referral to a paediatric ocular oncology service (d) Order an MRI and review in 2 weeks (e) Refer for strabismus surgery
4. Which two features would suggest heritable disease? (Select 2) (a) Unilateral single tumour (b) Bilateral involvement (c) Multifocal tumours (d) Age over 3 years (e) Absence of family history
5. Which gene and locus are responsible? (a) *PAX6*, 11p13 (b) *RB1*, 13q14 (c) *VHL*, 3p25 (d) *NF1*, 17q11 (e) *BEST1*, 11q12
6. Which second malignancy is a survivor of heritable disease at particular risk of? (a) Melanoma (b) Osteosarcoma (c) Colorectal carcinoma (d) Renal cell carcinoma (e) Thyroid carcinoma
7. What is the second most common presenting sign after leukocoria? (a) Nystagmus (b) Strabismus (c) Buphthalmos (d) Proptosis (e) Hyphaema

KEY

1. **(c)** Retinoblastoma.
2. **(b)** Intralesional calcification on CT or ultrasonography.
3. **(c)** Urgent, same-week referral to a paediatric ocular oncology service. Do not treat the amblyopia or the strabismus first.
4. **(b) and (c)** Bilateral and multifocal disease indicate a germline first hit.
5. **(b)** *RB1* at 13q14.
6. **(b)** Osteosarcoma (and pineoblastoma as trilateral retinoblastoma).
7. **(b)** Strabismus.

Case 11

SCENARIO

58-year-old woman. Chief complaint: routine examination; "my eyes water and burn all day and it's worse on the computer."
Review of systems: dry mouth, difficulty swallowing dry food; joint aches.
Medications: sertraline, loratadine, hydrochlorothiazide.
Findings: BVA `20/25` each eye, fluctuating. Tear break-up time `3 s` each eye. Tear osmolarity OD `318`, OS `324 mOsm/L`. Schirmer I OD `4 mm`, OS `3 mm`. Inferior and interpalpebral corneal punctate staining, grade 2. Lissamine green shows lid wiper epitheliopathy and conjunctival staining. Meibomian glands: capped orifices with turbid expression. Corneal sensation normal.

ITEMS

1. Which subtype best describes this dry eye? (a) Purely evaporative (b) Purely aqueous-deficient (c) Mixed, with both aqueous deficiency and meibomian gland dysfunction (d) Neuropathic (e) Neurotrophic
2. Which systemic diagnosis must be considered, and which two tests would you order? (Select 2) (a) Anti-SS-A (Ro) (b) HLA-B27 (c) Anti-SS-B (La) (d) ANCA (e) Aquaporin-4
3. Which three medications on her list contribute? (Select 3) (a) Sertraline (b) Loratadine (c) Hydrochlorothiazide (d) None (e) Only loratadine
4. Which management step should be *deferred* until inflammation is controlled? (a) Preservative-free lubricants (b) Warm compresses and lid hygiene (c) Punctal occlusion (d) Topical cyclosporine (e) Oral doxycycline
5. How long should you tell her topical cyclosporine `0.05%` may take to work? (a) 3 days (b) 2 weeks (c) 1 month (d) 3 to 6 months (e) 2 years
6. Which alternative prescription agent has a faster onset, and what is its characteristic side effect? (a) Loteprednol, cataract (b) Lifitegrast, dysgeusia (c) Doxycycline, photosensitivity (d) Azithromycin, QT prolongation (e) Varenicline nasal spray, sneezing

KEY

1. **(c)** Low Schirmer and raised osmolarity indicate aqueous deficiency; capped orifices with turbid meibum and a very short break-up time indicate MGD.
2. **(a) and (c)** Sjögren syndrome: anti-SS-A (Ro) and anti-SS-B (La), plus RF and ANA and a rheumatology referral.
3. **(a), (b), (c)** All three: antidepressant, antihistamine, and diuretic.
4. **(c)** Punctal occlusion, because occluding the puncta in an inflamed eye retains inflammatory tears.
5. **(d)** Three to six months.
6. **(b)** Lifitegrast `5%`, with dysgeusia (altered taste) as the characteristic side effect.

Case 12

SCENARIO

44-year-old man. Chief complaint: "everything looks smaller and slightly bent in my right eye for about three weeks."
Review of systems: snoring, unrefreshing sleep; work stress.
Medications: fluticasone nasal spray for allergic rhinitis, started 6 weeks ago.
Findings: BVA OD `20/30` improving to `20/25` with `+0.75 DS`; OS `20/20`. OD: a shallow serous elevation at the posterior pole with no haemorrhage and no drusen. OCT OD: subretinal fluid with an intact photoreceptor layer, a small pigment epithelial detachment, and a subfoveal choroidal thickness of `480 µm`. OS choroidal thickness `420 µm`.

ITEMS

1. What is the most likely diagnosis? (a) Neovascular AMD (b) Central serous chorioretinopathy (c) Vogt-Koyanagi-Harada disease (d) Optic pit maculopathy (e) Posterior scleritis
2. Which two features on the OCT support that diagnosis over neovascular AMD? (Select 2) (a) Subretinal fluid (b) Intact photoreceptor layer with no subretinal hyperreflective material (c) Thickened choroid (d) Pigment epithelial detachment (e) Absence of intraretinal fluid
3. Which two items in the history are most relevant? (Select 2) (a) Fluticasone nasal spray (b) Work stress (c) Age 44 (d) Male sex (e) Snoring
4. What is the most appropriate first management step? (a) Intravitreal anti-VEGF (b) Focal laser (c) Discuss stopping or substituting the corticosteroid with the prescriber, and observe (d) Oral prednisone (e) Immediate photodynamic therapy
5. If it persists at 4 months, which treatment has the best supporting evidence? (a) Full-fluence photodynamic therapy (b) Half-dose or half-fluence photodynamic therapy (c) Oral eplerenone (d) Intravitreal steroid (e) Observation indefinitely
6. Which systemic condition should you screen for? (a) Diabetes (b) Obstructive sleep apnoea (c) Hypertension (d) Thyroid disease (e) Coeliac disease

KEY

1. **(b)** Central serous chorioretinopathy: hyperopic shift, micropsia, smooth serous elevation, pachychoroid, no drusen, and a steroid exposure.
2. **(b) and (c)** An intact photoreceptor layer without subretinal hyperreflective material, and a thickened choroid.
3. **(a) and (b)** Corticosteroid exposure by any route, including intranasal, and psychological stress.
4. **(c)** Stop or substitute the steroid in discussion with the prescriber, and observe, since most acute cases resolve in 3 to 4 months.
5. **(b)** Half-dose or half-fluence photodynamic therapy has the best evidence (PLACE trial).
6. **(b)** Obstructive sleep apnoea, given the snoring and unrefreshing sleep.

Case 13

SCENARIO

52-year-old man. Chief complaint: "I got splashed with drain cleaner at work an hour ago and my left eye is agony."
Review of systems: nil.
Findings on arrival: severe blepharospasm; he is holding the eye closed. No eye protection was worn.

ITEMS

1. What is your first action? (a) Measure visual acuity (b) Instil anaesthetic and begin copious irrigation (c) Take a detailed history (d) Measure IOP (e) Photograph the eye
2. What is the irrigation endpoint? (a) One litre of saline (b) Ten minutes of irrigation (c) pH of 7.0 to 7.4 on recheck after a pause (d) When the patient is comfortable (e) When the cornea clears
3. Which single examination finding best predicts the visual outcome? (a) Corneal epithelial defect size (b) Extent of perilimbal (conjunctival) blanching (c) IOP (d) Anterior chamber reaction (e) Lid oedema
4. Which two oral agents are indicated, and why? (Select 2) (a) Doxycycline, for matrix metalloproteinase inhibition (b) Prednisone, for immunosuppression (c) Vitamin C, to support collagen synthesis (d) Acetazolamide, to lower IOP (e) Amoxicillin, for infection prophylaxis
5. Why is this agent's injury pattern more dangerous than that of a strong acid? (a) It coagulates protein and self-limits (b) It saponifies membrane lipids and penetrates progressively (c) It causes more pain (d) It is more likely to cause conjunctivitis (e) It has no difference
6. Which follow-up interval is appropriate for a moderate injury? (a) Daily (b) Weekly (c) In 1 month (d) As needed (e) In 3 months
7. Which occupational action should be documented? (a) Nothing is required (b) That eye protection was not worn, and counselling on ANSI Z87.1 protection (c) That the employer is at fault (d) A recommendation to change jobs (e) A referral to a lawyer

KEY

1. **(b)** Irrigation precedes everything, including acuity measurement.
2. **(c)** pH normalisation to 7.0 to 7.4, rechecked after a pause of 5 to 10 minutes, because pH can drift back up as retained material dissolves.
3. **(b)** Perilimbal blanching, which reflects limbal stem cell ischaemia.
4. **(a) and (c)** Doxycycline for MMP inhibition and vitamin C for collagen synthesis.
5. **(b)** Alkali saponifies lipids and penetrates deeply and progressively; acid coagulates protein, which limits penetration.
6. **(a)** Daily, for symblepharon lysis, IOP monitoring, and detection of melting.
7. **(b)** Document that protection was not worn and provide counselling on certified ANSI Z87.1 eyewear, both as clinical care and as a public health and medicolegal record.

Case 14

SCENARIO

48-year-old woman. Chief complaint: "my right upper eyelid has been drooping on and off for two months and I see double in the evenings."
Review of systems: tired chewing; occasional slurred speech late in the day; no pain.
Medications: levothyroxine.
Findings: BVA `20/20` each eye. Right MRD1 `1 mm`, left `4 mm`. **Pupils equal and briskly reactive with no RAPD.** Variable right hypotropia and limited elevation that changes during the examination. Cogan lid twitch present. Ptosis improves after 2 minutes with an ice pack. Occluding the right eye causes the left lid to fall slightly.

ITEMS

1. What is the most likely diagnosis? (a) Partial third nerve palsy (b) Horner syndrome (c) Ocular myasthenia gravis (d) Chronic progressive external ophthalmoplegia (e) Thyroid eye disease
2. Which single finding most reliably excludes a third nerve palsy? (a) Variability (b) A normal, briskly reactive pupil (c) The Cogan lid twitch (d) The ice pack response (e) Absence of pain
3. What explains the left lid falling when the right eye is occluded? (a) Sherrington law (b) Hering law of equal innervation (c) Bell phenomenon (d) Aberrant regeneration (e) Levator dehiscence
4. Which three investigations are most appropriate? (Select 3) (a) Acetylcholine receptor antibodies (b) MRI of the orbits (c) CT of the chest (d) Thyroid function tests (e) Carotid ultrasonography (f) Temporal artery biopsy
5. Why is the chest imaging indicated? (a) To exclude a Pancoast tumour (b) To exclude a thymoma (c) To exclude sarcoidosis (d) To exclude tuberculosis (e) To exclude lung metastasis
6. Which class of medication should be used cautiously in this patient? (a) Topical prostaglandins (b) Aminoglycosides, fluoroquinolones, macrolides, and β-blockers (c) Topical steroids (d) Antihistamines (e) Carbonic anhydrase inhibitors
7. Which management step is inappropriate now? (a) Prism for stable diplopia (b) Referral to neurology (c) Strabismus surgery (d) Occlusion for symptomatic relief (e) Patient education about generalisation risk

KEY

1. **(c)** Variable fatigable ptosis and motility with Cogan lid twitch, a positive ice pack test, and enhancement of ptosis.
2. **(b)** Myasthenia *always* spares the pupil, and a normal pupil with variability is the discriminator.
3. **(b)** Hering law: occluding the ptotic fixating eye removes the excess bilateral innervation, so the fellow lid falls.
4. **(a), (c), (d)** Acetylcholine receptor (and MuSK) antibodies, chest CT, and thyroid function.
5. **(b)** Thymoma, present in roughly 10 to 15% of myasthenia patients.
6. **(b)** These classes can worsen neuromuscular transmission.
7. **(c)** Strabismus surgery is inappropriate in a variable, unstable, myasthenic deviation.

Case 15

SCENARIO

69-year-old man. Chief complaint: routine glaucoma review.
Review of systems: nil.
Medications: latanoprost nightly both eyes for 4 years; tamsulosin; metformin.
Findings: BVA OD `20/30`, OS `20/40` (nuclear sclerosis both eyes). IOP OD `17`, OS `18 mmHg` at 10:40; CCT OD `505`, OS `498 µm`. Discs: vertical cup-to-disc `0.8` OD and `0.85` OS with inferior rim thinning and a **fresh inferotemporal disc haemorrhage OS**. Gonioscopy: open to the ciliary body band, with fine fibrillar material on the pupil margin and a Sampaolesi line both eyes. Visual fields (two reliable prior baselines available): OS shows a new superior arcuate deepening at three adjacent points, confirmed on repeat; MD OS has changed from `-4.2` to `-6.8 dB` over 2 years.

ITEMS

1. What is the most complete diagnosis? (a) Primary open-angle glaucoma (b) Pseudoexfoliation glaucoma, progressing in the left eye (c) Normal-tension glaucoma (d) Angle closure glaucoma (e) Pigmentary glaucoma

2. Which three findings establish progression? (Select 3) (a) A fresh disc haemorrhage (b) A repeatable new arcuate deepening (c) An MD change of `2.6 dB` over 2 years (d) IOP of 18 mmHg (e) Nuclear sclerosis (f) A CCT of 498 µm
3. Which single management step is most appropriate now? (a) Continue current therapy and review in 6 months (b) Lower the target IOP and escalate treatment (c) Reassure, since the IOP is under 21 mmHg (d) Repeat the field in 12 months (e) Stop latanoprost
4. Which treatment is likely to be particularly effective in this specific condition? (a) Oral acetazolamide (b) Selective laser trabeculoplasty (c) Pilocarpine (d) Cyclophotocoagulation (e) Peripheral iridotomy
5. Which trial quantified the benefit of further IOP reduction? (a) OHTS (b) EMGT, showing about a 10% reduction in progression risk per mmHg (c) CIGTS (d) LiGHT (e) ZAP
6. Which three surgical cautions must you communicate if he proceeds to cataract surgery? (Select 3) (a) Poor pupil dilation (b) Zonular weakness with capsule rupture risk (c) Late in-the-bag IOL subluxation (d) Higher rate of endophthalmitis (e) Corneal decompensation
7. Which additional medication history detail must be conveyed to the cataract surgeon, and why? (a) Metformin, for contrast risk (b) Tamsulosin, for intraoperative floppy iris syndrome (c) Latanoprost, for macular oedema (d) None (e) All three equally

KEY

1. **(b)** Fibrillar material on the pupil margin with a Sampaolesi line makes this pseudoexfoliation glaucoma, and the field and disc findings establish progression in the left eye.
2. **(a), (b), (c)** A disc haemorrhage is a strong predictor of progression; the repeatable field deepening and the rate of MD change confirm it.
3. **(b)** Lower the target and escalate. An IOP under 21 mmHg is irrelevant when the eye is demonstrably progressing.
4. **(b)** Selective laser trabeculoplasty is particularly effective in heavily pigmented angles, including pseudoexfoliation.
5. **(b)** EMGT, showing roughly a 10% reduction in progression risk per mmHg of IOP reduction.
6. **(a), (b), (c)** Poor dilation, zonular weakness with capsule rupture and dialysis risk, and late in-the-bag IOL subluxation.
7. **(b)** Tamsulosin causes intraoperative floppy iris syndrome, and the risk persists for years after stopping, so the surgeon must be told rather than the drug simply discontinued.

HOW TO USE YOUR ERROR LOG

1. Tabulate every miss by **condition area** and by **item type**. Two different weaknesses need two different remedies.
2. Misses concentrated in one **condition area** mean you have a knowledge gap: go back to that chapter and its retrieval blocks.
3. Misses concentrated in **diagnosis** items mean you are not drilling look-alike pairs. Use the differentiation tables.
4. Misses concentrated in **treatment** items mean you know the class but not the regimen. Use the formulary in Chapter 14.
5. Misses concentrated in **related to treatment** items mean you do not know follow-up intervals and prognosis. Use Chapter 15.
6. Misses on **multiple-response** items usually mean you did not count, or you included a plausible but unproven option. Re-read each selection and ask "is this definitely true for *this* patient?"
7. Misses caused by **time pressure** mean you are reading scenarios too slowly. Practise extracting demographics, complaint, and the one abnormal image finding in under 60 seconds.
8. If you missed an item because you did not know a **trial name**, add it to a single-page trial sheet and review it daily. Trials are named explicitly in PAM stems and answer options.